



ClearPath OS 2200 Software

Executive Control Language (ECL) and FURPUR Reference Manual

Level 33R5

ClearPath OS 2200 Release 22.0

June 2026

7830 7949-031

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Contents

Section 1. Introduction

Section 2. Using Control Statements

2.1.	Control Statement Format	2-1
2.2.	Transparent Control Statements	2-5
2.2.1.	Transparent ECL Statements (@@)	2-5
2.2.2.	Demand Symbiont Control Statements	2-6
2.2.3.	Guidelines	2-6

Section 3. Using Files and Elements

3.1.	Classification of Files	3-2
3.1.1.	Storage Classification—Mass Storage Versus Tape Files	3-2
3.1.2.	Life Classification—Cataloged Versus Temporary Files	3-3
3.2.	Additional Mass Storage and Tape File Topics	3-4
3.2.1.	Mass Storage File Topics	3-4
3.2.2.	Tape File Topics	3-5
3.2.3.	Virtual Tape Handler Files	3-6
3.2.4.	Omnibus File Format	3-7
3.2.5.	File Sharing	3-7
3.3.	Naming Files and Elements	3-7
3.4.	F-Cycles	3-12
3.4.1.	Numbering	3-12
3.4.2.	Referencing	3-12
3.4.3.	Deleting	3-13
3.4.4.	Internal Name	3-14
3.4.5.	Temporary Files	3-14
3.4.6.	Examples	3-14
3.5.	S-Cycles	3-15
3.5.1.	Numbering and the Element Cycle Word	3-16
3.5.2.	Referencing	3-18
3.6.	System-Created Files	3-19
3.6.1.	TPF\$	3-19
3.6.2.	READ\$	3-19
3.6.3.	PRINT\$	3-20
3.6.4.	DIAG\$	3-20
3.7.	Print Files	3-20
3.8.	File Formats	3-22
3.8.1.	System Data Format	3-22
3.8.2.	Track-Binary Format	3-23

3.8.3.	Element File Format	3-24
3.8.4.	Program File Format Structure.....	3-24
3.8.5.	Types of Program File Elements	3-25
3.8.6.	Table of Contents for a Program File	3-26
3.8.7.	Large Element Program Files	3-28
3.8.8.	Transfers Between Files of Different Formats.....	3-28

Section 4. File Security

4.1.	MAC and DAC.....	4-2
4.2.	Protecting Unowned Files.....	4-2
4.3.	Protecting Owned Files	4-3
4.3.1.	Mandatory Access Control	4-3
4.3.2.	Discretionary Access Control.....	4-5
4.4.	Changing Your File's Protection.....	4-10
4.4.1.	Modifying Mandatory Attributes	4-10
4.4.2.	Changing Access Lists	4-10
4.4.3.	Changing or Clearing the Read-Only or Write- Only Mechanism	4-11
4.4.4.	Changing Other File Attributes.....	4-11

Section 5. Getting On and Off the System

5.1.	Opening a Session (\$\$OPEN)	5-2
5.2.	Accessing the System	5-3
5.2.1.	Demand Sign-On Modes	5-3
5.2.2.	Signing On to the System in Demand.....	5-5
5.2.3.	Sign-On Security Measures	5-6
5.2.4.	Providing Compartment Information in Demand	5-7
5.2.5.	Providing an Account Number in Demand.....	5-9
5.2.6.	Providing a Project-ID in Demand	5-10
5.2.7.	System Response	5-11
5.2.8.	Authentication and Session Initiation Subsystem (ASIS)	5-11
5.2.9.	Signing On to TIP	5-13
5.3.	Changing Passwords in Demand.....	5-14
5.3.1.	Changing a Machine-Generated Password.....	5-14
5.3.2.	Changing a User-Specified Password	5-15
5.3.3.	Guidelines for Choosing a Password	5-15
5.3.4.	Changing Your Password at Sign-On	5-16
5.3.5.	Changing Your Password Using @@PASSWORD	5-16
5.4.	Run Termination.....	5-17
5.4.1.	Termination Actions.....	5-17
5.4.2.	Termination Task Scheduling.....	5-18
5.5.	Closing a Session (\$\$CLOSE).....	5-19

Section 6. Using Processors

6.1.	Calling a Processor (@processor).....	6-1
6.2.	Unformatted Processor Call Statements	6-3

6.3.	Basic Mode Programs	6-5
6.3.1.	Step 1. Writing the Program	6-5
6.3.2.	Step 2. Compiling the Program	6-5
6.3.3.	Step 3. Collecting the Program (Collection)	6-6
6.3.4.	Step 4. Executing the Program	6-6
6.4.	Extended Mode Programs	6-7
6.4.1.	Step 1. Writing the Program	6-7
6.4.2.	Step 2. Compiling the Program	6-7
6.4.3.	Step 3. Static Linking (Optional)	6-8
6.4.4.	Step 4. Executing the Program	6-9
6.5.	Calling a Compiler	6-10
6.5.1.	Calling a Basic Mode Compiler	6-10
6.5.2.	Calling an Extended Mode Compiler	6-13
6.6.	Correcting Symbolic Elements	6-18
6.6.1.	Correction Image Format Rules	6-19
6.6.2.	Deleting Lines	6-20
6.6.3.	Replacing Old Lines with New Lines	6-20
6.6.4.	Inserting New Lines	6-21
6.6.5.	Editing Lines	6-22
6.6.6.	EDIT 1100	6-24

Section 7. Control Statements

7.1.	@ADD—Add Partial Canned Runstreams and Data	7-5
7.2.	@ASG and @@ASG—Assign Files and Peripheral Devices	7-8
7.2.1.	Assigning Mass Storage Files	7-9
7.2.2.	Security Access Validation	7-19
7.2.3.	Assigning Files on Shared Devices	7-20
7.2.4.	Changing Assignment Fields for Mass Storage Files	7-22
7.2.5.	Assigning Tape Files to Your Run	7-24
7.2.6.	Assigning VTH Files to Your Run	7-51
7.2.7.	Assigning DVD Tape	7-52
7.2.8.	Assigning Absolute Devices	7-53
7.3.	@BRKPT and @@BRKPT—Breakpoint Primary Print Files	7-55
7.3.1.	Breakpointing Alternate Symbiont Files	7-58
7.3.2.	Using @BRKPT and @SYM Together	7-59
7.4.	@CAT and @@CAT—Create Permanent Files	7-60
7.5.	Cataloging Mass Storage Files	7-60
7.5.1.	Cataloging Tape Files	7-66
7.6.	@CHG—Change File or Element Attributes	7-80
7.6.1.	Changing Keys, Modes, and Names of Cataloged Files	7-81
7.6.2.	@CHG and File Security	7-85
7.6.3.	@CHG Examples for File Changes	7-86
7.6.4.	Changing Program File Element and Version Names	7-88
7.6.5.	@CHG Examples for Element Changes	7-89

7.7.	@CKPT and @@CKPT—Save a Complete or Partial Copy of a Run	7-93
7.8.	@CLOSE—Close Tape File.....	7-95
7.9.	@@CM—Send a Message to the Operator Console	7-96
7.10.	@@CONS—Enter Console Keyins	7-96
7.10.1.	Operating Considerations.....	7-97
7.10.2.	Console Modes	7-98
7.10.3.	Console Mode Input/Output Rules	7-100
7.10.4.	Console Mode Commands.....	7-101
7.11.	@@CONT—Continue to Send Output	7-108
7.12.	@COPIN—Copy from Tape to Program Files	7-108
7.13.	@COPOUT—Copy Program Files to Tape	7-111
7.14.	@COPY—Copy Files or Elements.....	7-114
7.14.1.	File-based Copy Options.....	7-116
7.14.2.	Element-based Copy Options	7-123
7.14.3.	Version Masking, Multiple Element Selection, and the V and X Options	7-125
7.14.4.	@COPY Examples	7-128
7.15.	@@CQUE—Queue Input in a Main Storage Buffer	7-131
7.16.	@CYCLE—Alter Cycle Retention Limit.....	7-132
7.16.1.	Set File Cycle Range	7-132
7.16.2.	Set Element Cycle Limit	7-135
7.17.	@@DCT—Change DCT Terminal Characteristics.....	7-136
7.18.	@DELETE—Delete Files and Elements	7-137
7.18.1.	Deleting Files	7-138
7.18.2.	Deleting Elements	7-139
7.18.3.	Undeleting Elements.....	7-141
7.19.	@ENABLE—Enable Disabled Files	7-144
7.20.	@END—End of Input for @DATA and @ELT	7-145
7.21.	@@END—Terminate Special Modes	7-145
7.22.	@ENDF—Terminate the @FILE Card File	7-146
7.23.	@ENDX—Terminate CLIST Mode.....	7-147
7.24.	@EOF—Indicate the End of Input	7-147
7.25.	@ERS—Empty (Erase) a File.....	7-148
7.26.	@@ESC—Pass Unaltered Input to a Program or Unaltered Output to a Terminal	7-150
7.27.	@FILE—Create a Mass Storage Card File.....	7-150
7.27.1.	Security Considerations	7-151
7.27.2.	Terminating @FILE	7-152
7.27.3.	Creating a Tape File from Card Images	7-152
7.28.	@FIN—Identify the End of a Run	7-153
7.29.	@FIND—Position Tape Within Element Files	7-154
7.30.	@FREE and @@FREE—Release Files and Peripheral Devices.....	7-155
7.31.	@@FUL—Enter Input a Full Screen at a Time.....	7-158
7.32.	@HDG and @@HDG—Provide a Heading for Printer Output.....	7-159
7.33.	@@HOLD—Hold Terminal Messages and Print Output.....	7-164
7.34.	@@INQ—Queue Input in a Mass Storage Buffer	7-165
7.35.	@@INS—Change the Insert Point on the Display.....	7-166

7.36.	@JUMP—Branch from Within the Runstream to Another Point.....	7-166
7.37.	@LOG and @@LOG—Place Information in the System Log.....	7-167
7.38.	@MARK—Mark an EOF on Tape	7-167
7.39.	@MODE and @@MODE—Change the Mode Settings for a Tape File	7-169
7.40.	@MOVE—Position Tape Files.....	7-176
7.41.	@MSG and @@MSG—Send a Message to Various Console Types.....	7-178
7.42.	@@NOPR—Terminate a Local Printer	7-181
7.43.	@PACK—Remove Deleted Elements	7-181
7.44.	@PASSWD and @@PASSWD—Use and Change Passwords.....	7-184
7.45.	@PCH—Punch Program File Elements.....	7-188
7.46.	@@PMOD—Control Output Sent to a Printer Connected to Your Terminal.....	7-190
7.47.	@PREP—Create Entry-Point Tables.....	7-190
7.48.	@@PRNT—Start a Local Printer.....	7-191
7.49.	@PRT—List Files, Elements, and Master File Directory	7-192
7.49.1.	@PRT,F Output Format.....	7-197
7.49.2.	Listing the Master File Directory	7-201
7.49.3.	Listing the Table of Contents	7-202
7.49.4.	Examples.....	7-211
7.50.	@QUAL and @@QUAL—Specify a Qualifier or Directory for File Names.....	7-215
7.50.1.	Initial and Default Values	7-217
7.50.2.	Single MFD Search	7-217
7.50.3.	Double MFD Search	7-218
7.51.	@REWIND—Rewind Tape Files.....	7-220
7.52.	@@RLD—Cause Screen Images to Scroll Down	7-220
7.53.	@@RLU—Cause Screen Images to Scroll Up	7-221
7.54.	@@RQUE—Stop an Output File and Place It in a Queue	7-221
7.55.	@RSTRT—Restart a Run.....	7-222
7.56.	@RUN—Identify a Run to the Exec	7-224
7.56.1.	Batch Run.....	7-225
7.56.2.	Demand Run	7-236
7.56.3.	How Runs and Files Work Together	7-240
7.57.	@@SEND—Receive a Print File Queued to Your User ID.....	7-241
7.58.	@@SETC—Store a Value in a Condition Word.....	7-242
7.59.	@@SKIP—Skip Lines of Output	7-244
7.60.	@@SOLI—Solicit for User-ID/Password.....	7-244
7.61.	@START and @@START—Start Canned Runstreams.....	7-245
7.62.	@SYM and @@SYM—Queue Print Files and @SYM CN Control Statement.....	7-255
7.63.	@@TERM—Sign Off from the System.....	7-260
7.64.	@TEST—Test a Condition Word to Control Execution.....	7-260
7.65.	@@TM—Send a Message to Another User	7-262
7.66.	@@TOUT—Change Your Timeout Period	7-263

7.67.	@@TRNS—Ignore Escape/EOT Characters	7-264
7.68.	@TTY—Change TTY Terminal Characteristics	7-264
7.69.	@USE and @@USE—Reference a File by Another Name.....	7-264
7.70.	@@X—Halt Processors	7-267
7.71.	@XQT—Execute Absolute Elements or Object Modules	7-270

Appendix A. ECL and Terminal Sign-on Messages

Appendix B. FURPUR Messages

B.1.	Functional and Error Messages	B-1
B.1.1.	Messages Prefixed with I/O Error Codes	B-1
B.1.2.	Messages Without Prefixed I/O Error Codes	B-2
B.2.	Quarter-Inch Tape Subsystem Operations.....	B-3
B.3.	Error Messages Associated with Element Subtype Selection	B-4
B.4.	Error Messages Associated with the Prep of Object Modules	B-4
B.5.	Error Messages Returned by PCFP	B-5
B.6.	Error Messages Associated with the @CHG Statement	B-11
B.7.	Error Messages Associated with the @CYCLE Statement	B-16
B.8.	Error Messages Associated with the @PRT Statement	B-18

Appendix C. Facility Request Status Codes

C.1.	Facility Inventory Informational Messages.....	C-2
C.2.	Facility Inventory Warning Messages.....	C-3
C.3.	Facility Inventory Error Messages.....	C-5
C.4.	PCN Fixed Messages.....	C-18
C.5.	User Profile Error Messages	C-19
C.6.	Facility Inventory Status Bits	C-20

Appendix D. Equipment Mnemonics

D.1.	Mass Storage Assign Mnemonics	D-1
D.2.	Tape Assign Mnemonics	D-5

Appendix E. Job and Summary Accounting Report (Tail Sheet)

Index		1
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Figures

3-1.	Multivolume Files and Multifile Volumes	3-2
3-2.	Normal Batch Output.....	3-21
3-3.	Normal Demand Output.....	3-21
3-4.	Breakpoint Output (Batch or Demand)	3-22
3-5.	Basic Formats Used to Transfer Data Between Files.....	3-29
3-6.	Statements for Copying Tape Files	3-31
3-7.	Data Paths, Data Formats, and FURPUR Commands	3-32
3-8.	FURPUR Data Transfer Command Options	3-33
6-1.	Generating and Executing a Basic Mode Program	6-6
6-2.	Generating and Executing an Extended Mode Program	6-9
7-1.	Assigning Files.....	7-9
7-2.	Positioning Tape Files.....	7-177
7-3.	Sending Messages.....	7-180
7-4.	Run and File Concepts.....	7-240
7-5.	Sample Printout Label	7-259

Figures

Tables

3-1.	End-of-Reel Configuration for Multireel Tapes.....	3-6
3-2.	Absolute Cycle Renumbering.....	3-17
3-3.	Referencing Element Cycles	3-19
3-4.	Program File Capacity	3-24
3-5.	Maximum Number of Items per Table	3-27
4-1.	File Protection.....	4-2
4-2.	Access List Type.....	4-5
5-1.	Terminal Sign-On and Session Establishment Procedures.....	5-2
5-2.	Compartment Sets	5-7
6-1.	Processor Calls with Symbolic Input, Processor Output, and Symbolic Fields.....	6-15
6-2.	Processor Calls with Symbolic Input and Symbolic Output Fields.....	6-17
7-1.	Statement Reference List	7-1
7-2.	Console Mode Commands.....	7-102
7-3.	FREE Options for Releasing Files and Devices	7-157
7-4.	@RUN,/B Following a Demand Sign-on.....	7-187
7-5.	Security Validations for @START Statements.....	7-250
7-6.	Actions Taken with Respect to an II Activity	7-269
	@XQT—Execute Absolute Elements or Object Modules.....	7-274
B-1.	ER SUMOD\$ Errors	B-14
C-1.	Facility Inventory Message Ranges.....	C-1
C-2.	Facility Inventory Status Bits	C-20
D-1.	Mass Storage Assign Mnemonics	D-1
D-2.	Default Mass Storage Assign Mnemonic Selection Sequences	D-3
D-3.	Tape Assign Mnemonics	D-5
D-4.	Default Tape Assign Mnemonic Selection Sequences	D-7

Section 1

Introduction

This manual provides a complete reference for all Executive Control Language (ECL) and File Utility Routines/Program File Utility Routines (FURPUR) statements and describes the purpose, format, options, and individual fields of each statement. Examples appear throughout the manual to help you understand the various uses of these statements.

The primary audience for this manual is any OS 2200 user. This includes system programmers, site administrators, and general users.

Before you begin using ECL, you should have some familiarity with the OS 2200 environment.

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<https://www.support.unisys.com/common/epa/DocumentationLibraries.aspx?PLA=OS2&NAV=OS2>

Notation Conventions

This manual uses the following conventions to describe messages and statements:

- Commas, periods, and parentheses must be keyed in as shown.
- Letters, words, and symbols in italics represent variable information that changes, depending on the circumstances. Letters, words, and symbols that are underlined represent user input in computer dialogue.
- Uppercase letters indicate a keyword or a required word. LEVEL in the following example is a required word:

```
LEVEL,product_level ;
```

Unless otherwise noted, uppercase letters can be coded in uppercase or lowercase.

- Lowercase italic text indicates a variable or information that the user must supply. *product_date* and *product_time* are user-supplied values in the following example:

```
TIME,product_date,product_time
```

- Ellipses (...) indicate a variable number of entries.
- Brackets ([]) surround an optional field, expression, or value. The following example uses brackets:

```
LOAD_FILE_PACK_ID type,pack[,pack,...]
```
- Braces indicate you have a choice between two or more values. A vertical bar separates the choices. The following example uses braces and vertical bars:

```
LOAD_FILE_SECURITY { NONE | KEYS | PRIVATE }
```
- The ► is the start-of-entry (SOE) character. The Exec places an SOE character on the screen to let you know that the system is ready for your input.
- Changes made since the last version of the manual are shown with yellow highlighting in the online version (and appears as shading in the printed version).

Section 2

Using Control Statements

Control statements are the means by which you make requests to the operating system to perform services. Control statements include Executive Control Language (ECL) control statements, FURPUR control statements, processor call control statements, and unformatted processor call control statements.

ECL control statements let you initiate and control demand and batch runs, create and use mass storage and tape files, print information, generate and execute programs, send messages to the system console and other terminals, and control certain terminal characteristics. ECL control statements are the interface between you and the Exec system software portion of the OS 2200 operating system.

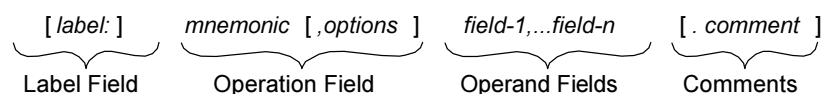
FURPUR control statements allow you to control and move data within OS 2200 files and elements.

Processor call statements are calls to compilers and system programs that perform various functions. See Section 6 for the processor call formats.

2.1. Control Statement Format

All control statements consist of a recognition character called the masterspace (@) and four major fields, as follows.

Format



Start all control statements in column 1 with the masterspace (@), which indicates that what follows is a control statement and not data. You can use spaces after the masterspace.

Label Field

The label is not necessary, but you can use it to name a control statement. The label is 1 to 6 characters from the set A through Z and 0 through 9. The first character must be alphabetic. If you specify a label, follow it with a colon (:). You can use spaces after the colon.

A label statement is any control statement containing a label but no operation or operand fields. A label statement consists of the masterspace (@), the label, the colon (:), and, optionally, a comment. Here is an example:

```
@TAG:
@MARK:
@LAST:XQT PROGX
```

In this example, the @XQT statement can be referenced by any of the labels TAG, MARK, or LAST, which are attached to the @XQT statement by label statements. As each label statement is encountered, no operation is performed and runstream processing continues until a control statement with an operation is found.

If the same label appears more than once within a run, the first forward occurrence from the point of reference is taken as the proper label.

You can specify a label for almost all control statements. The one exception to this is the @EOF statement, which cannot contain a label. The label provides the means to skip portions of the runstream and pass control to the statement having a particular label. However, you cannot pass control with labels on the @ENDF and @FILE statements because these statements are not entered in the runstream. Although control statements have only one label, you can attach additional labels to a control statement by using label statements.

Operation Field

Unless you plan to use the control statement only as a label statement, you must always specify the mnemonic name associated with the statement in the operation field. The mnemonic specifies what component of the operating system is to be called and what function is to be performed. The mnemonic on all ECL control statements is 1 to 6 characters from the set A through Z. Terminate the mnemonic with a space or (if you want to specify options) with a comma.

The options are alphabetic characters, which you can specify in any order. The option field must start with a comma (,) to separate it from the mnemonic.

You must separate the operation field from the operand field with a space or a series of spaces.

Operand Field

Use *field-1,...,field-n* to specify information associated with the operation field. Separate fields with commas. You can use a space or spaces after the comma.

The content and number of fields and whether they are required or optional depend on the statement. Some examples are

- Names of files or elements within files
- File specifications
- Messages
- Run-IDs
- Account numbers

- User-IDs

Fields, in turn, can contain subfields that are separated with a slash (/). You can use a space or spaces after a slash. For the most part, subfields are optional within a field.

The relative position of a field or subfield determines its meaning.

If you want to include a field that is preceded by optional fields you are not using, you must provide extra commas (,) to indicate that the preceding fields are blank. You do not need to include the extra commas if they occur after the last field you are including in the statement. Optional fields left blank assume system default values.

If you want to include a subfield that is preceded by optional subfields you are not using, you must provide extra slashes (/) to indicate that the preceding subfields are blank. You need not include the extra slashes if they occur after the last subfield. Any optional subfields not specified assume system default values.

Example

```
@CAT,P MAILSAVE,F///500
```

In this case, the extra slashes indicate the position of subfield 500; the subfields between F and 500 are not used and assume system default values for file allocation.

Comments Field

You can add comments to control statements. Comments follow the operand field. Start your comment with a space-period-space (.) and then enter your comment.

The comment can contain spaces and any character except the semicolon (;), which is the continuation character. The comments field is optional.

Examples

```
@RUN GHT,847120,ACCT-REC . Place comments after period
```

@RUN is the mnemonic; there are no options; and there are three operand fields: GHT (a run-ID), 847120 (an account number), and ACCT-REC (a project-ID).

```
@ASG,A ACCT-MASTER.
```

@ASG is the mnemonic; A is an option; and there is one operand field: ACCT-MASTER (a file name).

```
@COPY MO-TRANS.,AUG87-TRANS.
```

@COPY is the mnemonic; there are no options; and there are two operand fields: MO-TRANS and AUG87-TRANS (both are file names).

```
@FIN
```

There are no options or operand fields, just the mnemonic itself (@FIN).

Continuing Control Statements

The maximum length of a control statement is 84 characters per line for Fielddata and 80 characters per line for ASCII. In batch mode, statements exceeding this length cause your run to terminate. In demand mode, statements exceeding this length are truncated when input. A control statement requiring more than one line can be continued.

You can continue a control statement to another line by entering a semicolon (;) at the end of the line to be continued. You can split a statement at any point in the statement where spaces are allowed or following a delimiter.

You can start the next line in any column, with one exception: Do not place the masterspace (@) in column 1 of the continuation line.

You cannot continue the following control statements:

```
@END
@ENDF
@ENDX
@EOF
@FILE
@FIN
@RUN
```

A file name or element name is legal for certain control statement fields. But a file name cannot be continued at any of the statement subfields. If the period that distinguishes the file name is not encountered before the continuation character, an element name is assumed.

The following @ASG statements illustrate the possible coding combinations that result from applying general default rules. Since control statements are freeform (fields are not required to be in specific columns), you can align the label, operation, and operand fields to improve readability.

```
@ASG FILENAME
@ASG,A FILENAME,,REEL1
@ASG FILENAME,26N///4
@ASG,T FILENAME,26N//,REEL1;REEL/;
  REEL2/REEL3
@ASG,A FILENAME,///6
@TAG1:ASG FILENAME,,REEL1
```

2.2. Transparent Control Statements

Normally, control statements and data images in a runstream are processed sequentially and are processed only when directed to by the Exec, FURPUR, or an executing program. However, a special set of control statements (transparent control statements) are processed immediately after you enter them at a demand terminal.

Transparent control statements can be entered at any time. Since they are processed immediately, you cannot store them in a file as part of a runstream.

There are two types of transparent statements:

- Transparent ECL control statements
- Demand symbiont control statements

2.2.1. Transparent ECL Statements (@@)

Transparent ECL statements are double-masterspace (@@) versions of selected ECL statements. Transparent ECL statements follow the same general rules for syntax and format as their single-masterspace counterparts (with the exception of @@PASSWD).

These are the transparent ECL statements:

```
@@ASG
@@BRKPT
@@CAT
@@FREE
@@HDG
@@LOG
@@MODE
@@MSG
@@PASSWD
@@QUAL
@@START
@@SYM
@@USE
```

2.2.2. Demand Symbiont Control Statements

Another type of transparent control statement is the demand symbiont control statement. It is a special statement that controls the way the Exec handles your terminal. Unlike transparent ECL control statements, demand symbiont control statements have no corresponding single-masterspace (@) statements. Another difference is that you do not see a system response for some symbiont control statements.

The demand symbiont control statements are

@@CM	@@INQ	@@SKIP
@@CONS	@@INS	@@TERM
@@CONT	@@NOPR	@@TM
@@CQUE	@@PMOD	@@TOUT
@@DCT	@@PRNT	@@TTY
@@END	@@RLD	@@X
@@ESC	@@RLU	
@@FUL	@@RQUE	
@@HOLD	@@SEND	

2.2.3. Guidelines

Follow these guidelines for using transparent control statements:

- Use transparent control statements only in a demand run.
- Wait for the transparent control statement to finish processing before you enter another one from your terminal.

Transparent control statements can be entered at any time. To enter a transparent control statement, do the following:

1. Enter the transparent control statement following the start-of-entry (SOE) character (►) that the system provides. If the system is sending output to your terminal, press the MSG WAIT (break or interrupt) key first.
2. Press the transmit key again to get back an SOE character.

Section 3

Using Files and Elements

A file is an organized collection of data, treated as a unit, which is used to store programs or data. An element is a discreet entity within a special file called a program file.

This section includes the following information on files and elements:

- Classification of files
 - Mass storage and tape
 - Cataloged and temporary
- File size
- Naming files and elements
- File cycles (F-cycles)
- System-created files
- Print files
- File formats
- Types of program file elements
- The table of contents for a program file
- Multireel tape files

3.1. Classification of Files

Files are either on mass storage or on tape.

3.1.1. Storage Classification—Mass Storage Versus Tape Files

Mass storage files are files stored on magnetic disk. They can be read and written sequentially or randomly. Magnetic tape files can be only read and written sequentially.

The basic concepts that you must understand to create and manipulate tapes and tape files are as follows:

- A volume is the same as a reel.
- A multivolume file is a single tape file so large that it takes more than one reel to store it. A multifile volume is a single tape reel that has more than one tape file on it (see Figure 3–1).

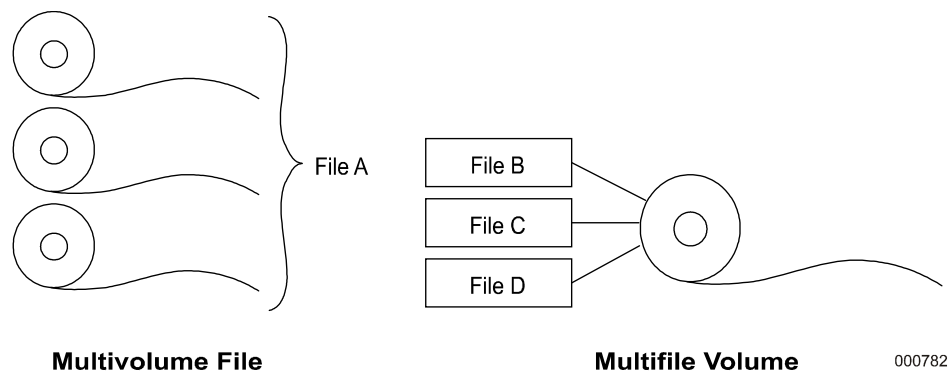


Figure 3–1. Multivolume Files and Multifile Volumes

- The load point is the position near the beginning of a tape where the reading or writing of data begins.
- When more than one file is on a tape, the files are separated by a hardware mark called an end-of-file mark or EOF mark.
- After the last file on a tape, two consecutive EOF marks are used to indicate that there is nothing else on the tape.

3.1.2. Life Classification—Cataloged Versus Temporary Files

Temporary files exist only for the duration of a run. When the run terminates, or if you release the file from your run, the file ceases to exist.

Cataloged files are permanent files. They exist whether or not a run is active. If you want to get rid of a cataloged file, you must specifically delete it with the @DELETE statement (see Section 7.17).

The technical difference between cataloged and temporary files is that cataloged files are recorded in a master file directory (MFD), whereas temporary files are not. Temporary files are only visible to the run that created them.

An MFD contains information about all permanent files. Permanent files are cataloged in a master file directory.

A master file directory contains the following information about each cataloged file:

- Complete file name
- Project number under which the file was cataloged
- Account number under which the file was cataloged
- Time the file was cataloged
- Time the file's current assignment started or last assignment ended
- Time the file was last backed up (if applicable)
- Time of the first write after the file was cataloged or after the file was backed up by FAS or time of unload (only one can apply)
- Number of times the file was assigned to a run
- Usage authorization
- Recording mode (tape files only)
- Size of a mass storage file, in tracks or positions (see Section 3.2 for a description of tracks and positions)
- Information that describes the physical location of the file on a device.
- Number of tape reels or cartridges used for the file, as well as the tape identifiers used (tape files only)
- Number of disk packs used for the file, as well as the pack-IDs of the packs used (removable mass storage files only)
- F-cycle numbers of the existing cycles of the file
- Position on the tape where the rolled-out file is stored (only for mass storage files that have been rolled out to tape)
- Security attributes (if your site has Security Level 1 or higher)
- Cartridge Tape Library (CTL) pool name (tape files only)

To examine the master file directory entry for a particular file, use the F option on the @PRT statement. See Section 7.48 for more information.

3.2. Additional Mass Storage and Tape File Topics

Various increments are used to describe the size of files. Most of these terms apply to mass storage files. The most common terms used to describe data are explained here.

A word is the smallest addressable unit of data; it consists of 36 bits. When used to store data, a word can contain either 4 or 6 bytes of data, depending on the format used. If the ASCII format is used, a word contains 4 bytes (9 bits each). If the Fieldata format is used, a word contains 6 bytes (6 bits each). (Fieldata is an older format for representing data; this format is used internally by the Exec. One disadvantage of Fieldata is that it does not let you represent lowercase letters.)

3.2.1. Mass Storage File Topics

The following units are used in reference to mass storage files. I/O operations to mass storage files specify the size of the operation in sectors. The size of a mass storage file when it is assigned or cataloged is specified in tracks or positions.

- A sector contains 28 words.
- A track contains 64 sectors or 1,792 words.
- A position contains 64 tracks, which is equivalent to 4,096 sectors or 114,688 words.

When you create a mass storage file, the system automatically sets the maximum file size to your site's default maximum (default_max_granules). If you think your file needs more than the maximum default, you can specify your own maximum, in either tracks or positions.

Your system can be configured to allow large files. The maximum allowable file size is determined by your site's installation (large_files_enabled). The values given are the maximum size the system can support, depending on the system configuration.

Maximum File Size (tracks)	Large_files_enabled	
	FALSE	TRUE
Track granular	262,143	262,143
Position granular	262,143	16,777,152

3.2.2. Tape File Topics

Tape files on unlabeled tapes are closed out with an end-of-file (EOF) mark written after the last data block in the tape file. The last tape file on a volume is closed out with two EOF marks, signifying the end-of-recorded data on the volume.

Tape files on labeled tapes are surrounded by groups of labels called label groups. Two or more header (HDR) labels comprise the HDR label group and precede the data in a tape file. Two or more trailer (TRL) labels comprise the TRL label group and follow the data in a tape file. The first file on a tape volume also contains one or more VOL labels comprising the VOL label group that precedes both the data in the first file and the HDR label group associated with that file. The HDR and TRL label groups are separated from the data by EOF marks. The VOL label group butts against the HDR label group with no separator. Tape labeling is presented in more detail in the *Exec Administration Reference Manual* (7831 0323). The end of a file and the end of recorded area on a labeled tape are marked exactly as on unlabeled tapes; in other words, with one and two EOF marks, respectively.

Upon receipt of an end-of-reel status, the statements that transfer files to tape (@COPUT, @COPY) deposit an end-of-file (EOF) mark and a file continuation (swap) indicator on the "unlabeled output reel" if the entire file contents has not been copied to the output reel. An exception to this procedure is the quarter-inch cartridge tape, which lacks the functionality to backspace and handles only unlabeled tapes. When you copy to a quarter-inch cartridge and you encounter an end-of-reel status, the swap indicator is written before an EOF mark. This is critical for a copy from quarter-inch tape.

The swap indicator has two forms:

- For labeled tapes, the operating system writes the information on the output reel in the form of end-of-volume tape labels.
- For unlabeled tapes, FURPUR writes one of two types of data patterns when it reaches a reel swap point, depending on the data format and type of copy.

For all COPY,G and all nonquarter-word tapes of other copy formats, a 16-word block is written as follows:

```
WORD 0          05416000000000
WORDS 1 - 7      0044105117122 ($EOR)
WORDS 8 - 15     00000000000000
```

For quarter-word format tapes, an 18-word block is written. This block is the same as the data pattern shown, but it is in quarter-word format.

The statements that read tapes (@COPIN, @COPY, @FIND, and @MOVE) check for a swap indicator after the EOF mark for each tape file. For open reel and ½ inch cartridge tapes, if a swap indicator is not found after the EOF, the open reel or cartridge tape is backspaced (repositioned) for the next operation. Because backspacing is not possible for quarter-inch cartridge tapes, the statements must check each data block for a swap indicator during a read operation. The absence of the backspace capability for the quarter-inch cartridge tape also means that it is possible to receive a 012 I/O status if attempts are made to copy or move the cartridge tape past the double EOF sequence after the last file.

Labeled tapes return a status code to FURPUR through the operating system when end-of-volume (EOV) labels are found.

For unlabeled tapes, the data pattern written to the tape is checked for a swap indicator. If this block is not a swap indicator, the reel is backspaced and repositioned for the next operation.

Table 3–1 summarizes the end-of-reel configuration for tapes that contain multireel files.

Table 3–1. End-of-Reel Configuration for Multireel Tapes

Tape	End-of-Reel Configuration
Labeled	File Blocks * EOV1 EOV2 * *
Unlabeled	File Blocks * swap indicator * *
Unlabeled, quarter-inch cartridge	File Blocks swap indicator *

* Indicates an EOF mark

When you execute a move across reels for a multireel file, the EOF preceding the swap information on the first reel (or cartridge) is not part of the EOF count for the MOVE since it does not delimit an entire file.

The @MOVE and @FIND statements are restricted when you use them with multireel files. They do not recognize reels preceding the one currently in use. However, they continue on succeeding reels. The @REWIND statement repositions the tape to the beginning of the first reel.

3.2.3. Virtual Tape Handler Files

Virtual tape handler (VTH) simulates tape by using mass storage. The @ASG statement lets you assign virtual tape files to a run. You assign them in the same manner as a standard tape file, using an assign statement. Your access and manipulation of virtual tape files is the same as to standard tape files. Your control statements remain identical for a virtual tape as for a standard tape. Programs cannot tell the difference between a virtual tape and a real tape. The only tape functionality that is not supported is the Fast Tape Access (FTA) functions of cartridge tape subsystems.

3.2.4. Omnibus File Format

The omnibus file format is the superstructure upon which the program file, SDF, and many other data reference structures are based. This is the level at which the Exec supports the mass storage devices. It can be assigned in 1,792-word increments (track) and addressed in 28-word increments (sector). The assign granules (track) are not necessarily all allocated in succession in an active file.

The @COPY and @COPY,G statements work with the omnibus format level and are not involved with the structures within the file. Those structures remain intact since the @COPY and @COPY,G statements view the substructure as data. This allows you to use the @COPY,G statement to transfer a program file to tape, or reload a program file on another OS 2200 system. MAP, MASM, and other processors work with the data directly.

3.2.5. File Sharing

The order in which master file directories are searched depends on the system configuration. Systems without the file sharing feature naturally search only the standard master file directory. Systems with the file sharing feature can search either one or both of the master file directories, depending on the system configuration.

3.3. Naming Files and Elements

The only part of a file name that is required is the basic file name, as follows:

filename.

All other fields are optional.

In some cases, the period is optional; if you do not include a period, the system assumes that it is the name of a file. In other cases, if you do not include a period, the system assumes that it is the name of an element. To avoid confusion, include the period except for file names specified on the @BRKPT statement.

The name originally given to a file is called the external file name. A file can also have an internal file name. See Section 7.68 for information on how to use the @USE statement to give a file an internal name.

File Name and Element Name Format

[[directory-id#]qualifier]filename[(F-cycle)/read-key/write-key.element-name/
version(S-cycle)]*

directory-id

Shared files exist exclusively on an OS 2200 mass storage device that is shared by two or more OS 2200 systems. Shared files are different from local files, which are files that are stored on a specific OS 2200 system.

The *directory-id* field is valid when file sharing is configured on the system. It indicates which master file directory contains the file. The directory-ID has two possible values: STD (standard) for local files on any system and SHARED for shared files.

The # is a delimiter to separate the directory-ID and qualifier.

This field is valid only for the following statements:

- @ASG
- @CAT
- @FREE
- @MODE
- @QUAL
- @USE

All other statements that have a *filename* field cannot include the directory-ID in the file name. You can indirectly use a directory-ID for file names on these statements by either giving them internal names with the @USE statement or by specifying directory-IDs with the @QUAL statement, as in the following examples:

```
@USE P1,STD#QUAL1*FILEX
@COPY P1.,X.
```

```
@QUAL,D SHARED#QUAL1
@COPY FILE1.,FILE2.
```

In the second example, the default qualifier (QUAL1) and directory-ID (SHARED) that were specified on the @QUAL,D statement are assumed in the file names on the @COPY statement.

For more information on the @USE statement, see Section 7.68. For more information on the @QUAL statement, see Section 7.49.

If you omit the directory-ID# from a file name and do not specify a directory-ID on a previous @QUAL statement, the system uses the director-ID value configured at your site. One exception to this is if a user is cataloguing a shared removable pack file, the system detects that it is a shared pack and uses the shared directory.

If you include the #, but not a directory-ID, the system uses the implied directory-ID you specified on a previous @QUAL statement. If you did not specify an implied directory-ID, the system uses the directory-ID value configured at your site.

The default directory-ID and default qualifier are the directory-ID and qualifier given on the last @QUAL,D statement performed in a run. The implied directory-ID and implied qualifier are the directory-ID and qualifier given on the last @QUAL with no options performed in a run.

qualifier

The qualifier helps to ensure that file names are unique. Different files can have the same file name, as long as their qualifiers or F-cycles are different. User-IDs and project-IDs are commonly used as qualifiers, but you can create your own. The qualifier can be up to 12 characters long and can include any combination of letters, numbers, hyphens (-), and dollar signs (\$). (The @QUAL statement, which also defines the qualifier, is described in Section 7.49.)

If you omit the qualifier and asterisk (*) from a file name and do not specify a qualifier on an @QUAL statement, the system uses the project-ID on your @RUN statement as the qualifier.

If you include an asterisk (*) but not a qualifier, the system uses the implied qualifier you specified on an @QUAL statement. If you did not specify an implied qualifier, the system uses the project-id on your @RUN statement.

filename

The basic name of a file can be up to 12 characters long and can include any combination of letters, numbers, hyphens (-), and dollar signs (\$).

Note: Do not use a processor name (ELT or DATA, for example) as a file name. The results will be unpredictable.

F-cycle

The F-cycle number identifies a file in a set of files that have the same qualifier and file name. Normally, you never have to include an F-cycle number with a file name. If you do not supply the F-cycle, you get the highest cataloged file cycle in the set. See Section 3.4 for more information on F-cycles.

read-key/write-key

Keys are like passwords that are required to read a file or write to it. Keys typically apply only to unowned files which are standard in Fundamental Security, but they can be used for owned files if the system is so configured. Keys are an easy way you can protect a file from unauthorized use. When you create a file, you can specify a read key, write key, both, or neither. If you use only a write key, you must precede it with two slashes to indicate that it is a write key, not a read key. Each key can be up to six characters long. You can use any characters but the period (.), comma (,), semicolon (;), slash (/), or blank.

To change the read or write key on a file, or to add a read or write key to an existing file, use the @CHG statement as follows:

```
@CHG filename/RK/WK, filename/RK/WK
```

where *filename1* is the file to be changed (including the current read/write keys), and *filename2* is the file with its new or changed read or write keys. See Section 7.5 for more information on the @CHG statement.

element-name

Is the name of the element. Element names can be up to 12 characters long and can include any combination of letters, numbers, hyphens (-), and dollar signs (\$).

version

Is used to distinguish between elements that have the same name and type, but contain different information. Using version names is one way you can save different renditions of the same program. Version names can be up to 12 characters long and can include any combination of letters, numbers, hyphens (-), and dollar signs (\$). If the element is an omnibus type, some processors can ignore the version.

A combination of element name and version need be unique only within a file. Thus, .ELT1/VER1 and .ELT1/VER2 are two different elements (note the leading period). Then with different files, Q*F1.CH2/VER1, Q*F1.CH2/VER2, Q*F2.CH2/VER1, and Q*F2.CH2/VER2 are four different elements.

S-cycle

Is a number identifying a cycle of a symbolic element. Only symbolic elements can use element cycling. Element cycles are different stages a symbolic element goes through as it is updated by certain processors (U-option). Normally, you do not need to be concerned about element cycling and S-cycle numbers. Some processors, including IPF 1100, do not use element cycling. Processors that accept symbolic element input always default to the most recent cycle of a symbolic element. See Section 3.5 for more information.

Only the basic element name is required; the version and cycle numbers are optional. Normally, different elements within the same file have different names. However, elements can have the same name, provided their element types are different (symbolic, relocatable, executable, or omnibus).

File Name Examples

ACCT-MASTER.

ACCT-MASTER is the basic file name. The system uses defaults for all other parts of the file name:

- It uses the project-ID from the @RUN statement as the qualifier.
- It uses the current cycle of the file.
- No read key or write key is used.

GL*ACCT-MASTER.

GL is the qualifier, and ACCT-MASTER is the basic file name.

GL*ACCT-MASTER/SESAME.

GL is the qualifier, ACCT-MASTER is the basic file name, and SESAME is the read key.

PR*ACCT-MASTER//POPPY.

PR is the qualifier, ACCT-MASTER is the basic file name, and POPPY is the write key. (The two slashes indicate that there is no read key and that POPPY is a write key, not a read key.) The current cycle of the file is used.

GL*ACCT-BAL/SESAME/##\$&!.

GL is the qualifier, ACCT-BAL is the basic file name, SESAME is the read key, and ##\$&! is the write key.

SHARED#DATA*BASE.

SHARED is the directory-id, DATA is the qualifier, and BASE is the basic file name.

Element Name Examples

COLORS.RED, .BLUE, .YELLOW

All three elements (RED, BLUE, and YELLOW) are assumed to be in file COLORS.

COLORS.RED, .BLUE, YELLOW

The elements RED and BLUE are assumed to be in file COLORS, while element YELLOW, which has no preceding period, is assumed to be in the TPF\$ file.

MAIN

In this example, only the basic element name is used. Since a cycle number is not specified, the current cycle of the element is used. The version name is blank. The file container for element MAIN is TPF\$.

SUB1/APR02

Here, SUB1 is the basic element name, and APR02 identifies the version of the element to be used. Since a cycle number is not specified, the current cycle of element SUB1/APR02 is used. The file container for element SUB1/APR02 is TPF\$.

When you specify an element name without a file name, the system usually assumes that the element specified is in TPF\$, a temporary file automatically assigned to your run. An exception occurs when you specify a series of elements on the same control statement. If you include a file name with the first element name and put a period before the subsequent element names, the system assumes that those elements are in the first file named.

3.4. F-Cycles

You can use file cycles (F-cycles) to manipulate a set of cataloged files.

Each *directory-id#qualifier*filename* constitutes an F-cycle set. Each file within one of these sets of cataloged files has the same read and write keys, as well as the same *directory-id#qualifier*filename*. Each master file directory on a system can contain an F-cycle set with the same *qualifier*filename*.

3.4.1. Numbering

When you create a file of a given F-cycle, a number is assigned to it. This number is called the absolute F-cycle number and uniquely identifies the file, along with its *qualifier*filename*. Absolute F-cycle numbers are unsigned integers that begin with 1 and continue through 999, at which point the numbering recycles to 1. The circular assignment of F-cycle numbers does not cause conflicts since a maximum of 32 consecutively numbered files can be retained in a set.

3.4.2. Referencing

You can reference a file within an F-cycle set by its absolute F-cycle number or by a relative F-cycle number.

Relative F-Cycles

Relative F-cycle numbers are signed integers in the range +1 to -31. Relative F-cycle number 0 (-0 or +0) refers to the cataloged file whose absolute F-cycle number is highest. Relative F-cycle -1 refers to the next highest, and so on. The highest is defined as having the highest absolute F-cycle number of any of the files in the F-cycle set, unless the absolute F-cycle numbers are at the point of recycling from 999 back to 1, in which case 1 is higher than 999.

When a cataloged file is assigned with a relative F-cycle, the list of currently assigned files is searched to determine whether that relative F-cycle is already assigned to the run. If it is, that F-cycle is used. If the specified relative F-cycle cannot be found, the master file directories are searched to determine whether that relative F-cycle exists. If the relative F-cycle does not exist, the assignment request is rejected.

If the relative F-cycle does exist, it is converted to the corresponding absolute F-cycle number and the list of currently assigned files is searched again to determine whether that absolute F-cycle is already assigned to the run.

If it is assigned and it already has a relative F-cycle number associated with it, the request is rejected because a naming conflict exists when one absolute F-cycle has two relative F-cycle values associated with it.

If it is assigned and no relative F-cycle number is associated with that absolute F-cycle, then the absolute F-cycle is associated with the relative F-cycle.

If it is not already assigned to the run, that relative F-cycle/absolute F-cycle is assigned to the run.

When you specify a relative F-cycle, the file is known to the assigning run by both its relative F-cycle and the appropriate absolute F-cycle number. However, if you specify an absolute F-cycle, the Exec does not attempt to convert it to a relative F-cycle. In this case, the file is known to the assigning run only by its absolute F-cycle.

The use of relative F-cycle numbers enables you to refer to a particular relative backup file. For example, assume a program required that a particular file always be created from an earlier cycle of the file. The relative F-cycle could be used to reference the earlier file each time the program is run. But, if an absolute F-cycle notation had been used, it would have to be changed each time the program was run.

Absolute F-Cycles

To create the next sequential absolute F-cycle, you can use the relative specification +1. If files have been deleted, the relative specification +1 creates the highest numbered deleted F-cycle. When this +1 file is cataloged (by freeing the file or by run termination), its relative F-cycle number is set to 0 and other existing files of the set have their relative F-cycle numbers decreased by 1, thus maintaining consecutive relative numbering.

If a file is created by its absolute cycle and the absolute cycle is not the next numerically sequential absolute F-cycle available, the sequence of F-cycles is updated to point at the newly created F-cycle. In addition, a buffer of noncataloged F-cycles exists between the newly cataloged F-cycle and the previously cataloged F-cycles. This increases the F-cycle range by more than 1.

When a cataloged file is assigned with an absolute F-cycle, the list of currently assigned files is searched to determine whether that F-cycle is already assigned to the run. If it is, that F-cycle is used. If the specified absolute F-cycle is not assigned to the run, the master file directories are searched to determine whether that absolute F-cycle exists. If it does not exist, the assignment is rejected. If it does exist, that absolute F-cycle is assigned to the run.

3.4.3. Deleting

Only one F-cycle can be deleted by the creation of another F-cycle. If the creation of an F-cycle causes more than one F-cycle to be deleted, the request is rejected.

For example, if absolute cycles 36, 6, and 5 (relative 0, -1, -2) are the only cycles of a file cataloged, then cycle 38 cannot be created because cycles 5 and 6 must be deleted first. The creation of absolute cycle 37 deletes absolute 5, but absolute cycles 37, 36, and 6 (relative 0, -1, -2) remain. Absolute cycles 38 through 68 can now be created because any cycle in this range causes at most one existing cycle to be deleted.

The deletion of the last remaining file of a set causes all recognition of the set to be removed from the system.

When the maximum number of files in one F-cycle set is exceeded, the record of the file with the most negative F-cycle number is deleted. If the file is stored on mass storage, the file's text is deleted. The maximum number is 32, but you can change it with the M\$CON\$ Executive request. See the *Exec Administration Reference Manual* (7831 0323) for more information on ER M\$CON\$.

3.4.4. Internal Name

An internal name can be attached (with @USE) to a file in the +1 condition. Then, if an @FREE,R is performed on this file, the resulting dummy item has no relative F-cycle value (null), but the absolute F-cycle value of the file is retained.

3.4.5. Temporary Files

F-cycles for temporary files do not serve the functions previously described, but are allowed and can be used to uniquely identify temporary files that have the same *qualifier*filename*. When F-cycles are used in this manner, there is no correlation between relative and absolute F-cycles.

3.4.6. Examples

Here are some examples using F-cycles.

Example 1

As an example of the correlation between the absolute and relative F-cycle numbering schemes, assume that the most recently cataloged cycle of a file has an absolute F-cycle number of 28. When referring to this file, the F-cycle could be either 28 (absolute) or 0 (relative). Now, assume that a new cycle of the file is to be cataloged. Since the file has not yet been cataloged, its relative F-cycle is +1 and its absolute F-cycle is 29. When the new file is freed (cataloged)

- The absolute F-cycle remains as 29, but the relative F-cycle becomes 0.
- The file whose absolute F-cycle is 28 is now given a relative F-cycle of -1.
- The file whose absolute F-cycle is 27 is now given a relative F-cycle of -2, and so on.

Example 2

Here is an example of creating and deleting F-cycles by using relative specification +1:

These statements create and catalog absolute F-cycles 1, 2, and 3 of the file QUAL*TEST, respectively.

```
@ASG,C QUAL*TEST(+1).  
I:002333 ASG complete.  
@FREE QUAL*TEST(+1).  
I:002333 FREE complete.
```

```
@ASG,C QUAL*TEST(+1).  
I:002333 ASG complete.
```

```
@FREE QUAL*TEST(+1).
I:002333 FREE complete.
```

```
@ASG,C QUAL*TEST(+1).
I:002333 ASG complete.
@FREE QUAL*TEST(+1).
I:002333 FREE complete.
```

This @CYCLE command reports that three F-cycles of the file exist. It displays their absolute F-cycles: 3, 2, and 1.

```
@CYCLE QUAL*TEST.
MAX-RANGE= 32      CUR-RANGE= 3
NO. OF F CYCLES= 3: 3, 2, 1
```

These @DELETE commands delete two of the F-cycles, beginning with the highest relative F-cycle (absolute F-cycle 3).

```
@DELETE QUAL*TEST.
END DELETE.

@DELETE QUAL*TEST.
END DELETE.
```

This @CYCLE command reports that F-cycles 2 and 3 were deleted, indicated by the *'s around the F-cycle number. F-cycle 1 is the only accessible F-cycle of the file.

```
@CYCLE QUAL*TEST.
MAX-RANGE= 32      CUR-RANGE= 3
NO. OF F CYCLES= 1: *3*, *2*, 1
```

These statements create another F-cycle of the file.

```
@ASG,C QUAL*TEST(+1).
I:002333 ASG complete.
@FREE QUAL*TEST(+1).
I:002333 FREE complete.
```

This @CYCLE command reports that F-cycle 2 is still deleted. The new F-cycle is F-cycle 3, taking the number of the highest deleted F-cycle.

```
@CYCLE QUAL*TEST.
MAX-RANGE= 32      CUR-RANGE= 3
NO. OF F CYCLES= 2: 3, *2*, 1
```

3.5. S-Cycles

Symbolic data is represented in system data format (SDF). Complete information about SDF data is contained in the *Data Structures Programming Reference Manual* (7833 3481). There are several types of SDF data, but only the "general symbolic" type can contain the element cycle information that allows different versions of the

symbolic data to be contained in a single symbolic element. Processors that maintain element cycle information, such as ED, ELT, MAP, MASM, and PDP, update the element cycle information as symbolic data records are added and deleted.

Note: *Not all processors that update symbolic elements update the element cycle information. IPF is an example of a processor that does not update cycle information.*

For symbolic elements that contain element cycle information, you can use S-cycles to select which version of the symbolic data to reference. S-cycles are often referred to as element cycles.

3.5.1. Numbering and the Element Cycle Word

Each symbolic element contains an element cycle word. This word contains three element cycle values:

- The maximum number of element cycles that can be contained in the element. It has a default value of 5 and can be changed to a value of from 1 to 63, inclusive. This value is referred to as *max* in the discussion of element cycles in this section.
- The current absolute element cycle number. It can have a value of from 0 to 62, inclusive. This value is referred to as *cur*.
- The number of element cycles present in the element. It can have a value of from 1 to *max*. This value is referred to as *num*.

The element cycle word can be displayed with the FURPUR @PRT,TL command (see Section 7.48.3). The following example is referred to throughout this section. It shows under the (CYCLE WORD) header the element cycle word values for an element ELT that has been initially created and then has been updated six times.

Example

```
@PRT,TL$ QUAL*FILE$.
STD#QUAL*FILE$(1) PF, 7/5000 elements, 29/256 tracks
ELEMENT TABLE
```

D	NAME	VERSION	TYPE	DATE	TIME	SEQUENCE NUMBER	PREAMBLE SIZE	TEXT SIZE	CYCLE WORD			FLAG	SECTOR LOCATION
									MAX	CUR	NUM	BITS	
*	ELT	VER	SYMBOLIC -Q	2014-10-01	16:14:58	1		2	5	0	1		1792
*	ELT	VER	ELT SYMB-Q	2014-11-12	12:11:38	2		3	5	1	2		1794
*	ELT	VER	ELT SYMB-Q	2014-11-12	12:11:38	3		3	5	2	3		1797
*	ELT	VER	ELT SYMB-Q	2014-11-12	12:11:38	4		4	5	3	4		1800
*	ELT	VER	ELT SYMB-Q	2014-11-12	12:11:38	5		5	5	4	5		1804
*	ELT	VER	ELT SYMB-Q	2014-11-12	12:11:38	6		4	5	5	5		1809
	ELT	VER	ELT SYMB-Q	2014-11-12	12:11:38	7		4	5	6	5		1813

```
NEXT AVAILABLE LOCATION-
*File contains 6 deleted elements and <1 track of deleted text
```

For the current ELT sequence number 7 *max* is 5, *cur* is 6, and *num* is 5.

When a symbolic element is initially created

- *max* is set to the default value of 5
- *cur* is set to 0
- *num* is set to 1

When a symbolic element is updated

- *max* is not changed
- If *cur* is less than 62, it is increased by 1. If *cur* is 62, it is set to the smaller of *max*-1 and *num*.
- If *num* is less than *max*, *num* is increased by 1. If *num* is equal to *max*, *num* is not changed.

The absolute cycle numbers contained in an element are *cur-num+1* to *cur*, inclusive. For convenience, the lowest absolute cycle number *cur-num+1* is referred to as *low* in the discussion of element cycles in this section. In the example, the current element ELT has the 5 absolute cycle numbers 2 (6-5+1) to 6.

Once *num* reaches *max*, the element contains the maximum number of element cycles. When the element is updated, data records that were deleted at cycle number *low* are removed, and data records that were added at cycle number *low* are changed to indicate that they were added at cycle number *low*+1. In the example, if the element ELT is updated, data records deleted at cycle 2 are removed, and data records added at cycle 2 are changed to indicate they were added at cycle 3.

Once *cur* reaches 62, the element is at the maximum absolute cycle number. When the element is updated, the absolute cycle numbers *low* to 62 are renumbered so that they fall in the new absolute cycle number range of 0 to *max*-1 (or 0 to *num* in the unusual case where *num* is less than *max* when *cur* is 62). Table 3–2 shows how absolute cycles are renumbered in the normal case when *max* is 5, *cur* is 62, and *num* is 5.

Table 3–2. Absolute Cycle Renumbering

Before Update	After Update
58	0
59	0
60	1
61	2
62	3
	4 (new data records)

After the element update, *max* is 5, *cur* is 4, and *num* is 5. This renumbering assures that element cycles always have only one range of absolute cycle numbers. This is unlike F-cycles (see Section 3.4.1), where there can be two ranges of absolute cycle numbers at the same time (lowest absolute F-cycle to 999 and 1 to highest absolute F-cycle).

The FURPUR @CYCLE command (see Section 7.15.2) can be used to change *max* for an element to any value between 1 and 63, inclusive. Changing *max* causes the element to be rewritten when the new *max* value is smaller than *num*.

In the example, the element `ELT` contains absolute cycle numbers 2 through 6. Note that sequence numbers 1 through 6 contain deleted versions of `ELT` that contain absolute cycle numbers 0 and 1. Until these deleted elements are removed from the file with the `FURPUR @PACK` command (see Section 7.42), the `FURPUR @DELETE,U` command (see Section 7.17.3) can be used to undelete any of the deleted elements. This can make additional versions of the symbolic data available for use.

3.5.2. Referencing

You can reference different versions of the symbolic data contained in a symbolic element by specifying an absolute S-cycle number or by specifying one of the two available formats of relative S-cycle number.

Absolute S-Cycles

Absolute S-cycles are specified as decimal numbers from 0 to 62 without a leading sign character (+ or -). The absolute cycle number that can be specified for an element is determined by the element cycle word values *cur* and *num*. The lowest absolute cycle number is *cur-num*+1 and the highest absolute cycle number is *cur*. If you do not specify an S-cycle, the default is the highest absolute cycle number *cur*.

Relative S-Cycles

Relative S-cycles can be specified in two formats:

Format 1

(-cycle)

Format 2

(+cycle)

Relative S-cycles are specified as decimal numbers from 0 to 62 with a leading sign character (+ or -). The relative cycle numbers that can be specified for an element are determined by the element cycle word value *num*. For both formats *cycle* is a decimal number from 0 to *num*-1.

For format 1 the relative S-cycle number (-0) is the element cycle with the **highest** absolute cycle number, *cur*. Relative S-cycle number (-1) is the element cycle with the next highest absolute cycle number, *cur*-1, and so on.

For format 2 the relative S-cycle number (+0) is the element cycle with the **lowest** absolute cycle number, *num-cur*+1. Relative S-cycle number (+1) is the element cycle with the next lowest absolute cycle number, *num-cur*+2, and so on.

Note: S-cycles 0, -0, and +0 can represent different element cycles. 0 is an absolute S-cycle, and -0 and +0 are relative S-cycles. This is unlike F-cycles, where 0, -0, and +0 are all relative F-cycles and all represent the same file cycle (see Section 3.4.2).

In the example element `ELT`, *cur* is 6 and *num* is 5. Table 3–3 shows the S-cycles that could be specified to select each of the five element cycles contained in `ELT`.

Table 3–3. Referencing Element Cycles

Element Cycle	Absolute S-Cycle	Relative S-Cycle Format 1	Relative S-Cycle Format 2
2	(2)	(-4)	(+0)
3	(3)	(-3)	(+1)
4	(4)	(-2)	(+2)
5	(5)	(-1)	(+3)
6	(6)	(-0)	(+4)

3.6. System-Created Files

When you start a run, the Exec automatically creates up to four files for you: TPF\$, READ\$, PRINT\$, and DIAG\$. READ\$ and PRINT\$ are created only if your run is a batch run.

3.6.1. TPF\$

TPF\$ is a temporary program file you can use as a working file during your run. Since, it is a temporary file, anything you store in it is lost as soon as your run terminates; use TPF\$ only for data and programs you do not want to save. If there is something in the file you do want to save, be sure to copy it into a permanent (cataloged) file before terminating your run.

Although the name implies that TPF\$ is a program file, it could actually be either a program file or a data file, depending on how it is first used. (This is true for all files.)

The system sometimes uses TPF\$ as a default file when you do not provide the name of a file on a control statement. If you do not provide a file and element name for the output produced by a compiler or the Collector, the system uses TPF\$.NAME\$ by default.

3.6.2. READ\$

READ\$ is used as the primary input file for batch runs. READ\$ serves as a holding area for the runstream. During the course of the run, the Exec retrieves the control statements, data, and other images from the READ\$ file and processes them in the order retrieved.

Demand runs do not use a READ\$ file. Instead, you type in all input directly from the keyboard of your terminal.

3.6.3. PRINT\$

PRINT\$ is used as the primary output file for batch runs. The PRINT\$ file contains the runstream statements from the READ\$ file, as well as all output produced by the run.

This output includes error and diagnostic messages, output generated by executing programs, and summary accounting information produced when the run terminates. After your run terminates, the PRINT\$ file is queued to a printer, and this output is printed. This process is automatic for batch runs.

Demand runs do not use a PRINT\$ file. All output that is normally stored in PRINT\$ is immediately displayed at your terminal. If you wish to direct output to a printer during a demand run, you must use the @BRKPT and @SYM symbiont statements.

See Section 3.7 for more information on print files.

3.6.4. DIAG\$

DIAG\$ is a diagnostic file used to store the parts of main storage that a program occupied when the program terminated. By analyzing this information, experienced programmers can determine what process has occurred. This can help them when a program terminates in error.

To produce a printout or dump of the information saved in the DIAG\$ file, use a PostMortem Dump (PMD) processor. For absolutes not collected with the TYPE NPEDIAG directive, use the @DAP or @PADS statement. For object modules, use the @PADS statement.

It is permissible to use the @PMD statement following the execution of any program regardless of type. The Exec determines which PMD processor must be used based on the type of the last program executed. That processor loads and, if not @PMD, executes as if the 'W' option were present on the @RUN image.

3.7. Print Files

When the system is configured, a default printer is assigned to each terminal. When no other printer is specified, the output from runs submitted from a particular input device is printed out on the default printer.

For batch runs, the output generated during the run is normally stored in the run's primary output file, PRINT\$. Then, when the run terminates, the output stored in the PRINT\$ file is queued to the default printer for the input device used. (See Figure 3-2)

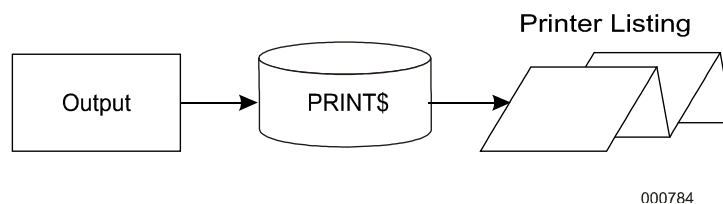


Figure 3-2. Normal Batch Output

For demand runs, output normally is neither stored in a file nor printed; it is sent directly to your terminal. (See Figure 3-3)

In spite of these differences in how output is handled for batch and demand runs, symbiont statements @BRKPT and @SYM perform the same basic function for both types of runs. The @BRKPT statement is used to direct output to a user-defined print file, and to redirect output back to the PRINT\$ file (batch runs) or to the terminal (demand runs).

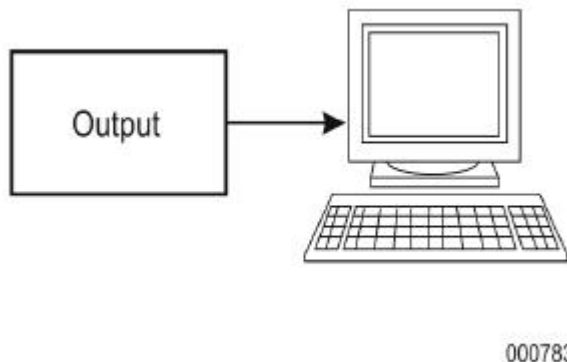


Figure 3-3. Normal Demand Output

The @SYM statement is used to queue a user-defined print file to a printer. Unlike the PRINT\$ file, user-defined print files are not queued automatically. You can also use the @SYM statement to queue the PRINT\$ file to a printer other than the default printer that is assigned to the input device from which you submitted your run.

Request output by using the @SYM statement (if the desired output is in a breakpoint file) rather than the SITE command from CTS, which would tie up your terminal.

Figure 3–4 represents these print file concepts.

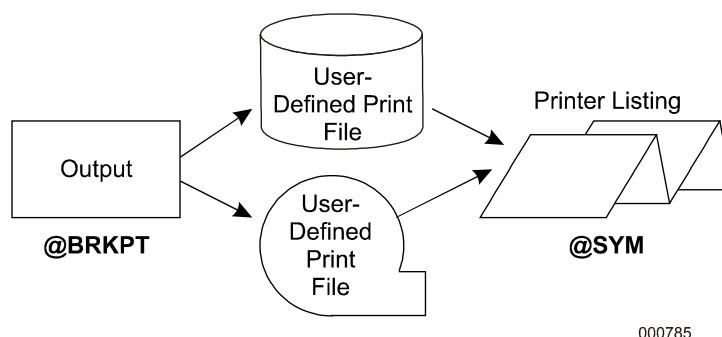


Figure 3–4. Breakpoint Output (Batch or Demand)

You can breakpoint to either a mass storage file or a tape file. When you are breakpointing to a tape file, there is a technique you can use to breakpoint several print files to the same tape file. In addition to specifying the tape file you want to breakpoint to, you also specify a label or part-name to identify each print file (part) on the tape. Then, when you want to queue those print files to a printer, you include the name of the tape file and the labels (part-names) on the @SYM statement, to specify which print files (parts) you want to print.

Although you usually queue a print file to a printer, you can also queue a file to a terminal or user-ID. If someone has queued a print file to your user-ID or terminal, you can receive the file by using the @@SEND statement after terminating your run with an @FIN statement. If a local printer is connected to your terminal, you can receive a printout of the queued file by entering the @@PRNT statement before you use the @@SEND statement. (See Section 7.47 for a description of the @@PRNT statement.)

3.8. File Formats

This section describes the following file formats:

- System data format (SDF)
- Track-binary
- Track-binary: revised variant
- Element
- Program

3.8.1. System Data Format

Symbolic data files are represented in system data format (SDF). Complete information about SDF data is contained in the *Data Structures Programming Reference Manual* (7833 3481). SDF files can be created by the symbiont complex, by products such as @DATA, and by your programs. SDF files can also be created from program file symbolic elements with the FURPUR @COPY,I command (see Section 7.13.2).

3.8.2. Track-Binary Format

The track-binary format is the representation of a mass storage file as a tape file.

The @COPY,G statement produces the track-binary format on tape. Before the file contents are copied, a 28-word label block is written to tape, giving details of the copied file and the time when the copy was made. The format of this 28-word block follows.

'COPYG'		
'BLKSEQ'		
Qualifier (2 words, left justified)		
File name (2 words, left justified)		
F-cycle		
Fielddata date in format month, day, year (mmddyy)		
Fielddata time in format hours, minutes, seconds (hhmmss)		
Revision-ID		Equipment Code
Highest track written		
0		Number of tracks in block
Always 0		
Always 0		
Always 0		
Always 0		

Subsequent data blocks are multiples of 1,794 words and consist of a two-word header followed by 1,792 data words, a mass storage track.

- When "Revision-ID" in the label block is zero, then "number of tracks in block" in the label block is zero and subsequent data blocks are 1,794 words.
- When "Revision-ID" in the label block is nonzero, then "number of tracks in block" in the label block is from 1 to 16 and subsequent data blocks are 1,794 words times this value. The default number of tracks in a block is configured to 8 in FURPUR. This format is written when data compression is enabled for the output tape and when the D option is included on the @COPY,G statement (see Section 7.13.1).

b	Sector address of first sector in track	
Checksum		Block sequence number
1 track: 1,792 words		

The "b" is a one-bit "last track" indicator. When "Revision-ID" in the label block is zero, this indicator is 0 for all tracks. When "Revision-ID" in the label block is nonzero, this indicator is 1 for the last track in the file; this indicator is 0 for all previous tracks in the file.

3.8.3. Element File Format

The element file format exists only on tape. It is created by the @COPOUT statement. You use the @FIND statement to locate specific elements within an element file. You use the @COPIN statement either to copy an element file to a program file or to bring specific elements or groupings of elements in an element file into a program file.

3.8.4. Program File Format Structure

Program files are files that are formatted for storing independent collections of information called elements. To keep track of the elements, each program file contains a table of contents (TOC). The table of contents identifies each element in the file and describes significant characteristics of each element.

There are three kinds of program files:

- Standard program files (PF) that have a variable-size table of contents and can contain up to 5,000 elements. Elements might have a maximum of 262,143 sectors of text.
- Large program files (LPF) that have a fixed size table of contents and might contain a guaranteed maximum of 26,146 elements. Elements might have a maximum of 262,143 sectors of text.
- A large element program file (LEPF) can be initialized with the small, variable-size table of contents of a standard program file, or with the large, fixed size table of contents of a large program file. Elements can have more than 262,143 sectors of text.

Table 3–4 shows the relationship between program file type, maximum number of elements and maximum element text length.

Table 3–4. Program File Capacity

Element Text Length	5,000 Element Maximum	26,146 Element Maximum
Element text less than 262,144 sectors	Standard PF	LPF
Element text of any size	Standard LEPF	Large LEPF

3.8.5. Types of Program File Elements

Program files can contain these types of elements:

- Symbolic (includes procedure elements)

Symbolic elements contain symbolic images or data. Symbolic elements can contain lines of a source program to be compiled, instructions (directives) for special processors (such as the Collector), data to be processed, or runstreams.

- Relocatable

Relocatable elements are machine-language elements produced by some compilers. The Collector resolves intercomponent references in relocatable elements and generates absolute elements.

- Omnibus

Omnibus elements are special-purpose elements that are not symbolic, relocatable, or executable. Omnibus elements are used to store information when a more flexible format is needed. For example, the Meta-Assembler (MASM) creates an omnibus element to contain the output of a definition mode assembly.

- Executable

Elements that are ready for execution on a processor. These subtypes are associated with executable elements:

- Absolute elements

Absolute elements are executable machine-language elements produced by the Collector, one of the system processors.

Absolute elements are created from user-defined relocatable elements, together with elements from relocatable subroutine libraries.

- Object modules

Universal Compiling System (UCS) compilers, MASM, and the static linker create these elements. They are directly executable; that is, you do not have to prelink or collect them before you execute them.

- Bound object modules

These elements are special object modules created only by the static linker. They cannot be linked to other bound object modules during a dynamic linking operation, and can be linked to other object modules only during a dynamic linking operation. For more information, see the *Linking System Programming Reference Manual* (7831 0505).

- Subsystem definition elements

These are definition elements created by the subsystem definition processor. They define the attributes of a logical group of object modules called a subsystem. For more information, see the *Linking System Subsystems Programming Guide* (7830 7451).

Symbolic procedure, relocatable, and absolute elements cannot be added to an LEPPF.

3.8.6. Table of Contents for a Program File

The table of contents for a program file consists of five tables: an element table, three procedure tables, and an entry point table. To examine the table of contents listing for a particular file, use the T and L options with the @PRT statement.

A table describing the entry points within object modules is located in the element text area. The table of contents contains a pointer to the start of this table, if it exists.

Element Table

The element name table contains an entry for each element in the program file. In addition to identifying the element name, version name, and type, these entries provide the following information about an element:

- Relative starting location within the file
- Size of the element
- Subtype of the element
- Time and date that the element was created
- Auxiliary information about the element

Procedure Tables

The table of contents also includes three procedure tables: one for COBOL procedures, one for FORTRAN and PLUS procedures, and one for assembly language (MASM) procedures.

A procedure is a set of instructions that is stored separately from the rest of the program and then copied into a program when needed. Procedures are used most often when the same instructions must be included in different programs. Entries in the procedure tables are created by the Procedure Definition Processor (@PDP). See the *PDP Operations Reference Manual* (UP-10070.1-A) for more information.

Each entry in the procedure tables contains

- The name of the procedure
- A link to the element that contains the procedure
- The location of the procedure
- Procedure name tables

Entry Point Table

The last table in the table of contents is an entry point table, which is used when a program is generated to link separate subprograms into a complete program. A program file can contain two entry-point tables. One table identifies the entry points for relocatable elements, and the other identifies the entry points for object modules.

The @PREP statement adds entry points to an entry-point table. Thus, when relocatable elements or object modules are created or copied into a program file, the entry points within these elements are not inserted into an entry-point table until you enter an @PREP statement. See Section 7.46 for more information on @PREP.

Large Program Files

Compared to a standard program file, a large program file has a larger capacity for all tables. For a standard program file, the table of contents is the first 28 tracks of the file; for a large program file, the table of contents is the first 629 tracks of the file. Table 3–5 shows the *maximum* number of entries that can be inserted in each table in a program file. The number of files can be less, depending on the types of elements contained in the file and the order in which they were created.

Table 3–5. Maximum Number of Items per Table

Table	Standard Program Files (Normal)	Standard Program Files (Maximum)	Large Program Files
Elements	2,671	5,000	26,146
Assembler Procedure	861	5,789	65,373
COBOL Procedure (4-word)	861	4,893	65,373
COBOL Procedure (8-word)	430	2,446	32,686
PLUS/FTN Procedure	861	3,997	65,373
Entry Point	3,101	3,101	20,125

To specify explicitly that a file is formatted as a large program file, you must use the @CHG,T command. The @CHG,T command can be used only on newly assigned or erased files. See Section 7.5 for more information on @CHG.

3.8.7. Large Element Program Files

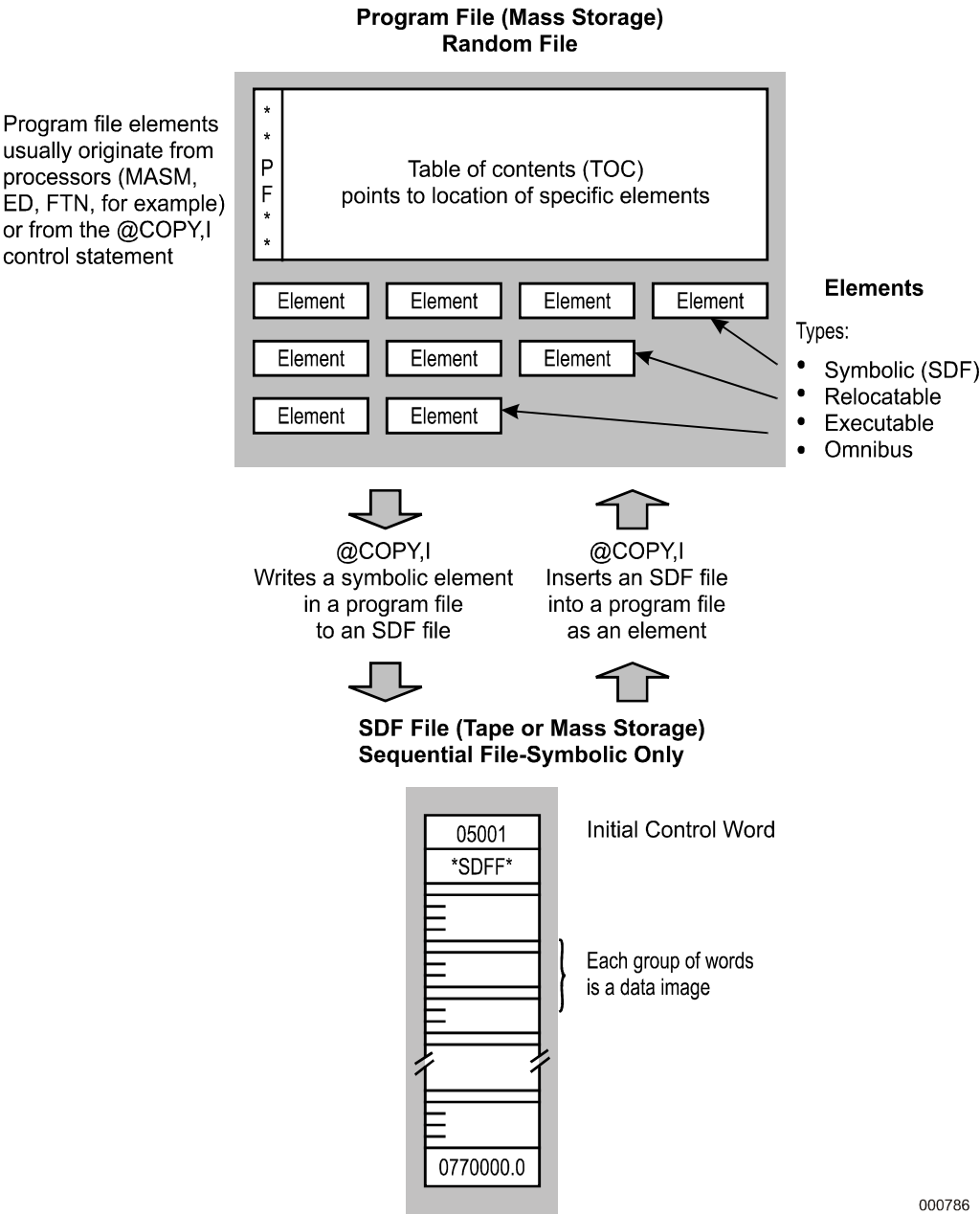
The large element program file (LEPF) format can contain symbolic and omnibus elements with text larger than 262,143 sectors. Relocatable, absolute, and symbolic procedure elements cannot be placed in an LEPF.

Creating, accessing, and updating LEPFs are supported by the SYSLIB Program File Basic Service Package (BSP) routines and by FURPUR and PCFP. LEPFs are not supported by the Exec Program File Package (PFP) or by the ECL commands that accept element names, such as @ADD, @CHKPT, @RSTRT, @START, and @XQT.

To specify that a file is formatted as an LEPF you must use the @CHG,E command. The @CHG,E command can be used only on newly assigned or erased files. See Section 7.5 for more information on @CHG.

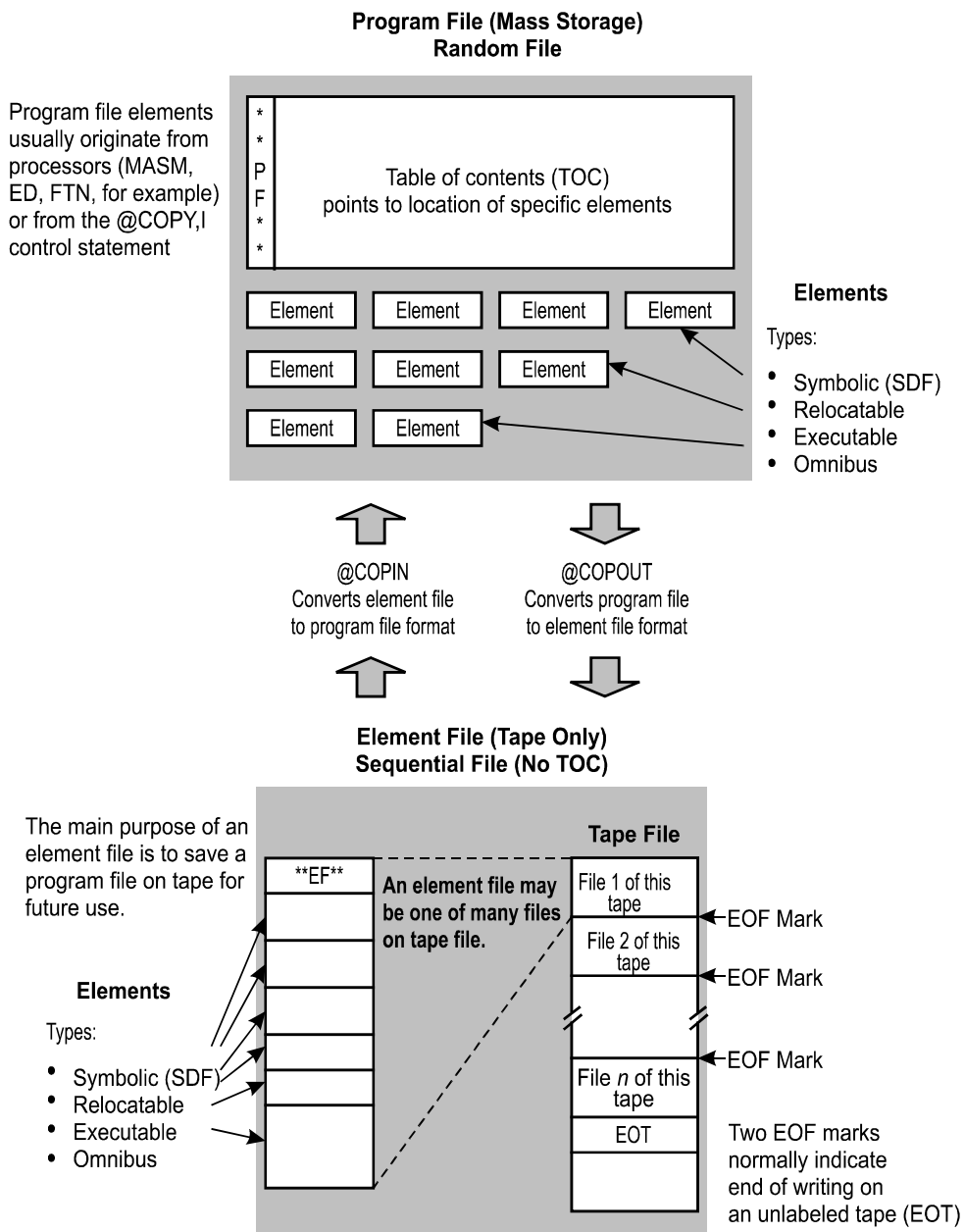
3.8.8. Transfers Between Files of Different Formats

Figure 3–5 illustrates the relationship of file formats to each other. The exact formats have been simplified for clarity. The statements illustrated are statements that transfer data between the indicated file types.



000786

Figure 3–5. Basic Formats Used to Transfer Data Between Files



000787

Figure 3-5. Basic File Formats Used to Transfer Data Between Files (cont.)

Figure 3–6 shows you the various ways to copy tape files.

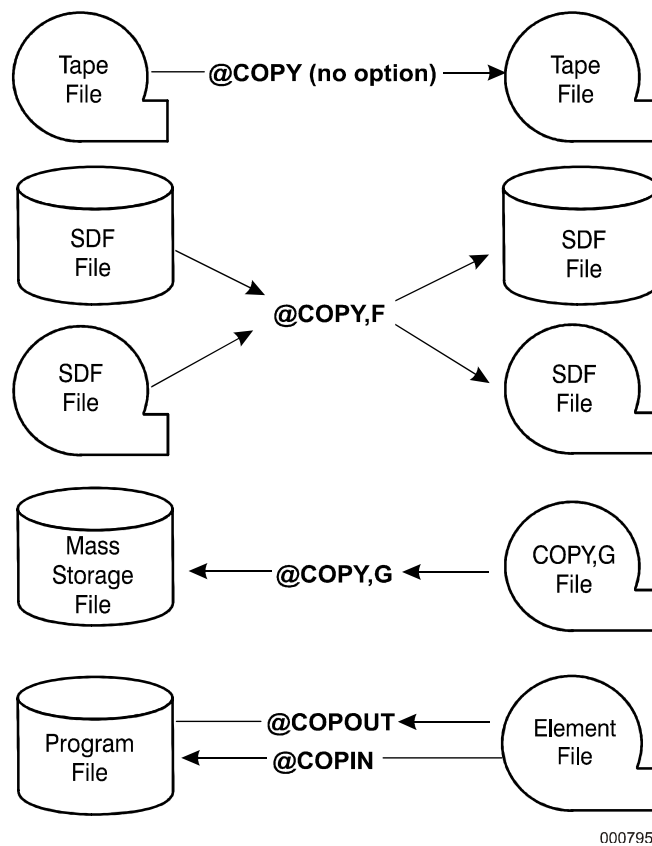


Figure 3–6. Statements for Copying Tape Files

Basic FURPUR Copy Operations

Figure 3–7 and Figure 3–8 summarize the various data formats, statements, and command objects that FURPUR used to make copies of a file or element on another (or the same) media.

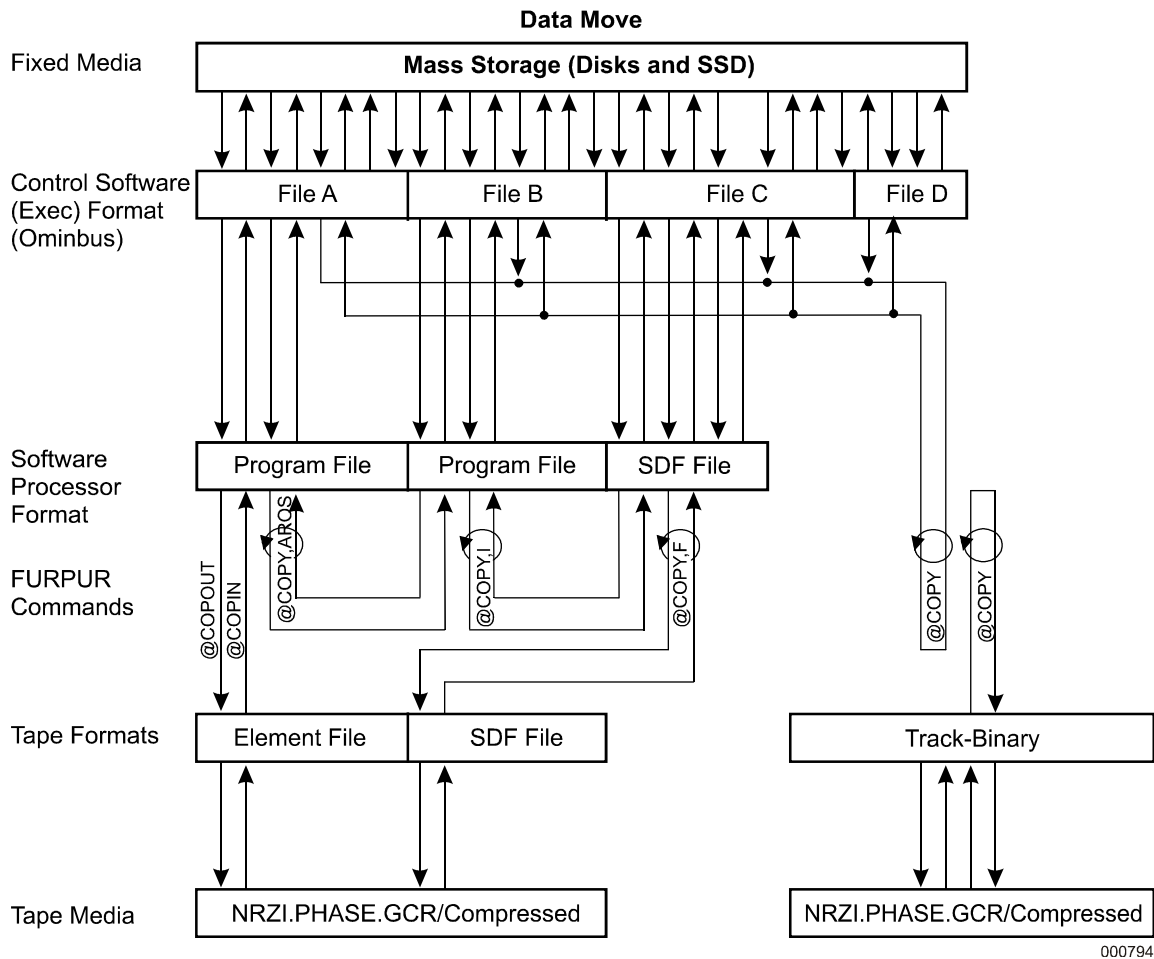


Figure 3–7. Data Paths, Data Formats, and FURPUR Commands

Transfer and Format Command	From								To							
	Mass Storage	Program File	SDF File	Omnibus	Tape	Element File	SDF File	Track Binary	Mass Storage	Program File	SDF File	Omnibus	Tape	Element File	SDF File	Track Binary
@COPY,G	X			X									X			X
@COPY,G					X			X	X			X				
@COPY,F	X		X										X		X	
@COPY,F					X		X		X		X					
@COPIN					X	X			X	X						
@COPOUT	X	X											X	X		
@COPY,AROS	X	X							X	X						
@COPY,I	X	X							X		X					
@COPY,I	X		X						X	X						
@COPY,IO	X	X							X		X					
@COPY,IO	X		X						X	X						
@COPY	X			X					X			X				
@COPY					X								X			

000790

Figure 3-8. FURPUR Data Transfer Command Options

Section 4

File Security

The OS 2200 security system ensures that only authorized users have access to information in your computer system. It protects sensitive data from both unauthorized use and inadvertent or unintentional disclosure.

Unisys offers security software in several levels:

- Fundamental Security
- Security Level 1
- Security Level 2
- Security Level 3
- Security Level 3 Controlled Access Protection Environment
- Security Level 3 Trusted Environment

This section gives you instructions on how to apply security mechanisms to your files. For an overall understanding of OS 2200 Security, see *ClearPath OS 2200 Software Security Overview* (8205 7282). For more details, refer to the *ClearPath OS 2200 Apex Help* (8207 4154).

The term "security management product" implies one of the following products:

- Apex is a web application for managing the OS 2200 operating system. It consolidates administrative tasks that have traditionally been performed using various Unisys provided and user developed utilities and processes, such as Security Administration for ClearPath OS 2200 (no longer supported) and Repository for ClearPath OS 2200 (UREP). Apex includes a command line utility, SECMGR, that was previously released as part of Security Administration for ClearPath OS 2200.

As an administrative user, you can manipulate the current state of the OS 2200 system or request historical reports of data from the OS 2200 system. From the Apex web application, you can perform the following:

- Administer all aspects of user, account, and file security.
- Run reports that support internal security audits and check for users who are out of compliance with your security policies.
- View and modify the configuration of the Apex OS 2200 agent, the Configured Password Profiles authentication module, the OS 2200 Exec, the security environment, and network authentication.

- o View information about the installed OS 2200 software products and the OS 2200 Exec.

As a non-administrative end user, you can use Apex to view your own environment.

As a database administrator, you can view the following:

- o Current UDS application group configuration
- o RDMS objects such as schema, table, and storage-area definitions.

For more information, see the *ClearPath OS 2200 Apex Help* (8207 4154).

- Site Management Complex (SIMAN) is the TeamQuest interactive software product that provides a tool with a screen-oriented, menu-driven interface for creating and maintaining security and resource control in an OS 2200 system. You can set up security for the whole system, for a particular application, or for personal security. For more information, see the *TeamQuest Site Management Complex (SIMAN) Administration and End Use Reference Manual* (TQ-01151).

4.1. MAC and DAC

File protection is divided into two categories: mandatory access control (MAC) and discretionary access control (DAC). The MAC and DAC security mechanisms you can use for a file depend on whether the file is owned or unowned, as shown in Table 4–1.

Table 4–1. File Protection

File Type	DAC Protection
Owned	Public or private access list or ACRs Read-only/write-only files
Unowned	Public/private by project-ID or account number Read-only/write-only files Read/write keys Owned or unowned ACRs (Options 1, 2, and 3 only)

4.2. Protecting Unowned Files

Files can be unowned in systems with Fundamental Security or Security Level 1, 2, and 3, except in a Security Level 3 Trusted Environment or a Security Level 3 Controlled Access Protection Environment, where the files must be owned. Unowned files can be protected using

- Read/write keys

Keys are similar to passwords that you need to know to read from a file or write to it. For example, if a file has a write key on it, you cannot delete it, erase it, or overwrite it unless you know the write key.

If your site has Fundamental Security or Security Level 1, 2, or 3 and the file is unowned, the following applies:

- When the file is cataloged, the G option guards against having its read key, write key, privacy mode, or all three overridden by privileged runs. Only the unconditional delete file system privilege overrides the G option.
- If any F-cycle of a file set is cataloged with the G option, the read or write keys for the file set are obscured by '/////////' from all users. If a particular F-cycle of a file set is cataloged with the G option, the privacy mode and either the project-ID or account number under which the file was created are also guarded.

- Read-only or write-only mechanism

You can create your files as read only or write only. If a file is read only, you can only read it; if a file is write only, you can only write to it. To make a file read only, specify the R option when you create it with the @CAT, @ASG,C or @ASG,U statement; to make a file write only, specify the W option. (See Section 7).

- Private by project-ID or account number

An unowned cataloged file can be private by project or account, depending on how your system is configured (check with your security officer or site administrator). Note that the definition of private for unowned files is different from the definition of private for owned files (private by owner).

To make a file public, use the P option on the @CAT, @ASG,C, or @ASG,U statement. To make a file private, omit the P option. A private file takes on the project-ID or account number that you are executing with during file creation.

Only users who have either the same account number or project-ID as the private file can access it.

- Owned or unowned access control records (ACR) (Security Levels 1, 2, and 3 only)

Unowned files are made ACR-controlled through SECMGR, Apex, Security Admin Client, or SIMAN, which must be used to attach an ACR automatically. See the *ClearPath OS 2200 Apex Help* (8207 4154) for more information.

4.3. Protecting Owned Files

If your system has Security Level 1, 2, or 3, files usually have owners. The user creating the file is its owner.

4.3.1. Mandatory Access Control

Files have security records that contain their security attributes. Those attributes include the file's security level, which is the combination of its clearance level and its compartment set.

When you create a file, it automatically takes on your executing clearance level and compartment set. (Compartment sets are available only in systems having Security Level 2 or higher).

The types of access a user is granted to a file depend on the comparison of the user's mandatory attributes with the file's mandatory attributes. For this reason, you must carefully consider your security level for a session before you sign on and create or access files (see the *ClearPath OS 2200 Apex Help* (8207 4154)).

The security system enforces the following validation rules for files:

- Read and execute access

The mandatory access rules allow read and execute access to a file if

- Your executing clearance level is greater than or equal to the clearance level of the file, and the file's clearance level lies within your clearance level range; or the clearance level of the file is zero.
- The compartment set of the file is contained in your executing compartment set. Note that NULL is always a member of a compartment set. This rule applies only to Security Levels 2 and 3.

- Write and delete access

The mandatory access rules allow write and delete access to a file if

- Your executing clearance level equals the clearance level of the file.
- Your executing compartment set equals the compartment set of the file. This rule applies to Security Levels 2 and 3.

Note: When a file has a clearance level of 0 and a null compartment set, it is considered "unclassified." Every user has mandatory read and execute access to unclassified files. You still must have read access through the access list attached. To have write and delete access, your executing clearance level must also be equal to zero and your executing compartment set must be null. This applies only to sites with Security Level 2 or higher. For sites with Security Level 1, an unclassified file is a file with a security clearance level of 0.

Examples of MAC Validation

1. Your clearance level range = 20-25
Your executing clearance level = 22
Your executing compartment set = personnel, payroll
File's clearance level = 18
File's compartment set = personnel, payroll

You are denied access because the file's clearance level is less than your minimum clearance level.

2. Your clearance level range = 20-25
Your executing clearance level = 22
Your executing compartment set = personnel, payroll
File's clearance level = 22
File's compartment set = personnel, payroll

You have read, execute, delete, and write access to the file because your executing clearance level equals the file's clearance level, and the file's compartment set is the same as your executing compartment set.

3. Your clearance level range = 20-25
 Your executing clearance level = 25
 Your executing compartment set = personnel, payroll, financial
 File's clearance level = 20
 File's compartment set = personnel

You have read and execute access to the file because your executing clearance level is greater than the file's clearance level, which is equal to your minimum clearance level. The file compartment set is a subset of your executing compartment set.

4. Your clearance level range = 20-25
 Your executing clearance level = 22
 Your executing compartment set = personnel, payroll
 File's clearance level = 20
 File's compartment set = financial, accounting

You are denied access to the file because its compartment set is not a subset of your executing compartment set.

4.3.2. Discretionary Access Control

Over and above the protection provided by mandatory controls, you can give your files protection with discretionary controls, either with default or explicit action.

When creating a file, you must decide whether to give it a public, private, or ACR-controlled access list. ACR-controlled files can be considered as having semiprivate access as they can typically be accessed by a limited number of people apart from the owner. The type of access list you choose depends on who needs access to your file. Table 4–2 shows the three access list types, how to give them to a file, and who has access.

Table 4–2. Access List Type

Access List Type	How to Catalog	Who Has Access	Allowed Access
Public	@CAT,P @ASG,CP, or @ASG,UP	Any valid user	Read, write, execute, and delete
Private	Omit P option on @CAT, @ASG, or @ASG,U	Owner only	Read, write, execute, and delete
ACR-controlled	Catalog public or private and attach an ACR	Owner	Read, write, execute, and delete

Table 4–2. Access List Type

Access List Type	How to Catalog	Who Has Access	Allowed Access
		Other users as you specify in the ACR	You determine the types of access given to specified users by using an ACR.

Access to a file is allowed only if a user passes both mandatory and discretionary control validation.

Always create files with the least amount of access permitted, and relax restrictions only if necessary. That is, always create private files and then change to ACR-controlled or public if necessary.

Access Control Records

An access control record (ACR) allows you to restrict access to your files or other objects. An ACR identifies the users that have access to a file and the access types permitted. For example, an ACR could be used to limit user-ID XYZ or Account ABC to read access.

An ACR can specify one or more of the following access types:

- Read
Allows full read access to the file and automatically grants execute access.
- Write
Allows write access to the file.
- Delete
Allows deletion of the file.
- Execute
Permits access to programs in the file.
- Delete ACR
Allows deletion of the ACR if it is unowned.
- Modify
Permits modification of the ACR if it is unowned.

The ACR also identifies the users who can access the file using one or more of the following types:

- User-ID
- Account
- Project
- Time of day

The relation parameters EQ (equal) and NE (not equal) are used to include or exclude users from the group that is allowed to access the file. The following expression includes users assigned to account 123456 and excludes all others:

```
ACCOUNT EQ 123456
```

The relation parameter NE excludes account 123456 and includes all others:

```
ACCOUNT NE 123456
```

Caution

Be careful when placing the project-ID Q\$Q\$Q\$ in any ACR. This project-ID is available to all users on the system. You can easily compromise the security on your file even though you create other restrictions.

The key words AND and OR are used to connect the parts of complex expressions.

Separate the parts of a complex expression with OR when only one of two or more conditions must be true. The expression

```
ACCOUNT EQ 123456 OR PROJECT EQ PROJECT-X
```

grants access to any user associated with account 123456 or project X. This group consists of all users who meet one or both conditions.

Use the word AND to connect expressions that must both be true. The expression

```
ACCOUNT EQ 123456 AND PROJECT EQ PROJECT-X
```

grants access to users who meet both conditions. It excludes users who meet only one or neither of the conditions.

Parentheses are required in some complex expressions. The expression

```
PROJECT EQ X OR PROJECT EQ Y AND USERID EQ BJSMITH
```

is ambiguous and requires parentheses for precise interpretation.

The parentheses are placed as follows if user BJSMITH is to be given access only if using project X or project Y:

```
(PROJECT EQ X OR PROJECT EQ Y) AND USERID EQ BJSMITH
```

In this instance, both conditions must be true. Users of project X or project Y that have a user- ID other than BJSMITH are excluded.

A different placement of parentheses alters the meaning of the expression:

```
PROJECT EQ X OR (PROJECT EQ Y AND USERID EQ BJSMITH)
```

In this instance, there are two conditions that are independent of one another. All users of project X are included, as are all users of project Y who have the user-ID BJSMITH.

You can also designate a file public, which lets any user-ID have the access types associated with the condition.

Creating an ACR

You can create an ACR for your own user-ID and, if you have sufficient privilege, for other user-IDs. The security officer for your site determines the number of ACRs that can be created for your user-ID. When you create an ACR, the number allocated to your user-ID is decreased by one. You must always have at least one ACR available for your user-ID.

Use your security management product. See the *ClearPath OS 2200 Apex Help* (8207 4154) or *SIMAN Administration and End Use Guide* (TQ-01151) for additional information on creating ACRs.

Attaching an ACR to a New File

To attach an ACR to a new file, specify the ACR on the @CAT, @ASG,C, or @ASG,U statement.

For example, attach the ACR named RDPUB to the file DEPT*PERSONNEL:

```
@CAT DEPT*PERSONNEL,,,,,RDPUB
```

For more information on attaching an ACR see the @ASG (Section 7.2) and @CAT (Section 7.4) statements.

Attaching an ACR to an Existing File

You can attach an ACR to an existing file or replace an ACR with a new ACR by using the following:

- @CHG statement, as in the following example:

```
@CHG,K filename.,ACR-name
```

- SECMGR keyword syntax interface as in the following example:

```
@SECMGR
Update file = filename.
Attached_ACR = ACR-name
ACR_Owner = your-userid
```

- Apex Security → Files → <select file> → General → Access
- Security-Admin Client Update Catalogued File Security
- SIMAN keyword syntax interface as in the following example:

```
@SIMAN,B
Update file = filename.
Attached_ACR = ACR-name
ACR_Owner = your-userid
```

- SIMAN full-screen mode, Cataloged File Security Maintenance menu.

See the *ClearPath OS 2200 Apex Help* (8207 4154) and your security management product documentation for additional information on attaching ACRs.

Deleting an ACR

You cannot physically remove an ACR from the system. Deleted ACRs are marked as not in use. Deletion of an ACR does not increase the number allocated to your user-ID. You can recover a deleted ACR by retrieving it with your security management product and updating its contents.

Updating an ACR

The ACR name and owner cannot be changed, but there are no other restrictions for owned ACRs. You cannot update an unowned ACR. You can recover an ACR that is marked for deletion by updating the ACR.

See the *ClearPath OS 2200 Apex Help* (8207 4154) or the *SIMAN Administration and End Use Guide* (TQ-01151) for additional information on updating an ACR.

Note: *Modifications to the content of an ACR have no effect on users who have current access to a file or other object. If the changes to the ACR remove a current user's access, the restriction does not take place until the file is reassigned. The only exception is a TIP user who has accelerated access to a TIP file. If the user's access is revoked by changing the ACR attached to the TIP file, access is denied the next time the user attempts to access the file.*

Making Files Read Only or Write Only

The read-only/write-only mechanism, when applied to files at the time they are created, does not protect files from deletion, but it does prevent someone from inadvertently or intentionally reading or overwriting them. Create files as read only or write only by using the R or W option on the @CAT, @ASG,C, or @ASG,U statements.

Do not use the read-only/write-only mechanism as a substitute for ACRs. The read-only/write-only mechanism protects only a single cycle of a file set, but ACRs protect an entire file cycle set. Read-only mode prevents users from writing to a file cycle. Write-only mode prevents users from reading a file cycle.

You might want to access a previously created file as read only, so you do not accidentally destroy data in it or delete it.

Use @ASG,AR to assign a previously cataloged file as read only. The file could be originally cataloged as not read only, but you can still assign it as read only by using the R option. The A option indicates that the file is already cataloged and ensures that it is not treated as a temporary file. The R option write protects the file and prevents its deletion.

The read-only assign mechanism merely reinforces read accesses specified in an ACR or the read access usually allowed by the owner of the file.

4.4. Changing Your File's Protection

You might need to change a file's protection so that other users can see the data, or because the security sensitivity of the file's data has changed.

You can change some security attributes of a file you own; you must have a privilege to change other attributes.

4.4.1. Modifying Mandatory Attributes

To modify a file's clearance level or compartment set, or to set the ALL flag (gives the file the system compartment set and all future system compartment sets), you must be the file's owner and have the privileges to change a file's security record.

See your security officer if you want to make any of these modifications.

4.4.2. Changing Access Lists

To change a file's access list, use the @CHG statement with the K, P, or Q option, as follows:

- @CHG,K
Sets ACR-controlled access list (you must include the *ACR-name* parameter).
- @CHG,P
Sets public access list.
- @CHG,Q
Sets private access list.

Another method of setting the ACR-controlled access list is by using your security management product.

4.4.3. Changing or Clearing the Read-Only or Write-Only Mechanism

To change or clear the read-only or write-only mechanism on a file, use the @CHG statement with the appropriate option:

- @CHG,V
Sets the file as read only; clears write only.
- @CHG,W
Sets the file as write only; clears read only.
- @CHG,Z
Clears the file as read only and write only.

4.4.4. Changing Other File Attributes

To give your file a different owner, the security attributes of the new owner must be a superset of the file's attributes and a subset of your attributes. In other words, the file's attributes must be a subset of the new owner's attributes. To set special handling flags, you must have certain privileges. See your security officer if you want to make these modifications.

Section 5

Getting On and Off the System

This section explains how to

- Open a session
- Access the system, with information on
 - The three application mode types
 - The three sign-on modes
 - The sign-on request and your response to it (entering a user-ID/password and clearance level, if your system requires you to do so)
 - Entering an account number in demand
 - Entering a project-ID in demand
 - Entering compartment set information in demand
 - Working with an Authentication and Session Initiation Subsystem (ASIS)
- Select enter, and change passwords in demand runs
- Start a demand run
- Start a batch run
- Use passwords within batch runs (also includes entering clearance level and compartment set in batch)
- Identify the end of a run
- Sign off from the system and close a session

Notes:

- *Read the ClearPath OS 2200 Software Security Overview (8205 7282) for an overview of the security system.*
- *Consult your UTS terminal emulator documentation on how to configure a host and access it.*

This section describes the following ECL statements.

Statement	Purpose	Reference
\$\$OPEN	Establishes a session with the system you specify	5.1
\$\$CLOSE	Closes an application session	5.5

5.1. Opening a Session (\$\$OPEN)

You can start a communications network session in any of several ways, depending on the configuration of your UTS terminal emulator. Table 5–1 lists the possible setups and describes how to sign on for each setup.

Table 5–1. Terminal Sign-On and Session Establishment Procedures

Terminal Setup	To Sign On Terminal	To Establish Session
Automatic sign-on with manual session establishment	Transmit.	Type \$\$OPEN and transmit.
Automatic sign-on and session establishment	Transmit.	Session automatically established to the default application program.
		Type \$\$OPEN and transmit only when you want to change sessions or establish a new session.
Immediate sign-on	Sign-on occurs at initialization time.	Session automatically established to the configured application program.
		Type \$\$OPEN and transmit only when you have configured the application program.

Your terminal emulator automatically connects to the communications network. Depending upon its configuration, you may skip opening a session path and go directly to the login prompt, or you might see this message:

```
ENTER SESSION ESTABLISHMENT REQUEST:
```

Enter the \$\$OPEN statement in response to this message.

\$\$OPEN Statement

To open a session with an application, enter

```
$$OPEN system-id
```

The system-ID identifies the system that you want to access. Your supervisor or site administrator can tell you the system-ID. After typing the \$\$OPEN statement, the system asks you for your user-ID and password.

When the communications network establishes your session, you might get this message:

```
SESSION PATH OPEN TO: application-name
```

or

```
SESSION PATH OPEN
```

5.2. Accessing the System

The amount and type of information you supply the system when signing on depends on how your system and your user-ID are configured. The following subsections tell you what kinds of information the system can solicit and how you supply the information.

Before you sign on the system, you need to know the type of applications you can perform. The application modes are as follows:

- Demand (interactive) run
- Batch runstream
- Transaction Processing (TIP)

Your user-ID can be configured to access the system only in certain modes. Your site administrator or security officer determines which application modes your user-ID is allowed to access.

5.2.1. Demand Sign-On Modes

When your security officer or site administrator registers your user-ID with the system and lets you work in demand, you are assigned one of three sign-on modes for accessing the system in demand. The three modes are just different ways to begin a terminal session. The difference lies in how much information you supply the system on the @RUN statement and how much information the system already has at hand.

These are the three sign-on modes:

- User entered @RUN statement

You must enter an @RUN statement after signing on. See Section 7.55 for information on the @RUN statement.

- System generated @RUN statement

The Exec automatically uses the @RUN statement configured for your user-ID. You do not have the option of starting your own run by entering an asterisk before your user-ID. If you try to start your own run, the system ignores the asterisk.

- User entered or system generated @RUN statement

You do not have to enter an @RUN statement to start a demand run; the system does that for you, using the information configured for your user-ID. You can bypass the automatic @RUN statement if you want to enter your own @RUN statement. To bypass the automatic @RUN statement, enter an asterisk (*) before your user-ID when responding to the sign-on request.

Your site administrator can configure your user-ID for automatic @RUN statement generation so that a program such as Interactive Processing Facility (IPF) is automatically started when you sign on, or so that your run-ID is different from your user-ID.

If your site requires you to have a user-ID and password, the system displays one of the following messages, depending on whether or not your system requires a clearance level:

```
Enter your user-id/password and clearance level:
```

```
Enter your user-id/password:
```

Precede your user-ID with an asterisk (*) if you want to prevent the system from automatically generating an @RUN statement for you.

For example, this is how you bypass the automatic @RUN statement if your user-ID has a configured @RUN statement but you want to enter a different @RUN statement:

```
Enter your user-id/password and clearance level:
*user-id/password clearance-level
*UNISYS 1100 Operating System Lev. 49R3
@RUN,/B run-id,acct-id/user-id,...
*BATCH RUN MODE - ENTER RUN STREAM*
```

This example shows a batch mode run starting, but bypassing the automatic @RUN statement is done the same way when signing on to the system to use demand mode. At this point, enter the rest of your runstream.

5.2.2. Signing On to the System in Demand

Your user-id might be configured for single sign-on so that you do not have to enter a user-ID and password combination. In this case, the multi-factor authentication that established your identity at your workstation combines with Kerberos, NTLM, or TLS/SSL certificate authentication to securely identify you to the OS 2200 system. If your user-ID is not configured for single sign-on, you must sign on to the system..

The operating system issues the following sign-on requests:

Enter your user-id/password and clearance level:

Enter your user-id/password:

Format

The format of your response to the sign-on prompt is

```
[*]user-id/password[/new password] [clearance level]
```

Enter an asterisk (*) before the user-ID only if you are allowed to enter your own @RUN statement instead of using the one configured for your user-ID.

user-id

The user-ID is a unique identifier that your site administrator or security officer assigns to you. It can be up to 12 characters in length and can consist of the characters A through Z, a through z, 0 through 9, the hyphen (-), and the period (.).

The slash (/) is the delimiter character used to separate the user-ID from the password.

password

Your password is a character string that you must specify when you sign on to the system.

The first time you sign on to the system, use the password assigned to you by the site administrator or security officer. Change your password immediately after the initial sign-on to prevent unauthorized use of your user-ID.

The procedure for changing your password depends on the configuration of your system. If the machine-generated password feature is implemented at your site, the system generates a new password for you when you request a change. If this feature is not in use, you can specify your new password.

See Section 5.3 for more information on changing your password.

clearance level

If your system has Security Level 1 or higher, the security officer establishes the clearance levels for your system. They can consist of numbers between 0 and 63 or of symbolic names. (Symbolic names are available only with Security Levels 2 and 3.)

You can supply a clearance level after you enter your password (or a new password). You have a clearance level range associated with your user-ID, but you can run with only one clearance level at a time.

If you do not specify a clearance level, the system gives you the lowest level in your clearance level range, called the minimum clearance level.

If you enter an illegal clearance level, you receive an error message and the system gives you the following prompt for a new clearance level:

```
Enter your clearance level (0-63; <XMIT>=default):
```

5.2.3. Sign-On Security Measures

If you enter an illegal user-ID or password or both when signing on, the system response depends on how the system is configured:

- You can continue receiving the sign-on request.
- The system ignores any sign-on requests after a configured number of incorrect user-ID/password combination attempts occurs, but the system continues soliciting entries.
- You can experience an increasing delay time between each successive invalid attempt to sign on with each successive prompt taking about 4 times as long as the previous one, and after the maximum allowed attempts, the system disables your terminal. Once a site-ID has been disabled, the sign-on prompt is not displayed to the terminal. To use the terminal again, the system operator must re-enable it.

Approximate delay times are displayed in the following table:

Number of Invalid Attempts	Delay Time
1-3	Almost unnoticeable
4	64 seconds
5	4 minutes
8	Terminal disabled

Depending on the configuration of the terminal emulator you are using, when you type in your sign-on response, the characters you type at the keyboard might not show up on your screen. This is a security measure to prevent someone from looking over your shoulder and stealing your user-ID and password.

For demand sign-on, the following message indicates that you have successfully completed the sign-on process:

```
*UNISYS 1100 Operating System Level 49R3 ...
```

5.2.4. Providing Compartment Information in Demand

If your site has Security Level 2 or higher, your system can have an added security feature called *compartments*. The compartment feature places groups or categories of information from the same interest or topic area into compartments, and then limits access to those compartments. For example, compartments on your system can include medical, legal, personnel, financial, or payroll. Several compartments can be associated with one file.

Compartment Sets

Table 5–2 shows the four types of compartment sets you need to know about.

Table 5–2. Compartment Sets

Compartment Set	Definition
System	All the compartments on the system.
User	All the compartments listed in your security record. The user compartment set is a subset of the system compartment set.
Executing	The compartment set with which you are executing. This can be your entire user set or a subset of it.
Default	The compartment set the system automatically uses if you do not specify any compartments when you sign on. Either you or your security officer can set up your default set.

Your security officer specifies your user compartment set in your user-ID's security record. The system checks this record whenever you sign on.

You can access data associated with any compartment that is specified in your user compartment set, as long as the compartment is also in your executing set.

Rules for Entering a Compartment Set

Your system can be configured so that you do not receive a solicitation for compartments, but instead, you automatically receive a default set established for you. Your administrator can change your default to a different set by using your security management product, as long as the default is a subset of your user compartment set.

Your system can be configured so that you receive the following solicitation for your compartment set after you sign on:

```
Enter compartment set (or NULL, MAX, <XMIT>=default):
```

You can respond in several ways:

- Enter the compartment set you choose as your executing compartment.
- Enter NULL if you only want to access system resources that have no compartments associated with them.
- Enter MAX if you want your user compartment set as your executing compartment set. Your security officer might choose to define your user compartment set as the entire system compartment set.
- Press the <XMIT> key if you want your default compartment set.

Follow these rules and examples when entering your user compartment set:

- A comma must separate the names. Spaces are allowed before or after the commas. You can use uppercase or lowercase characters or a combination.
- If the names all fit on one line, transmit after the last name. The system then checks the compartment names you entered against your security record. If the names you listed are included in your record, validation passes.
- If the names do not fit on one line, use a semicolon (;) as the continuation character. Press XMIT after entering each line of compartment names. Each time you transmit, the system checks the names you listed against your security record.

As long as you keep entering the continuation character, and the compartment names on each line are in your record, the system responds with the following prompt:

```
Continue entering compartment set (or <XMIT>):
```

- When you are finished entering compartment set names, press XMIT after the last line (do not include the continuation character). If you discover, after you have already transmitted a line with the continuation character, that you are through entering compartment names, press XMIT without any names in response to the "Continue entering compartment set (or <XMIT>)" prompt. Compartment set validation is now complete.
- If you enter a name incorrectly, or if the compartment you listed is not in your security record, you receive the following message:

```
Input was ignored. An illegal compartment was entered.
```

Either the "Enter compartment set (or NULL, MAX, <XMIT> = default)" prompt or the Continue entering compartment set (or <XMIT>): prompt is displayed, depending on whether you have just entered your first line of compartment names or were transmitting a continuation line. You can then correct the line that was in error by retyping all the compartment names on that line. You do not need to retype compartments validated on previous lines.

Example

In this example, the user executes with FINANCIAL, PAYROLL, LEGAL, MEDICAL, and PROCUREMENT, but not with PERSONNEL.

```
Enter compartment set (or NULL, MAX, <XMIT> = default):  
FNANCIAL, PAYROLL, LEGAL  
Input was ignored. An illegal compartment was entered.  
Enter compartment set (or NULL, MAX, <XMIT> = default):  
FINANCIAL, PAYROLL, LEGAL;  
Continue entering compartment set (or <XMIT>):  
PERSONNEL, MEDICAL, PROCUREMENT  
Input was ignored. An illegal compartment was entered  
Continue entering compartment set (or <XMIT>):  
MEDICAL, PROCUREMENT
```

- If you enter a compartment name and it is validated, but your security officer happens to change the name while you are entering other compartment names, the following error message is displayed:

```
A compartment name you entered is no longer in use. Please start over.
```

You are then given this prompt:

```
Enter compartment set (or NULL, MAX, <XMIT> = default):
```

and you must start over your entering of compartment names.

Once you complete the sign-on process, any file or tape volume you create is labeled with your executing compartment set. You and any other user accessing the file or tape volume must be executing with at least this set of compartments to access the information.

For batch runs, you provide compartment set information on the @PASSWD statement (see Section 7.43).

5.2.5. Providing an Account Number in Demand

After you sign on and provide compartment information (if required), you might be prompted for an account number before you or the system generates the @RUN statement.

If you are not prompted for an account number, the system automatically takes the account number the site administrator or security officer has placed in your user-ID security record, or it takes the default account number 000000; however, you can still override the account number in a demand session.

When you are prompted for an account number, your response depends on whether you are allowed to use only one account number or can choose from several or all of the valid account numbers on your system.

If you can use only one account number

- You cannot get any prompts, and the system automatically supplies your default account number.
- You can receive the following prompt:

```
Enter your account number:
```

Either type in the number and press XMIT, or just press XMIT to receive your default account number.

If you can choose from several account numbers

- You can receive the account number prompt. Enter any valid account number. For example

```
Enter your account number:
```

```
599210
```

If a default account number resides in your user-ID security record, you can accept your default by pressing XMIT. You might have the ability to change your default, depending on the capabilities given to you by your security officer.

- You can be asked to select one from an index of up to five account numbers. The index number corresponds to the account numbers as displayed on your security management product screen. If you receive the account number prompt, press XMIT, and then you receive another prompt that asks you to choose a number from the account index. For example

```
Enter your account number:
```

```
(XMIT)
```

```
Choose your account index (1-5):
```

```
3
```

If you receive only the account index prompt, answer it by typing an index number. See Section 7.55 for information about supplying an account number on the @RUN statement.

5.2.6. Providing a Project-ID in Demand

A project-ID is a method of classifying a run for file access. If your system is configured for project-IDs, how you supply a project-ID depends on whether your @RUN statement is automatically generated by the system or whether you enter the @RUN statement yourself.

If you sign on, but are entering your own @RUN statement, you must provide the project-ID the @RUN statement (see Section 7.55). You must also supply a project-ID on the @RUN statement if you are entering a batch run using @RUN with the B option.

If your @RUN statement is automatically generated, you might have to supply a project-ID when signing on the system. Depending on how your user-ID is configured, you supply the project-ID by using one of the following methods.

- Select a project-ID from a list of up to five project-IDs. For example

```
1. PROJA 2. PROJB 3. PROJC 4. PROJD 5. PROJE
Choose your project-identifier index
(1-5; <XMIT> = Q$Q$Q$):
```

The range of indicators on the second line corresponds to the number of project-IDs listed on the first line. Your response to the prompt must be the one-digit number of your selected project-ID (spaces are ignored).

If you respond to the project-ID index prompt incorrectly, the project-ID list and prompt are displayed again.

If you press XMIT without entering a number, the system gives you default project-ID Q\$Q\$Q\$.

- If you receive a project-ID prompt that displays only the default project-ID your security officer has established for your user-ID, either accept the default by pressing XMIT or enter any valid project-ID on the system. For example

```
Enter your project-identifier (<XMIT> = PROJA):
ABC
```

The project-ID displayed after <XMIT>= is the default (PROJA in the example). If you have no project-IDs assigned, the project-ID displayed is Q\$Q\$Q\$. You can respond by either transmitting a blank line (accepting the default) or entering a project-ID (ABC in the example).

If you respond to the project-ID prompt incorrectly, the project-ID prompt is displayed again.

5.2.7. System Response

After you complete the sign-on process, you might receive one or more "messages of the day" on your terminal; this is one method the site administrator uses for contacting all the users of your system to inform you of special developments concerning your system, or other appropriate information.

Following the messages of the day, the system informs you of your current session number and the date and time of your last sign-on to the system.

Your current session number is the total number of valid sign-on attempts to the system for your user-ID. Initially 0, this value can be reset to 0 after you have made 999 valid sign-on attempts.

5.2.8. Authentication and Session Initiation Subsystem (ASIS)

Demand sign-ons may be affected if ASIS is used. See the *User Authentication Administration Guide* (7850 4586) for more information.

Site Does Not Have ASIS Registered

If ASIS is not registered, session initiation proceeds as normal, unless a user-ID with a nonzero authentication type is encountered. If the Exec is set to allow traditional authentication passwords, all users with traditional passwords can sign on.

If the Exec is set to not allow traditional authentication passwords, only the security officer is allowed to sign on using a traditional authentication password. Other user receives this error message:

```
The Authentication Subsystem is not available. This sign-on is rejected.  
Contact your site administrator.
```

Site Has ASIS Registered

If ASIS is registered, it processes the sign-on attempt with the help of authentication modules (AMs) that have been linked into ASIS. ASIS can respond with several different statuses:

- If ASIS declines to authenticate the user-ID/password, then the Exec processes the user-ID/password text that ASIS has solicited.
- If ASIS processed a successful authentication, then it has several ways to indicate the clearance level to use. If ASIS returns a valid clearance level, then there is no further solicitation. ASIS can also ask the Exec to supply the default (minimum) clearance level for the user-ID. ASIS can also supply no clearance level, in which case the Exec prompts the user with the message "Enter your clearance level."

The Exec and ASIS cooperate to make sure that the user-ID exists, that it has not been disabled previously and that the user-ID has access to demand.

- If ASIS processed the sign-on attempt and returns a failed authentication with no retry, the Exec performs the usual hacker frustration on the user-ID and the terminal, and then may terminate the session.
- If ASIS processed the sign-on attempt and returns a failed authentication with retry, then the Exec performs the usual hacker frustration on the user-ID and the terminal. The Exec does not terminate the session unless the maximum attempts have been reached, but allows ASIS to process the next sign-on attempt.

If the user authentication failed, the Exec terminates the session. If it has failed due to bad user-ID, the Exec generates this console message:

```
Sign-on security error at site: terminal-id attempt # nnnnn userid: uuuuuuuuuuuu
```

and generates a log entry for this bad user-ID sign-on.

If the ASIS returned a message saying that authentication failed for any other reason and the Exec is configured to allow terminal messages, then the Exec displays the following message:

```
You have entered invalid authentication data.
```

If authentication retry is allowed, the Exec logs the authentication failure and institutes hacker frustration as configured.

If session initiation processing requires additional information, then the user receives the regular solicits for clearance level, compartment set, account number or index, and project-ID or index.

ASIS Not Available

If a demand mode user tries to sign on while ASIS is temporarily not available, the Exec sends to the terminal the message

```
The Authentication Subsystem is not available. This sign-on is rejected.  
Contact your site administrator.
```

The session is not immediately terminated, but session timing continues and the session might eventually time out. The demand user can enter a @@SOLI command that causes a traditional solicit for a user-ID/password. All users who are allowed to enter traditional authentication passwords can sign on. If traditional authentication is not allowed, only the security officer is allowed to sign on using a traditional user-ID/password.

5.2.9. Signing On to TIP

If you are a TIP user, you might need to sign on to TIP before running any application programs. Signing on to TIP is similar to signing on to demand. If your user-ID is not configured for TIP access, you see the following message after entering your user-ID and password:

```
Your user-id has no access to TIP.
```

When you sign on to TIP, the system establishes a TIP session during which you can enter input to application programs. The session continues until you sign off the system or until you time out by failing to enter input. If your system requests that you sign on with a user-ID/password, then TIP Session Control is configured for your system. While TIP Session Control is active, you cannot override the default account number for that TIP session.

If you have the appropriate security attributes in your security record, you can change your timeout period with the @@TOUT statement (see Section 7.65). During a TIP session, @@TOUT is the only transparent statement that you can enter.

ASIS

ASIS affects TIP sign-ons the same way as demand sign-ons. See the earlier subsections "Site Does Not Have ASIS Registered" and "Site Has ASIS Registered". For more information, see the *User Authentication Administration Guide* (7850 4586).

ASIS Not Available

If a user tries to sign on to TIP while ASIS is temporarily not available or not accepting new authentication requests, the Exec sends to the terminal the message:

```
The Authentication Subsystem is not available. This sign-on is rejected.  
Contact your site administrator.
```

The session is immediately terminated.

5.3. Changing Passwords in Demand

Passwords are an important mechanism in your security system. Your password restricts the use of your user-ID and helps prevent unauthorized access to the system. You must specify a password when you sign on to the system unless your user-ID has been configured to use single sign-on, in which case you have already been authenticated to your PC and your authorization and identity are securely passed to the OS 2200 host.

At most sites, you must change your password at regular intervals. Changing your password is an important part of maintaining a secure environment. You can receive a message indicating that your password has expired and a prompt to choose a new password. Your user-ID can be disabled if you fail to change your password when prompted to do so. Only you or an administrator with permission to change your user-ID record or the security officer for your site can change your password.

Your system is configured for one of the following options:

- Machine-generated passwords
- User-specified passwords

The procedure for changing your password depends on which option has been implemented at your site. Check with your security officer if you do not know which option is implemented at your site.

5.3.1. Changing a Machine-Generated Password

If the machine-generated passwords feature is configured, your system generates a password when you request a new password at sign-on.

Use the following statement to request a new password:

```
userid/old_password/
```

The second slash in the statement signals the system that you want a new password. The system ignores any characters after the second slash because only system-generated passwords are permitted.

The system responds to a request for a new password as follows:

```
Your new password is: xxyyzzzz (xx-yy-zzzz)
Enter your new password (or transmit to request another password):
```

The new password can consist of 1 to 18 characters, depending on system configuration.

You can now enter the new password to verify the change, or press the transmit key, to request generation of another password. If the new password is successfully entered at the prompt, the system displays the following message:

```
Your password has been replaced.
```


If the user enters the new password incorrectly in response to the prompt, the following message is displayed:

Re-enter your generated password (or transmit to request another password):

If a user-ID is allowed to ignore solicitation for a new password, the following message is displayed:

Do you wish to change your password?

If the user enters Y or presses the transmit key, a new password is displayed. If N is entered, sign-on continues with no change to the password.

Note: You cannot use the @@PASSWORD statement to change your password if the machine-generated passwords feature is configured at your site.

5.3.2. Changing a User-Specified Password

If the machine-generated passwords feature has not been configured at your site, you must specify a new password when you make a change. You can change your password when you sign on to the system, or you can use the @@PASSWORD statement (see Section 7.43), unless the system configuration prohibits the use of @@PASSWORD.

Your security officer specifies the length of passwords on your system (1 to 18 characters if the Exec passwords are used, or 1 to 32 if your user-id uses Configured Password Profiles, one of the ASIS AMs) and how often you must change your password.

5.3.3. Guidelines for Choosing a Password

Follow these guidelines when choosing a new password:

- Choose a password you have never used before.
- Avoid repetition of characters.
- You can use any characters on the keyboard for your password combination except the slash (/), comma (,), and blank () characters.
- Use variable lengths for your series of passwords, which makes them harder to guess.
- Remember that long passwords are more secure than short ones.
- You cannot use the same sequence of characters as your user-ID for your password.
- Avoid initials and names that others might connect with you.
- If you must record your password, lock it up.
- Do not reveal your password to anyone.
- Change your password if you think some else knows it.
- Do not sign on if someone is looking over your shoulder.

If your user-ID's authentication method is Configured Password Profiles, you must comply with additional rules established by your security officer. For instance, you might not be allowed to reuse a number of past passwords or your new password might have to have a combination of upper and lower case letters, numbers, and special characters.

5.3.4. Changing Your Password at Sign-On

You can change your password when you sign on to the system by specifying a new password in the statement as follows:

```
userid/old_password/new_password
```

If the system accepts the new password, the following message is displayed:

```
Your password has been replaced.
```

If your user-ID uses Configured Password Profiles authentication, you might have to enter your new password twice, as a safeguard against a typing mistake becoming your new password.

5.3.5. Changing Your Password Using @@PASSWORD

The @@PASSWORD statement lets you change your password in demand mode on all systems where the system configuration allows it. Attempts to change your password on a system that does not allow @@PASSWORD returns the error

```
System configuration disallows password changes by @@PASSWORD.
```

The format is

```
@@PASSWORD old-password/new-password
```

For example

```
@@PASSWORD QMKE/PAX01
```

changes your password from QMKE to PAX01.

Caution

The password specified in an @@PASSWORD statement is visible to observers. Security can be compromised if you execute this statement. The most secure way to change your password is during sign-on.

You can change your password when you are working in batch mode by using the @@PASSWORD control statement (see Section 7.43).

5.4. Run Termination

Run termination is a state that can be reached either normally or abnormally.

5.4.1. Termination Actions

A number of actions occur at either normal or abnormal run termination, including the following:

- Summary accounting information is entered in the print file, including run start and termination time, standard units of processing (SUP) used, pages printed, and pertinent console messages.
- Symbiont output files are closed and queued for printing/punching in batch mode.
- All facilities allocated to the run are released back to the system facilities pool, with the exception of mass storage space being retained in a cataloged file.
- Temporary files are freed and disappear from the system.
- Cataloged files are freed and are retained, unless you specify assignment options to the contrary. Any exclusive use interlocks are released.

A demand run terminates normally when you enter @FIN. A demand run terminates abnormally when you enter @@TERM or \$\$CLOSE without entering @FIN first, or as a result of an operator keyin.

A batch run terminates normally when it reaches the end of the runstream (@FIN). An abnormal termination occurs in a batch run when one or more activities terminate in error.

Batch runs that terminate because of an @FIN statement can have files assigned with the C or D options, even though there are error contingencies set in the control word for the run. See the *ER Programming Reference Manual* (7830 7899) for an explanation of error termination contingencies.

If you execute the @SETC,I control statement before run termination, the @SETC,I makes the run termination process act as if the run were terminating without errors. The C option causes the files to be cataloged; the D option causes the files to be deleted. See Section 7.57 for more information on @SETC.

Caution

If a system stop occurs before run termination, the run does not terminate either normally or abnormally. Files in a to-be-cataloged state (created with @ASG,C or @ASG,U) are not cataloged.

5.4.2. Termination Task Scheduling

A site can configure the Exec to schedule a task (in other words, a program) that executes in a batch or demand run just prior to the termination of the run.

The scheduling of the termination task overrides forced run terminations. For example, if an X keyin terminates a batch run prior to the scheduling of the termination task, the termination task is still scheduled. Once the task begins execution, the run can again be subject to forced termination.

The termination task can be thought of as a FIN processor. It does not replace any of the standard run termination process performed by the Exec, but rather executes and terminates the task, and then the Exec begins its regular final termination actions.

No control statements are processed after the termination task is done (including PMD, PADS, @ASG, @XQT, processor calls, and so on).

The same task is scheduled for all batch and demand runs. The first of two configuration parameters (run_termination_task) is a program name. The program must reside in SYS\$*RUN\$ or SYS\$LIB\$*RUN\$. When the program is scheduled, the Exec searches SYS\$LIB\$*RUN\$ first. The Exec executes the program as a program rather than a processor.

The second configuration parameter (sys_termination_task_schedule) allows a site to configure whether or not the task must be scheduled for the SYS run. This provides a way to avoid system hangs caused by SYS accessing files on removable packs that cannot be registered until after SYS FIN.

The configuration of a termination task affects which part of a breakpoint (BRKPT) PRINT\$ file contains the tailsheet. In the default condition, when a termination task is not configured, the BRKPT file is closed and the tailsheet is printed in the standard PRINT\$ file.

But if a termination task is configured, then the tailsheet (and all other PRINT\$ output from the termination task) is printed in the BRKPT file, and then the BRKPT file is closed. The final part of the PRINT\$ file is not queued and is deleted because it always be empty. The termination task never receives images from the primary READ\$ file.

Facility control statements can be included in a termination task, but not in an addstream (because no control statements are processed after the termination task ends). A demand user might submit transparent facility ECL statements.

The termination status of the termination task does not affect the C or D conditional cataloging options on the @ASG control statement. The cataloging and dropping of files still occur after the termination tasks ends, but the options are enforced using the value of the run status just prior to the scheduling of the termination task. This applies even to files that are named on transparent facility control statements entered after the termination task is scheduled.

Accounting for system resource usage continues while the termination task is executing. (QUOTA limits also continue to be enforced.) Thus the system accounting file, written after the termination task ends, includes charges accumulated throughout the execution of the termination task.

Because SUP accumulations continue during the execution of a termination task, it is possible for the termination task to receive a "max time" contingency.

5.5. Closing a Session (\$\$CLOSE)

To close the application session you opened with the \$\$OPEN communications network statement (see Section 5.1), enter

```
$$CLOSE
```

When your session is closed, you receive this message:

```
*SESSION PATH CLOSED*
```

It is not necessary to issue a \$\$CLOSE statement when you want to change sessions. To establish a session with another application, enter another \$\$OPEN statement. This closes the current session and opens a new one. Use this procedure even when your system administrator specifies automatic session establishment for your terminal.

Section 6

Using Processors

This section covers processor calls and other statements that are used with processors. It describes how to generate, correct, and execute basic and extended mode programs.

6.1. Calling a Processor (@processor)

A processor is a system program that performs a function. A processor call statement is the instruction to execute a processor. An example is a language processor that translates a source program into machine-readable form.

Format

The general format of the processor call statement is

```
@[label:]processor[,options] field-1,field-2,field-3,..., field-n
```

label

Can be used to name the processor field, but is not required.

processor

Identifies the processor to be called.

The processor call is actually an instruction to execute the absolute element specified in the processor field. For processors supplied by Unisys, such elements are contained in the system library files.

To locate the absolute element on a processor call, the system uses these rules:

- If you specify a file name (for example, @UTL.PROOFER), that file is searched for the absolute element.
- If you do not specify a file name but do use a leading period (for example, @.PROOFER), the Exec searches the default file, TPF\$, for the element.
- If you do not specify a file name and there is no leading period (for example, @SPELL), the Exec searches the system library files first; if the element is not found, TPF\$ is searched.

The abbreviations for the standard processors (for example, SSG for Symbolic Stream Generator) are the names of the respective absolute elements.

options

Provide specific instructions about how the processor is to be used.

In general, the options field has meaning only for a particular processor, although some options have the same meaning for most processors supplied by Unisys. Refer to the reference manual of a particular processor for a list and explanation of options for that processor.

field-n

The fields (*field-1,field-2...field-n*) provide information to a processor; they are placed in an internal table and are supplied to the processor when the processor executes the first read instruction. These fields are numbered from 1 through 63.

Note: *Although you can use more than 63 fields on the call line, you must not do so. The fields after 63 are put in the internal table following the first 63 fields and are numbered from 1 to n again (up to 63), giving some fields duplicate numbers.*

Use a semicolon (;) at the end of a line to continue the fields on the processor call.

The fields are assumed to be in the system standard file name or element name notation, although they do not have to be names of elements. For example, since the DATA processor works only with system data format (SDF) files, it assumes file names rather than element names. The meaning of the fields depends upon the processor.

The section discusses the rules about fields and the processors that are supplied by Unisys:

- If a processor call has a field for the name of a program file and the program file is not specified, the Exec uses file TPF\$, unless the file name is taken from a previous field in that call statement.
- If a field must contain an element name and you specify an element name without the file name and the leading period, the Exec uses TPF\$ as the file.
- If a field must contain an element name and you specify an element name without the file name but with the leading period, either of the following happens:
 - The file name is taken from the previous field (if the field exists and is intended to name an element or a program file)
 - The file TPF\$ is used.

Examples

```
@SPELL,AHPUV PROG-DOC.SEC3
```

Calls Spelling Checker to check the spelling of the text within element SEC3 of program file PROG-DOC.


```
@UFTN FILE26.TEST,,,,OPTIONS
```

Calls UCS FORTRAN to compile a FORTRAN element called TEST in file FILE26. This statement produces an object module of the same name in the same file. Compiler options are printed due to the OPTIONS keyword.

```
@FTN ICARUS.CALC
```

In this example, the ASCII FORTRAN language processor compiles symbolic element CALC in file ICARUS. Since the processor-output field is not used, the resulting element has the same name as the element in the symbolic-input field (ICARUS.CALC). In other words, program file ICARUS has two elements named CALC, one symbolic and one relocatable, after execution of the FTN processor.

Processor Load Error

Refer to the *ER Programming Reference Manual* (7830 7899) for more information.

6.2. Unformatted Processor Call Statements

The unformatted processor call statement is used to load a processor and then to pass the complete unformatted text of the image verbatim to the processor. It is identified by the character sequence '@' in the first two columns of the image. Not all processors can be called with an unformatted processor call statement. Only those processors that explicitly document its usage (for example, CIFS utility @'ZIPUT) can be called with an unformatted processor call statement.

Format

```
@'[label:]processor[,options] [arbitrary-char-string]
```

label

Can be used to name the processor field, but is not required.

processor

Identifies the processor to be called.

The processor call is actually an instruction to execute the absolute element specified in the processor field. For processors supplied by Unisys, such elements are contained in the system library files.

To locate the absolute element on a processor call, the system uses these rules:

- If you specify a file name (for example, @UTL.PROOFER), that file is searched for the absolute element.
- If you do not specify a file name but do use a leading period (for example, @.PROOFER), the Exec searches the default file, TPF\$, for the element.
- If you do not specify a file name and there is no leading period (for example, @SPELL), the Exec searches the system library files first; if the element is not found, TPF\$ is searched.

The abbreviations for the standard processors (for example, SSG for Symbolic Stream Generator) are the names of the respective absolute elements.

options

Provide specific instructions about how the processor is to be used.

In general, the options field has meaning only for a particular processor, although some options have the same meaning for most processors supplied by Unisys.

Refer to the reference manual of a particular processor for a list and explanation of options for that processor.

The following text discusses the rules about fields and the processors that are supplied by Unisys:

- If a processor call has a field for the name of a program file and the program file is not specified, the Exec uses file TPF\$, unless the file name is taken from a previous field in that call statement.
- If a field must contain an element name and you specify an element name without the file name and the leading period, the Exec uses TPF\$ as the file.
- If a field must contain an element name and you specify an element name without the file name but with the leading period, either of the following happens:
 - The file name is taken from the previous field (if the field exists and is intended to name an element or a program file)
 - The file TPF\$ is used.

Additional Unformatted Processor Control Statement Syntax

Throughout this discussion, the @'[label:]processor[,options] portion of the unformatted processor control statement is referred to as the *first part* of the image. The following are key discussion points that are unique to the unformatted processor control statement syntax:

- The *first part* of the image can be up to 80 characters long. It is scanned according to the standard control statement syntax rules.
- When the end of the *first part* of the image is detected, no further scanning of the image is done.
- Traditional continuations of an unformatted processor control statement are not allowed, so a semicolon encountered as part of this initial scan is treated as a space and ends the scan.
- The maximum length of unformatted processor control statements is not restricted to the 80-character maximum of other control statements. The maximum length is the same as for data images such as traditional noncontrol statements and varies by run type (demand, batch, or demand batch) and file type (@ADD file, @@INQ, or demand interactive).

- For batch runs, the Exec normally prints the entire control statement. For unformatted process control statements, the Exec prints only the portion of the image scanned to produce the INFOR. This is done because of the potentially larger size of the image. The remainder of the image could also contain sensitive security information such as read or write keys.
- An INFOR table is created for the *first part* of the image and is used by the Exec to load the appropriate program or processor. The INFOR is then released and no INFOR is returned to the program. Once loaded, the program is expected to issue the appropriate ER SYMB\$ to read the unformatted processor control statement. The complete text of the image is passed verbatim including the @' identifier and any semicolons. See the *ER Programming Reference Manual* (7830 7899) for more information on ER SYMB\$.
- In addition to ASCII and Fielddata, unformatted processor calls in the Unicode character set are accepted by the Exec. Characters in the *first part* of the image (used to load the appropriate processor) must be in the ASCII/ISO range 000 through 0177. The first character not in this range terminates the Exec scan for the processor and options information.

6.3. Basic Mode Programs

Generating and executing a basic mode program is a four-step process (see Figure 6–1):

6.3.1. Step 1. Writing the Program

The first step in generating a program is to write a source program in a programming language supported on the system.

To enter the lines of the program into the system, you can use a text editor, such as EDIT 1100. Once entered, your program is stored as a symbolic element in a program file and is ready to be compiled.

Note: Do not choose a processor name such as DATA or ELT as the name for your program. The Exec recognizes these names in a processor call (@DATA or @ELT,D). If you use a processor name, the results are unpredictable.

6.3.2. Step 2. Compiling the Program

Before the computer can read and interpret your program, it must be translated into a form the computer can understand. This form is known as machine language, and a compiler performs the translation. To start a compiler, you use a compiler call statement, such as @ACOB for ASCII COBOL or @FTN for ASCII FORTRAN.

The compiler produces a machine-readable element called a relocatable element, which is stored in a program file. However, this relocatable element is still not ready to be executed; first it must undergo a process called collection.

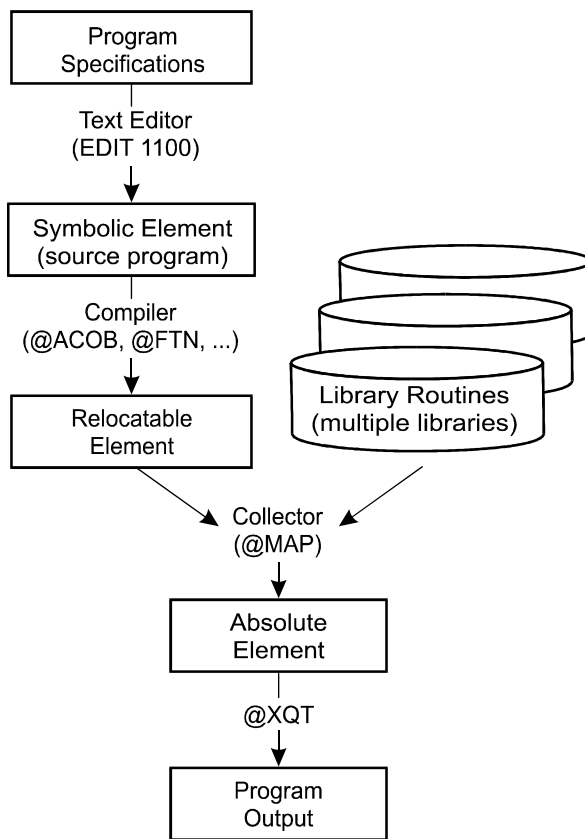
6.3.3. Step 3. Collecting the Program (Collection)

A processor called the Collector finishes the job of preparing a program so that it can be executed. The main function of the Collector is to join your relocatable elements with relocatable elements from user and system relocatable subroutine libraries.

The Collector joins these relocatable elements to form a single new element called an absolute element, which is ready to be executed. You call the Collector with the @MAP statement. Refer to the *Collector Programming Reference Manual* (7830 9887) for instructions on how to use the Collector.

6.3.4. Step 4. Executing the Program

Once you have produced an absolute element, execute it using the @XQT statement (see Section 7.70).



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Figure 6–1. Generating and Executing a Basic Mode Program

6.4. Extended Mode Programs

Figure 6–2 shows the process of generating and executing an extended mode program. It involves four steps, eliminating the intermediate step of using the Collector, although you can optionally use a process called static linking, which is similar to using the Collector.

6.4.1. Step 1. Writing the Program

Just as you would for a basic mode program, you first write a source program in a programming language supported on the system. The four UCS language products supported are UCS COBOL (@UCOB), UCS FORTRAN (@UFTN), UCS Pascal (@UPAS), and UCS C (@UC).

Refer to these reference manuals for detailed information on each UCS product:

- *COBOL Compiler Programming Reference Manual, Volume 1* (7831 0448)
- *COBOL Compiler Programming Reference Manual, Volume 2* (7831 0455)
- *FORTTRAN Compiler Programming Reference Manual, Volume 1* (7831 0489)
- *FORTTRAN Compiler Programming Reference Manual, Volume 2* (7830 7477)
- *C Compiler Programming Reference Manual, Volume 1* (7831 0422)
- *C Compiler Programming Reference Manual, Volume 2* (7831 0430)

Enter the lines of the program into the system by using a text editor. Once entered, your program is stored as a symbolic element in a program file and is ready to be compiled.

6.4.2. Step 2. Compiling the Program

Use the Universal Compiling System (UCS) compiler for your programming language. A UCS compiler produces a type of compiled element called an object module. This object module can be executed immediately after compilation; you do not have to explicitly combine a program's separately compiled object modules before execution, as with collection.

6.4.3. Step 3. Static Linking (Optional)

Although object modules can be executed immediately after compilation, this can be inefficient for some programs. If one object module of a program calls another, execution is interrupted while the Linking System searches the file system for the called object module. This process is called dynamic (execution time) linking.

To avoid this interruption of execution (and for other reasons, see the *Linking System Programming Reference Manual* (7831 0505)), you can choose to link a program's object modules together before execution. This process is called static (preload) linking and is similar to collection. Static linking creates another object module, which is the same as an object module produced by compilation. This object module has all external references removed, thus removing interruptions of execution.

You can link some or all of a program's object modules together during static linking. Object modules not linked during static linking are linked during execution.

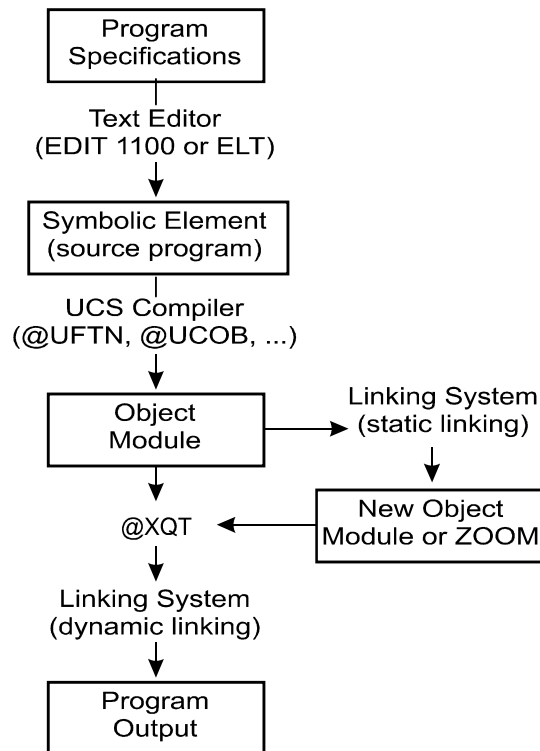
Static linking also lets you create special kinds of object modules called bound object modules and zero overhead object modules (ZOOM). A ZOOM is a type of object module that does not contain unresolved external references and cannot be called by other object modules. Object modules for a transaction environment must be ZOOMs. Bound object modules and ZOOMs execute more efficiently than standard object modules but lose the ability to link with other object modules. (Bound object modules can still call other object modules.)

See the *Linking System Programming Reference Manual* (7831 0505) for details on the Linking System.

6.4.4. Step 4. Executing the Program

Use the @XQT statement to execute an object module, bound object module, or ZOOM, just as you do with an absolute element produced by the Collector (see Section 7.70).

Figure 6–2 displays how to generate and execute an extended mode program.



000791

Figure 6–2. Generating and Executing an Extended Mode Program

6.5. Calling a Compiler

A language processor, often referred to as a compiler, is used to translate a source program from a programming language, such as ASCII FORTRAN or ASCII COBOL, into machine-readable form.

6.5.1. Calling a Basic Mode Compiler

Language processors for basic mode programs have a common interpretation of several options, as well as the first three fields.

Format

The typical language processor call statement is

```
@processor[,options] [symbolic-input,processor-output,symbolic-output ]
```

processor

Identifies the processor to be called. See the description and rules for the general processor call statement in Section 6.1.

options

Provide specific instructions about how the compiler is to be used.

The two major options are

I

Indicates that you are providing the source program to be compiled in the runstream, immediately after the compiler call statement.

U

Indicates that you are updating a source program that already exists in a symbolic element on mass storage. The resulting updated source program is stored as the next S-cycle of the original symbolic element.

If neither the I nor the U option is used on a compiler call statement, it indicates that you are updating a source program that already exists, but without S-cycling. The updated program is saved only if the symbolic-output field is used.

These are other options that are common to the standard compiler call statements:

A

Allows the runstream to continue in spite of errors detected in the compilation.

L

Provides a long listing. The exact meaning of this option varies from compiler to compiler, but it usually includes a listing of the object program generated. This option can be used instead of the S option.

N

Provides the shortest possible listing of the compilation. This option can be used in place of the S option.

P

Indicates that the symbolic element produced by the compiler must be in Fielddata format.

Q

Indicates that the symbolic element produced by the compiler must be in ASCII format.

P and Q

Indicates that the symbolic element produced by the compiler can be a combination of Fielddata format and ASCII format.

neither P nor Q

Indicates that the symbolic element produced by the compiler can be a combination of Fielddata format and ASCII format. When the I option is included, the format is determined by the format of the runstream images. When the I option is not included, the format is determined by the format of the symbolic input element images.

S

Provides a summary listing of the compilation. The resulting listing gives the average user the information needed.

Note: *If you do not specify the L, N, or S option, the compiler produces a listing that is less than the S option listing, but slightly more than the N option listing.*

W

Lists the correction images entered. This option can be used when correction images are used with the U option. It can also be used with neither the I nor the U option.

X

Terminates the run, if errors are detected in the compilation.

Symbolic-input

This can be

- The name of the file.element where the source program given in the runstream is to be saved as a symbolic element (I option).

- The name of the symbolic element to be updated and then compiled (U option, or neither I nor U). If the U option is used without the symbolic-output field, the updated element is saved as an S-cycle of the symbolic-input element.

The I and U options, (along with the *symbolic-input* field) determine how the compiler call statement is interpreted. Table 6–1 and Table 6–2 describe the valid possibilities. You receive an error message if you deviate from these rules.

Table 6–1 is valid for the language processors that use the symbolic-input, processor-output, and symbolic-output fields (FTN, for example). Table 6–2 is valid for the processors that use only the symbolic input and symbolic output fields (for example, ELT).

processor-output

Specifies the name of the file.element into which the processor output is to be placed. There is no restriction on the type of element produced. For example, most of the processors produce relocatable binary elements; the Collector produces either absolute or relocatable binary elements.

symbolic-output

Is the name of the file.element to be used for the updated symbolic element, if any. Table 6–1 and Table 6–2 describe the use of the symbolic output field, along with the I and U options, and the symbolic-input and processor output fields.

The compiler automatically assigns to your run those files that are specified in the output fields. This requires exclusive use of the output files. If another run has any of the files exclusively assigned, you receive an error message and the compilation terminates. You can prevent a compiler termination by placing @ASG,AX statements for the cataloged files named in the runstream before you execute the compiler call statement.

Compilers do not check the input and output field to determine whether the names are unique or duplicated. If an element name in an output field matches an already existing element in name, version, and type, the old element is replaced with the new element.

Examples

```
@FTN ICARUS.CALC
@EOF
```

In this example, the ASCII FORTRAN compiler compiles symbolic element CALC in file ICARUS. Since the processor-output field is not used, the resulting element has the same name as the element in the symbolic-input field (ICARUS.CALC). In other words, program file ICARUS has two elements named CALC, one symbolic and one relocatable.

```
@ACOB,S VIDEO.EXAMPLE,VIDEO.REL,NEWFILE.NEWCOPY
correction image
correction image
correction image
.
.
.
@EOF
```

This example calls the ASCII COBOL compiler, uses the correction images that follow to update symbolic element EXAMPLE of program file VIDEO, and produces new symbolic element NEWCOPY in program file NEWFILE. This new symbolic element is compiled to produce relocatable element REL in program file VIDEO.

6.5.2. Calling an Extended Mode Compiler

Format

The standard processor call format for a UCS compiler is as follows.

```
@processor[,letter-options] [source-element][,[object-element]
[,,[keyword-options-element][,keyword-options]...]]
```

letter-options

The options are as specified for each UCS compiler. See the documentation on the compiler.

If the effect of a letter option conflicts with the effect of a keyword option specified on the processor call, the keyword option takes precedence. If the effect of a letter option conflicts with the effect of a keyword option specified in the *keyword-options-element*, the letter option takes precedence.

source-element

Specifies the name of the program file symbolic source element that contains the program to compile. If you do not specify the source element name, the compiler source program must directly follow the processor call in the runstream, and you must use letter option I.

If only the I option is used, the @EOF statement (or another control statement) must follow the source program to signal termination of the source.

object-element

Specifies the name of the program file object module that receives the compiled program. If you do not specify the object element name, the file and element name of the source program is used. If you do not specify either name, temporary file TPF\$ and element name NAME\$ are used.

keyword-options-element

Specifies a source element that contains compiler keyword options.

keyword-options

Specifies the keyword options as described in the compiler documentation. Separate *keyword-options* with commas.

General Rules

These general rules apply to UCS compilers:

- Correction lines are not accepted. You must apply correction lines before compiling the source program. Use the ELT processor or a text editor to do this.
- In most cases, UCS compilers do not produce symbolic-output. Therefore, field 3 of the processor call line is normally blank. Some special processing, such as the UFTN CONVERT routine, accepts a value in field 3. See the *UCS FORTRAN Programming Reference Manual Volume 1* (7831 0489) and *Volume 2* (7830 7477).

Examples

```
@UCOB PEGASUS.FRONT-END3, . FRONT-END
```

In this example, the UCS COBOL compiler compiles source element FRONT-END3 in program file PEGASUS. The resulting object element is named FRONT-END. Since a file is not specified, the object element is placed in PEGASUS, which is the file given in the source element field.

```
@UFTN, S PROGFIL.SYM, TPF$.OM, , SITE*SITE.UCS-OPTS, NO-WARNING
```

In this example, the UCS FORTRAN compiler compiles FORTRAN element SYM from program file PROGFIL. During compilation, the compiler uses keyword option NO-WARNING to suppress warning messages. Keyword options in element SITE*SITE.UCS-OPTS are also used. In addition, a listing of the source program is printed as a result of the S option letter. Object module OM is produced in TPF\$.

Table 6–1. Processor Calls with Symbolic Input, Processor Output, and Symbolic Fields

*I or U Option	Symbolic Input Field	Processor Output Field	Symbolic Output Field	Elements Produced
Neither or I only	Not specified	If this field is not specified, NAME\$ is assumed as the element name and TPF\$ is assumed as the file name. It is invalid to specify an S-cycle.	Illegal to use this field.	New processor output element.
Neither	Field is completely specified.	If this field is not completely specified, the information from the symbolic-input field is used. It is invalid to specify an S-cycle.	If void, no symbolic output is produced. If this field is not completely specified, information from the symbolic-input field is used.	New processor output element. Also symbolic output element if symbolic output field is specified.
I only	Field is completely specified but without an S-cycle.	If not completely specified, the information from the symbolic input field is used. It is invalid to specify an S-cycle.	Illegal to use this field.	New processor output element and symbolic output element.

Table 6–1. Processor Calls with Symbolic Input, Processor Output, and Symbolic Fields

*I or U Option	Symbolic Input Field	Processor Output Field	Symbolic Output Field	Elements Produced
U only	Field must be completely specified.	If this field is not completely specified, the information from the symbolic-input field is used. It is invalid to specify an S-cycle.	Not required; information from the symbolic-input field is used. When symbolic-output is not specified, the symbolic input element is updated to produce the next higher S-cycle from the symbolic input cycle specified, or the latest cycle (if no cycle is specified). If symbolic output is specified, the next higher S-cycle is created and transferred, along with all previous cycles up to the cycle maximum, to the next symbolic output element, symbolic-input element remains unchanged. In either case, it is invalid to specify a cycle for symbolic-output.	New processor output and symbolic output element.

Table 6–2. Processor Calls with Symbolic Input and Symbolic Output Fields

I or U Option	Symbolic Input Field	Symbolic Output Field	Element Type Produced
Neither	Not specified.	Illegal to use this field.	The runstream input is processed, but no element is produced.
Neither	Field is completely specified.	If this field is void, no symbolic output is produced. If it is not completely specified, then the information from the symbolic-input field is used. It is invalid to specify an element-cycle.	Update (no cycling).
I only	Not specified.	Illegal to use this field.	No element is produced.
I only	Field is completely specified, but without the element-cycle.	Illegal to use this field.	New element.
U only	This field must be completely specified.	If this field is not completely specified, then the information from the symbolic-input field is used. When symbolic-output is not specified, the symbolic-input element is updated to produce the next higher element-cycle from the symbolic input cycle specified, or the latest cycle if no cycle was specified. If symbolic output is specified, the next higher element cycle is created and transferred, along with all previous cycles up to the cycle maximum, to the new symbolic-output element; symbolic input element remaining unchanged. In either case, it is invalid to specify a cycle for symbolic output.	Update (with cycling).

6.6. Correcting Symbolic Elements

To correct a symbolic element in batch mode, you can use statements called correction images. You can change source basic mode programs by inserting correction images immediately following a processor call statement.

Correction images can be used only to update symbolic elements and cannot be used with processor call statements that use the I option.

To either create or update symbolic elements, you can use correction images with the ELT processor. See the *ClearPath OS 2200 ELT Processor Operations Reference Manual* (8206 2365) for more information on ELT.

Note: *Correction images are not supported for extended mode programs. You can use ELT before compilation by a UCS compiler as an alternative.*

You can use correction images to do the following:

- Delete one or more lines
- Insert one or more lines
- Replace old lines with new lines
- Replace one character string with another
- Replace all characters in certain columns with a character string
- Replace partial lines

The correction images are applied to the element identified in the symbolic-input field of the processor call statement. A symbolic element consists of a sequence of implicitly numbered lines or images. The lines or images are sequentially numbered, beginning with line 1, and increment by 1 for each successive line. The numbers used in correction images refer to the line numbers within the element you are correcting.

Images from the symbolic-input element and from the runstream have traditionally been in the Fielddata and ASCII coded character sets (CCS). For international use there are now available an additional 39 ASCII-like CCSs. The alphabetic, numeric, and most special characters in the ASCII-like CCSs are identical to the traditional ASCII CCS. Each ASCII-like CCS has additional characters that are unique for a country or language. Both basic mode and extended mode programs can be written with any desired combination of Fielddata, ASCII, and ASCII-like CCSs for the symbolic-input element images and for the runstream correction images.

There are an additional 22 CCSs defined that are not Fielddata, ASCII, or ASCII-like. Programs cannot be written using images in these CCSs. However, elements containing images in these CCSs could be used as data to be read by your programs. You can use the ELT processor to either create or update symbolic elements containing images in these 22 "other" CCSs. Note the documented restrictions on the use of images in these CCSs as runstream correction images.

6.6.1. Correction Image Format Rules

Follow these rules when using correction images:

- A change indicator must appear in the first column. This is normally a minus sign (-). If you need to use minus signs within the correction images, you must redefine the change indicator. To do so, enter

`--x`

where *x* is the new change indicator, which can be from 1 to 3 characters in length. If you do not define *x*, the statement is ignored. You can redefine the change indicator as often as you like, but only one change indicator is recognized at any one time.

- All correction images must be arranged in numerical order according to line number.
- When one correction image follows another, the line numbers specified in the second correction image must be greater than the line numbers specified in the first correction image; they cannot overlap. The only exception is when you are inserting new lines after replacing lines or changing partial lines. In this case, you can insert lines at the last line replaced or changed.
- Any lines inserted between the processor call statement and a correction image are placed before the first line of the element being updated.
- All correction images must be Fielddata, ASCII, or ASCII-like.

Use the W or V option on the processor call statement to see a listing of the corrections that were made by using the correction image format.

The following subsections give the format and explain how to use correction images to perform these tasks:

- Deleting lines
- Replacing old lines with new lines
- Inserting new lines
- Replacing partial lines

6.6.2. Deleting Lines

Format

-n,m

Deletes specified lines from the symbolic element you are updating, where *-n,m* indicates the range of lines to be deleted. *m* must be greater-than-or-equal-to *n*. If you use an exclamation point (!) for *m* (*-n,!*), everything is deleted between *n* and the end of the symbolic element.

Example

```
@ACOB,US ACCT-SUMM.MAIN,ACCT-SUMM.MAIN
-240,!
@EOF
```

The compiler deletes line 240 through the end of symbolic element MAIN of file ACCT-SUMM. The U option causes the updated element to be stored as the next S-cycle of element MAIN. The S option generates a summary listing.

6.6.3. Replacing Old Lines with New Lines

Format

```
-n,m
new line
new line
new line
.
.
.
```

Deletes lines *n* through *m* from a symbolic element and replaces them with new lines. If you use an exclamation point (!) for *m* (*-n,!*), everything is deleted between *n* and the end of the symbolic element.

All lines between the correction image (*-n,m*) and the next correction image or control statement are considered replacement lines.

Any number of lines can follow the correction image. You can replace the deleted lines with fewer or more new lines. The symbolic element is implicitly renumbered to number the lines consecutively, beginning with 1.

The new lines can be Fielddata, ASCII or ASCII-like images, or images in any of the 22 "other" CCSs.

Example

```

@ACOB,SW LEDGER.SUB1,LEDGER.SUB1R,LEDGER.SUB1/NEW
-62,65
DATA DIVISION.
FILE SECTION.
-138,138
MOVE A TO B.
PERFORM LOOK-UP.
@EOF

```

The S option generates a summary listing; the W option causes the six correction images to be listed (everything between the @ACOB and @EOF statements). Lines 62 through 65 are deleted and replaced with two lines, DATA DIVISION and FILE SECTION. Line 138 is also deleted and replaced with two lines, MOVE A TO B. and PERFORM LOOK-UP.

The changes shown are applied to symbolic element SUB1 of file LEDGER. The updated element is saved as symbolic element SUB1/NEW. The updated element SUB1/NEW is compiled to produce relocatable element SUB1R.

6.6.4. Inserting New Lines**Format**

```

-n
new line
new line
new line
.
.
.

```

Inserts new lines in a symbolic element following the line number that you specify, where *n* is the line number after which the new lines are inserted, and new line represents a line to be inserted.

The new lines can be Fielddata, ASCII or ASCII-like images, or images in any of the 22 "other" CCSs.

To insert new lines before line 1, place the new lines immediately after the processor call statement without specifying any insertion line numbers, or use zero (0) as the line number.

To insert new lines after the last line, use an exclamation point (!) as the line number.

The compiler inserts all lines between the preceding correction image (-*n*) and the next correction image or control statement.

After the element has been completely updated, it is implicitly renumbered to number the lines consecutively, beginning with 1.

Example

```
@FTN,UL FILEX.SYM,FILEY.REL
-31
WRITE(6,10)
FORMAT(1X,'HI,I')
-56
CALL PLOTTER (XXX,100,500,INTRVL)
@EOF
```

The changes shown are applied to symbolic element SYM of file FILEX. Because the U option is used, the updated element is saved as the next S-cycle of element SYM. The L option generates a complete listing. Two new lines, WRITE(6,10) and FORMAT(1X,'HI,I'), are inserted between lines 31 and 32 of element SYM. One new line, CALL PLOTTER (XXX,100,500,INTRVL), is inserted after line 56 in element SYM (after the line that was 56th before the compiler was called).

6.6.5. Editing Lines

Format

```
-n,m-
editing statement
editing statement
editing statement
.
.
.

or

-n-
editing statement
editing statement
editing statement
.
.
.
```

Replaces portions of the specified lines in the symbolic element you are updating, where *n,m* indicates the range of lines to be partially replaced. The line number specified by *n* must be greater than the number specified on any previous correction image. The line number specified by *m* must be greater than or equal to *n*. All lines *n* through *m* must be in Fielddata, ASCII, or an ASCII-like CCS.

The minus sign (-) immediately following the *m* is an editing indicator. You must always use a minus sign to designate the editing indicator. You must follow the *-n,m-* with one or more editing statements, with the number of editing statements corresponding to the number in the *-n,m-* range.

For example, if

`-2,7-`

is the correction image, you must follow it with six editing statements corresponding to lines 2 through 7. If the number of editing statements is greater than the number in the `-n,m-` range, the excess statements are ignored and an error message is issued.

The `-n-` indicates that the following editing statements are applied successively to line *n*. The changes are applied until another correction image appears or an error occurs. If an error occurs, the remaining editing statements are ignored.

Note: *Replacement of partial lines is possible, but avoid doing so because it is difficult to apply changes later.*

Editing statements must be in Fielddata, ASCII, or an ASCII-like CCS.

Editing statements can take one of the following forms:

c/new-data

Replaces characters from column *c* to the end of the line. The new-data field contains the replacement characters.

c,d/new-data/

Replaces a specified number of characters on a line. The columns specified by *c,d* indicate the range of characters to be replaced; the new-data field contains the replacement characters. If the number of replacement characters is greater than the specified range, all characters on the line are shifted to the right. If the number of characters is fewer than the range, all characters are shifted to the left to adjust the spacing.

/old-data/new-data

Replaces the character string indicated by the old-data field with the characters in the new-data field to the end of the specified line. The old-data field recognizes both uppercase and lowercase letters.

/old-data/new-data/

Replaces one or more characters specified by the old-data field with the characters in the new-data field. If the fields are not of equal length, the line is shifted to the right to accommodate extra characters, or to the left to eliminate extra spaces. The old-data field recognizes both uppercase and lowercase characters.

For each of the editing statement formats, the slash (/) is commonly used as a separator character. You can change the separator character to any character other than a digit or a blank space in any of the editing statement formats. In the *c/new-data* format, the separator also cannot be a comma. In either of the *old-data/new-data* formats, the character cannot be the same character that you use as a change indicator.

Regardless of the separator character you choose, you must never use the same character within the old-data or new-data fields in any of the formats.

Example

```
@FTN,LU FILEX.ABC
-1-
/ABC/abc/
/GHI/ghi/
/MNO/mno/
-2,5-
9/ccccdddd
5,8/ffff/
/KKKK/kkkkl1111/
/NNNN/nnnn/
```

Updates element ABC in FILEX and creates a new cycle of element ABC. The following corrections are made to the element:

Line 1

Uppercase characters ABC, GHI, and MNO are replaced with their lowercase counterparts.

Line 2

Starting at column 9, characters cccccddd replace the original characters. Any remaining characters are cleared to the end of the line.

Line 3

Between columns 5 and 8, characters ffff replace the original characters.

Line 4

Lowercase characters kkkkl1111 replace uppercase characters KKKK. Any remaining characters are cleared to the end of the line.

Line 5

Lowercase characters nnnn replace uppercase characters NNNN.

6.6.6. EDIT 1100

To correct a symbolic element, you can also use EDIT 1100, the IPF full screen editor for manipulating text, source code, or other data. The editor provides an easy way to move, copy, reformat, insert, and delete (among many other editing functions) data within files and elements.

You can use EDIT 1100 to correct any symbolic element, whether it contains a source program, a canned runstream, general text, or other symbolic images. You can also use it to correct data files.

To learn more about EDIT 1100, refer to the *IPF 1100 EDIT 1100 User's Guide* (7833 3812).

Section 7

Control Statements

Table 7-1 alphabetically lists the ECL and FURPUR control statements.

Table 7-1. Statement Reference List

Statement	Purpose
@ADD	Adds partial canned runstreams
@ASG/@@ASG	Assigns special I/O and symbiont devices Changes assignment fields for a mass storage file Assigns a tape file and one or two tapes to a run Assigns a mass storage file to a run
@BRKPT/@@BRKPT	Diverts output to a user-defined print file instead of to the terminal or the PRINT\$ file
@CAT/@@CAT	Catalogs a mass storage file in a master file directory without having the file assigned to the run Catalogs a tape file in a master file directory without allocating tape facilities to your run
@CHG*	Changes element name, version name, read key, write key, and mode of a file
@CKPT	Saves a complete or partial copy of a run's environment
@CLOSE*	Writes a hardware EOF mark on a magnetic tape file and rewind the tape
@@CM	Sends a message to the operator's console
@@CONS	Allows you to enter console keyins
@@CONT	Continues sending output
@COPIN*	Copies elements from an element file located on magnetic tape into a program file on sector-formatted mass storage.
@COPOUT*	Copies a program file or selected elements from a program file located on sector-formatted mass storage onto a magnetic tape file in element file format
@COPY*	Copies a file or element from one file to another

Table 7-1. Statement Reference List

Statement	Purpose
@@CQUE	Queues input in a main storage buffer
@CYCLE*	Sets the maximum range of absolute F-cycle numbers to be retained for cataloged files or sets the maximum number of element cycles to be retained for the specified symbolic element
@@DCT	Changes DCT terminal characteristics
@DELETE*	Deletes cataloged files Changes program file elements from current to deleted Changes program file elements from deleted to current
@ENABLE*	Clears the disable flags for cataloged files
@@END	Terminates special modes
@ENDF	Marks the end of the images for a file created by the @FILE statement
@EOF	Indicates the end of input to a processor
@ERS*	Returns to the system all sector-formatted mass storage granules allocated to a file
@@ESC	Passes unaltered input to a program, or unaltered output to a terminal
@FILE	Creates a tape file at input time and assigns it to the run as images are read
@FIN	Identifies the end of a run
@FIND*	Locates an element in a magnetic tape file (file must be in element file format) and positions the file before the element's label block
@FREE/@@FREE	Releases files, reels, internal file names, and the exclusive use of files from a run
@@FUL	Enters input a full screen at a time
@HDG/@@HDG	Specifies a heading for a printout and changes certain print parameters, such as page size, number of lines per inch, and margins
@@HOLD	Holds terminal messages or print output
@@INQ	Queues input in a mass storage buffer
@@INS	Changes the insert point on the display
@JUMP	Branches from one point within the runstream to another point
@LOG/@@LOG	Places a message in the system log
@MARK*	Writes a hardware EOF mark on a magnetic tape file.

Table 7-1. Statement Reference List

Statement	Purpose
@MODE/@@MODE	Changes the mode settings that were specified for a tape file when the file was assigned to the run
@MOVE*	Moves a magnetic tape file forward or backward over a specified number of EOF marks.
@MSG/@@MSG	Sends a message to an I/O, communications, hardware-confidence, or system console
@@NOPR	Terminates a local printer
@PACK*	Rewrites an entire program file, removing certain types of elements (depending on the options specified) and all elements marked as deleted
@@PASSWD	Lets you change your password in demand mode and provides passwords within batch runs
@PCH*	Creates 80-column card images from program file elements
@@PMOD	Controls output sent to a printer connected to your terminal
@PREP*	Creates relocatable and object module entry-point tables within a program file
@@PRNT	Starts a local printer
@PRT*	Provides a listing of the master file directory items for one, a group, or all cataloged files; information regarding temporary files; information concerning currently assigned files; the table of contents of a program file; or the text of a symbolic element
@QUAL/@@QUAL	Specifies a qualifier or a directory (with the D option)
@REWIND*	Rewinds magnetic tape files back to the load point of the first reel
@@RLD	Causes screen images to scroll down
@@RLU	Causes screen images to scroll up
@@RQUE	Stops an output file and places it in a queue
@RSTRT	Restarts a run
@RUN	Identifies a run to the Exec, and provides the system with accounting and scheduling information
@@SEND	Tells the system to send you the print files queued to your terminal's site-ID
@SETC	Stores a value in a condition word
@@SKIP	Skips lines of output
@@SOLI	Forces a solicit for a user-ID /password

Table 7-1. Statement Reference List

Statement	Purpose
@START/@@START	Starts full canned runstreams
@SYM/@@SYM	Sends the output of a print file to a specified printer, terminal, or user-ID
@TEST	Tests a condition word to control execution
@@TERM	Signs off your terminal from the system at the end of a run
@@TM	Sends a message to a specified run-ID, user-ID, or site-ID
@@TOUT	Changes your timeout period
@@TRNS	Causes the escape and end-of-transmission characters to be ignored
@@TTY	Changes TTY terminal characteristics
@USE/@@USE	Specifies an internal file name
@@X	Stops the current processing
@XQT	Initiates the execution of an absolute element

* Command handled by FURPUR.

7.1. @ADD—Add Partial Canned Runstreams and Data

The @ADD statement lets you include canned runstreams or data for repetitive functions. Control statements or data are entered only once, but can be used in many subsequent runs.

When the system processes an @ADD statement, the contents of the added file or element replace the @ADD statement. Control then returns to the run that contained the @ADD statement.

If the file that contains the added runstream is assigned to the run when the Exec reads the @ADD statement, the file is still assigned when control returns to the main runstream. If the file is not assigned to the run when the Exec reads the @ADD statement, the file's condition when control returns to the main runstream depends on whether the file has an internal file name attached to it. An added file that is not assigned, but has an internal file name, is assigned. An added file that is not currently assigned and does not have internal file name is not assigned, even if the added runstream contains an @ASG statement to assign the file. The F option can be used to explicitly free the file upon completion of @ADD processing.

Format

```
@ADD[,options] name
```

options

D

Allows the insertion of files or elements when operating under the DATA or the ELT,D processors.

E

Provides an implied @EOF statement at the end of the added file or element. If the DATA or the ELT,D processors are executed at the time of the @ADD with the E option, the EOF return is ignored.

F

Frees the file containing the added runstream upon completion of @ADD processing. All @USE names attached to the file are retained, with the exception of any @USE names attached by @ADD processing.

L

Lists all control statements in an added file or element at the demand terminal until the runstream returns control to the demand terminal input. Also lists control statements from other files or elements added by any @ADD statements nested within the overall addstream. This option is ignored in batch mode. Control statements in an addstream are always listed in batch.

P

Prints the @ADD statement in the program listing.

R

Gives the control statements in the @ADD file execution priority over the top entry of the deferred control statement execution (DCSE) stack.

If ER QECL\$ is used with @ADD files and the @ADD files are to be interpreted first, the R option must be used. The following restrictions apply when ER QECL\$ and @ADD are used together:

- If something is on the DCSE stack and you submit an @ADD statement with the R option, all subsequent @ADD files must have the R option to maintain priority.
- You must not execute @IPF from an @ADD,R file if something is on the DCSE stack. Execution of these statements might result in IPF reload errors.

See the *ER Programming Reference Manual* (7830 7899) for a description of ER QECL\$.

name

Identifies the file or element to be added, which must be stored in either an SDF file or a symbolic element of a program file. The symbolic element cannot reside in a Large Element Program File (LEPF). The file or element can be created by any of the following:

- The SSG processor
This is a method to generate runstreams to be added with @ADD. The SSG processor can add (@ADD) them automatically. See the *SSG Programming Reference Manual* (7831 1784).
- The EDIT 1100 processor
See the *IPF 1100 EDIT 1100 User's Guide* (7833 3812).
- The @FILE statement
See Section 7.27.
- The DATA processor
See the *ClearPath OS 2200 ELT Processor Operations Reference Manual* (8206 2365).
- The ELT processor
See the *ClearPath OS 2200 ELT Processor Operations Reference Manual* (8206 2365).
- Symbiont output (for example, alternate print or @BRKPT)
- A user program

A file name must be followed by a period; otherwise, it is interpreted as an element name. If a file name is not specified, TPF\$ is assumed. If an element name is specified without a version name, the system uses the symbolic element with a blank version name, if one exists. If one does not exist, the system uses the first nondeleted version of the symbolic element in the program file.

Notes:

- *Do not include statements such as @FREE, @PACK, or @ERS that could change the structure of the file that contains the statements.*
- *Do not use the @PASSWD statement within an added file. It compromises the secrecy of your password, and the system ignores the @PASSWD statement anyway.*

You can correct the prestored, partial runstreams before they are added by using the EDIT 1100 processor. Do not type directly into batch-type processors such as ASCII COBOL, MASM, or MAP. Instead, introduce your line corrections with the @ADD,E statement or use EDIT 1100 to make source corrections.

Example 1

Suppose you have several files that you always use together. Instead of assigning them individually, you can store @ASG statements for the files in an element. Whenever you want to assign the files to your run, you can add the element with an @ADD statement.

To illustrate this, suppose you create an element called PJT*UTL.STD-ASGS and insert the following statements into it:

```
@ASG,A PJT*WORKFILE.  
@ASG,A PJT*ACCT-PROGS.  
@ASG,A PJT*ACCT-DATA.
```

You can assign all three files at once using the following statement:

```
@ADD PJT*UTL.STD-ASGS
```

Example 2

```
@ADD,P PROFITS.
```

The statements in file PROFITS replace the @ADD statement. Because of the P option, the @ADD statement appears in the program listing.

7.2. @ASG and @@ASG—Assign Files and Peripheral Devices

The @ASG statement is used to name a file, state its I/O facility requirements, and assign it to the requesting run under the given external file name. If the file is cataloged, the facility requirements are known and need not be specified when assigning the file.

You can use the @ASG statement for the following device types:

- Sector-addressable mass storage
- Word-addressable mass storage
- Magnetic tape
- Absolute devices

You must assign files to your run before you can reference them for I/O operations.

The only time you can reference a file without first assigning it with an @ASG statement is when you specify the file on some control statements or processor call statements (the Collector, for example). The actual assignment is made when the program handles the given statement. Information concerning the assignment is taken from the master file directory (MFD).

The @FREE statement releases files and peripheral devices from your run. After you release a file or device, you can no longer reference it without reassigning it with another @ASG statement.

You can place the @ASG and @FREE statements anywhere in the runstream. Dynamic @ASG and @FREE statements can appear anywhere in the user program. These features let you assign and free files as required, without tying up the files and facilities from the beginning of the run until its completion.

However, you might have to wait until the facility or file is available. Examples of when this can happen are as follows:

- A magnetic-tape unit is being used by another run.
- An arbitrary device is being used by another run.
- A cataloged magnetic-tape file is being used by another run.
- A cataloged file is being used exclusively by another run.
- A magnetic-tape reel is being used by another run, except when the reel number is for a scratch tape.
- Insufficient mass storage is available.
- A reel number is currently in use for another file.
- A cataloged file is rolled out to tape.

When you assign a file to your run, you are adding an entry for the file to your run's program control table (PCT). The PCT contains information about the files assigned to your run. If a file is not listed in your run's PCT, you cannot read from it, write to it, or execute it.

Similarly, when you release a file from your run (using an @FREE or @@FREE statement), you are actually removing the file from your run's PCT (see Section 7.30).

You can examine the files assigned to your run using @PRT,I (see Section 7.48).

Figure 7-1 shows you how to assign files.

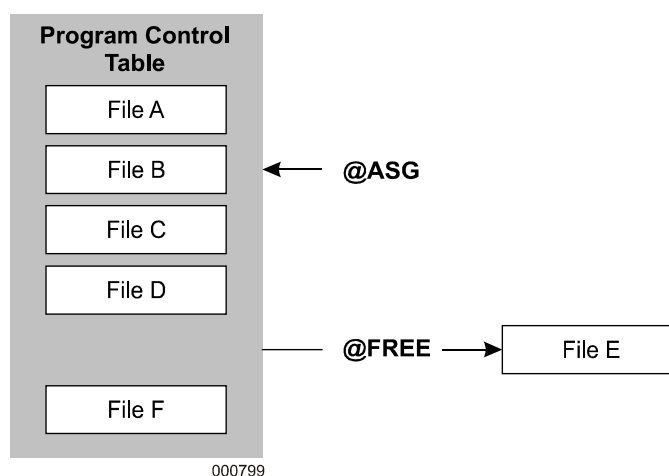


Figure 7-1. Assigning Files

7.2.1. Assigning Mass Storage Files

The @ASG statement assigns a random-access, mass storage file to your run.

Note: When you assign a previously created file, you must pass security validation in order to gain access to the file. See the discussion in 4.3 on security access validation.

Format

```
@ASG[,options] filename[,type/reserve/granule/maximum  
/placement][,pack-id-1/.../pack-id-n,,,ACR-name/encrypt-option]
```

options

General Options

I

Automatically frees the file at the next task termination regardless of how the task terminates. (A task is a discrete processing step in a run, involving the execution of an absolute element; for example, a system processor or user program.)

Z

Specifies that this statement is not to cause a hold condition. Control is returned to the user, indicating whether a hold state would have resulted from the assignment. Also, this option requests that this file be reloaded if it is unloaded.

Options for Assigning Files to Be Cataloged

B

Indicates that when this file is cataloged and subsequently assigned, it is saved and reloaded as part of any full checkpoint or restart.

C

Catalogs the file at run termination if the run terminates normally. (See Section 5.4 for a description of run termination.) If the file is freed before termination, it is cataloged at that time. If a file by this file name already exists in the master file directory, the request is rejected. Use @CAT instead of @ASG,C to catalog a file if you want the file cataloged without having it assigned to your run.

G

The meaning of this option depends on how security and Quota are configured at your site. Only the unconditional delete file system privilege overrides the G option.

- **Unowned files**

If your site has fundamental Security or Security Level 1, 2, or 3 and the file is unowned, the following applies:

- When the file is cataloged, the G option guards against having its read key, write key, privacy mode, or all three overridden by privileged runs.
- If any F-cycle of a file set is cataloged with the G option, the read keys or write keys for the file set are obscured by '/////////' from all users. If a particular F-cycle of a file set is cataloged with the G option, the privacy mode and either the project-ID or account number under which the file was created are also guarded.
- The G option means the file cannot be backed up.

- **Owned files**

If your site has Security Level 1, 2, or 3, the file is owned, and the dynamic configuration parameter (owned_file_read_write_keys) is FALSE:

- G-option files are no longer protected from privileged runs.
- The G option means the file cannot be backed up by FAS. When you need to back up a file, you must use another product such as FURPUR or IRU.

If your site has Security Level 1, 2, or 3 and the file is owned and the dynamic configuration parameter (owned_file_read_write_keys) is TRUE, the following applies:

- When the file is cataloged, the G option guards against having its read key, write key, privacy mode, or all three overridden by privileged runs.
- If any F-cycle of a file set is cataloged with the G option, the read keys or write keys for the file set are obscured by '/////////' from all users. If a particular F-cycle of a file set is cataloged with the G option, the privacy mode and either the project-ID or account number under which the file was created are also guarded.
- The G option means the file cannot be backed up.

- **Quota system**

The use of the G option is also controlled by the Quota system on an account basis, as follows:

- If the Exec is configured to maintain quota limits for each account, this option can be used only if it is allowed in the quota limits for the run.
- If the Exec is not maintaining quota limits, this option can be used by all runs.

P

Catalogs the file as a public file. If you create unowned files (only for sites with Fundamental Security or Security Level 1, 2, or 3 and you omit the P option (default), the file is cataloged as a private file and can be accessed only by runs that have the same project-ID or account number as your run.

If you create an owned file and you omit the P option, the file can be private (only you can access it) or ACR-controlled (if you attach an ACR to it).

For owned files, a file is cataloged as public when the P option is specified on the @ASG statement and an access control record (ACR) is not specified.

The file is cataloged as private if the P option and ACR are not specified on the @CAT statement (default).

R

Indicates that it is a read-only file. You cannot write to a read-only file; you can only read or delete it.

If your site has Security Level 1 or higher, you can assign a cataloged file as read-only, regardless of how the file was cataloged.

S

Indicates store-through file. Data written to this file is stored directly on a nonvolatile mass storage medium. Absence of this option indicates data can be stored on volatile areas of mass storage, such as cache storage. Using this option does not prevent the storage of data on solid-state disk. (Although this option no longer has any effect, it is allowed since it might still be in user runstreams.)

U

Indicates the same conditions as the C option, except that the file is cataloged unconditionally when the run terminates (excluding a system stop) or when you release the file from your run with an @FREE statement.

V

Indicates that, when the file is cataloged, its text is not to be unloaded to tape at any time. If the Exec is configured to maintain quota limits for each account, this option can be used only if it is specifically allowed in the quota limits for the run. If the Exec is not maintaining quota limits, this option can be used by all runs.

W

Indicates write-only file. No user can read a file cataloged with the W option. Anyone who tries to read the file receives an error message.

Options for Assigning Previously Cataloged Files

A

Indicates that the file is already cataloged and ensures that it is not treated as a temporary file. The request is rejected if the name cannot be found in the master file directory.

If the file is owned, the following applies:

- An ACR cannot be specified with this option.
- If this option is combined with the R option, read-only access is given.
- The user must have access privileges specified in the file's ACR.

If the file is application private, you must be running in the application-private common name section. Refer to the *MHFS Planning, Installation, and Operations Overview* (7830 7469).

If the file is unowned, you cannot use the A and R options together. The combination of these options results in an error.

D

Deletes a cataloged file from the master file directory if the run terminates normally (excluding a system stop), or when you release the file from your run with the @FREE statement. (See Section 5.4 for a description of run termination.)

If the file is owned, you must have delete access to the file in order to delete it.

If your site has Fundamental Security, you must include any read or write keys that are attached to the file in order to delete the file.

If your site has Security Level 1, 2, or 3 and the dynamic configuration parameter `owned_file_read_write_keys` is TRUE, then you must include read or write keys to delete the file.

E

Assigns this file with only the directory information accessible. The file is assigned as read- and write-inhibited and the file statuses of disabled, rolled out, and exclusive use are ignored. In addition, if the file is a removable disk file, a unit is selected and the pack or packs mounted. This option is intended for use primarily by file security and maintenance programs.

K

Indicates the same conditions as the D option, but the file is deleted from a master file directory when the run terminates regardless of how the run terminates (excluding a system stop). An @FREE statement releases the file before run termination.

If the file is owned, you must have delete access to the file in order to delete it.

If your site has Fundamental Security, you must include any read or write keys that are attached to the file in order to delete the file.

If your site has Security Level 1, 2, or 3 and the dynamic configuration parameter `owned_file_read_write_keys` is TRUE, then you must include read or write keys to delete the file.

M

Saves and reloads any cataloged file assigned as part of any full checkpoint or restart.

Q

Assigns the file even if the system has disabled it; does not inhibit read/write because of the disable. This option does not override R or W. It lets you recover what remains of the file after a disable.

R

On systems with Security Level 1 or higher, assigns the file as read only. You cannot write to the file or delete it. You can assign a file as read only with this option, regardless of how the file was originally cataloged.

T

Specifies that the file is temporary. The file can have any name, as long as the name is not assigned to your current run.

X

Specifies that this run is to have exclusive use of the file until the run terminates, the file is released by an @FREE statement, or exclusive use is released through @FREE. The X option is ignored for read-only files.

Y

Indicates the same conditions as the E option, except disks are not selected. If you specify a pack-ID with the Y option, the statement is rejected.

If you are creating a new cycle for an existing file cycle set, and you omit the type, the type is taken from the latest existing cycle of the file. If all cycles of the file are in a to-be-cataloged or to-be-dropped state, then the type from the highest cycle of this type is used.

Options for Assigning Temporary Files

Do not use a processor name (ELT or DATA, for example) as a file name. The results can be unpredictable.

You cannot use the shared directory-ID (for MHFS).

The *options* field, if left blank, defaults to the A option unless the file is already assigned, in which case the options on the previous assign are used. If the file specified is not cataloged, the blank options field defaults to a T option, thus creating a temporary file. Any option not allowed on a statement causes the statement to be rejected.

If your site has Fundamental Security, the cataloging options B, C, G, P, S, R, U, V, and W are valid only when you are assigning a file to be cataloged. The C and U options indicate that a file is to be cataloged. The G, P, R, S, V, and W options place restrictions on the file after it is cataloged.

If you are assigning a file that was previously cataloged, be sure to use the A option. If you assign a file without the A, C, T, or U option, the Exec must search the master file directory for it. If it finds the file, the Exec assigns the file and honors any other options. If it cannot find the file, the Exec assigns the file as a temporary file and might ignore some of the options.

If you are assigning an owned file, you can assign a cataloged file as read only, regardless of how the file was cataloged, using

```
@ASG,AR filename
```

Do not attempt to do this if the file is unowned.

To delete a cataloged file by using the D, K, E, or Y option, you must include any read or write keys for the file on the @ASG statement if your site has Fundamental Security. If your site has Security Level 1 or higher, you must have delete access to the file.

filename

Identifies the file being assigned to the run.

type

Specifies a combination of letters and numbers that identify the addressing mode of the file (sector- or word-addressable) and the equipment type required for the file. The *type* field is also called the assign mnemonic.

Most types supplied by Unisys start with D or F, but sites can configure any mnemonics they wish. D represents word-addressable and F represents sector-addressable mass storage. If you omit *type*, the Exec uses sector-addressable mass storage. You cannot use the shared directory-ID for memory files (FMEM and DMEM) assign types. Appendix D, Table D-1 shows the standard type supplied by Unisys entries.

Refer to Appendix D, Table D-1 for a list of the available mass storage assign mnemonics and Table D-2 for the selection sequences supplied by Unisys.

Local fixed files can be placed in memory rather than on channel. FMEM (sector-addressable) and DMEM (word-addressable) can be used for this purpose; your site can configure additional assign mnemonics. Placing files in memory reduces access time, but consumes memory space that might otherwise be available for programs. Also, data in memory files is lost when the system is rebooted. Use of memory files assign mnemonics on systems that are not or cannot be configured to support memory files results in the request being satisfied on channel-connected mass storage.

Most removable disk files can be assigned using any assign mnemonic. However, for physically - the pack is mounted on a device that does not support physically removable packs, the following error message appears:

```
E:200633 pack xxxxxx cannot be mounted -- equipment type does not
permit interchanging of packs.
```

If you attempt to assign a file using a physically removable assign mnemonic and there are no units configured, the following error message appears:

```
E:256033 requested unit(s) not available
```

reserve

Gives an estimate of the space needed to create or update the file. The reserve can be 1 to 12 numbers. If you omit *reserve*, the Exec uses 0.

For sector- and word-addressable files, the initial reserve is an integer specifying the number of granules required by the file. See Section 3.2. You must provide a reasonable estimate of the space needed to create or update the file. The value used for a file update must include those granules already in use. Specifying a reserve reduces system overhead, as the space is allocated in contiguous granules, if possible. Omission causes the Exec to dynamically allocate the granules as they are required by the file.

The system places the first reserve value you specify for a cataloged file in the master file directory. Subsequent changes to the reserve subfield value are then treated according to the type of access established for a particular file. For cataloged files that are write enabled, the system overrides and replaces the recorded value with the new value you specify in your @ASG statement.

You cannot assign a new reserve value greater than a file's maximum. For files cataloged as read-only or assigned without an existing write key, the reserve subfield value must be the same as the master file directory value; otherwise, the request is rejected.

The facility that enables a user to expand an initial reserve (on a cataloged or temporary file) following its first specification is referred to as static expansion. With this facility, an initial reserve can be expanded, both on the device that was specified on the first @ASG statement and on other units or device types.

Expansion of a file assignment across a set of units is accomplished by changing the initial reserve and the type or placement fields. Changing the initial reserve field initiates allocation if the specification is an amount greater than the current initial reserve or the highest granule assigned. The extent of allocation corresponds to the increment above the mass storage currently allocated to the file. For files created with absolute placement, the allocation occurs on the device originally specified if the placement field is empty. If the mass storage is unavailable to do this, the request is rejected.

The maximum number of words allowed for the initial reserve is 262,143 times the number of words-per-granule, based on the granularity (track or position) of the file. Word-addressable files are allocated on a granule basis, and the relative sizes of reserve and maximum subfields are compared on a granule basis, not on a word basis.

Your system can be configured so that the unused portion of the initial reserve area allocated to cataloged files is released when the file is freed. This area is reallocated when the file is reassigned.

granule

Specifies the granule size. It can be as follows:

TRK

One software track (64 software sectors).

POS

One software position (64 software tracks). File space is allocated and released in 64-track increments or file-relative 64-track boundaries. The 64-track increments are not guaranteed to be contiguous on the physical medium.

If you omit granule, the Exec uses TRK. Once a file is created, the granularity can never be changed. The granularity is recorded in the master file directory.

maximum

Is the maximum allowable length of the file, in granules (for sector-addressable files) or words (for word-addressable files), given in 12 characters or less. Values for maximum are the same as for reserve. The maximum overrides the system default. If you did not specify maximum when you created the file, the reserve or system default is used, whichever is larger.

If a maximum was supplied when the file was cataloged, its value and the number of granules currently in use are recorded in the master file directory and used whenever the file is assigned. If a maximum is specified on a subsequent @ASG of a cataloged file, it replaces the previous maximum in the master file directory only if the file is assigned write-enabled. If the file is not assigned write-enabled and a new maximum is specified, the request is rejected. If you specify a maximum that is less than the highest granule assigned, the request is rejected.

This field is used to give the run an I/O error if the length of the I/O request exceeds the number of granules specified. It is used primarily to ensure that a runaway-file situation does not occur during debugging. However, you can also use this field to override the system default for all files.

placement

Specifies the placement of the file on a controller or device. The specification can be a logical or absolute request.

The *placement* can be any of the following:

**cu*

Is a controller name (absolute request).

**device*

Is a device name (absolute request).

i

Is a letter, A to Z, representing logical subsystems A through Z.

in

n is a number, 1 to 15, representing a logical device on controller *i*.

If an absolute placement cannot be done as specified, it is rejected. The Exec attempts the following file assignments:

- Files with the same logical specifications (that is, the same controller and device number) are assigned to the same physical device.
- Files with the same logical controller but different device numbers are assigned to different physical devices on the same control unit.
- Files with different logical controllers and device numbers are assigned to physical devices that are reachable by completely different CUs within the CONV selection string for the assign type specified.

Since logical placement is driven off of logical CU selections, specifying logical placement causes the Exec to avoid placing this file into memory space, even if MEMFL is specified for the assign type. Consequently, do not use the logical placement field if you want a file placed in memory on systems with MEMFLSZ configured. If a logical specification cannot be honored on a device within the CONV selection string, processing of the assign request ignores the placement field.

Placement on removable disk is not supported. The pack-ID enables you to specify allocation on a removable disk.

When the file is cataloged, placement information is placed in the directory to indicate the type of selection (logical or absolute, controller or device) and the device chosen for initial allocation. The directory cell containing this information is not affected by subsequent assignment of the file.

On reassignment, a placement specification is taken as a placement change overriding the last device used for allocation. A maximum of six characters, seven in the case of absolute placement, is allowed for specification of this subfield.

Logical placement specifications have meaning only within the current run. For example, logical controller "A" can represent two different physical controllers in two different runs.

Logical placement applies only to the reserve specified. If the file dynamically expands beyond the initial reserve, the Exec attempts to allocate on the same device as the last placement specified. If this is not possible, the expansion occurs on devices other than those specified in the placement field.

If the file is expanded through static expansion (reserve specified) and the file was created with absolute placement, the Exec allocates mass storage only on the device originally specified, unless a different device is specified in the placement field on the assignment. If there is not enough room on the specified (either implicitly or explicitly) device to satisfy the request, the request is rejected.

pack-ID

Specifies the removable disk packs required for the file. Pack-IDs consist of from one to six characters of the set A through Z, 0 through 9. The pack-IDs for cataloged files are recorded in the master file directory and need not be specified on reassignments.

A pack's master file directory must have sufficient space available to record the file otherwise a generation request is rejected with the following message:

```
E:207433 File cannot be created due to lack of removable directory  
space on pack pack-id.
```


A pack-ID cannot be specified unless the type subfield is also specified. If pack-ID is omitted on requests for noncataloged file space, fixed disk is assumed if disk equipment is requested. For cataloged removable disk files, specification of a pack-ID causes the pack to be mounted and registered. If already registered, the pack information is compared with the image for compatibility.

All cycles of a removable disk file must be on the same set of packs.

The maximum number of pack-IDs that can be specified on an @ASG statement is 510.

Many jobs can specify the same set of removable disk packs for unique files.

Pack-IDs can be added to unassigned single file cycle files by specifying the original pack-ID list with the new pack-ID s appended on an @ASG,A request.

Packs can only be added if the A option is specified (but not the Y option).

Packs cannot be added to currently assigned files or to files that have more than one file cycle.

If packs are being added, you must have delete access to the file.

For important information about using removable pack files, see the removable disk packs section in the *Exec System Software Administration Reference Manual* (7831 0323).

ACR-name

Specifies the access control record (ACR) for the cataloged file (if your system has Security Level 1 or higher). This must be an ACR that you have previously created with your security management product. See Section 4.3 for more information on creating and attaching ACRs.

encrypt-option

This subfield is used to specify encryption actions. For @ASG options C and U, DOENCR specifies that the file must be catalogued as encrypted, and DNENCR specifies that it must not be catalogued as encrypted. When these mnemonics are omitted, the default flag setting in the file owner's userid record is used to determine encryption state.

For @ASG with the A and X options, the MTENCR (maintain encryption) option can be specified when assigning encrypted files. This option allows the file to be written to and read from without encrypting or decrypting the data. This option is required by FAS (the File Administration System) to successfully save and restore encrypted files. ***Warning***This option is not recommended for general use.

7.2.2. Security Access Validation

When you assign a previously created file, you must pass security validation in order to gain access to the file. You receive access according to the validation rules for mandatory access control and discretionary access control. See Section 4 for more information.

7.2.3. Assigning Files on Shared Devices

If your system has Multi-Host File Sharing (MHFS), the following applies:

- You can assign cataloged files to shared devices. When you assign the file, it is allocated to either shared or local devices, but not both.
- You cannot assign a temporary file to a shared device.
- You cannot assign a shared device absolutely.
- You might not be able to assign a shared mass storage file if its assignment is restricted by the application-private feature. Refer to the *MHFS Planning, Installation, and Operations Overview* (7830 7469) for further information.

Examples

Assume that the following examples are all in the same run:

```
@ASG NEWBASE1.
```

The system initially assumes that NEWBASE1 is a cataloged file. If it cannot find the file, the system assigns the file as a temporary file.

```
@ASG,A NEWBASE2.
```

This statement assigns the file NEWBASE2 to the run. The A option means the file has been previously cataloged. The characteristics of the file were determined when the file was created.

```
@ASG,AZ NEWBASE3/NEON.
```

This statement requests that file NEWBASE3 be assigned to the run. NEON is the read key for the file. The A option means the file was previously cataloged. Because of the Z option, the system cannot assign the file if this would cause the run to be held.

The Z option is useful when you suspect a file has been rolled out to tape and you do not want to wait for it to be rolled back in.

```
@ASG,AX NEWBASE4.
```

File NEWBASE4 was previously cataloged (A option). The X option indicates that the file must be assigned exclusively to this run.

```
@ASG,T NEWBASE5.,F///400
```

This statement creates temporary file NEWBASE5. It is a sector-addressable file (F) with a maximum size of 400 tracks.

```
@ASG,T NEWBASE5.,F94/100//300
```

NEWBASE5 refers to the same file as the one in the previous example. This shows how you can use the @ASG statement to change the equipment type (F to F94) and initial reserve (10 to 100 tracks) from what you specified previously. If the file had already been cataloged, you would have used the A option.

```
@ASG,C NEWBASE6.,F/10//300
```

This statement assigns file NEWBASE6 to the run. The C option means the file must be cataloged if the run terminates normally, or if you release the file (@FREE) before the run terminates. NEWBASE6 is a sector-addressable file (F) with an initial reserve of 10 tracks and a maximum size of 300 tracks. It is a private file (no P option).

```
@ASG,UPR NEWBASE7.,F///3000/A1
```

This statement assigns file NEWBASE7 to the run. The U option means the file is cataloged when the file is released from the run (@FREE) or when the run terminates, even if the run terminates in error. When the file is cataloged, it is public (P option) and read only (R option). NEWBASE7 is given logical control unit placement A1. The file is on sector-addressable mass storage (F) and has a maximum size of 3000 tracks.

```
@ASG,CP NEWBASE8.,F///100/A2
```

This statement assigns the file NEWBASE8 to the run. The file is cataloged as a public file (P option) if the run terminates normally or if the file is released (@FREE) before the run terminates. NEWBASE8 is on sector-addressable mass storage (F) and has a maximum size of 100 tracks. NEWBASE8 is given the logical unit placement A2. Since NEWBASE7 was given the logical unit placement A1 in the previous example, the system tries to place the two files on disk drives that are connected on the same subsystem.

```
@ASG,AD NEWBASE9//ARGON.
```

This statement assigns file NEWBASE9 to the run. ARGON is the file's write key (ignored if the site has Security Level 1 or higher and the file is owned). The A option means the file has been previously cataloged.

If the site has Fundamental Security, the D option deletes the file if the run terminates normally, or when you release the file from the run (@FREE).

If the site has Security Level 1 or higher, you must have delete access to the file.

```
@ASG,CP SHARED#FILEM.
```

FILEM is cataloged in the shared master file directory. If the system does not have the file-sharing feature configured, the request is rejected. If the assignment is successful, then all hosts in the file-sharing system are able to assign the file after the file is freed.

7.2.4. Changing Assignment Fields for Mass Storage Files

You can change the type, reserve, maximum, or placement fields for a file with an @ASG statement when the following applies:

- You have already assigned the file with the A option. Change by adding the X and I options. You can assign the file again with the same options and specify a new type, reserve, maximum, or placement.
- You have already assigned the file with a C or U option. You can assign the file again with the same option and specify a new type, reserve, maximum, or placement.
- You have already cataloged the file but not assigned it. You can assign the file with an @ASG,A statement and give new specifications for the type, reserve, maximum, or placement.

To change the subfields that can be changed, the file must be write enabled. You cannot change the granule subfield.

The format for changing assignment fields is

```
@ASG[,options] filename[,type/reserve//maximum/placement]
```

options

A

Indicates that the changes apply to an already cataloged file. This option is required.

C

Indicates that the file is to be cataloged if the run terminates normally or if the file is released from the run. Use this option if the file was initially assigned with the C option.

I

Automatically releases the file from the run the next time a task terminates, regardless of the type of termination.

U

Indicates that the file is to be cataloged unconditionally when the run terminates (excluding a system stop) or when the file is released from the run. Use this option if the file was initially assigned with the U option.

X

Specifies that this run is to have exclusive use of the file until the run terminates, the file is released from the run, or the exclusive use of the file is released.

Z

Specifies that this statement is not to cause a hold condition. Control is returned to the user, indicating whether a hold state would have resulted from the assignment.

This option also requests that this file be reloaded if it is unloaded.

filename

Identifies the file on which you are changing fields. See Section 3.3 for a complete description of the *filename* field.

type

Specifies the addressing mode of the file (sector- or word-addressable) and the equipment type requested for the file. For more information about the type field, refer to the explanation of the @CAT statement.

You can use *type* to spread a file across different equipment types. However, the new type must be compatible with the old type. For example, if the old type is mass storage, the new type cannot be tape.

reserve

The number that you specify here changes the value of your initial reserve. This lets you increase the space for your file. For more information about the reserve, refer to the explanation of the @CAT and @ASG statements. The reserve cannot be expanded beyond the size given in the maximum subfield.

maximum

The number you specify changes the previous maximum allowable length of the file in granules (sector-addressable files) or words (word-addressable files). For more information about the maximum subfield, refer to the explanation of the @CAT and @ASG statements. The maximum can be expanded or reduced, but not below the highest granule assigned.

placement

Specifies the placement of the file on a controller or device. For more information about the placement field, refer to the explanation of the @ASG statement.

You can change the placement for the current assignment of the file. However, placement is not changed in the master file directory unless the placement is an absolute placement request.

7.2.5. Assigning Tape Files to Your Run

The @ASG statement lets you assign a tape file and one or two tape drives to a run. Using the subfields and letter options, you specify the type of hardware required and the mode in which it is to be used. The information recorded in the master file directory (MFD) and the specifications from the @ASG statement are used to select a tape unit for the file.

You must be aware of the following terms that are used in this section.

- The Half-Inch Cartridge (HIC) tape that can be used to record 18-track or 36-track formatted data.
- The Digital Linear Tape (DLT)
- Linear Tape Open (LTO): This device is released as different generations of the Ultrium technology. To denote the different generations, the generation number is appended to LTO. Therefore, a third generation LTO device is referred to as LTO3.

If the file is a shared file, then the file's MFD items are in the shared MFD, but only tape drives on the host on which the assign is done can be allocated.

You can specify all subfields of the @ASG statement when you are assigning a tape file that was previously cataloged, and the subfields are honored for the duration of the run. However, the values stored in the MFD are not changed.

A tape file or tape unit can be assigned to only one user at a time. Subsequent requests for the assigned file or unit are held until the file or unit is released from its current assignment, so you must free them as soon as possible.

If you enter an @ASG statement for a tape file that is already assigned to your run, the statement is ignored. You are warned that the file is already assigned; fields or options are not changed.

A temporary tape file is a file that exists only during the duration of a run and is accessible only by the run that created the temporary file. A temporary file can have the same file name as an unassigned cataloged file.

You might have to wait until a tape unit or file becomes available if your request is for one of the following:

- A tape unit that is being used by another run
- A cataloged tape file that is being used by another run
- A tape that is being used by another run except when the tape identifier is for a scratch tape (This is true only if the Tape Automatic Volume Recognition (TAVR) feature is on.)
- A tape currently in use for another file (only if TAVR is on)

Note: If your site has Security Level 2 or higher, you must pass security validation to access a tape with a security label.

Format

```
@ASG, options filename, type[/units/log/noise/processor/tape/  
format/data-converter/block-numbering/data-compression/  
buffered-write/expanded-buffer/, reel-1/reel-2/.../  
reel-n, expiration/mmspec, ring-indicator, ACR-name, CTL-pool ]
```

options

General Options

B

Specifies that the tape identifiers associated with the file being assigned must be considered nonunique on a system and run basis.

I

Automatically releases the file from the run the next time a task terminates, regardless of the type of termination. (A task is a discrete processing step in a run, involving the execution of an absolute element, such as a system processor or user program.)

E

Indicates even parity. Available only on 7-track tape units. This option is not allowed for any other tape types.

N

Specifies that a tape unit is to be assigned to the user, but no volume is assigned to the unit, and load messages to the operator are inhibited. In a program that uses the N option, you must issue ER TINTL\$ or ER TSWAP\$ before the tape file can be accessed for requests requiring that a tape be mounted (ER IOW\$, ER TLBL\$, and so forth). If you do not, the request is not processed and it hangs, waiting for a tape to be mounted.

If you use the N option for a cartridge tape within a cartridge tape library (CTL), the Exec searches the CTL for the cartridge-ID (or the BLANK identifier if a cartridge-ID is not specified). If the cartridge is found or if the CTL allocates a blank tape for this @ASG statement, a CTL-controlled tape drive is selected. When you issue ER TSWAP\$, a request is issued to the CTL to mount the tape indicated on ER TSWAP\$. Therefore, you need to ensure that the tape drive allocated on an N option assignment is the correct one to handle ER TSWAP\$.

No general statement can be made about the result of a processor call that accesses a tape file assigned with the N option because of the various ways tape files are handled. See the *ER Programming Reference Manual* (7830 7899) for information on Executive requests.

R

Indicates a read-only file. A file assigned with the R option cannot be overwritten. The file can only be read or deleted. Any activity requesting to write in the file is put in error mode. If your system has Security Level 1 or higher, you can assign a previously cataloged file as read only, regardless of how the file was originally cataloged.

W

Indicates a write-only file. The file can be only written into and extended in the process. Files created on labeled volumes are labeled as write-only.

Z

Specifies that the run must not be held, waiting for a tape unit or tape. If the required type of tape unit is configured in the system but is not currently available for assignment to the requesting run, a reject status is returned to the user.

In a CTL system, the Z option bypasses the database cartridge-ID search if the CTL is unavailable and no CTL pool name is specified. The system then attempts to allocate a freestanding tape drive. If the Z option is specified on an @ASG when requesting a blank tape and the CTL is unavailable when a CTL pool name is specified, the @ASG fails.

Options for Assigning Tape Files to Be Cataloged

C

Catalogs the file name and the file characteristics in the master file directory when the run terminates normally or when the file is freed. If a file by this file name already exists in the master file directory, the @ASG statement is rejected. Use @CAT instead of @ASG,C to catalog a file if you want the file cataloged without having it assigned to your run.

G

The meaning of this option depends on how security and Quota are configured at your site. Only the unconditional delete file system privilege overrides the G option.

- Unowned files

If your site has Fundamental Security or Security Level 1, 2, or 3 and the file is unowned, the following applies:

- When the file is cataloged, the G option guards against having its read key, write key, privacy mode, or all three overridden by privileged runs.
- If any F-cycle of a file set is cataloged with the G option, the read keys or write keys for the file set are obscured by '/////////' from all users. If a particular F-cycle of a file set is cataloged with the G option, the privacy mode and

either the project-ID or account number under which the file was created are also guarded.

- The G option means the file cannot be backed up.

- Owned files

If your site has Security Level 1, 2, or 3, the file is owned, and the dynamic configuration parameter (owned_file_read_write_keys) is FALSE:

- G-option files are no longer protected from privileged runs.
- The G option means the file cannot be backed up by FAS. When you need to back up a file, you must use another product such as FURPUR or IRU.

If your site has Security Level 1, 2, or 3 and the file is owned and dynamic configuration parameter (owned_file_read_write_keys) is TRUE, the following applies:

- When the file is cataloged, the G option guards against having its read key, write key, privacy mode, or all three overridden by privileged runs.
- If any F-cycle of a file set is cataloged with the G option, the read keys or write keys for the file set are obscured by '/////////' from all users. If a particular F-cycle of a file set is cataloged with the G option, the privacy mode and either the project-ID or account number under which the file was created are also guarded.
- The G option means the file cannot be backed up.

- Quota system

The use of the G option is also controlled by the Quota system on an account basis, as follows:

- If the Exec is configured to maintain quota limits for each account, this option can be used only if it is allowed in the quota limits for the run.
- If the Exec is not maintaining quota limits, this option can be used by all runs.

P

Catalogs the file as a public file. If your site has Security Level 1, 2, or 3 and you create unowned files and omit the P option (default), the file is cataloged as a private file and can be accessed only by runs that have the same project-ID or account number as your run.

If your site has Security Level 1 or higher and you create an owned file and omit the P option, the file can be private (only you can access it) or ACR-controlled (if you attach an ACR to it).

For owned files, a file is cataloged as public when the P option is specified on the @ASG statement and no access control record (ACR) is specified. The file is cataloged as private when the P option and ACR are omitted from the @CAT statement (default).

Use the TLABEL processor to change the public, private, or ACR-controlled access list of a tape file.

U

Indicates the same conditions as the C option, except that the file is cataloged unconditionally when the run terminates (excluding a system stop) or when you release the file from the run with an @FREE statement.

Options for Assigning Previously Cataloged Tape Files

A

Indicates that the file is already cataloged and ensures that it is not treated as a temporary file. The request is rejected if the name cannot be found in the master file directory.

For an owned file, the following applies:

- An ACR cannot be specified with this option.
- If this option is combined with the R option, read-only access is given.
- The user must have access privileges specified in the file's ACR.

D

Deletes a cataloged file, in addition to releasing it from the run. If it is a temporary file, it is just released from the run. If the file was assigned (@ASG) with the C or U option, it is not cataloged as it normally would be.

For systems with Fundamental Security, to delete cataloged files assigned with the C or U option, you must have included the correct read and write keys (if any) when you assigned the file to the run.

If the file is owned, you must have delete access to the file in order to delete it.

If your site has Fundamental Security, you must include any read or write keys that are attached to the file in order to delete the file.

If your site has Security Level 1, 2, or 3 and the dynamic configuration parameter `owned_file_read_write_keys` is TRUE, then you must include read or write keys to delete the file.

K

Indicates the same conditions as the D option, except that the file is deleted when the run terminates (regardless of the manner of termination) or when you release the file from the run with an @FREE statement.

If the file is owned, you must have delete access to the file in order to delete it.

If your site has Fundamental Security, you must include any read or write keys that are attached to the file in order to delete the file.

If your site has Security Level 1, 2, or 3 and the dynamic configuration parameter `owned_file_read_write_keys` is TRUE, then you must include read or write keys to delete the file.

Q

Requests that this file assignment be honored even if the system has disabled the file, and that read/write inhibit must not be set as a result of the file being disabled.

X

For tape assignments, the X-option can be used when the open reel block count checking feature is enabled by configuration tag `block_count_check` (BLOKCK). The X-option has no effect for cartridge tapes. The X-option ensures that I/O is not inhibited after the following two errors have occurred:

- Volume has no EOF label.
- Block count found in the EOF1/EOV1 label does not match the actual block count found on the tape.

Y

Assigns the file only for the purpose of examining the master file directory entry for the file. A PCT file item is created. Exclusive use, disabled, rolled out, and private mode are ignored. The file is read- and write-inhibited and the inhibit can never be overridden. No tape units are selected.

Options for Assigning Temporary Tape Files

R

Specifies that files written to labeled volumes are labeled as read-only files. They cannot be written to or deleted. If your system has Security Level 1 or higher, you can assign a previously cataloged file as read only, regardless of how the file was originally cataloged.

T

Specifies that the file is a temporary file. If one of the options from the group A, C, T, or U is not specified and the file name is not found in the master file directory, the system defaults to this option. The shared directory-ID cannot be used.

You cannot use the T option with the A, C, or U option; it is considered an error and causes the @ASG statement to be rejected.

W

Specifies that files written to labeled volumes are labeled as write-only files.

Hardware Options

E

Indicates even parity.

H

This option is available on 9-track and U9840D tape subsystems. If the density option "H" is specified with the general assign mnemonic "T", then one of the devices supporting the highest density that allows "H" density option is assumed by the Exec. This option is not available for other tape subsystems.

With the "H" density option allowed for two different tape types, such as 9-track and U9840D. These tapes are selectable through the general assign mnemonic "T" and the Exec's selection sequence criteria remains the same except for either 9-track or U9840D would get assigned. The selection sequence is dependent on what is configured, available, premounted, and what is in the CONV default list for tape assignment mnemonics and how it is ordered.

Note: *The potential for a run stream compatibility issue, for files assigned using the general assign mnemonic T option and the H density option, exists if the CONV default has both U9840D and U9H devices on the system.*

The following error message is returned if an invalid density option is used:

```
E:207133 Assign mnemonic [asg_mnemonic] does not support density option [n].
```

L

This option is available on all 18-track half-inch cartridge, U9840, and UT10KA tape subsystems. If the density option "L" is specified with the general assign mnemonic "T", then one of the devices supporting the highest density that allows "L" density option is assumed by the Exec. This option is not available for other tape subsystems.

With the "L" density option now allowed for three different tape types, such as 18-track half-inch cartridge, U9840, and UT10KA; and these tapes selectable through the general assign mnemonic "T", the Exec's selection sequence criteria remains basically the same except that 18-track half-inch cartridge, U9840, or UT10KA, would get assigned. The selection sequence is dependent on what is configured, available, premounted, and what is in the CONV default list for tape assignment mnemonics and how it is ordered.

Note: *The potential for a run stream compatibility issue, for files assigned using the general assign mnemonic T option and the L density option, exists if the CONV default has both UT10KA and HICL devices on the system.*

The following error message is returned if an invalid density option is used:

```
E:207133 Assign mnemonic [asg_mnemonic] does not support density option [n].
```

M

This option is a density option for the U9840C cartridge subsystems. This M option is also currently used for 36-track half-inch cartridge and CTS9940B tape subsystems. All other density options (for example, L and H) are not allowed for the U9840C subsystem.

The M option is available on all 36-track half-inch cartridge, U9940B, and U9840C tape subsystems. If the M option is specified with the general assign mnemonic T option, then one of the devices supporting the highest density that allows the M density option is assumed by the Exec. This option is not available for other tape subsystems.

With the M option now allows three different tape types (36-track half-inch cartridge, U9940B, and U9840C), and these tapes selectable through the general assign mnemonic T, the Exec's selection sequence criteria remains basically the same, except that 36-track half-inch cartridge, U9940B or U9840C, would get assigned. The selection sequence is dependent on what is configured, available, premounted, and what is in the CONV default list for tape assignment mnemonics and how it is ordered.

Note: *The potential for a run stream compatibility issue, for files assigned using the general assign mnemonic T option and the M option, exists if the CONV default has both U9840C and HICM devices on the system.*

The following error message is returned if an invalid density option is used:

```
E:207133 Assign mnemonic [asg_mnemonic] does not support density option [n].
```

O

Indicates odd parity (assumed for 9-track equipment). This option is allowed for 7-track, 9-track, and 18-track tapes.

S

Indicates density mode of 6250 bpi. This option is available only for UNISERVO 26, 28, 32, 34, 36, and 45N subsystems.

V

Indicates density mode of 1600 bpi. This option is available on all 9-track tape units and specifies a density of 85,937 bpi for DLT subsystems. If the density option is not specified for DLT subsystems, then the highest density supported by the DLT tape device is assumed. This option is not available for other tape subsystems.

Tape Labeling Options

F

Writes information on a labeled output tape to indicate that any reassignment of the tape for input requires only verification of the tape identifier unless security for tapes is configured. If you omit this option when you create a labeled output tape, the qualifier and file name are written into the label blocks. This forces all later assigns to use the same qualifier and file name that were used to create the tape and pass any security checks enforced at your site.

The F option has meaning only at file creation, not at file input time. Specifying the F option for an unlabeled tape does not cause a reject.

For systems with autolabeling configured, an unlabeled tape assigned with the F option is labeled automatically. For systems without autolabeling, the F option for an unlabeled tape is ignored.

Note: The F-option is implied for J-option tape assigns when the tape volume security level equals 2 (TVSL = 2). See the following table for tape volume security level descriptions.

J

Your system is configured at one of four tape volume security levels (0-3); the J option meaning depends on which level your system has:

Level	Meaning
0	Indicates the tape volume is to be unlabeled. Read and write accesses are allowed.
1	Indicates the tape volume is to be unlabeled. Read and write accesses are allowed.
2	For non-privileged users, all tapes written with the J option are automatically given security labels and those labels are created as if the user had also specified to F-option. Read of unlabeled tapes is permitted. For privileged users, write of unlabeled tapes requires executing with the system compartment set and clearance level 63 (system high) and possession of the bypass volume check privilege (which requires that the file SYS\$*DLOC\$ be assigned to the run).
3	To read or write to an unlabeled tape, you must be executing with the system compartment set and clearance level 63 (system high) and possess the bypass volume check privilege. If you are not executing under these conditions you receive an error message and the J option is ignored.

F and J

When the Media manager is loaded, F and J assign options are allowed together. If you use the F option along with the J option, the Media Manager does not enforce Qual*filename matching, but does provide all other access controls. However, these options are not allowed together when the Media Manager BYPASS, is used.

Subsequent tape accesses must use the matching Qual*filename with the Media Manager when J is used without the F.

If you do not specify an option, the master file directory is searched for a cataloged file with this name. If the file cannot be found, it is assigned as a temporary file.

The @ASG statement options for tape files affect either the creation of a new tape file or the assignment of an existing tape file. If you use an option that is not allowed, the statement is rejected.

The options for assigning a file to be cataloged cause the name and characteristics of the file to be placed in a master file directory, and the file to be assigned to your run. The C and U options cause the file to be cataloged; the G, P, R, and W options place restrictions on the file when it is cataloged. If your system has Security Level 1 or higher, the G, P, R, and W options work as listed in the mass storage assign.

For cataloged tape files, the J option is recorded in the master file directory; however, the F option cannot be used when cataloging tapes.

The hardware options are used with the temporary, cataloging, or cataloged options. All hardware options specified on the assignment of a tape file are recorded in the master file directory upon cataloging of the file.

If you specify more than one density option (H, L, M, S, and V), the @ASG statement is rejected. When you use a density option (E, H, S, or V), make sure it is compatible with the type. If a density option and the type conflict, the @ASG statement is rejected.

For 18-track tape assignments, the E, H, M, S, and V options are invalid. For HICL, the L option is assumed. These assumed options are used in the absence of overriding mode options.

General Mnemonics

The assign mnemonic used is the one supplied by the user, unless it is one that is flagged as a general mnemonic. Assign mnemonics are designated as general mnemonics by the site at system generation time. General mnemonics are refined to a more specific assign mnemonic by the Exec before cataloging the file. The Exec, as supplied by Unisys, has U9, HIC, and T flagged as general mnemonics.

You must use the most general assign mnemonic for your requirements in order to give the system more freedom in assigning units. To ensure that the options are compatible with all tape types accessible with that assign mnemonic, be careful

when you use options with the type field. For example, when using assign mnemonic T, you must specify only those options compatible with all types of units.

Selection Sequences

All assign mnemonics have a selection sequence associated with them. In some instances, only one choice is defined for an assign mnemonic. In those cases where more than one equipment type is associated with an assign mnemonic, the selection sequence supplied by Unisys always starts with the fastest equipment and then moves to slower equipment. This selection sequence can be changed by a site at system generation time. See Table D-4 for the selection sequences supplied by Unisys.

Assign Mnemonics and the CTL

The assign mnemonics associated with a CTL are determined at your site. The Exec, as supplied by Unisys, recognizes HICL, HICM, HIC, HIC40, HIC51, HICSL, LTO, LTO3, LTO4, and T as CTL-related assign mnemonics. When you use an assign mnemonic that is associated with a CTL, the Exec searches the CTL for the specified cartridge-ID (*reel-ID*).

If that cartridge is found in the CTL, a tape drive controlled by the CTL is allocated for the @ASG statement. If not found in the CTL, the Exec attempts to allocate a freestanding tape drive. If not found and if no freestanding devices are configured, the assignment is rejected. (The CTL is not searched if NONCTL is specified on the CTL-pool field.)

See your site administrator for CTL information specific to your site.

Assign Mnemonics and the ACS

When a system has a StorageTek Automatic Cartridge System (ACS) installed, there are special considerations to take into account when using certain assign mnemonics. Mnemonics such as U47, U47M, HICL, HICM, HIC, T, HIC51, HIC52, DLT70, HIS98, HIS99B, LTO, LTO3, and LTO4 select the type of device to assign, but do not determine whether the device should be selected from an ACS or a freestanding device.

For specific reel-ID assignments, an ACS-query determines the location of the cartridge. Once the location of the specific reel-ID is known, the Exec can select an appropriate device. An assignment requesting a specific cartridge that is found in the ACS results in a tape drive controlled by the ACS being allocated for the @ASG statement. An assignment requesting a specific cartridge that does not reside in an ACS results in a freestanding device selected.

For BLANK tape assignments, two CSC configuration tags (with the names `prefer_scratch_in_acs` and `query_for_scratch`) control whether a freestanding drive or a drive in the ACS library is assigned. See your system administrator for information about your site configuration. Also, see the *CSC Site Administrator's Guide* for more information on the ACS.

Mnemonics like U47L, U47NL, U47M, U47LM, NLCART, or LCART not only select the type of device, but also determine the location of the device—in other words,

whether it must be inside an ACS or outside an ACS. The mnemonics U47NL, U47M, and NLCART always select a freestanding device. The mnemonics U47L, U47LM, and LCART always select an ACS device.

Notes:

- *Assign mnemonics that can select an ACS drive are T, HIC, HICL, HICM, HIC51, HIC52, HIS98, U47, U47L, U47LM, LCART, DLT70, LTO, LTO3, and LTO4. Assign mnemonics HICL, U47, DLT70, HICM, HIC, LTO, LTO3, and LTO4 result in either an ACS query or cartridge tape library (CTL) poll. If the Automated Cartridge System (ACS) query is positive, an ACS drive is assigned. If the CTL poll is positive, a CTL drive is allocated. If the ACS query and CTL poll are negative, a free standing drive is allocated. NONCTL can be used on ACS or CTL assignments.*
- *In any configuration, absolute device assignment selects the requested device.*
- *For an ACS, the pool has no bearing on device assignment. The pool has bearing only during BLANK tape allocation and mounting.*

ACS names can also be supplied in place of the assigned mnemonic field. See your system administrator for these names.

Tape Density

Unless otherwise specified by assign mnemonic or mode option, the highest possible density for a tape is assumed. For example, 6250 bpi is the assumed density for a UNISERVO 36 tape unit.

Note: *In some cases, you are not able to append data to a tape (write past loadpoint).*

Compatibility Between Devices

Half-inch Serpentine(HIS) tape devices and LTO tape devices cannot read U9 (9-track) or DLT formatted tapes. HIC, U9(9-track), HIS, DLT, and LTO device types can only read tapes created for those specific types of drives. Tapes for one device type are not compatible with any other device type. Also, a two-unit assign cannot be mixed between HIC, DLT, HIS, LTO and U9 type devices.

Tape Automatic Volume Recognition (TAVR)

When the TAVR feature is installed, the assign request can be satisfied by a unit that is premounted with the requested tape and is compatible with the request.

If there is no premounted unit compatible with the request, the assignment is satisfied by a unit that is selected according to the selection sequence dictated by the type field. This unit can be overridden by an aftermount of the tape by the operator, if the aftermount unit is compatible with the request.

If the requested tape is being used by another run, the request is held except when the tape identifier is for a scratch tape. If the requested tape is being used by the requestor's run and if the run has only one activity, the request is rejected.

If this is a multiactivity run, the request is held except for scratch. However, if this file is assigned with a B option (subject to security check), the request overrides the tape-ID conflict.

When all available units are in a premount-only status, the Exec cannot select a unit except when the request is for an absolute device/control unit or the requested tape is premounted on one of the units. For more information on TAVR, refer to the *Exec Administration Reference Manual* (7831 0323).

units

Specifies the number of tape units required; it can be integers 1 or 2. If omitted, the Exec assumes that one unit is required. A nonnumeric character or a numeric value greater than 2 causes the statement to be rejected. The number of units assigned is not retained in the master file directory upon cataloging. When two units are specified, the Exec guarantees that both units are compatible.

Prior to SB4R6, the two cartridges specified on a two-unit @ASG could not reside in different cartridge library units (CLU), nor could one of the cartridges be in a CLU while the other was nonresident. Now (with TAVR), a two-unit tape @ASG completes successfully even if the two cartridges specified reside within different CLUs, or if one cartridge resides in a CLU and one cartridge is not resident in any CLU. The nonresident cartridge is assigned to a freestanding drive.

With absolute device specification, a two-unit @ASG can explicitly allocate cartridge tape drives in different CLUs, or mix freestanding and CLU-controlled drives. In addition, the specified cartridge-IDs might reside in any CLU or outside the CLUs. The cartridges must be available to the explicitly specified drives for a successful @ASG.

You cannot request a BLANK tape on both a CTL-drive and a freestanding drive on the same two-unit @ASG request.

If the system cannot immediately grant the user's request but has the potential to do so, the user can be put on hold and sent an appropriate message. If the user's request cannot be granted now or later, the request is rejected. For example, a request for two tape servos is rejected if the user's quota limit is set to one servo, regardless of how many servos are currently available. But if a user with a quota limit of two servos makes the same request with one servo currently available, that request is put on hold.

log

Assigns a single letter indicating a logical control unit. The Exec attempts to assign all files with the same letter to the same physical control unit, and those with different letters to different control units. The letter specified is not placed in the master file directory. This subfield permits more efficient I/O operation because of the separate control units.

Logical control unit specification has meaning only within the current run.

noise

Specifies an integer from 1 to 63 that overrides the standard system noise constant. If you omit noise, either the standard system noise constant is used or

the value is stored in the master file directory, in the event of the assignment of a cataloged file (@ASG,A). Any 9-track tapes written in a density of 1600 bpi and 6250 bpi have 0 as the system standard value. The noise value is recorded in the master file directory upon cataloging. Do not specify a noise subfield if you are assigning a cartridge tape.

processor/tape

Specifies the type of translation required. Processor defines the format of the data in the processor, and tape defines the format of the data on tape. You must specify both processor and tape or omit both. If you omit them, no translation occurs.

Data translation is file independent and the translation mode remains in effect for the duration of the reading or writing of the tape. You can change data translation from file to file, but you must track this information yourself.

The supported processor/tape mnemonics are

Processor Code	Tape Code
ASCII	EBCDIC
ASCII	FLDATA
EBCDIC	ASCII
EBCDIC	BCD
FLDATA	ASCII
FLDATA	BCD
FLDATA	EBCDIC

Your site can define additional processor/tape mnemonics, provided neither code exceeds 6 characters. When the file is cataloged, the values given in processor/tape are stored in the master file directory.

All translation takes place in the hardware.

Note: *If you are using the data compression feature to read and write to half-inch cartridge tape, you cannot use the data translation feature.*

format

Specifies the data transfer format for the word-to-byte conversion.

Permissible format entries are as follows:

Q

Quarter word

8

8-bit packed (default on 9-track, 1/2-inch cartridge tape)

The data transfer format is recorded in the master file directory upon cataloging, to be used on subsequent assigns unless specifically overridden.

data-converter

This field only applied to 7-track tape and now must be left blank. (Although this option no longer has any effect, it is allowed since it might still be in user runstreams.)

Its allowable value is as follows:

OFF

Turns off the data-converter if assign is made from the master file directory (@ASG,A) and the file was cataloged with data conversion. In all other cases, OFF is ignored.

If you do not specify data-converter, no action is taken in relation to the data converter. Furthermore, the OFF mnemonic is not recorded in the master file directory, except as implied by the absence of data-converter.

block-numbering

Determines whether block numbering is on. Block numbering is available on HIC, HIS, DLT, and LTO tape subsystems. For the HIC, HIS, DLT, and LTO cartridge tape subsystems, the user cannot turn on block numbering with this subfield because it is performed by the hardware and is transparent to the user.

Specify one of the following values:

BLKON

Turns block numbering on (invalid for HIC ,DLT, and LTO cartridge tape subsystems).

BLKOFF

Indicates that blocks must not be numbered. (This does not actually turn off block numbering.)

BLKOPT

Gives the user the option to turn block numbering on at a later date (invalid for HIC, DLT, and LTO cartridge tape subsystems).

data-compression

Reduces the amount of physical tape required to record data by removing repetitive data before recording it (available for HIC, HIS, DLT, and LTO cartridge tape subsystems only). Data compression is generally performed on a file basis.

For half-inch cartridge tape subsystems, these are the valid data-compression values and their meanings:

CMPON

Enables 8- or 9-bit data compression depending on the format parameter. When Q is specified on the format subfield, enables 9-bit data compression; when either 8 or nothing is specified on the format subfield, enables 8-bit data compression. Enables LTO-DC for an LTO cartridge tape device.

CMPOFF

Disable data compression.

CMP8ON

Enables 8-bit mode of data compression.

CMP9ON

Enables 9-bit mode of data compression.

CMPOPT

Capable of LTO-DC data compression. Gives you the option to enable data compression at a later date; must be used if there is any possibility that the cartridge tape contains an LTO-DC compressed file.

CMPEON

Enables enhanced data recording capability (EDRC) mode of data compression.

CMEOPT

Capable of EDRC data compression. Gives you the option to enable EDRC data compression at a later date; must be used if there is any possibility that the cartridge tape contains an EDRC compressed file.

For DLT, HIS98, and HIS99B tape subsystems, these are the valid data-compression values and their meaning:

CMPLON

Enables LZ1 data-compression. DLT and HIS98 cartridge tapes are not compatible with each other or with future Unisys tape products that can use this compression value.

CMPOFF

Disable data compression.

Examples

```
@ASG,TF TAPE.,HICL////////CMPEON/BUFON,111C,,RING
```

The half inch cartridge tape is assigned as labeled with EDRC data compression explicitly requested and buffered-write capability.

```
@ASG,TJ TAPE.,HICL////////CMEOPT,111C,,RING
```

The half-inch cartridge tape is assigned as unlabeled, requesting a tape device that has the hardware capability of supporting EDRC data compression.

Support for 4780 18- and 36-track tapes is provided by standard Exec. Compression on the 4780 subsystems is called improved cartridge recording capability (ICRC) and is totally compatible with the 5073 enhanced data recording capability (EDRC). It is selected by using the CMPEON value.

Device characteristics for compressed tape readability are

- Unisys proprietary compressed cartridge tapes cannot be read on 4780 drives, but EDRC compressed cartridges from 5073 drives can be successfully read.
- A cartridge drive on a 4780 that does not have compression hardware can read only uncompressed cartridge tapes.
- The 4780 drives that have compression hardware can read EDRC compressed tapes without having to explicitly request CMPEON.

Also, 36-track tapes have more capacity on current cartridges because data is written in both directions on the tape, and more capacity with longer tapes can hold up to four times more data than current cartridges (1,100 feet [E-carts]).

The following tape assign and catalog modes are available for 4780 drives:

```
@ASG,TJ T.,HICM////////CMPEON/BUFON/EBCOPT,REELID
```

The following tape assign and catalog modes are available for 5136 devices:

```
@ASG,TJ T.,HIC51////////CMPEON/BUFON/EBCOPT,REELID
```

Notes:

- *Although you can explicitly specify 8- or 9-bit data compression with CMP8ON and CMP9ON, you can allow the system to select the best compression ratio by specifying the format of the data.*
- *You might experience a loss in performance if you use the read backwards function (the RB\$ function of the ER IOW\$) of the 8- or 9-bit data compression feature on a half-inch cartridge tape subsystem.*
- *Tapes created by using 8- or 9-bit data compression cannot be used with Exec levels lower than 41R6/41R6M.*
- *An I/O error status of 024 (012 for MR\$ operation) results if you use the read backward function on an EDRC compressed data block.*
- *When you are using EDRC data compression on a half-inch cartridge tape, user data block sizes are limited to 128K bytes or smaller. An I/O error status results if user data block sizes are greater than 128K bytes when writing with EDRC data compression used.*
- *When you are reading a tape with EDRC compressed data, EDRC mode need not be enabled, as long as an EDRC capable device has been assigned (for example, CMEOPT was used on the assignment).*

- *When you are reading a tape with LZ1 compressed data, LZ1 mode need not be enabled, as long as an LZ1 capable device has been assigned (such as a DLT70 assign mnemonic was used on the assignment).*
- *Tapes created using EDRC data compression cannot be used with Exec levels lower than SB4R7.*
- *Tape devices that support EDRC or LZ1 data compression do not support data translation (ASCII/EBCDIC) or 8- or 9-bit data compression.*
- *If you write data that is not highly compressible to USR5073, a compression error occurs in the subsystem and is reported to the Exec. This compression error is logged in the 1252 log entry by the Exec and your I/O is reissued as part of recovery action. The USR5073 then writes the data uncompressed into the tape. If, however, the compression error occurs too often, you might experience a loss in performance. A solution would be to turn compression off.*
- *CMPEON, CMEOPT, and EBCOPT are valid values for the USR5073 device when using HICSL.*
- *When using the HICL mnemonic in combination with CMPEON, CMEOPT, and EBCOPT, a SCSI USR5073 or a BMC 5073 device might be selected.*

buffered-write

Specifies whether buffered writes are to be used when a tape is mounted on a device controlled by a control unit that has buffering capability.

These are the values:

- BUFTAP

Enables buffered write mode for user data blocks and file marks for the entire tape.

- BUFFIL

Enables the buffering of write data including label blocks and tape marks for a file written to labeled tapes. The system ensures that writing of the data and file marks occurs at the close of a file.

- BUFON

Enables buffered write mode for user data blocks.

- BUFOFF

Disables any buffered write mode.

The @ASG statement is rejected if it contains the BUFTAP or BUFFIL value and there are no tape units up and available that support tape mark buffering (TMB). Even though TMB is supported on the CTS5236, the BUFTAP value is not. The rejection message is:

```
E:256033 requested unit(s) not available.
```

The following rules apply when using BUFTAP, BUFFIL, BUFOFF, and BUFON:

- When the user switches modes from BUFFIL to BUFON or BUFOFF, writing the buffer to tape is postponed until the next I/O to the tape subsystem. This also applies when switching modes from BUFTAP to other modes.
- If BUFTAP, BUFFIL, BUFOFF, or BUFON is not specified, and the configuration tag TAPBUF (dynamic nametape_buffering) is zero, the system uses the default value BUFON.
- If BUFFIL is specified for an unlabeled tape, the data blocks for writes are buffered and one synchronization per file is done by a WEF\$ function just as it is without the BUFFIL specification.
- When using buffering, it is the user's responsibility to retain the original data for possibly performing error recovery, if that is the user's requirement. The Exec provides no means for the user to recover this data. With the use of the BUFTAP option, it is possible for the user to have received good status on previous writes of thousands of files, when in fact none of the files has made it to tape, and if an error occurs while they are being written to tape, the remaining blocks in the tape subsystem's buffer might never get written to tape.

The Block ID SAFE (BSAFE\$) I/O function provides a means for the user to determine what block has been actually written to tape by the tape subsystem.

- When using BUFTAP, a FSAFE\$ can be performed at any point if a user wants to insure that buffered data/tape marks have been written to tape. It is highly recommended that an FSAFE\$ be performed and its status checked after all user data/tape marks have been written to the tape device, but before releasing the tape. This insures that the user is notified of any errors in writing any remaining buffered data/tape marks to the tape. This also allows the user to know that an error on a @FREE, TSWAP\$, TINTL\$, or TLBL\$ was not the result of writing any user buffered data/tape marks to the tape.

expanded buffer

The expanded buffer lets you explicitly select a tape unit that has the hardware capability of supporting the data block sizes up to 256K bytes. If the tape unit supporting the selection is not available, then the request is rejected with the error message, "Requested unit(s) not available." If EBCOPT is not specified in the expanded buffer subfield, then any one of the first available units is selected.

Notes:

- An I/O error status of 06 results if user data block sizes are greater than 128K bytes when writing with EDRC data compression.
- The CTS5136, CTS5236, CTS9840, DLT7000, CTS9940B, LTO3, and LTO4 devices always have expanded buffering, so the EBCOPT is not needed to get a CTS5136, CTS5236, CTS9840 DLT7000, CTS9940B, LTO3, or LTO4 which supports expanded buffering.

Examples

```
@ASG,TF TAPE.,HICL/////////EBCOPT,111C,,RING
```

The half-inch cartridge tape is assigned as labeled, requesting a tape device that has the hardware capability of supporting block sizes of up to 256K bytes.

```
@ASG,TJ TAPE.,HICL/////////BUFON,111C,,RING
```

The half-inch cartridge tape is assigned as unlabeled with buffered-write capability. The selected drive can be capable of supporting block sizes of up to 256K bytes.

reel-n

The reel-n specifies the identifier for each tape reel or cartridge required. Each tape identifier is limited to 6 characters from set A through Z and 0 through 9. Do not specify an identifier with a single A, E, Y, or N. These characters correspond to valid responses to tape labeling messages. Using them as identification could cause tape assign problems.

When an assignment uses a specific reel-ID, its location is determined by an ACS query or a CTL poll. The reel-ID's location, together with the assign mnemonic, determines the resultant device assigned. When a BLANK is assigned, no ACS query is done. However, BLANK assignments result in a CTL poll.

The special mnemonics BLANK and SCRTCH have these meanings:

BLANK

Designates a tape volume that can be overwritten. In a CTL system, blank tapes can be allocated automatically from a named pool (CTL-pool). A tape with the BLANK identifier can be retained after the assign for the duration of the label's expiration period, or until you release the tape.

Obtain the tape reel number from the system operator or from operator messages logged in your run's tailsheet. Your site must have procedures established for allocating blank tapes.

SCRTCH

Designates a tape volume for temporary use during this assignment. The tape can be used by anyone else after it is freed (@FREE). The label might be written, but if you use the SCRTCH reel-ID, anyone is allowed access.

If you want a BLANK tape request to be satisfied by the CTL, specify a CTL-pool name on the CTL-pool field. If you do not want a BLANK tape request to be satisfied by the CTL, specify NONCTL on the CTL-pool field.

Note: *Your system can be configured to allocate BLANK cartridge tapes from the CTL even if you do not specify a CTL-pool name (for example, the default CTL pool can be used).*

A maximum of 2,047 reels or cartridges can be specified with the @ASG statement, although only two can be accessed at any one time.

If the file is to be cataloged, all tape identifiers are recorded in the master file directory. Tapes are used and cataloged in the order specified.

If you omit this field, the Exec directs the operator to mount blank tapes on the appropriate tape units and to provide identifiers for each tape that is unlabeled. (Omitting this field is the same as entering BLANK.)

If additional tapes are requested by ER TSWAP\$, the operator is asked to load the required blank tapes and provide identifiers for each unlabeled tape. On a tape file with SCRTCH for an identifier, ER TSWAP\$ cannot swap tapes beyond the total units count (1 for one-unit assignment and 2 for two-unit assignment). On an attempt to swap tapes beyond the total units count, the first scratch tape is rewound and used again.

For currently cataloged files, the reel field is normally void, indicating that the reels or cartridges listed in the master file directory are used in the order created.

If you supply tape identifiers, they must be of the set listed in the master file directory, but can be a subset and listed in any order, allowing the user to omit or access them in any order. If you enter an invalid tape identifier, the @ASG statement is rejected. The Exec does not detect the case of a legal identifier for a tape that does not exist.

In either case, when the known tapes are exhausted and additional ones are requested, blank tapes are used, and their numbers are added to the master file directory. (This is not allowed if the file is cataloged in the read-only state.)

When an unlabeled tape is being mounted, the following prompt is displayed on the system console to ask for the tape's reel number:

```
n ANSWER WITH REEL NO ON unit E,REEL
```

When this tape is @FREE and the next one assigned, the same scenario is repeated.

However, a special case occurs when an autoloader is installed and the S option is used on the @FREE statement: For the first tape, the operator is prompted for the reel number as described above. If this tape is released using @FREE,S, the next unlabeled tape in the autoloader is not prompted for the reel number.

This is a useful feature that can significantly improve efficiency, especially when a bunch of unlabeled tapes are to be processed.

expiration-period

This field has meaning only for labeled tape volumes and is ignored in all other cases. It specifies the number of days that this file is to be retained.

The maximum number allowed is a site configuration field that can never exceed 4,095 days. Specification of a value greater than the maximum causes the @ASG statement to be rejected. The system standard value, which is also site configurable, is used if no expiration period is specified.

In addition, you can use the F option to determine whether the tape can be overwritten prior to the expiration date.

The Exec guarantees that the expiration period on all files beyond the first file on a volume can be less than or equal to the first file. If a subsequent assign has an expiration period greater than that remaining on the first file, the Exec automatically changes the expiration period of the newly added file to the remaining expiration period of the first file. The expiration period is stored on the tape and not in the master file directory.

Assign Tape	File name match	File names do not match
BLANK tape requested	read - > error write - > only if expired	read - > error write - > only if expired
BLANK tape not requested	expired - read or write is allowed. (Access code no longer has meaning.) not expired - set read/write inhibits based on HDR1 access code	expired write - > OK not expired if 'F' set read/write inhibits based on access code. Not 'F' error.

The expiration period also specifies how long a file is protected by

- File name verification (F option on the @ASG statement)
- Restricted read access (R option on the @ASG statement)
- Restricted write access (W option on the @ASG statement)

mmspec

A Media Manager specification subfield of mmspec is relevant when the Media Manager is being used. The subfield must be 6 characters or less, and the variable portion must be less than 100 for xx, yy, and zz for specifications, USEDxx, DCYCyy, and YEARzz. *mmspec* can be one of the following:

BYPASS

Bypass the Media Manager, subject to site restrictions.

USEDxx

Expire if not used for xx days.

MFD

Expire if not cataloged in the MFD.

DCYCyy

Retain yy copies with unique create dates.

FCYCyy

Retain yy copies regardless of create dates.

READ

Do not expire and do not allow to be written.

NOSCR

Do not expire but allow write by owner.

YEARzz

Retain zz years.

DAYS

Interpret expiration as a number of days regardless of the Media Manager parameter STRXCODE.

You can code any legal value for the expiration field when you use it with the Media Manager specification for the HDR1 label. The expiration of the tape by the Media Manager is subject to the following rules:

- READ, NOSR, and YEAR take precedent over the expiration value
- USED, MFD, DCYC, and FCYC are considered in addition to the expiration.

You cannot use mmspec on an @CAT statement, but you can enter it on the @ASG,A statement for a cataloged tape file.

You can overwrite the files on the tape before the end of the expiration period (assuming you pass the security checks) only if one of the following sets of conditions is true:

Condition 1:

- You created the tape with the F option and without the R option.
- You specified the reel-ID on the @ASG call or ER TSWAP\$ function 2 request that matches the reel-ID in the tape volume label that you want to overwrite.

If additional tapes are requested by ER TSWAP\$, the operator is asked to load the required blank tapes and provide identifiers for each unlabeled tape. On a tape file with SCRTCH for an identifier, ER TSWAP\$ cannot swap tapes beyond the total units count (1 for one-unit assignment and 2 for two-unit assignment). On an attempt to swap tapes beyond the total units count, the first scratch tape is rewound and used again.

Condition 2:

- You created the tape without either the F or R options.
- You access the file with the same qualifier and file name.
- The reel-ID matches the one in the tape volume label.

If you used both the F and R options when you created the tape, you cannot overwrite the files until the end of the expiration period.

ring-indicator

This field specifies whether the media for this file (open reel or cartridge) is write protected or write enabled. For 10-1/2 inch open reels, a write ring allows/disallows write operations. For cartridge tapes, a file-protect switch on the cartridge itself allows/disallows write operations.

The allowable values (independent of tape equipment used) are as follows:

WRITEPROTECT or NORING

Specifies that write operations (cartridge or open reel media) are not allowed.

WRITEENABLE or RING

Specifies that write operations (cartridge or open reel media) are allowed.

If you do not specify a value for ring-indicator, the operating system ensures that the tape is write enabled if the first I/O to the tape at load point is a write function.

Note: *You must always specify an allowable value in this field to ensure that the tape is properly protected.*

Enforcement of the ring-indicator values in a CTL is accomplished in a different manner. The CTL sets the tape drive allocated for the assignment in a write-protect or write-enable mode. The cartridge is not required to have the write-protect switch set when the tape drive is set to write-protect mode. However, if the cartridge has write-protect set, no write is possible without changing the physical cartridge switch.

ACR-name

Specifies the access control record (ACR) for the cataloged file if your site has Security Level 1 or higher. The *ACR-name* is 1 to 6 characters in any combination of letters (A through Z), numbers (0 through 9), \$, and - (dash). The ACR must already exist and must be owned by the user doing the assign.

To attach an ACR to a temporary tape, you must use @TLABEL.

CTL-pool

Can be one of the following:

CTL-pool-name

Directs the Exec to allocate BLANK tapes from the designated

CTL pool of cartridge tapes. *CTL-pool-name* can be 1 to 12 characters, consisting of the alphanumeric characters, \$, and - (hyphen). Using a CTL pool name might place your assignment in a hold condition while the Exec waits for an available tape drive (except when you specify the Z option, which returns a reject status). You can remove the wait status with an @@X T statement.

NONCTL

Directs the Exec to use freestanding cartridge tape drives. The Exec cannot search the CTL for a cartridge-ID because the cartridge cannot be inserted into the CTL.

Your site might have a default CTL pool for assignments that specify BLANK tapes but do not specify a CTL pool.

You can override a previously specified CTL pool name for a cataloged tape file by specifying a different CTL pool name on an @ASG,A statement. If you do not specify a CTL pool name on an @ASG,A statement, the CTL pool name is taken from the master file directory. You cannot change the CTL pool name for a cataloged file in the master file directory.

If the CTL-pool field is blank, the Exec searches the CTL for the cartridge (reel-ID) and uses the CTL if the cartridge is found there; otherwise, it allocates a freestanding cartridge tape drive and issues the LOAD message. If the cartridge is not found in the CTL or if no freestanding drives are configured, the assignment is rejected.

If you specify the Z option when the CTL is not available, and you do not specify a CTL pool name, the Exec attempts to use freestanding tape drives; if you do specify a CTL pool name, the request is rejected. If you specify NONCTL on the CTL-pool field, the assignment request is not impacted when the CTL is unavailable. See your site administrator for CTL information specific to your site.

Examples

```
@ASG,A FREMONT1.
```

This statement assigns file FREMONT1 to the run. The characteristics of the file, facility requirements, and reel numbers were determined when the file was cataloged. The project-ID of the current run is used as a qualifier.

```
@ASG,ADS FREMONT2.,28N//A////////,N212,,,ACR5
```

This statement assigns file FREMONT2 to the run. The file is currently cataloged but is deleted if the run terminates normally. The UNISERVO 28 nine-track tape unit is to be on logical channel A and reel N212 is to be used. The density is 6250 bpi (S option).

```
@ASG,U FREMONT3./ECTO/ENDO,U9V/2,707/708/709/710
```

This statement assigns file FREMONT3 to the run. Because of the U option, the file is cataloged when the file is released from the run (@FREE), or when the run terminates, even if it terminates in error. U9V/2 means the file requires two 1600-bpi 9-track tape units on any channel. Reels 707 through 710 are to be used. The file is protected by a read key (ECTO) and a write key (ENDO). (Read and write keys are available only on systems with Fundamental Security.)

```
@ASG,T FREMONT4.,T///36
```

This statement assigns temporary file FREMONT4 to the run. The file requires one tape unit selected by the Exec; one or more blank reels can be used. The noise constant is set to 36 words.

```
@ASG,T FREMONT5.,T////////,SCRTCH
```

This statement assigns temporary file FREMONT5 to the run. SCRTCH (you must spell it this way) is a reserved word that indicates you want any unlabeled tape. SCRTCH can be used multiple times by multiple users simultaneously; it is a generic designation for an unlabeled tape. The Exec selects a type of tape unit and tape density for the file.

```
@ASG,U FREMONT6.,30////////DC
```

This statement assigns file FREMONT6 to the run. The file is cataloged when the file is released from the run (@FREE), or when the run terminates, even if it terminates in error. The file is assigned to a UNISERVO 30 seven-track tape unit.

The 6-bit packed data transfer format is the default. Because of the DC value for the data-converter subfield, three 8-bit bytes are converted from four 6-bit tape characters. The data-converter status is recorded in the master file directory upon cataloging.

```
@ASG,TJ FREMONT7.,U9V///ASCII/FLDATA
```

This statement assigns temporary file FREMONT7 to the run. The file is assigned to a 1600-bpi 9-track tape drive that has ASCII-to-Fielddata hardware translation. An unlabeled blank volume is required.

```
@ASG,T FREMONT8.,U9S////////Q////////,100
```

This statement assigns temporary file FREMONT8 to the run. The file is assigned to a 6250-bpi 9-track tape drive (U9S) by using quarter-word data transfer format (Q). An expiration period of 100 days is specified. One or more blank reels are used.

```
@ASG,V FREMONT9.,45N////////8////////,WRITEENABLE
```

This statement assigns file FREMONT9 to the run. It is assigned to a 2145 tape unit, using a density mode of 1600 bpi. The 8-bit packed data transfer format is used.

```
@ASG,C FREMONT10.,HICL/////////BUFON,,WRITEEPROTECT
```

This statement assigns file FREMONT10 to the run. It is assigned to an 18-track cartridge tape unit. The file is cataloged when the file is freed or when the run is terminated normally. Buffered write mode is enabled by specifying BUFON. The cartridge tape is write protected.

```
@ASG FREMONT11.,36/////////BLKON/CMPON
```

This statement assigns file FREMONT11 to the run. It is assigned to a UNISERVO 36 nine-track tape unit on a 5061 tape control unit. Block numbering and data compression are turned on.

```
@ASG FREMONT12.,HICL////////Q///CMPON/BUFON,,WRITEENABLE
```

This statement assigns file FREMONT12 to the run. It is assigned to an 18-track cartridge tape unit. The Q value in the format subfield and the CMPON value in the data-compression subfield enable 8-bit data compression on the half-inch cartridge tape subsystem. BUFON enables buffered write mode.

```
@ASG FREMONT13.,HICL,BLANK,,,,CTL-AUDIT1
```

This statement assigns file FREMONT13 to the run and allocates a BLANK tape from CTL pool CTL-AUDIT1.

```
@ASG FREMONT14.,HICL,,,,CTL-AUDIT2
```

This statement assigns file FREMONT14 to the run and allocates a BLANK tape from CTL pool CTL-AUDIT2.

```
@ASG,T FREMONT15.,HICL,9286A
```

This statement assigns temporary file FREMONT15 to the run. The Exec searches the CTL for cartridge 9286A and, if found, requests automatic mounting. If this cartridge is not found in the CTL, the Exec allocates a freestanding drive and issues a LOAD message.

When TAVR is installed, a two-unit tape @ASG successfully completes even if the two specified cartridges reside in different cartridge tape libraries, or if one cartridge resides in a CLU when another cartridge is not resident in any CLU.

The following examples show you how @ASG is handled in a multi-CLU environment with TAVR installed.

```
@ASG,TJ T1.,H1CL/2,REEL1/REELX
```

This example shows the @ASG,TJ with Reel1 in a CLU-A and ReelX outside of the CTL. A tape drive in CLU-A is assigned and Reel1 is mounted on it by the CTL. ReelX is mounted by the operator onto a freestanding drive.

```
@ASG,TJ T1.,*D1/*D5,REEL1/REEL3
```

This example shows an absolute @ASG,TJ where each tape is in a separate CLU. Reel1 is a cartridge which resides in CLU-A. Reel3 is a cartridge that resides in CLU-B. If drive D1 resides in CLU-A and drive D5 resides in CLU-B, then the assignment completes successfully. Reel1 is mounted by the CTL on drive D1 and Reel3 is mounted by the CTL on drive D5.

```
@ASG,T file,H1CL/2,BLANK,,,,FAS-POOL
```

```
@ASG,T file,*CLU2-drive1/*CLU2-drive2,BLANK,,,,FAS-POOL
```

These two statements are examples of using FAS with CTL-pool. The second shows how drives are allocated in CLU2 only. If the FAS-POOL is depleted, the @ASG is rejected.

7.2.6. Assigning VTH Files to Your Run

Virtual tape handler (VTH) simulates tape by using mass storage. The @ASG statement lets you assign virtual tape files to a run. You assign them in the same manner as a standard tape file, using an assign statement. Your access and manipulation of virtual tape files is the same as standard tape files. Your control statements remain identical for a virtual tape as for a standard tape. Programs cannot tell the difference between a virtual tape and a real tape. Virtual tapes have the characteristics of most open reel types and therefore do not support fast tape access (FTA), block numbering, data compression, buffered writes, or expanded buffer.

The options and parameters on the assign statement for a virtual tape are generally the same as for a standard file.

Format

```
@ASG,options filename,type[/units/log/noise/processor/tape/  
format/data-converter/,reel-1/reel-2/.../reel-n,expiration/  
mmspec,ring-indicator,ACR-name,CTL-pool]
```

options

Because there is no physical reel to mount and no way to identify to the handler a different file to assign, the B option is illegal for virtual tape assignments.

type

Installation of the virtual tape feature adds two new type mnemonics (VT and VTH) to the standard set used for assigning tape files.

reel-n

The reel number specifies the identifier for each virtual tape assignment. Each reel identifier is limited to 6 characters from the set A through Z and 0 through 9.

7.2.7. Assigning DVD Tape

The DVD tape simulates tape by using DVDs as the recorded media. You assign them the same way as a standard tape file, using a tape assign statement.

Your access and manipulation of DVD tape files is the same as standard tape files except that you only have read capability. DVD tapes have the same characteristics of most open reel types, and therefore do not support fast tape access (FTA), block numbering, data compression, buffered writes, or expanded buffer. DVD tapes do not support ACS or CTL; therefore assignments with CTL-pool are rejected.

Because the DVD tape is read-only, use of the W option, use of RING or WRITEENABLE in the ring field, and use of BLANK or SCRTCH in the reel-ID field are all rejected.

The options and parameters on the assign statement for DVD tape are generally the same as for a standard tape file but are restricted to the functionality described above.

Format

```
@ASG, options filename, type[/units/log/noise/processor/tape/format/  
data-converter/, reel-1/reel-2/.../reel-n, expiration/mmspec, ring-indicator,  
ACR-name, CTL-pool]
```

where:

options

Because the DVD tape is read-only, the W option is rejected for DVD tape assignments.

type

The DVD tape feature adds the type mnemonic "DVDTP" to the tape assignment set.

reel-n

The reel number specifies the identifier for each DVD tape assignment. Each reel identifier is limited to 6 characters from the set A through Z and 0 through 9. Because the DVD tape is read-only, a reel-ID of BLANK or SCRTCH is rejected.

ring-indicator

Because the DVD tape is read-only, the values RING or WRITEENABLE are rejected.

7.2.8. Assigning Absolute Devices

Use an absolute device assign to assign special I/O devices and symbiont devices, and for online maintenance of all system peripherals. If the device is configured as a mass storage device or a symbiont device, then it must be put in reserved state by the system operator before you assign it.

You cannot catalog (@CAT) a file as an absolute device except in the case of tape files.

Note: *You must have the privilege to assign arbitrary devices in order to assign absolute devices.*

Absolute Device Assign for Mass Storage

The format is

```
@ASG[,options] filename,*name,pack-id
```

options

The only allowed options are H, I, T, and Z in any combination. The H option assigns cache mode capability on the mass storage subsystem where the device is connected. Successful assignment of cache mode lets you use cache mode hardware commands through the arbitrary device interface. See the *ER Programming Reference Manual* (7830 7899) for more information on the arbitrary device interface.

filename

Specifies the external name of the file.

name

The asterisk preceding the name field is mandatory for an absolute device assign. The name field can be one of the following:

**device-name*

Must be reserved by the operator (RV keyin). The Exec considers only the specified device for allocation to the run.

**controller-name*

The Exec considers only devices that are attached to the specified controller and allocates one device to the run. There must be at least one device on the control unit that is reserved.

pack-ID

Specifies that the disk pack is required for the file. A pack-ID consists of from 1 to 6 characters from set A through Z and 0 through 9. This field is required.

Absolute Device Assign for Tape

The Exec normally controls the units for tape file assignment. However, you can direct the tape assignment to a particular control unit or device by specifying that control unit or device. Unisys does not recommend an absolute control unit or device specification for assigning tapes used in normal operations. However, maintenance routines, real-time programs, or special hardware requirements might make it necessary for you to use these specifications.

When a tape device is put in reserved state by an RV unsolicited console keyin, it can be assigned only by specifying one of the following on the @ASG statement:

- The absolute control unit name
- The tape device name itself

However, it is not necessary to have a tape device in a reserve state for an absolute assign.

The format is

```
@ASG[,options] filename[,*type/*units/noise/processor/  
format,data-converter/block-numbering/data-compression  
/buffered-write,reel-1/.../reel-n,expiration-period,  
ring-indicator,ACR-name,CTL-pool]
```

All fields except type and units have the same meaning as the fields described in 7.2.5. The *ACR-name* field is valid only if you specify the C or U option.

Specify one of these combinations of values for **type/*units*:

**dev1/*dev2*

Use this combination to explicitly choose two devices on the assignment.

**cuname/units*

Use this combination to select one or two units on the control unit *cuname*. The integers 1 and 2 are the allowed values for units. If you do not specify a value, the system uses the default value of 1.

Devices Other Than Mass Storage or Tape

The format is

```
@ASG[,options] filename,*type
```

options

The only options allowed are I, T, and Z in any combination.

filename

Specifies the external name of the file.

type

The asterisk (*) preceding the type field is mandatory for an absolute device assign. The type can be either of these:

**device-name*

The Exec considers only the specified device for allocation to the run.

**controller-name*

The Exec considers only devices that are attached to the specified controller and allocates one device to the run.

The devices can be in a reserved state or in an UP state. However, you can assign symbionts only if they are reserved. Input/output to these devices must be through the arbitrary device interface (ER IOAID\$).

7.3. @BRKPT and @@BRKPT—Breakpoint Primary Print Files

For demand runs, the @BRKPT statement diverts output to a user-defined print file instead of to the terminal.

For batch runs, it can divert several print files to the same tape file.

This statement also diverts output back to the terminal (demand) or the PRINT\$ (batch) file.

Note: The @BRKPT statement also diverts output from PUNCH\$, the primary output file for 80-column card images, to a user-defined file. The description of @BRKPT PRINT\$ also applied to @BRKPT PUNCH\$.

Format 1

```
@BRKPT,L PRINT$[/part-name]
```

L

Provides a method of stacking multiple files on a single magnetic tape file and provides a label for each file by which the file can be referenced on a subsequent @SYM statement. The label is the *part-name* from the @BRKPT,L statements.

PRINT\$

Is the primary print file being breakpointed.

part-name

Is an internal file name identifying a new user-defined file to which subsequent primary output is to be written. If you omit the *part-name* field, subsequent output is directed back to the PRINT\$ file. If the L option is used, *part-name* is required; *part-name* provides a label, which identifies a new part to be written on a previously breakpointed magnetic tape file. This subfield is applicable only with Format 1 of the @BRKPT statement.

A slash (/) is the only legal punctuation mark for Format 1. You cannot use a period after *symbiont-name* or *part-name*. The @BRKPT statement closes the previous file part.

If you do not specify *part-name* in a batch run, the Exec creates a new part for the PRINT\$ file. The previous part is queued for printing.

If the part name or file name is a user-defined file, the file is closed by writing an EOF mark. In the case of a magnetic tape file, the tape is positioned so that a new file can be started.

If *part-name* is not assigned to your run, the Exec assigns the file for you with an implied @ASG,AZ statement and performs the breakpoint. When the user-defined file is breakpointed back to PRINT\$, the file's assignment condition depends on whether the file had an internal file name attached to it. A breakpoint file that was not assigned, but had an internal file name, is assigned. A breakpoint file that was not assigned and did not have an internal file name is not assigned. If the Exec cannot assign the file, the @BRKPT statement is not executed and a warning message is displayed.

You must use a user-defined mass storage file as a *part-name* only once, unless you save the contents of the file. If you attempt to write multiple parts into a mass storage file, the previous parts are overwritten.

User-defined breakpoint files are not automatically printed. You must use the @SYM statement to queue such files for printing.

Format 2

```
@BRKPT PRINT$[,filename]
```

PRINT\$

Is the primary symbiont output file being breakpointed.

filename

Specifies the user-defined internal or external file to which subsequent primary output is to be written. If you omit *filename*, output is again directed to the terminal or PRINT\$ file.

The comma and any file-name punctuation are allowed on Format 2. You cannot use a period after *sybiont-name*.

If the file name is a user-defined file, the file is closed by writing an EOF mark. In the case of a magnetic tape file, the tape is positioned so a new file can be started. You can have a maximum of 511 @BRKPT or @@BRKPT statements in each demand run.

If you do not specify the file name in a batch run, the Exec creates a new part number for the PRINT\$ file. The file name created with the previous part number is queued for printing. A run can breakpoint to a maximum of 511 print files using part numbers. For a batch run, this includes user-defined print files and Exec-defined print file parts.

If the file name is not assigned to your run, the Exec assigns the file for you with an implied @ASG,AZ statement and performs the breakpoint. When the user-defined file is breakpointed back to PRINT\$, the Exec releases the file from the run (@FREE), assuming the Exec assigned the file implicitly. If the Exec cannot assign the file, the @BRKPT statement is not executed and a warning message is displayed.

User-defined breakpoint files are not automatically printed. You must use the @SYM statement to queue such files for printing.

You can use the @BRKPT statement within a batch run or a demand run. After you breakpoint to a user-defined file within a demand run, all interactive responses are written to the file until output is redirected back to the terminal with another @BRKPT statement.

Example

```
@BRKPT PRINT$,POUT.  
@PRT,S WORKFILE.NBR-CRUNCH  
@BRKPT PRINT$
```

After the first @BRKPT statement, output is diverted from the terminal (assuming it is a demand run) to the user-defined print file POUT. The @PRT statement lists the contents of the symbolic element NBR-CRUNCH in program file WORKFILE. Since output has been breakpointed, this listing is written to print file POUT instead of being displayed at the terminal. After the second @BRKPT statement, output is again directed to the terminal.

7.3.1. Breakpointing Alternate Symbiont Files

The @BRKPT statement closes or partitions alternate print and read files that you define.

An @FREE of the alternate file must not be done before an @BRKPT. An @FREE does not close the alternate file, and a successive @BRKPT or alternate Executive request to the freed file results in an I/O error.

Format 1

```
@BRKPT[,E] internal-fname
```

E

Inhibits EOF positioning for alternate read files on magnetic tape.

internal-fname

Identifies the alternate read or print file being breakpointed. Alternate symbiont files assigned as temporary must use this field. The *internal-filename* corresponds to the *internal-filename* used in the ER packet for the alternate read and print requests.

Format 2

```
@BRKPT[,E] ,fname
```

E

Inhibits EOF positioning for alternate read files on magnetic tape.

fname

Is an optional field that specifies an external or internal file name identifying the alternate read or print file being breakpointed. If you are using Format 2, the comma preceding the file name is required.

Do not use Format 2 for temporary assigned alternate files.

The discussion on primary file breakpointing generally applies to alternate file breakpoints. The differences are as follows:

- The alternate file is closed and is no longer known to the symbionts.
- In the case of input tape files, breakpoint is normally used to prematurely terminate reading of the file. In the absence of the E option, the input tape is positioned forward to the next EOF mark so that a subsequent file on the tape can be initiated as a new alternate read file. Using the E option avoids needless tape movement when you are finished with the tape.
- Stacking of output on tape is possible, but you must provide the part labeling by varying the internal-filename for each alternate file written to a tape.

7.3.2. Using @BRKPT and @SYM Together

The following example illustrates how you can use the @BRKPT and @SYM statements together. This run partitions its primary print output into five parts. The example shows line numbers from 1 to 13, which are reference points for the description that follows.

```
1. @RUN
2. @ASG, CP MTAPE, T, 1234
3. @ASG, CP SYMFILE, F
4. @BRKPT PRINT$/MTAPE
   .
   .
   .
5. @BRKPT, L PRINT$/LABEL1
   .
   .
   .
6. @BRKPT PRINT$/SYMFILE
7. @FREE MTAPE
8. @SYM MTAPE,, PR1, MTAPE/LABEL1
   .
   .
   .
9. @BRKPT PRINT$

10. @FREE SYMFILE
11. @SYM, U SYMFILE
12. @SYM, U SYMFILE, 3, RMSITE
13. @FIN
```

The first print part (all output generated up to line 4) is stored in primary print file PRINT\$ and is printed on the default printer for the input device used for the run.

The second print part (output generated between lines 4 and 5) and the third print part (output generated between lines 5 and 6) are stacked on magnetic tape file MTAPE. The second print part is given the label (part-name) MTAPE, and the third print part is given label LABEL1.

Line 7 releases print file MTAPE from the run, causing the file to be cataloged. (The C option is in line 2.) A user-defined print file must be cataloged before you can queue it to a printer.

Line 8 queues the second and third print parts to the printer PR1. Labels (part-names) MTAPE and LABEL1 are specified. After being printed, file MTAPE is deleted.

The fourth print part (all output generated between lines 6 and 9) is stored in mass storage file SYMFILE.

Line 10 releases SYMFILE from the run, causing it to be cataloged (because the C option was used in line 3).

Line 11 queues SYMFILE to the default printer for the input device used for the run.

Line 12 queues the same print file (SYMFILE) to the printer at remote site RMSITE; three copies are printed. The U option is used to prevent SYMFILE from being deleted after either @SYM statement (line 11 or 12) is processed

The last print part (all output generated after line 9) is again stored in primary print file PRINT\$ and printed at the default printer.

7.4. @CAT and @@CAT—Create Permanent Files

The @CAT statement catalogs a file in a master file directory without having it assigned to the run.

7.5. Cataloging Mass Storage Files

Format

```
@CAT[,options] filename[,type/reserve/granule/maximum,pack-id-1/...  
/pack-id-n,,,ACR-name/encrypt-option]
```

To use @CAT as a transparent statement, enter two masterspace (@) characters instead of one. (For more information about transparent statements, see Section 2.2)

options

B

Saves and reloads the file as part of any full checkpoint or restart.

G

The meaning of this option depends on how security and Quota are configured at your site.

- Unowned files

If your site has Fundamental Security or Security Level 1, 2, or 3 and the file is unowned, the following applies:

- When the file is cataloged, the G option guards against having its read key, write key, privacy mode, or all three overridden by privileged runs.
- If any F-cycle of a file set is cataloged with the G option, the read keys or write keys for the file set are obscured by '/////////' from all users. If a particular F-cycle of a file set is cataloged with the G option, the privacy mode and either the project-ID or account number under which the file was created are also guarded.
- The G option means the file cannot be backed up.

- **Owned files**

If your site has Security Level 1, 2, or 3, the file is owned, and the dynamic configuration parameter (owned_file_read_write_keys) is FALSE:

- G-option files are no longer protected from privileged runs.
- The G option means the file cannot be backed up by FAS. When you need to back up a file, you must use another product such as FURPUR or IRU.

If your site has Security Level 1, 2, or 3, the file is owned, and the dynamic configuration parameter (owned_file_read_write_keys) is TRUE, the following applies:

- When the file is cataloged, the G option guards against having its read key, write key, privacy mode, or all three overridden by privileged runs.
- If any F-cycle of a file set is cataloged with the G option, the read keys or write keys for the file set are obscured by '/////////' from all users. If a particular F-cycle of a file set is cataloged with the G option, the privacy mode and either the project-ID or account number under which the file was created are also guarded.
- The G option means the file cannot be backed up.

- **Quota system**

The use of the G option is also controlled by the Quota system on an account basis, as follows:

- If the Exec is configured to maintain quota limits for each account, this option can be used only if it is allowed in the quota limits for the run.
- If the Exec is not maintaining quota limits, this option can be used by all runs.

P

Catalogs the file as a public file. For systems with Fundamental Security, if you omit the P option (default), the file is cataloged as a private file and can be accessed only by runs that have the same project-ID or account number as the run that created the file.

For systems with Security Level 1 or higher, the file is cataloged as public if the P option is specified on the @ASG statement and an ACR is specified. The file is cataloged private when the P option and ACR are omitted from the @CAT statement (default). You can even be given the option to create unowned files with Security Level 1, 2, or 3.

R

Catalogs a file as a read-only file. A file cataloged with the R option cannot be overwritten. The file can be only read or deleted. Any nonprivileged activity requesting to write in the file is placed in error mode.

S

This option only applied to 7-track tape and now must be left blank (although this option no longer has any effect, it is allowed since it can still be in user runstreams.) It indicated that this file is a store-through file. Any writes done to this file result in the data being stored directly on a nonvolatile mass storage medium. Absence of this option indicates that data can be stored on volatile areas of mass storage such as cache storage. Presence of the option does not prevent the storage of data on solid-state disk. If the file is XPC cached, this S option is ignored because the data for the file is kept in the XPC.

V

Indicates that this file is never unloaded to tape at any time. If the Exec is configured to maintain quota limits for each account, you can use this option only if it is specifically allowed in the quota limits for the run. If the Exec is not maintaining quota limits, all runs can use this option.

W

Catalogs a file as a write-only file. A file cataloged with the W option cannot be read. Any nonprivileged activity requesting to read the file is placed in error mode.

Z

Indicates that the Exec must not hold the run if it cannot process this statement right away. The Exec must process the next statement instead. If the Exec goes into a hold situation, the statement is rejected and the appropriate message is displayed on the screen.

filename

Identifies the file being assigned to the run.

type

Specifies a combination of letters and numbers that identify the addressing mode of the file (sector- or word-addressable) and the equipment type required for the file. The *type* field is also called the assign mnemonic. When a system is configured, assign mnemonics are defined to correspond to a prioritized list of specific equipment types.

Most type field entries supplied by Unisys start with either D or F, but sites can configure any mnemonics they wish. Typically, D represents word-addressable and F represents sector-addressable mass storage. If you omit type, the Exec uses sector-addressable mass storage. Appendix D, Table D-1 shows the standard type entries supplied by Unisys.

If you are creating a new cycle for an existing file cycle set, and you omit the type, the type is taken from the latest existing cycle of the file. If all cycles of the file are in a to-be-cataloged or to-be-dropped state, then the type from the highest cycle of this type is used.

When space is not available for the specified device type, or if the specified device type is no longer available on your system, another device type is substituted that satisfies the request. The order in which device requests are satisfied is configured by your site addressable or word addressable, and if necessary, what type of equipment must be administrator. Requests are satisfied from fastest to slowest device. See Appendix D, Table D-2 for selection sequences supplied by Unisys.

The selection sequences pertain only to fixed mass storage. For removable disk file assignments, only the first equipment type is searched or selected.

Local fixed files can be placed in memory rather than on channel-connected mass storage. The standard mnemonics FMEM (sector-addressable) and DMEM (word-addressable) can be used for this purpose; however your site can configure additional assign mnemonics. Placing files in memory reduces access time, but consumes memory space that might otherwise be available for programs. Also, data in memory files is lost when the system is rebooted. Use of memory file assign mnemonics on systems that are not or cannot be configured to support memory files results in the request being satisfied on channel connected-mass storage.

For physically removable disk file assignments, you must use the assign mnemonic that corresponds to the pack type that is in use.

F30	selects equipment type 8430
F33	selects equipment type 8433
D30	selects equipment type 8430
D33	selects equipment type 8433

If any other mnemonic is used to catalog and assign removable disks, the file cannot be assignable after changes to equipment configurations.

When specifying a physically removable assign mnemonic, the first equipment specified in the configuration statement CONV's selection sequence must be for a physically removable device. If the first equipment specified in the CONV statement is not a physically removable device, the following error message appears:

```
E:200633 pack xxxxxx cannot be mounted -- equipment type does not permit
interchanging of packs.
```

If you specify a physically removable assign mnemonic but no units are configured, the following error message appears:

```
E:256033 requested unit(s) not available
```

reserve

Specifies the amount of space you want to reserve. The reserve consists of 1 to 12 numbers. For most files, you do not need to use reserve.

When you specify a reserve, you must provide a reasonable estimate of the space needed to create the file. When you specify a reserve, you reduce system overhead by allowing the Exec to find and allocate space in contiguous granules, if possible. When you do not specify a reserve, the Exec must find and allocate the space dynamically, as it is needed by the file. The amount of space given as the initial reserve is not actually allocated until you assign the file, and even then, only if the file can be written to, as follows:

- For sector-addressable files, the initial reserve is the number of granules (tracks or positions) required by the file. Refer to 2.2 for file size information.
- For word-addressable files, the initial reserve is the number of words required by the file. The maximum number of words allowed for the reserve is 262,143 times the number of words-per-granule, based on the granularity (track or position) of the file. Word-addressable files are allocated on a granule basis.

granule

Specifies granule size. A granule is the increment by which storage space is allocated to your file (initially and when more space is needed).

The granule can be either of these:

TRK

One software track (64 software sectors)

POS

One software position (64 software tracks)

If you omit granule, the Exec uses TRK. The granularity is recorded in the master file directory.

maximum

Specifies the maximum allowable length of the file. Permissible values are the same as the reserve subfield.

- For sector-addressable files, this maximum is given in granules (tracks or positions).
- For word-addressable files, the maximum is specified in words.

If an I/O request exceeds the number of granules specified, the run is given an I/O error.

When you specify maximum, it overrides the system default. If you omit maximum, the reserve subfield value or system default is used, whichever is larger. The maximum must be at least as large as the reserve. This lets your file dynamically expand to the maximum length when it is needed.

pack-ID

Specifies the removable disk packs required for the file. A pack-ID consists of from 1 to 6 characters in any combination of letters and numbers. The pack-ID s for cataloged files are recorded in the master file directory and need not be specified on reassignments.

A pack's master file directory must have sufficient space available to record the file otherwise a generation request is rejected with the following message:

```
E:207433 File cannot be created due to lack of removable directory space
on pack pack-id.
```

A pack-ID is applicable only to removable disks. You cannot specify a pack-ID unless you specify the equipment type. If you omit the pack-ID on requests for noncataloged file space, fixed disk is assumed if you request disk equipment. For cataloged removable disk files, specification of a pack-ID causes the pack to be registered. If the pack is already registered, its information is compared with the image for compatibility.

All cycles of a removable disk file must be on the same set of packs. The maximum number of pack-IDs that you can specify on an @ASG statement is 510.

For important information about using removable pack files, see the removable disk packs section in the *Exec System Software Administration Reference Manual* (7831 0323).

ACR-name

Specifies the ACR for the cataloged file (for Security Level 1 or higher). It must be an ACR that was previously created using your security management product.

encrypt-option

This subfield is used to specify encryption actions. For @CAT, DOENCR specifies that a file must be catalogued as encrypted. DNENCR specifies that a file must not be catalogued as encrypted. When these mnemonics are omitted, the default flag setting in the file owner's userid record is used to determine encryption state.

Examples

```
@CAT PISCES1.
```

Since only the file name is given (PISCES1), all of the default values are used.

It is private, sector-addressable mass storage file, with a maximum track size configured by the system. A default qualifier is used as the qualifier for the file name.

```
@CAT,P FRS*PISCES2.,FMD///500,PK12
```

This statement catalogs a public file (P option) named FRS*PISCES2. It requests sector-addressable mass storage on a 9720 or 9494 disk (FMD), with a maximum size of 500 tracks. The removable disk pack requested is PK12.

```
@CAT,W FRS*PISCES3.,D/10//300,,,,ACR1
```

This statement creates a ACR-controlled file named FRS*PISCES3. It has an ACR attached to it called ACR1.

This file is word addressable (D), with an initial reserve of 10 words and a maximum size of 300 words.

```
@CAT,PR PISCES4/RK54.,F
```

This statement catalogs a file that is public (P option) and read only (R option).

If your site has Fundamental Security, anyone who wants to read from the file must know the read key, RK54. If your site has Security Level 1 or higher, the read key is ignored.

The F means that the file is sector addressable. Since no maximum size is specified, the file cannot exceed the maximum track size configured by the system. The run project-ID is used as the qualifier for the file name, since none is specified.

```
@CAT,W FRS*PISCES5.,F//POS/20
```

This statement catalogs a file named FRS*PISCES5 that is private (no P option) and write only (W option). The F indicates that the file must be sector addressable. Space is allocated to the file in positions (POS) instead of tracks. The maximum size of the file is 20 positions, which is equal to 1280 tracks.

7.5.1. Cataloging Tape Files

The @CAT statement catalogs a tape file without allocating tape facilities to your run. You can even catalog a tape file on equipment that is down, provided that you designate a specific equipment type (if equipment is down, you cannot use T or U9 on the type field).

Note: If your site has Security Level 2 or higher, your security officer chooses one of four levels of protection for tape volumes (TVSL). If TVSL is set to 1, 2, or 3, you can give a tape an access list. Please see the discussion on tape security in Section 7.3 for more information.

You must be aware of the following two tape terms that are most important when cataloging tape files:

- The half-inch cartridge (HIC) tape that can be used to record 18-track or 36-track formatted data.
- The digital linear tape (DLT)

Format

```
@CAT, options filename, type[/units/log/noise/processor/tape/  
format/data-converter/block-numbering/data-compression/  
buffered-write/expanded-buffer, reel-1/reel-2/.../reel-n,  
expiration-period/mmspec,, ACR-name, CTL-pool]
```

options

The @CAT options let you put restrictions on the file being cataloged and specify hardware modes that apply to the file being cataloged.

If you use an option that is not allowed, or if you use more than one density option (H, S, L, M, and V), the Exec rejects the statement.

If you do not specify a density, either with an option or in the type field, the system uses the highest possible density for the equipment type selected.

These are the options:

E

Indicates even parity. Available only on 7-track tape units. This option is not allowed for any other tape types.

G

The meaning of this option depends on how security and Quota are configured at your site.

- Unowned files

If your site has Fundamental Security or Security Level 1, 2, or 3 and the file is unowned, the following applies:

- When the file is cataloged, the G option guards against having its read key, write key, privacy mode, or all three overridden by privileged runs.
- If any F-cycle of a file set is cataloged with the G option, the read keys or write keys for the file set are obscured by '/////////' from all users. If a particular F-cycle of a file set is cataloged with the G option, the privacy mode and either the project-ID or account number under which the file was created are also guarded.
- The G option means the file cannot be backed up.

- Owned files

If your site has Security Level 1, 2, or 3, the file is owned, and the dynamic configuration parameter (owned_file_read_write_keys) is FALSE:

- G-option files are no longer protected from privileged runs.
- The G option means the file cannot be backed up by FAS. When you need to back up a file, you must use another product such as FURPUR or IRU.

If your site has Security Level 1, 2, or 3, the file is owned, and the dynamic configuration parameter (owned_file_read_write_keys) is TRUE, the following applies:

- When the file is cataloged, the G option guards against having its read key, write key, privacy mode, or all three overridden by privileged runs.
- If any F-cycle of a file set is cataloged with the G option, the read keys or write keys for the file set are obscured by '/////////' from all users. If a particular F-cycle of a file set is cataloged with the G option, the privacy mode and either the project-ID or account number under which the file was created are also guarded.
- The G option means the file cannot be backed up.

- Quota System

The use of the G option is also controlled by the Quota system on an account basis, as follows:

- If the Exec is configured to maintain quota limits for each account, this option can be used only if it is allowed in the quota limits for the run.
- If the Exec is not maintaining quota limits, this option can be used by all runs.

H

This option is available on 9-track and U9840D tape subsystems. If the density option H is specified with the general assign mnemonic T, then one of the devices supporting the highest density that allows H density option is assumed by the Exec. This option is not available for other tape subsystems.

With the H density option now allowed for two different tape types, such as 9-track and U9840D; and these tapes selectable through the general assign mnemonic T, the Exec's selection sequence criteria remains basically the same except that either 9-track or U9840D would get assigned. The selection sequence is dependent on what is configured, available, premounted, and what is in the CONV default list for tape assignment mnemonics and how it is ordered.

Note: The potential for a run stream compatibility issue, for files assigned using the general assign mnemonic T option and the H density option, exists if the CONV default has both U9840D and U9H devices on the system.

The following error message is returned if an invalid density option is used:

```
E:207133 Assign mnemonic [asg_mnemonic] does not support density option [n].
```

J

Your system is configured at one of four tape volume security levels (0 through 3); the J option meaning depends on which level your system has.

Level	Meaning
0	Indicates the tape volume is to be unlabeled. Read and write accesses are allowed.
1	Indicates the tape volume is to be unlabeled. Read and write accesses are allowed.
2	All tapes written with the J option are automatically given security labels. Read of unlabeled tapes is permitted. Write of unlabeled tapes requires executing with the system compartment set and clearance level 63 (system high) and possession of the bypass volume check privilege (which requires that the file SYS\$*DLOC\$ is assigned to your run).
3	To read or write to an unlabeled tape, you must be executing with the system compartment set and clearance level 63 (system high) and possess the bypass volume check privilege. If you are not executing under these conditions, you receive an error message and the J option is ignored.

L

This option is available on all 18-track half-inch cartridge, U9840, and UT10KA tape subsystems. If the density option L is specified with the general assign mnemonic T, then one of the devices supporting the highest density that allows L density option is assumed by the Exec. This option is not available for other tape subsystems.

With the L density option now allowed for three different tape types, such as 18-track half-inch cartridge, U9840, and UT10KA; and these tapes selectable through the general assign mnemonic T, the Exec's selection sequence criteria remains basically the same except that 18-track half-inch cartridge, U9840, or UT10KA, would get assigned. The selection sequence is dependent on what is configured, available, premounted, and what is in the CONV default list for tape assignment mnemonics and how it is ordered.

Note: The potential for a run stream compatibility issue, for files assigned using the general assign mnemonic T option and the L density option, exists if the CONV default has both UT10KA and HICL devices on the system.

The following error message is returned if an invalid density option is used:

```
E:207133 Assign mnemonic [asg_mnemonic] does not support density option [n].
```

M

This option is a density option for the U9840C cartridge subsystems. This M option is also currently used for 36-track half-inch cartridge and CTS9940B tape subsystems. All other density options (for example, L and H) are not allowed for the U9840C subsystem.

The M option is available on all 36-track half-inch cartridge, U9940B, and U9840C tape subsystems.

If the M option is specified with the general assign mnemonic T option, then one of the devices supporting the highest density that allows the M density option is assumed by the Exec. This option is not available for other tape subsystems.

With the M option now allows three different tape types (36-track half-inch cartridge, U9940B, and U9840C), and these tapes selectable through the general assign mnemonic T, the Exec's selection sequence criteria remains basically the same, except that 36-track half-inch cartridge, U9940B or U9840C, would get assigned. The selection sequence is dependent on what is configured, available, premounted, and what is in the CONV default list for tape assignment mnemonics and how it is ordered.

Note: *The potential for a run stream compatability issue, for files assigned using the general assign mnemonic T option and the M option, exists if the CONV default has both U9840C and HICM devices on the system.*

The following error message is returned if an invalid density option is used:

```
E:207133 Assign mnemonic [asg_mnemonic] does not support density option [n].
```

O

Indicates odd parity (assumed for 9-track equipment). Allowed for all non-7-track equipment.

P

Catalogs the file as a public file. For systems with Fundamental Security: if you omit the P option (default), the file is cataloged as a private file and can be accessed only by runs that have the same project-ID or account number as the run that created the file.

For systems with Security Level 1 or higher, a file is cataloged as public when the P option is specified on the @CAT statement and no ACR is specified. The file is cataloged as private when you omit the P option and ACR from the @CAT statement (default).

R

Catalogs the file as a read-only file. A file cataloged with the R option cannot be overwritten. If labeled tape volumes are used, the files on the tape are labeled as read-only. The file can only be read or deleted. Any activity requesting to write in the file is placed in error mode.

S

Indicates density mode of 6250 bpi. Available only for UNISERVO 26, 28, 32, 34, 36, and 45N subsystems.

V

Indicates density mode of 1600 bpi. This option is available on all 9-track tape units and specifies a density of 85,937 bpi for DLT subsystems. If the density option is not specified for DLT subsystems, then the highest density supported by the DLT tape device is assumed. This option is not available for other tape subsystems.

W

Catalogs the file as a write-only file. A file cataloged with the W option cannot be read. If labeled tape volumes are used, the files on the tape are labeled as write-only. The file can only be written to. Any activity that requests to read the file is placed in error mode.

Z

Specifies that this statement is not to cause a hold condition. Control is returned, indicating whether a hold state would have resulted from the statement.

filename

Specifies the external name of the file.

type

A combination of letters and numbers that specifies what type of magnetic tape equipment must be used for the file. If you do not specify type and the file is not already cataloged, the Exec catalogs the file as a mass storage file.

The type field is also called the assign mnemonic. The assign mnemonic is stored in the MFD upon cataloging the file. See Table D-3 for the allowable assign mnemonics.

When a system is configured, assign mnemonics are defined to correspond to a set of specific equipment types. Table D-3 shows the permissible type entries.

If you use a density option (H, S, or V), make sure that it is compatible with the assign you use. For example, when you use the assign mnemonic T, specify only those options compatible with all types of units.

Based on the assign mnemonic you specify, the system selects an equipment type for your file. If the assign mnemonic corresponds to a particular equipment type, that type is used. If you use a more general type, however, the Exec selects an equipment type before cataloging the file, based on a standard selection sequence. The selection sequence supplied by Unisys is from the fastest equipment to the slowest, but this can be changed when the system is configured. See Table D-4 for the selection sequences supplied by Unisys.

Note: Use the most general assign mnemonic that still meets your requirements, because it gives the system more freedom in assigning units.

Unless otherwise specified by an assign mnemonic or mode option, the highest possible density for the device is assumed. The assumed density for UNISERVO 26 and 28 tape units is 6250 bpi (bits per inch).

noise

Specifies an integer from 1 to 63 that overrides the standard system noise constant. If you omit noise, the Exec uses the system standard noise constant. The system standard value is 3 for all 9-track tapes written in a density of 800 bpi. For 9-track tapes written in a density of 1600 bpi or 6250 bpi, the system standard value is 0. The noise value is recorded in the master file directory when the file is cataloged. The value for noise must be zero for all cartridge type tape assigns.

processor/tape

Specifies the type of translation required. Processor defines the format of the data in processor, and tape defines the format of the data on tape. You must specify both processor and tape or omit both. If you omit them, no translation occurs.

Data translation is file independent and the translation mode remains in effect for the duration of the reading or writing of the tape. You can change data translation from file to file, but you must track this information yourself.

The supported *processor/tape* mnemonics are

Processor Code	Tape Code
ASCII	EBCDIC
ASCII	FLDATA
EBCDIC	ASCII
EBCDIC	BCD
FLDATA	ASCII
FLDATA	BCD
FLDATA	EBCDIC

Additional processor/tape mnemonics can be defined by your site, provided neither code exceeds six characters. When the file is cataloged, the values given in processor/tape are stored in the master file directory.

All translation takes place in the hardware.

Note: If you are using the data compression feature to read and write to a 5073 cartridge tape, you cannot use the data translation feature.

format

Specifies the data transfer format for the word-to-byte conversion.

The format subfield is also used in determining the data compression mode for write operations for the 5073 tape subsystem.

Permissible format entries are as follows:

Q

Quarter word

8

8-bit packed (default on 9-track and 1/4-inch tape cartridge)

The data transfer format is recorded in the master file directory upon cataloging, to be used on subsequent assigns unless specifically overridden.

data-converter

This option only applied to 7-track tape and now must be left blank (although this option no longer has any effect, it is allowed since it can still be in user runstreams.) The allowable *data-converter* values were as follows:

DC

Three 8-bit bytes are converted to or from four 6-bit tape characters. The data-converter (DC) status is recorded in the master file directory upon cataloging.

OFF

Turns off the data-converter if an assign is made from the master file directory (@ASG,A) and the file was cataloged with data conversion. In all other cases, OFF is ignored.

If you do not specify data-converter, no action is taken in relation to the data converter. Furthermore, the OFF is not recorded in the master file directory, except as implied by the absence of data-converter.

block-numbering

This field must be left blank (although it no longer has any effect, it is allowed since it can still be in user runstreams). It determines whether block numbering is on. Block numbering is available only on 5061, HIC, DLT, and LTO tape subsystems. For the HIC, DLT, and LTO tape subsystems, you cannot turn on block numbering with this subfield because it is performed by the hardware and is transparent to the user. Specify one of the following:

BLKON

Turns block numbering on (invalid for HIC, DLT, and LTO tape subsystems).

BLKOFF

Indicates that blocks must not be numbered.

BLKOPT

Gives you the option to turn block numbering on at a later date (invalid for HIC, DLT, and LTO tape subsystems).

data-compression

Reduces the amount of physical tape required to record data by removing repetitive data before recording it (available for 5061, HIC, DLT and LTO and tape subsystems only). Data compression is generally performed on a file basis.

For half-inch cartridge tape subsystems, these are the *data-compression* values:

CMPON

Enables 8-bit data compression when Q is specified on the *format* subfield; enables 9-bit data compression when either "8" or nothing is specified on the format subfield. Enables LTO-DC for an LTO cartridge tape device.

CMPOFF

Indicates no data compression.

CMPOPT

Capable of 8- or 9-bit data compression. Gives you the option to enable 8- or 9-bit data compression at a later date must be used if there is any possibility that the cartridge tape contains an 8- or 9-bit compressed file. Capable of LTO-DC data compression. Gives you an option to enable data compression at a later date; must be used if there is any possibility that the cartridge tape contains an LTO-DC compressed file.

CMPEON

Enables Enhanced data recording capability (EDRC) mode of data compression.

CMEOPT

Capable of EDRC data compression. Gives you the option to enable EDRC data compression at a later date; must be used if there is any possibility that the cartridge tape contains an EDRC compressed file.

For DLT and HIS98 tape subsystems, these are the valid data-compression values and their meanings:

CMPLON

Enables LZ1 data-compression. DLT and HIS98 cartridge tapes are not compatible with each other or with future Unisys tape products that can use this compression value.

CMPOFF

Disable data compression.

Notes:

- Although you can explicitly specify 8- or 9-bit data compression with *CMP8ON* and *CMP9ON*, you can allow the system to select the best compression ratio by specifying the format of the data.
- *CMPEON*, *CMEOPT*, and *EBCOPT* are valid values for the *USR5073* device when using *HICSL*.
- When using the *HICL* mnemonic in combination with *CMPEON*, *CMEOPT*, and *EBCOPT*, a *SCSI USR5073*, *BMC 5073*, or a *4780* device can be selected.
- The *CTS9840*, *DLT7000*, *U9840C*, *CTS9940B*, *LTO3*, and *LTO4* devices always have expanded buffering, so the *EBCOPT* is not needed to get a *CTS9840*, *DLT7000*, *U9840C*, *CTS9940B*, *LTO3*, or *LTO4* which supports expanded buffering.
- The *CTS9840*, *DLT7000*, *U9840C*, *CTS9940B*, *U9840D*, and *UT10KA* tapes always have expanded buffering, so the *EBCOPT* is not needed to get a *CTS9840*, *DLT7000*, *U9840C*, or *CTS9940B*.
- Assign mnemonics that can select an ACS drive are *T*, *HIC*, *HICL*, *HICM*, *HIC51*, *HIC52*, *HIS98*, *U47*, *U47L*, *U47LM*, *LCART*, *DLT70*, *LTO*, *LTO3*, and *LTO4*. Assign mnemonics *HICL*, *U47*, *HICM*, *HIC*, *DLT70*, *LTO*, *LTO3*, and *LTO4* result in either an ACS query or cartridge tape library (CTL) poll. If the Automated Cartridge System (ACS) query is positive, and ACS drive is assigned. If the CTL poll is positive, a CTL drive is allocated. If the ACS query and CTL poll are negative, a free standing drive is allocated. *NONCTL* can be used on ACS assignments.

Examples

```
@CAT, P TAPE., HICL////////CMPEON, 111C
```

The tape is cataloged as a public file with EDRC data compression requested.

```
@CAT, P T., HICM, REELID
```

This assign mnemonic selects a 36-track drive.

```
@CAT,J TAPE.,HICL/////////CMEOPT,111C
```

The tape is cataloged as a private file requesting a tape device that has the hardware capability of supporting EDRC data compression.

Support for 4780 18- and 36-track tapes is provided by the standard Exec. Compression on the 4780 subsystems is called improved cartridge recording capability (ICRC) and is totally compatible with the 5073 enhanced data recording capability (EDRC). Compression can be selected by using the CMPEON value.

Device characteristics for compressed tape readability are:

- Unisys proprietary compressed cartridge tapes cannot be read on 4780 drives, but EDRC compressed cartridges from 5073 drives can be successfully read.
- A cartridge drive on a 4780 that does not have compression hardware can read only uncompressed cartridge tapes.
- The 4780 drives that have compression hardware can read EDRC compressed tapes without having to explicitly request CMPEON.

Note: *In some cases, you are not able to append data to a tape (write past loadpoint). Appending is not allowed on 18-track half-inch cartridge, U9840C, and UT10KA tape subsystems.*

buffered-write

Specifies whether to use buffered writes when a tape is mounted on a device controlled by a control unit that has buffering capability.

These are the values.

BUFTAP

Enables buffered write mode for user data blocks and file marks for the entire tape.

BUFFIL

Enables buffering of write data including label blocks and tape marks for a file written to labeled tapes. The system ensures that writing of the data and file marks occurs at the close of a file.

BUFON

Enables buffered write mode for user data blocks.

BUFOFF

Disables any buffered write mode.

The @CAT statement is rejected if it contains the BUFTAP or BUFFIL value and there are no tape units up and available that support tape mark buffering (TMB).

Even though TMB is supported on the CTS5236 the BUFTAP value is not. The rejecting message is:

E:256033 requested unit(s) not available.

The following rules apply when using BUFTAP, BUFFIL, BUFOFF, and BUFON:

- When the user switches modes from BUFFIL to BUFON or BUFOFF, writing the buffer to tape is postponed until the next I/O to the tape subsystem. This also applies when switching modes from BUFTAP to other modes.
- If BUFTAP, BUFFIL, BUFOFF, or BUFON is not specified, and the configuration tag TAPBUF (dynamic name_tape_buffering) is zero, the system uses the default value BUFON.
- If BUFFIL is specified for an unlabeled tape, the data blocks for writes are buffered and one synchronization per file is done by a WEF\$ function just as it is without the BUFFIL specification.
- When using buffering, it is the user's responsibility to retain the original data for possibly performing error recovery if that is a requirement. The Exec provides no means for the user to recover this data. With the use of the BUFTAP option, it is possible to have good status on previous writes of thousands of files, when in fact none of the files has made it to tape. If an error occurs while they are being written to tape, the remaining blocks in the tape subsystems buffer can never get written to tape. The Block ID SAFE (BSAFE\$) I/O function provides a means for the user to determine what block has been actually written to tape by the tape subsystem.
- When using BUFTAP, a FSAFE\$ can be performed at any point if a user wants to ensure that buffered data/tape marks have been written to tape. It is highly recommended that the FSAFE\$ be performed and its status checked after all user data/tape marks have been written to the tape device, but before releasing the tape. This ensures that the user is notified of any errors in writing any remaining buffered data/tape marks to the tape. This also allows the user to know that an error on a @FREE, TSWAP\$, TINTL\$, or TLBL\$ was not the result of writing any user buffered data/tape marks on the tape

Examples

```
@CAT,P TAPE.,HICL/////////EBCOPT,111C
```

The tape is cataloged as a public file, requesting a tape device that has the hardware capability of supporting block sizes of up to 256K bytes.

```
@CAT,J TAPE.,HICL/////////BUFON,111C
```

The tape is cataloged as a private file with buffered-write capability requested. The requested drive cannot be capable of expanded buffer operation.

reel-n

Specifies the identifier for each tape reel or cartridge required. Each tape identifier is limited to six characters from set A to Z and 0 to 9. Do not specify an identifier with a single A, E, Y, or N character. These characters correspond to valid responses to tape labeling messages. Using them as identification could cause tape assign problems.

Tapes are used in the order specified, and all reel or cartridge identifiers are recorded in the MFD. A maximum of 2047 reels or cartridges can be specified with the @CAT statement.

Also, 36-track tapes have more capacity on current cartridges because data is written in both directions on the tape, and more capacity with longer tapes can hold four times more data than current cartridges (1,100 feet [E-carts]).

The @CAT statement accepts the mnemonic. For example:

```
@CAT,P T.,U47L////////CMPEON/BUFON/EBCOPT,REELID
```

This statement can use the mode values.

When an assignment uses a specific reel-ID, its location is determined by an ACS or a CTL poll. The reel-ID's location, together with the assign mnemonic, determines the resultant device assigned. When a BLANK is assigned, no ACS query is done. However, BLANK assignments result in a CTL poll.

ACR-name

Specifies the access control record (ACR) for the cataloged file (for sites with Security Level 1 or higher). It must be an ACR that you have previously created by using your security management product.

CTL-pool

Can be one of the following:

CTL-pool name

Allocates a BLANK tape from the CTL pool of BLANK cartridge tapes. *CTL-pool-name* can be 1 to 12 characters, consisting of the alphanumeric characters, \$, and -.

NONCTL

Associates the file with freestanding cartridge tape drives. The Exec does not use the CTL for this file.

See your site administrator for CTL information specific to your site.

Examples

```
@CAT TAPE-STORE1.,T
```

This statement catalogs tape file TAPE-STORE1 as a private file (no P option). The T means the file can be on any type of tape unit that is available. Default values are used for all other fields. The project-ID is used as the qualifier for the file name.

```
@CAT,PJ GCH*TAPE-STORE2.,U9
```

This statement catalogs tape file GCH*TAPE-STORE2 as a public file (P option) on an unlabeled tape (J option). The U9 means that the file can be on any 9-track tape unit that is available.

```
@CAT,PR TAPE-STORE3/KIWI.,U9V/////6
```

This statement catalogs file TAPE-STORE3 as a public (P option) and read-only (R option) tape file.

If your site has Fundamental Security, or Security Level 1, 2, or 3 and the file is unowned, anyone who wants to read from the file must know the read key (KIWI). If your site has Security Level 1 or higher and the file is owned, the read key is ignored.

U9V means that the file must be on a 9-track tape unit, and that very high tape density (1600 bpi) must be used. The 6 indicates that 6-bit packed format must be used for the word-to-byte conversion.

```
@CAT,S TAPE-STORE4//FRITZ.,28N,NC18
```

This statement catalogs the file TAPE-STORE4 as a private file (no P option).

If your site has Fundamental Security or Security Level 1, 2, or 3 and the file is unowned, anyone who wants to write to the file must know the write key (FRITZ). If your site has Security Level 1 or higher and the file is owned, the write key is ignored.

28N means that the file must be on a UNISERVO 28 tape unit. The S option means that the file must be written with super high density (6250 bpi). Reel number NC18 is used for the file.

```
@CAT,PRV TAPE-STORE5.,45N//////////,,,ACR2
```

This statement catalogs file TAPE-STORE5 as a read-only public file on a 9-track tape in the 2145 tape subsystem, using the required density mode of 1600 bpi. The O option indicates odd parity. Access control record ACR2 is attached to the file. Because the data-converter, block-numbering, and data-compression subfields are not specified, they take on the system default values, which are required by the 2145 tape subsystem.

```
@CAT,P TAPE-STORE6.,HICL,,,,,CTL-AUDIT1
```

This statement catalogs file TAPE-STORE6 as a public file on 18-track cartridge tape. The BLANK tape is allocated from the CTL pool named CTL-AUDIT1.

```
@CAT,P T.,HIC51/////////CMPEON/BUFON,EBCOPT,REELID
```

This statement specifies data compression, buffering, and expanded block capability for a U5136 36-track cartridge tape.

7.6. @CHG—Change File or Element Attributes

In a single @CHG statement, you can change either file attributes or program file element attributes.

File attributes changes are described in Section 7.6.1 and Section 7.6.2. Section 7.6.3 contains examples of each kind of file attribute change. File attributes that can be changed using @CHG are as follows:

- Keys
 - Read
 - Write
- Modes
 - Read-only
 - Write-only
 - V-OPTION FLAG
- File Names
 - Qualifier
 - Filename
- Security
 - public
 - private
 - ACR-controlled
- Caching attributes
- File type
 - Program file
 - Large Program file
 - Large Element Program file

Program file element attributes changes are described in Section 7.6.4. Section 7.6.5 contains examples of each kind of program file element attribute change. Program file element attributes that can be changed using @CHG are as follows:

- Element name
- Version name

7.6.1. Changing Keys, Modes, and Names of Cataloged Files

You can use the @CHG statement to change the following cataloged file attributes:

- Change the privacy state to public (P option) or private (Q option). Change the privacy state to ACR-controlled (K option) for systems with Security Level 1, 2 or 3. Refer to Section 7.6.2 , “@CHG and File Security”, for information on whether the change applies to one or all members of an F-cycle set. Include the M option to change the privacy state for all members of an F-cycle set for all security configurations and file types.
- Change the read key and write key for all members of an F-cycle set.
- Change the read-only mode and write-only mode (V, W, and Z options) for one member of an F-cycle set. Include the M option to change the modes for all members of an F-cycle set. The change does not take effect until the file is freed by all runs that currently have it assigned.
- Change the file caching attributes (F option) for one member of an F-cycle set.
- Change the qualifier and filename (N option) for all members of an F-cycle set.
- Change an empty, sector-addressable, temporary or cataloged mass storage file to a program file (E and T options) for one member of an F-cycle set. For a description of program file types and attributes, refer to Section 3.8.4 through 3.8.7. This is the only type of file @CHG that applies to temporary files.
- Change the V-option flag mode (I and U options) for one member of an F-cycle set. Include the M option to change the modes for all members of an F-cycle set. I option requires user to have SSDNK and SSDBACK privileges while U option can be used by all users.

You can use a single @CHG statement to change the first three change types (privacy state, read and write keys, and read-only and write-only modes). There is one exception: You cannot change the privacy state to ACR-controlled and the read and write keys on the same @CHG statement. The three change types have different rules on which members of an F-cycle set are changed. Include the M option if you want all the change types to apply to all members of an F-cycle set.

Format

```
@CHG[,options] filename-1[,name-1]
```

options

C

Used to direct FURPUR not to exit through ERR\$ if an error is encountered.

Use of the C option on any FURPUR command will prevent execution of ER ERR\$ on error exit if FURPUR is running in breakpoint or batch. The C option should be used with caution because a bad result of an erroring FURPUR command could be input to a process later on in the runstream.

The C option is valid for all FURPUR commands.

E

Initialize an empty, sector-addressable file as a large-element program file (LEPF) that can contain symbolic and omnibus elements with over 262,143 sectors of text. Specify the E option by itself to create a standard LEPF that has a maximum of 5,000 elements. Specify the ET options to create a large LEPF that has a maximum of 26,146 elements. The change applies only to the F-cycle member specified by *filename-1*. No other options can be specified. Do not include *name-1*.

F

Change the file-caching attributes as specified by *name-1*. The change applies only to the F-cycle member specified by *filename-1*. No other options may be specified.

I

Change the V-option mode of file to set the unload inhibit flag.

K

Change the privacy state to ACR-controlled for all members of the F-cycle set specified by *filename-1*. The ACR is specified by *name-1*. The M option and one mode change option (V, W, or Z) may be specified.

M

Change the privacy state as specified by the K, P, or Q options, change the read-only and write-only modes as specified by the V, W, or Z options and change the V-option flag for all members of the F-cycle set specified by *filename-1*.

N

Change the qualifier and filename for all members of the F-cycle set specified by *filename-1* to the qualifier and filename specified by *name-1*. No other options may be specified.

P

Change the privacy state to public. Refer to Section 7.6.2, “@CHG and File Security”, for information on which members of the *filename-1* F-cycle set are changed. The M option may be specified to change all members of the F-cycle set. One mode change option (V, W, or Z) may be specified. A read and write key change may be specified by *name-1*.

Q

Change the privacy state to private. Refer to Section 7.6.2, “@CHG and File Security”, for information on which members of the *filename-1* F-cycle set are changed. The M option may be specified to change all members of the F-cycle set. One mode change option (V, W, or Z) may be specified. A read and write key change may be specified by *name-1*.

T

Initialize an empty, sector-addressable file as a program file that can contain up to 26,146 elements. Specify the T option by itself to create a large program file (LPF) that has a maximum element text size of 262,143 sectors. Specify the ET options to create a large LEPF that can have an element text size greater than 262,143 sectors. The change applies only to the F-cycle member specified by *filename-1*. No other options may be specified. Do not include *name-1*.

U

Change the V-option mode of the file to unset (clear) the unload inhibit flag.

V

Set read-only mode and clear write-only mode for the F-cycle member specified by *filename-1*. The M option may be specified to change all members of the F-cycle set. One privacy change option (K, P, or Q) may be specified. If the K option is not specified, then a read and write key change may be specified by *name-1*.

W

Clear read-only mode and set write-only mode for the F-cycle member specified by *filename-1*. The M option may be specified to change all members of the F-cycle set. One privacy change option (K, P, or Q) may be specified. If the K option is not specified, then a read and write key change may be specified by *name-1*.

Z

Clear read-only mode and clear write-only mode for the F-cycle member specified by *filename-1*. The M option may be specified to change all members of the F-cycle set. One privacy change option (K, P, or Q) may be specified. If the K option is not specified, then a read and write key change may be specified by *name-1*.

filename-1

Specifies the file to be changed. *Filename-1* is required.

name-1

Specifies additional change information for some @CHG statement types, as follows:

Change Read Key and Write Key

When the E, F, K, N and T options are not specified, *name-1* specifies the filename, new read key, and new write key for all members of the *filename-1* F-cycle set. The specified *filename-1* must include the current read key and the current write key. The qualifier, filename, and cycle specified on *filename-1* must be the same as the qualifier, filename, and cycle specified on *name-1*. If you do not specify a new read key on *name-1*, then the read key on the file will be removed (changed to spaces). If you do not specify a new write key on *name-1*, then the write key on the file will be removed. The formats for *name-1* are as follows:

<i>filename-1.</i>	Remove both read key and write key.
<i>filename-1/new-read-key.</i>	Change read key and remove write key.
<i>filename-1//new-write-key.</i>	Remove read key and change write key.
<i>filename-1/new-read-key/new-write-key.</i>	Change both read key and write key.

Change File Caching Attribute (F Option)

When the F option is specified, *name-1* specifies the file caching attribute to be applied to the F-cycle member specified by *filename-1*. The values for *name-1* are as follows:

D	XPC default
DC	XPC cache
DNC	XPC not cached
XRP	XPC resident permanent

Change Privacy State to ACR-controlled (K Option)

When the K option is specified, *name-1* specifies the ACR name to be applied to all members of the *filename-1* F-cycle set. Do not follow the ACR with a period. The ACR must be owned by the user-id associated with the run.

Change File Name (N Option)

When the N option is specified, *name-1* specifies the new name for all members of the *filename-1* F-cycle set. The qualifier, filename, or both may be changed. The formats for *name-1* are as follows:

<i>new-filename.</i>	Do not change the qualifier. Change the filename.
<i>*new-filename.</i>	Change the qualifier to the implied qualifier and change the filename.
<i>new-qualifier*.</i>	Change the qualifier. Do not change the filename.
<i>new-qualifier*new-filename.</i>	Change both the qualifier and the filename.

7.6.2. @CHG and File Security

Section 4 discusses the topic of file security in detail, including references to using @CHG to change discretionary access control (DAC) file protection attributes.

The DAC attributes that can be changed are as follows:

- File privacy state (K, P or Q option)
- Read-only and write-only modes (V, W or Z option)
- Read and write keys (*name-1* specified without the K option)

The following is a summary of the DAC attribute changes available with @CHG:

- Change File Privacy State
- Change Read-Only and Write-Only Modes
- Change Read Key and Write Key

Change File Privacy State

The members of the *filename-1* F-cycle set changed depend on which of the six security software options is configured on the system, whether the file is owned or unowned, and which options are specified on @CHG, as follows:

- For systems configured with Fundamental Security, all files are unowned and only the P or Q option can be specified to change the file privacy state. The privacy change applies to the F-cycle member specified by *filename-1*. The M option may be specified to apply the privacy change to all members of the F-cycle set.
- For systems configured with Security Level 1, 2 or 3, files may be owned or unowned. Special security privileges are required to create and maintain unowned files.
 - If the file is owned, then the privacy change applies to all members of the F-cycle set.
 - If the file is unowned and if the file is currently public or private and if the P or Q option is specified, then the privacy change applies to the F-cycle member specified by *filename-1*. The M option may be specified to apply the privacy change to all members of the F-cycle set.
 - If the file is unowned and if either the file is currently ACR-controlled or the K option is specified, then the privacy change applies to all members of the F-cycle set.
- For systems configured with Security Level 3 Controlled Access Protection Environment or Security Level 3 Trusted Environment all files are owned and the privacy change applies to all members of the F-cycle set.

Change Read-Only and Write-Only Modes

Mode changes specified by the V, W, or Z option apply to the F-cycle member specified by *filename-1*. The M option may be specified to apply the changes to all members of the F-cycle set.

Change Read Key and Write Key

Key changes specified by *name-1* apply to all members of the *filename-1* F-cycle set.

7.6.3. @CHG Examples for File Changes

The file changing @CHG examples are presented in the following order:

- Master File Directory Attributes (names, keys, and modes)
- File Caching
- File Security
- Program File Initializing of Program Files

Master File Directory Attributes (name, keys, and modes)

```
@CHG,N FN1.,FN2.
```

Changes the file name from FN1 to FN2.

The qualifier of the file is not changed.

```
@CHG FN/KEY1/KEY2.,FN/KEYA.
```

The read key of cataloged file FN is changed from KEY1 to KEYA.

The write key is removed (changed from KEY2 to spaces).

```
@CHG,V FN.
```

For cataloged file FN, set read-only mode and clear write-only mode.

```
@CHG,W FN.
```

For cataloged file FN, set write-only mode and clear read-only mode.

```
@CHG,ZM FN.
```

For all members (M option) of the FN F-cycle set, clear both read-only and write-only modes (Z option).

```
@CHG,I FN.
```

For cataloged file FN, set V-option flag, used along with M option to effect all members of FN F-cycle set.

```
@CHG,U FN.
```

For cataloged file FN, unset (clear) V-option flag, used along with M option to effect all members of FN F-cycle set.

File Caching changes

```
@CHG,F FN.,D
```

Sets the file-caching specification of file FN to the configured system default.

@CHG, F FN., DC

Specifies that the file FN must be cached.

Security changes

@CHG, KM FN., ACR17

Change the privacy state (K option) of all members (M option) of the FN F-cycle set. By adding the ACR ACR17 to the security record, the files become ACR-controlled.

For systems configured with Fundamental Security, ACR-controlled files are not allowed. Therefore, this request will produce an error message.

@CHG, Q FN.

Change the file privacy state for file FN to private. Whether the privacy change applies to just FN or to all members of the FN F-cycle set depends on the system security software configuration and the file type. Refer to Section 7.6.2 for more information.

@CHG, PWM FN/R1/W1., FN/R2/W2.

For all members (M option) of the FN F-cycle set

- Change the file privacy state to public (P option).
- Set write-only mode and clear read-only mode (W option).
- Change the read key from R1 to R2, and change the write key from W1 to W2.

Initializing program file structures within an empty file.

By default, a program file is initialized as a standard program file with a table of contents capacity of a standard program file. The @CHG with combinations of E and T options can change the file internally to allow other program file types. Only newly cataloged files, or erased files (@ERS) can be initialized as program files by the @CHG command.

@CHG, T LPF-FILE

The file table index of FN is formatted as a standard program file with the table of contents capacity of a large program file.

@CHG, E SLEPF-FILE.

The file table index of FN is formatted as a LEPP with the table of contents capacity of a standard program file.

@CHG, ET LLEPF-FILE

The file table index of FN is formatted as a LEPP with the table of contents capacity of a large program file.

7.6.4. Changing Program File Element and Version Names

You can use the @CHG statement to change the element names and version names of program file elements. You can also change element names and version names when you copy elements with the @COPIN, @COPOUT, and @COPY statements.

Format

```
@CHG[,options] element-name-1,element-name-2[,subtype-list]
```

options

Recognizes the A, C, O, R, S, V, X, and Y options for changing program file element and version names.

You must specify one or more of the A, O, R, and S options. Only the elements of types you specify by options have their names changed. You cannot specify element cycles.

When you use the V option for changing an element version, you must also include one or more of the A, O, R, and S options. The V option allows the changing of element/version names for all elements with the same version name as that specified in *element-name-1*. Only the element types specified by the options are changed. You can also use the Y option with the A and V options to limit changing element/version names to elements with subtypes that match the subtypes specified in the third specification field. See the @COPY statement description (see Section 7.14.3) for the definition of the V option, version mask, and X option.

element-name-1

Specifies the program file element.

element-name-2

Specifies the same program file, the new element, and version names.

subtype-list

Is a required parameter when the combined AY options are used to restrict selected elements to those with subtypes as listed. Each of the executable subtypes is defined with a mnemonic.

ABS	Absolute
OM	Object Module
SD	Subsystem Definition
ZM	Zero Overhead Object Module
BOM	Bound Object Module

Up to three subtypes can be named in a subtype-list. These examples model the acceptable options for specifying 1, 2, and 3 subtypes. A maximum of 3 subtypes can be specified on a single command.

1 subtype:	ABS	or ABS.	or ABS//.	or ABS//
2 subtypes:	ABS/ZM	or ABS/ZM.	or ABS//ZM.	
3 subtypes:	OM/ZM/BOM	or OM/ZM/BOM.		

7.6.5. @CHG Examples for Element Changes

The element changing @CHG examples are presented in the following order:

- Change a specific element or a version name.
- Change a specific element or version using the Y option to restrict subtypes of executable elements.
- Change matching elements with a selection criteria governed by use of the V option, version masking and variations on *name-1* specifications.
- Change matching element names and versions with the X option affecting the interpretation of *name-2*.

Change a specific element name or version name.

Specification of the A, O, R, and S options allows you to limit the changes to a selected set of element types. Any combination of these four element type options can be specified.

```
@CHG,OS F1.ELT,F1.NEWELT
```

Change the element name only.

The omnibus (O option) and symbolic (S option) elements named ELT in the program file F1 are changed to element name NEWELT.

```
@CHG,S F1.ELT1/VER1,F1.ELT2/VER2
```

Change both the element name and version name.

The symbolic (S option) element ELT1/VER1 in program file F1 is changed to an element name of ELT2 and a version name of VER2.

```
@CHG,A F1.ELT1/VER1,F1.ELT2/VER1
```

Change the element name, but not the version name.

The executable (A option) elements named ELT1/VER1 in program file F1 are changed to an element name of ELT2. The version is retained as VER1. This may change more than one element.

Without the Y option to restrict the change to a list of subtypes, any matching executable elements of all subtypes, (Absolute, OM, BOM, ZOOM, and SSD) will undergo the requested changes.

```
@CHG,R F1.ELT1/VER1,F1.ELT1/VER2
```

Change the version name only.

The version name of relocatable (R option) element ELT1/VER1 in program file F1 is changed from VER1 to VER2.

The Y option

Change only a select subtype of executable elements.

To limit the search for matching executable elements (A option) to only certain subtypes, add the Y option and provide a subtype-list in specification 3. Presence of the Y option requires the A option and it requires a subtype-list in specification 3.

```
@CHG,AY F1.ELT1,F1.ELT2,OM/BOM
```

Executable (A option) elements named ELT1 of the OM or ZM subtypes in file F1 will be changed to ELT2. Elements named ELT1 of other subtypes (ABS, BOM, or SSD) will not be renamed.

The V option

Change multiple elements by using version masking or by omitting the element name.

Using V option allows you to select multiple elements with a common version name or with version names containing common specified parts. The V option is described in detail in Section 7.14.3.

Multiple elements can be selected using the V option by:

- Using version masking with the * character.
- Omitting the element name.
- Using version masking AND omitting the element name.

The selected elements can be of the same element type or of varying element types. If any selected elements are of the same element type and are renamed with the same element and version name, all but one of the renamed elements will be marked as deleted.

```
@CHG,SV F1./VER1,F1./VER2
```

The input element name is omitted. Therefore, all input element names match.

The output element name is omitted. Therefore, the input element name is carried forward as the output element name.

All symbolic (S option) elements with any element name and a version name of VER1 in the program file F1 are given the version name of VER2, retaining the original element name.

```
@CHG,AV F1.ELT1/V*****,,F1.ELT2
```

Version masking selects executable (A option) elements with the name ELT1 and a version name of 1 to 12 characters, beginning with the character V. All selected elements are renamed ELT2. No output version name is specified. Therefore, the version names are removed (changed to blank).

Using masking character (*) can result in selection of multiple elements. In this case, many elements could be renamed ELT2, resulting in the deletion of previous ELT2 instances.

```
@CHG,SV F1.ELT1/V**,F1.
```

All symbolic (S option) elements with the element name ELT1 and a 1 to 3-character version name beginning with V are all given blank version names.

ELT1/V	changed to	ELT1
ELT1/V2	changed to	ELT1
ELT1/V23	changed to	ELT1
ELT1/V234	not selected	
ELT1/X23	not selected	
ELT2	not selected	

Because all the changed elements end up with the same element and version name, and because they all have the same element type (symbolic), all but one of the elements will be marked as deleted. The single remaining element will be the one with the highest sequence number.

To be compatible with lower levels of FURPUR, this form (*name-2* consisting of only a file name) assumes the X option, retaining the original element name when no replacement is specified.

```
@CHG,AVY F1./V**,F1.,OM/ZM
```

No element name is specified in this command. Therefore, all element names match.

All executable (A option) elements of the OM or ZM subtypes (Y option) in the F1 file with any element name and a 1- to 3-character version name starting with V keep their element names, but are given a blank version name.

<any>/V	changed to	same element name with no version
<any>/V2	changed to	same element name with no version
<any>/V23	changed to	same element name with no version
<any>/V234	not selected	
<any>/X23	not selected	

In this example, because of the omitted element name and the loss of version names, there is a high probability that duplicate element names will result. Any previously existing elements for which a name was duplicated will be marked as deleted.

The X option

Interpreting the destination (*name-2*) element name.

The X option affects the interpretation of the output element specification and determines the resulting element name and version name. A detailed description of the X-option is in Section 7.14.3 @COPY statement.

```
@CHG, SXV F1.ELT1/V**, F1.ELT2
```

All symbolic elements with the name ELT1 and a 1- to 3-character version name beginning with V in program file F1 are selected and changed to have the element name ELT2.

Because the X option is specified, the original version names are retained.

ELT1/V	changed to	ELT2/V
ELT1/V2	changed to	ELT2/V2
ELT1/V23	changed to	ELT2/V23
ELT1/V234	not selected	
ELT1/X23	not selected	
ELT2	not selected	

```
@CHG, SV F1.ELT1/V**, F1.ELT2
```

All symbolic elements with the name ELT1 and a 1- to 3-character version name beginning with V in program file F1 are selected and changed to have the element name ELT2.

Without the X option, the version names will be removed (changed to blanks). In this case, ELT2 will be impacted, although it does not meet selection criteria. It will be deleted during the processing of name changes from ELT1 to ELT2 and the removal of version names.

ELT1/V	changed to	ELT2
ELT1/V2	changed to	ELT2
ELT1/V23	changed to	ELT2
ELT1/V234	not selected	
ELT1/X23	not selected	
ELT2	not selected, but deleted as a consequence of name changes	

7.7. @CKPT and @@CKPT—Save a Complete or Partial Copy of a Run

The @CKPT statement lets you store information about a programming environment so that you can halt processing and continue at a later time. The checkpoint/restart (CKRS) program consists of two routines: a checkpoint routine and a restart routine. Complete information about the CKRS program is in the *CKRS Programming Reference Manual* (7830 7360).

Format

```
@CKPT[,options] [filename.element,,eqmnt-type,reel-1/.../reel-n,,,CTL-pool]
```

options

A

Does not terminate the run for certain nonfatal errors.

B

Does not save the absolute element being executed. A checkpoint always saves the current state of the program.

C

Indicates checksum of the checkpoint data.

J

Assigns the checkpoint tape as unlabeled.

M

Prints full diagnostic messages without diagnostic snapshots while processing the full checkpoint data.

N

Does not print error messages or diagnostics in the run's print file.

P

Displays error messages on the console.

Q

Prints diagnostic snapshots while processing full checkpoint data. Diagnostic snapshots consist of messages and dumps of specific CKRS data areas at various points in the checkpoint process.

R

Saves the mass storage symbiont files. (Tape symbiont files are always saved.)

S

Performs only a partial checkpoint.

T

Terminates the current batch run if the checkpoint is successfully taken. For demand runs, only the program terminates.

Z

Prints diagnostic snapshots while processing partial checkpoint data.

filename

Identifies the file for the checkpoint. The file name is optional if you specify the reel number and the J option. A file name not followed by a period is treated as an element in TPF\$ if you specify a version or element cycle or both, if this is a partial checkpoint. A file name not followed by a period is treated as a file name if this is a full checkpoint. A file name can include the directory-ID.

element

Identifies the element for the checkpoint. This element cannot reside in a Large Element Program File (LEPF).

If you specify an element, CKRS writes the checkpoint information to that element.

If you do not specify an element and the file is not a tape file, the checkpoint is written from the beginning of the file. More details and examples are given in the *CKRS Programming Reference Manual* (7830 7360).

eqpmnt-type

This optional field identifies the equipment type if you are using a tape and if filename is neither cataloged nor assigned.

reel-n

Specifies the reel number of the tape the checkpoint is written to. This is optional. You cannot specify expiration-period, ring-indicator, and *ACR-name*, but you must include the commas for these fields if you use the *CTL-pool* field.

CTL-pool

Can be one of the following:

CTL-pool-name

Directs the Exec to allocate a BLANK cartridge from the designated Cartridge Tape Library (CTL) pool of cartridge tapes.

NONCTL

Directs the Exec to use freestanding cartridge tape drives.

Examples

Examples of @CKPT are given in the *CKRS Programming Reference Manual (7830 7360)*.

7.8. @CLOSE—Close Tape File

The @CLOSE statement closes the specified magnetic tape files by writing two hardware EOFs on the associated magnetic tape and rewinding it. The @CLOSE statement performs the same operations as the @MARK statement followed by the @REWIND statement.

Only the *filename-1* parameter is required in the @CLOSE statement.

Format

```
@CLOSE[,options] filename-1 [,filename-2,...,filename-n ]
```

*options**C*

Do not take ERR\$ exit if an error is detected.

Use of the C option on any FURPUR command will prevent execution of ER ERR\$ on error exit if FURPUR is running in breakpoint or batch. The C option should be used with caution because a bad result of an erroring FURPUR command could be input to a process later on in the runstream.

The C option is valid for all FURPUR commands.

I

Rewind with interlock. If the I option is omitted, the tape is rewound without interlock.

filename-n

Specifies the tape files to be closed.

7.9. @@CM—Send a Message to the Operator Console

The @@CM statement is another way you can send a message to the operator console.

One of the main differences between the @MSG statement and this statement is that the @MSG statement must be used within an active run, whereas the @@CM statement can be used outside of a runstream (before an @RUN statement or after an @FIN statement). In addition, the @@CM message is prefixed by your site-ID instead of your user-ID. You can think of the @@CM statement as a console message statement.

Format

```
@@CM message
```

message is the message you are sending and can be up to 57 characters long.

Example

```
@@CM PLEASE CHECK IF TAPE 3556A IS THERE
```

Sends the message following @@CM to the operator console.

7.10. @@CONS—Enter Console Keyins

The @@CONS statement allows you to enter console keyins. This subsection shows you two ways to enter console keyins. It also supplies a table of console mode keyins that can be used and the console mode required for each keyin (Table 7-2).

Format 1

Use this format when you want to enter a series of unsolicited keyins:

```
@@CONS  
keyin 1  
keyin 2  
.  
.  
.  
keyin n  
@@END CONS
```

The @@END CONS statement terminates console mode. If you do not enter an @@END CONS statement, you stay in console mode until you type in an @TERM or @FIN statement.

If you have display-and-keyin mode or response-and-keyin mode, you can use option letter K on the N @@CONS statement to inhibit automatic displaying of message groups, as follows:

```
@@CONS, K
```

Format 2

Use this format when you want to enter only one keyin:

```
@@CONS[,host-id] keyin
```

host-ID

Can be A through L. The *host-ID* is valid only if your system has Multi-Host File Sharing. Using this option lets you submit a keyin on one host for processing on another host in the file-sharing environment. The response is routed back to the original host. The output display does not indicate the host-ID of the processing host.

keyin

Includes the keyin and the text of the keyin.

Note: If you enter @@CONS keyin between an @@CONS and an @@END CONS statement (see Format 1), and you do not include any other keyins between those statements, the @@CONS and @@END CONS are extraneous (unless a read-and-reply message is possible as a result of the keyin).

7.10.1. Operating Considerations

The following operating considerations apply to the @@CONS statement:

- As with all previous Exec levels, the start-of-entry (SOE) solicitation of input at a demand terminal (or through the ER RSI\$ interface) is unpredictable after a user enters the @@CONS statement. That is, solicitation cannot occur.

See the *ER Programming Reference Manual* (7830 7899) for more information on ER RSI\$.
- Read-only console messages and the message groups display can be discarded when they build up. You receive a warning message when console output is lost. When you terminate console mode, passing of message groups and read-and-reply messages to your terminal ends. The output (normal output from a keyin routine) is displayed until the keyin routine sends the last line of output.
- For Asian messages sent to remote consoles (@@CONS)
 - The Exec Console Message Handler does not check a remote console to make sure that it can translate and display 2-byte Asian characters. If the console cannot, what is displayed is uncertain.
 - A 2-byte Asian character can be truncated in the middle of the 2 bytes, causing that character to be displayed incorrectly. (The Exec Console Message Handler does not check for this.)

7.10.2. Console Modes

As a demand terminal operator, you can perform many of the console commands used by a system console operator. Console commands, sometimes called keyins, are instructions that tell the Exec to perform certain tasks. The console commands you are allowed to use are determined by the remote console mode capability level given to you by your site administrator. There are five levels of console mode capabilities:

- Basic
- Limited
- Full
- Display
- Response

Following are descriptions of each console mode. Each console mode is described in terms of the maximum console command capabilities possible for that console mode level. Whether you actually have the maximum, or a subset of the maximum, console commands allowed in a particular mode depends on how the site administrator sets up your user-ID. For example, a limited mode user might be restricted to doing only the symbiont console commands. See the *ClearPath OS 2200 Apex Help* (8207 4154) or the *SIMAN Administration and End Use Guide* (TQ-01151) for more information on how your site administrator configures your user-ID for console commands.

Basic

This mode lets you manipulate or request information on your run, on any run started by you, or on any run controlled by your user-ID.

Limited

In this mode, you can use all the commands specified in basic mode, and you can use many of the status commands.

Full

With full console capability, you can use most of the unsolicited commands. However, you are not allowed to use the security, performance analysis, and common bank reload keyins, and a few others.

Display

This mode lets you use the same console commands as in full mode and also displays message groups at your terminal. You can see all message groups or a subset of the message groups defined at your site, depending on how your user-ID is configured. For more information on message groups, see the *Exec Operations Reference Manual* (7831 0281).

Display console mode gives you capabilities similar to those of an onsite system console operator, with the following exceptions:

- Your terminal cannot be a boot console.
- Your terminal cannot be a response console for a group. In other words, you cannot reply to read-and-reply messages. The system operator cannot use an AC keyin to make your terminal a response console.
- The system status continuous display is not sent to your terminal.
- No special scrolling procedures are used for messages coming to your terminal.

Response

This mode lets you use all unsolicited commands except COLOR and AC. This mode also lets your terminal act as a display or response console, or both, for the message groups configured for your user-ID. This mode has all the privileges and limits of display mode, with the following exceptions:

- Your terminal can be the response console for one or more specified message groups. Up to 10 read-and-reply messages can be displayed at the same time; their message numbers range from 0 to 9. The message number is not removed from the message once the message is answered.
- You can issue console commands without having answered any outstanding read-and-reply messages.
- If no command groups are selected for your user-ID, you can perform all the keyins except AC. If specific command groups are selected, you can perform only the commands in those groups and the MR command.

Refer to the *Exec Operations Reference Manual* (7831 0281) for the actual console command descriptions and other operator procedures.

The dynamic configuration parameter, *echo_remote_console*, can be used to specify that messages for the system console must be displayed on the local system console in addition to any active remote response console.

7.10.3. Console Mode Input/Output Rules

Following are various rules associated with using console modes.

Terminating Console Mode

Entering an @@END CONS statement terminates console mode. Once you have terminated console mode, any input you enter is treated as regular input. This includes unsolicited commands and responses to read-and-reply messages resulting from the processing of unsolicited commands.

When you are not in console mode and you type in an @@CONS *command* statement, console mode is terminated immediately after the command has been accepted. If this statement contains a command that results in a read-and-reply (such as a DN command), the read-and-reply is not displayed at the terminal and you cannot respond to the read-and-reply.

If a read-only message is outstanding or occurs after you have terminated console mode, the message is discarded.

Message Group Display Termination

Termination of display mode or response mode also terminates all message groups going to your terminal. If you initiate display mode or response mode with an @@CONS *command* statement, no message groups are displayed because console mode is terminated before routing of the message groups is set up.

Read-and-Reply Messages

Read-and-reply messages and responses to these messages are handled in the following ways:

- If a read-and-reply message is outstanding or occurs after you have terminated console mode, the message, along with a header message if the read-and-reply message was the result of a previous unsolicited command, is sent to an onsite console. The format of the header is as follows:

```
RESULT OF KEYIN FROM site-id
```

If you have response mode, no header is sent if the read-and-reply message was not the result of a previous unsolicited command.

If you do not have response mode, once a read-and-reply message has been displayed at your terminal, no additional input other than a response to that message is allowed. If you enter other input, the following message appears:

```
*ANSWER MESSAGE*
```

A redisplay of the read-and-reply message appears next. The following cycle continues until you answer the message, the terminal times out, you leave console mode, or you type in @@TERM:

1. The system accepts your input.
2. It verifies the input as a response.
3. It sends the reject message and the original read-and-reply message if the answer is unacceptable.

If the input is acceptable, it is sent to the routine that issued the read-and-reply message, and the console handler deletes the original read-and-reply message. The routine then determines whether the response is correct and reissues the message if it is not.

The read-and-reply message sent to a terminal has the following format:

```
# read-and-reply-message
```

where # is the read-and-reply message number.

The answer (also called a solicited keyin) to the read-and-reply message must be in this format:

```
# answer
```

- If you have response mode, some read-and-reply messages associated with a particular message group can already be displayed on a local console. These messages are not redisplayed at your remote console; they must be answered at the local console.

Print Output

If you have limited console mode, you cannot terminate print files (using the SX keyin) from runs other than your own. If you have full, display, or response mode, and you are allowed to enter the symbiont group of console commands, you can terminate any print files.

Note: *The Exec and communications software do not attempt to segregate print output from console output (while in console mode); or console output from print output (once console mode has terminated).*

7.10.4. Console Mode Commands

The @@CONS transparent commands allow you to perform a subset of the commands used by a system console operator. Most users can use only those commands that affect their own runs. For descriptions of these commands, refer to the *Exec Operations Reference Manual* (7831 0281) for a list and explanation of all console messages associated with these commands.

Table 7-2 lists commands that can be performed by demand users and shows the console mode required for each command.

Table 7-2. Console Mode Commands

Keyin	Console Mode
\$!	Response
AP ALL FS	Full
AP <i>n</i> FS	Full
AP <i>n</i> DN	Full
AP <i>n</i> UP	Full
AR	Full
ARCMR <i>type</i>	Response
AT <i>n function parameters</i>	Full
B <i>nbr</i>	Full
BL	Limited
BP	Response
C	Response
CJ A	Full
CJ ALL	Full
CJ <i>number,number</i>	Full
CP	Response
CK	Response
CS A	Full
CS A <i>run-id</i>	Basic (5)
CS ALL	Full
CS <i>run-id</i> *F	Basic (5)
CS H	Full
CS H <i>run-id</i>	Basic (5)
CS AD	Full
CS AT	Full
CS HD	Full
CS HT	Full
CTL	Full
D	Basic
D <i>date</i>	Full
D <i>date time</i>	Full
DC <i>devname1,devname2</i>	Full

Table 7-2. Console Mode Commands

Keyin	Console Mode
DISPLVL	Full
DJ	Full
DN	Full
DP	Response
DU MP	Response
E run-id	Basic (5)
EU	Full
EX	Full
FA	Full
FB	Full
FB bit	Full
FB LST	Full
FB ON bit	Full
FB OFF bit	Full
FC	Response
FF [dir-id#]qual*filename(f-cycle).	Limited
FS	Full
HK	Response
HM	Response
HU	Full
II	Response
IN	Full
IT	Full
LB	Full
LB LST	Full
LB OFF bit	Full
LB ON bit	Full
LB bit	Full
LC OFF	Full
LC ON	Full
LG*run-id text	Basic (5)
LG text	Full

Table 7-2. Console Mode Commands

Keyin	Console Mode
MD	Full
MH	Response
MH,AD	Response
MR RC [,subfunction] [identifier]	Full (10)
MR TR [,subfunction] [site-id]	Full (10)
MS[,options]	Limited
MU	Limited
MU,ALL	Full
MU run-id	Basic (11)
MV	Full
PL	Response
PM	Full
PR	Full
RC run-id	Basic (3)
RD	Full
RE run-id	Full
RL bankname	Response
RM run-id	Basic (5)
RP pack-id/directory name	Response
RS	Response
RV	Full
SB command	Response
SC action/item index	Response
SEC, SAVE	Full
SEC, RESTORE	Full
SF [device]/[reeln],filename	Response
SH	Full
SJ A	Full
SJ ALL	Full
SJ number,number, ...	Full
SJ O xxxxxxxxxxxxxx	Full
SLB	Response

Table 7-2. Console Mode Commands

Keyin	Console Mode
SLC	Response
SM (All)	Full
SP run-id	Full
SQ	Limited
SQ run-id file QTO sname2 or SQrun-id file QTO user-id/U	Basic (8)
SQ run-id file x or SQ run-id x	Full
SQ run-id *R	Basic (7)
SQ sname	Basic (1)
SQ sname *	Basic (2)
SQ sname1 QTO sname2	Basic (4)
SQ sname1 TO sname2	Basic (4)
SQ user-id *U	Basic (6)
SQ USR*ID	Limited
SQ USR*ID *	Limited
SR ,unit/reel A/F	Full
SR ,unit/reel A/F sname	Full
SR ,unit/reel run-id1 ,..., run-idn	Full
SR ,unit/reel run-id file	Full
SS	Limited
ST	Full
SU	Full
SV ,unit/reel,option A/F	Full
SV ,unit/reel,option A/F sname	Full
SV ,unit/reel,option run-id1 ,..., run-idn	Full
SV ,unit/reel,option run-id file	Full
SX A/F	Full
SX A/F sname	Basic (4)
SX run-id1,..., run-idn	Basic (8)
SX A/F user-id *U	Basic (9)
SX A/F USR*ID	Full
SX run-id file	Basic (8)

Table 7-2. Console Mode Commands

Keyin	Console Mode
T	Limited
T,B	Limited
T,B D	Limited
T,D	Limited
T,D D	Limited
T D	Limited
T H	Limited
T S	Limited
TB text	Full
TF	Full
TIPSS	Full
TM addressee text	Limited
TP A [transaction code] or [#application number]	Full
TP CLR	Full
TP DTQ	Full
TP D RTPS\$,program#,program-name	Full
TP I RTPS\$,program#,program-name, #-of-copies	Full
TP MAX [n]	Full
TP QUE	Full
TP RA [transaction-code] or [#application number]	Full
TP RC*[run-id]	Full
TP RH [number-of-seconds]	Full
TP RL	Full
TP RP	Full
TP RS [transaction-code] or [#application number]	Full
TP S RTPS#	Full

Table 7-2. Console Mode Commands

Keyin	Console Mode
TP T [<i>program name</i>] or [#application number]	Full
TP TA [<i>transaction-code</i>] or [#application number]	Full
TP TIP [<i>transaction-code</i>] or [#application number]	Full
TP TQ [<i>transaction-code</i>] or [#application number]	Full
TP TR [<i>transaction-code</i>] or [#application number]	Full
TP TS [<i>transaction-code</i>] or [#application number]	Full
TP XL	Full
TS	Response
TU	Full
UD	Full
UL	Full
UP	Full
X <i>run-id</i>	Full

Notes:

1. You can use this command in basic console mode only if *sname* matches *site-ID*; otherwise, it is restricted to limited console mode.
2. You can use this command in basic console mode only if *sname* matches *site-ID*, and then only for output queue entries that have *user-IDs* matching the current *user-ID* at the terminal; otherwise, it is restricted to limited console mode.
3. You can use this command in basic console mode only if the *user-ID* of *run-ID* matches the *user-ID* at the terminal; otherwise, it is restricted to limited console mode.
4. You can use this command in basic console mode only if *sname1* matches *site-ID*, and then only for output queue entries that have *user-IDs* matching the current *user-ID* at the terminal; otherwise, it is restricted to full console mode.
5. You can use this command in basic console mode only if the *user-ID* of *run-ID* matches the *user-ID* at the terminal; otherwise, it is restricted to full console mode.
6. You can use this command in basic console mode only if the *user-ID* matches the currently active *user-ID* at the terminal; otherwise, it is restricted to limited console mode.

7. *You can use this command in basic console mode only for those output queue entries with run-IDs that also have user-IDs matching the current user-ID at the terminal; otherwise, it is restricted to limited console mode.*
8. *You can use this command in basic console mode only for those output queue entries with run-IDs that also have user-IDs matching the current user-ID at the terminal; otherwise, it is restricted to full console mode.*
9. *You can use this command in basic console mode only if user-ID matches the currently active user-ID at the terminal; otherwise, it is restricted to full console mode.*
10. *This command is implicitly allowed only for response console mode.*
11. *This command requires limited console mode if the user-ID specified is not the user-ID of the user entering the command.*

The maximum length for commands entered through @CONS, not including the command prefix or read-and-reply number, is 66 characters for unsolicited commands or 68 for replies to read-and-reply messages. The Exec ignores any additional characters.

7.11. @@CONT—Continue to Send Output

The @@CONT statement tells the system to resume sending output to the terminal after you have interrupted it with the MSG WAIT key. (Although you can continue output by just pressing the XMIT key, it can generate an error message, or the blank line might be accepted by the executing program.)

Format

@@CONT

7.12. @COPIN—Copy from Tape to Program Files

The @COPIN statement copies one or more elements from an element file located on magnetic tape into a program file located on sector-formatted mass storage. Large Element Program Files (LEPFs and LLEPFs) cannot be used as output program files for COPIN.

When a relocatable element is added to a program file, any entry-point table that can have existed for the file before the execution of the @COPIN relocatable statement is destroyed. You can use the @PREP statement (see Section 7.46) to recreate the relocatable entry-point table. If a tape error occurs, only those elements completely transferred before the error occurred are entered in the program file and listed in its table of contents.

Format

@COPIN[,options] name-1,name-2

options

Can be file options or element options (listed in the following "File Options" and "Element Options").

File Options

Following are the @COPIN statement options which are specified when *name-1* is a file name:

None

Used to copy elements from the input magnetic tape file (in element file format) into the program file located on sector-formatted mass storage. The tape file must be positioned at the label block of the first element being copied (use the @FIND statement, see Section 7.29) and continues until a hardware EOF mark is encountered. The elements retain the element names used in the element file.

A,O,R,S

Same operation as if no options were specified, except that the A, O, R, and S options can be used to designate the type of element to be copied. Any combination of A, O, R, and S can be used. All elements of the types specified are transferred. The elements retain the names they had in the element file.

C

Used to direct FURPUR not to exit through ERR\$ if an error is encountered.

Use of the C option on any FURPUR command will prevent execution of ER ERR\$ on error exit if FURPUR is running in breakpoint or batch. The C option should be used with caution because a bad result of an erroring FURPUR command could be input to a process later on in the runstream.

The C option is valid for all FURPUR commands.

V

Same as for elements.

X

Same as for elements.

Element Options

Following are the @COPIN statement options which are specified when *name-1* is an element name:

A,O,R,S

Used to copy one element and insert it into the output program file. The element name remains the same unless renamed in *name-2*.

V

Used to copy elements having the same version name from an element file on magnetic tape into a program file on sector-formatted mass storage. The input file must be positioned at the label block of the first element to be copied. Copying continues until a hardware EOF mark is encountered. See

the @COPY statement description in Section 7.14.3 for a definition of the V option and version mask.

X

Used to exchange *name-2* form interpretation for V option processing. See the @COPY statement description in Section 7.14.3.

name-1

Specifies the input tape file or input element to be copied.

name-2

Specifies the output program file or output program file and element name. If *name-1* is an element name and *name-2* is a file name, the element retains its name in the new file.

Examples

The following examples illustrate the use of the @COPIN statement:

```
@COPIN HOLDPROG., PROGRAM.
```

The element file located on magnetic tape file HOLDPROG is copied into program file PROGRAM on sector-formatted mass storage.

```
@COPIN, R TEMP.ELTA, PF1.
```

Relocatable element ELTA in element file on tape file TEMP is copied into program file PF1. The relocatable entry-point table of file PF1 is destroyed (an @PREP statement is needed to rebuild the relocatable entry-point table; see Section 7.47).

```
@COPIN, RV A. /B, C.
```

All relocatable elements in element file on tape file A with the version name B are copied into program file C. They retain the version name B in the program file.

```
@COPIN, SV A., C.
```

All symbolic elements in element file on tape file A with a blank version name are copied into program file C. These elements added to the program file retain blank version names.

```
@COPIN, R PET.ELT3, REPET.VERSG
```

The relocatable element in element file on tape file PET with the element name ELT3 and a blank version name is copied into program file REPET and given the element name VERSG with a blank version name.

@COPIN, O T., F.

All omnibus elements in element file on tape file T are copied into program file F. The element and version names remain the same.

@COPIN, SV TAPE./V*****, FAST.

All symbolic elements in element file on tape file TAPE having a version name beginning with V are copied into program file FAST. The element and version names remain the same.

@COPIN, AV TAP.OLDABS/VERS*****, FAS.

All executable elements named OLDABS having a version name beginning with VERS are copied into program file FAS. The element and version names remain the same. See the description of the @COPY statement (Section 7.14.3) for examples of how to use the X option.

7.13. @COPOUT—Copy Program Files to Tape

The @COPOUT statement copies a program file or selected elements from a program file located on sector-formatted mass storage into a magnetic tape file in element file format. Large Element Program Files (LEPF and LLEPFs) cannot be used as input program files for COPOUT.

Procedure name entries are saved but relocatable entry points are discarded. Tape files must be in element file format in order to use the @COPIN and @FIND statements (see Sections 7.12 and 7.29).

Note: *If you just want to save a program file on tape, it is faster and more efficient to use the @COPY,G statement instead of the @COPOUT statement.*

All parameters of the @COPOUT statement are optional except *name-2*.

Format

@COPOUT[,options] [name-1], name-2

options

The choices are either file or element options.

File Options

The following are the @COPOUT statement options that can be specified when *name-1* is a file name:

None

Used to write all nondeleted elements onto the magnetic tape output file in element file format. For unlabeled tapes, two EOF marks are written at the end of the file and the tape is backspaced one EOF mark. For labeled tapes, one EOF mark is written and the tape is positioned after the EOF mark. Elements retain the element name they had in the program file.

A, O, R, S

Used to write all nondeleted elements of the types specified by the options onto the magnetic tape output file in element file format. The elements retain the names they had in the program file. Any combination of A, O, R, and S can be used. No EOF mark is written.

C

Used to direct FURPUR not to exit through ERR\$ if an error is encountered.

Use of the C option on any FURPUR command will prevent execution of ER ERR\$ on error exit if FURPUR is running in breakpoint or batch. The C option should be used with caution because a bad result of an erroring FURPUR command could be input to a process later on in the runstream.

The C option is valid for all FURPUR commands.

E

Used for unlabeled tapes to improve write performance. At the end of the file, write a single EOF mark instead of writing two EOF marks and backspacing over one of them. Before finishing with the tape (for example @FREE, @MOVE,B or @REWIND), you must close the output tape with @CLOSE (see Section 7.8) or @MARK (see Section 7.38). The E option cannot be used with the A, O, R, S, V, or X options, which do not write an EOF mark. The E option is ignored for labeled tapes, which always write a single EOF mark.

V

Same as for elements.

W

Used when the buffered-write mode BUFTAP is enabled on the output tape to assure that all buffered data has successfully been written to tape. After the write of the EOF mark, a synchronize (FSAFE\$) I/O function is issued, if necessary. Note that the synchronize increases the time to write this file, as if the buffered-write mode BUFFIL was enabled. To maximize performance the W option must be used only periodically, to establish error recovery "checkpoints." The W option cannot be used with the A, O, R, S, V, or X options, which do not write an EOF mark. The W option is ignored for unlabeled tapes when the E option is not specified. In the latter two cases, the tape is implicitly synchronized by the Exec or by the tape unit.

X

Same as for elements.

Element Options

The following are the @COPOUT statement options that can be specified when *name-1* is an element name:

A, O, R, S

Used to write all specified element types for the element names given into the output magnetic tape file in element file format. Only nondeleted elements are transferred. If the *name-1* and *name-2* element names are different, all elements copied have the *name-2* element name. One or more options must be specified. No EOF mark is written.

V

Used to write all nondeleted elements, selected by version name and type, onto the magnetic tape output file in element file format. The V option can be used in combination with the A, O, R, and S options. When it is used alone, all element types are considered. See the description of the @COPY statement in 7.14.3 for a definition of the V option and version mask.

X

Used to exchange *name-2* parameter form interpretation for V option processing. See the description of the @COPY statement in 7.14.3.

Note: If you have specified any of the A, O, R, S, or V options on the @COPOUT statement, FURPUR does not write an EOF mark. Thus, you need to add a final @MARK statement (see Section 7.38).

name-1

Specifies the input program file or element to be copied. If you do not specify *name-1*, the default (TPF\$) is used.

name-2

Specifies the output element file, or output element file and element name.

Examples

The following examples illustrate the use of the @COPOUT statement:

```
@COPOUT PROGRAM.,HOLDPROG.
```

The contents of program file PROGRAM located on sector-formatted mass storage are copied to magnetic tape file HOLDPROG as an element file. Since no options are specified, a hardware EOF mark is written following the output file.

@COPOUT,ARS C.,D.

The nondeleted executable, symbolic, and relocatable elements of program file C located on sector-formatted mass storage are copied into magnetic tape file D as an element file. No EOF marks are written following the file because options were specified.

@COPOUT,R A.,B.

All nondeleted relocatable elements in program file A located on sector-formatted mass storage file A are copied into magnetic tape file B as an element file. No EOF marks are written following the file.

@COPOUT,S A.B,C.D

Symbolic element B in program file A is copied into magnetic tape file C in element file format and given element name D.

@COPOUT,SV A./B,C.

All nondeleted symbolic elements in program file A with a version name of B are copied into magnetic tape file C and retain the version name B.

@COPOUT,AV A.,C.

All nondeleted executable elements in program file A with a blank version name are copied into magnetic tape file C.

@COPOUT,V A./*B*****,C.

All nondeleted elements in program file A with a version name containing a B as the second character are copied to magnetic tape file C. The version names are unchanged.

See the description of the @COPY statement (Section 7.14.3) for examples of how to use the X option.

7.14.@COPY—Copy Files or Elements

The @COPY statement copies between SDF files, program files, and program file elements. The @COPY functions can be divided into these categories:

File to File (see Section 7.14.1)

- SDF files into a destination SDF file
- All program file elements into a destination program file
- Program file elements of selected types into a destination program file

Program File element(s) (see Section 7.14.2)

- A single program file element into a destination program file element
- SDF file contents into a program file element
- A program file element into an SDF file
- Sets of program file elements into a destination program file

The format of the basic @COPY statement is the same for either of the category.

All parameters of the @COPY statement are optional.

Format

```
@COPY[,options] [name-1,name-2,{ number-of-files | subtype-list } ]
```

options

The options possible depend on which type of COPY is requested.

- For options for File to File copy requests, see options list in Section 7.14.1.
- For options for Program File element(s) copy requests, see options list in Section 7.14.2.
- For C option, see Appendix B.1.

name-1

Specifies the input file or element to be copied.

name-2

Specifies the output file or element into which the input file or element is to be copied.

number-of-files

This specification can only be used for file-to-file copy requests.

When used, it indicates the number of files to copy; if omitted, 1 is assumed. When used for tape-to-tape copying of entire files, it specifies the number of input files to copy onto the output tape. The input tape remains positioned between the two EOF marks. The number of blocks in each file copied and the number of files copied are indicated in the output listing.

When used in conjunction with the B option, it specifies the number of OS 2200 FORTRAN files to copy from one file to another. The number of files copied is indicated in the output listing.

When an attempt is made to copy an empty file (two hardware EOFs), the copy operation is immediately terminated regardless of the contents of the *number-of-files* parameter.

subtype-list

In addition to specifying the number of files, you can also use the third specification to specify the mnemonics of executable element subtypes.

This is a required parameter when the combined AY options are used to restrict selected elements to those with subtypes as listed. Each of the executable subtypes is defined with a mnemonic.

ABS	Absolute
OM	Object Module
SD	Subsystem Definition
ZM	Zero Overhead Object Module
BOM	Bound Object Module

Up to three subtypes can be named in a subtype-list. These examples model the acceptable formats.

1 subtype: ABS or ABS.
2 subtypes: ABS/ZM or ABS/ZM. or ABS//ZM.
3 subtypes: OM/ZM/BOM.

7.14.1. File-based Copy Options

Mass storage files can be either word-addressable or sector-formatted. When copying from one mass storage to another, any combination is allowed (sector to word, word to word, word to sector, or sector to sector).

The following are the options that you can supply for @COPY statements to copy a source file into a destination file:

None

- Copying from mass storage to mass storage—Used to overwrite one mass storage file (name-2) with the contents of another mass storage file (name-1), without regard to the file format and size. If name-2 is smaller than name-1, copying is done partially (till end of name-2).
- Copying from tape-to-tape—Used to copy one or more files (depending on number-of-files parameter) from the input tape to the output tape without regard to the file format. No hardware EOF marks are written. Block size is limited to 65,000 words.
- Copying from mass storage to tape—Used to copy the contents of a mass storage file to an output tape. The mass storage file is copied to tape a track at a time with blocks written to tape to represent any unassigned (whole) tracks in the mass storage file. At the conclusion of the copy, no EOF is written to the tape. This allows appending of additional files to the tape. @MARK must be used to place an EOF at the end of the file. This is a very simple, unformatted copy and in most cases another copy command (COPY,G) would probably be much more useful for depositing single mass storage file contents to tape.

A, O, R, S

- Used to restrict the copy of the file contents to the elements of the type specified. Both program files must be located on sector-formatted mass storage. Only non-deleted elements are copied. Any combination of A, O, R, and S can be used.
- When the S option is specified, all element S-cycles are copied.
- When the A option is used, the Y option can be added to further restrict the scope of the file copy. When the AY options are used together, a subtype-list must be provided in specification 3.

B

- Used only with OS 2200 FORTRAN formatted data files to copy the FORTRAN-specified data files from the input file to the output file. The input and output files cannot both be mass storage files, nor can they both be tape files.
- When the input file is on mass storage, each software EOF encountered designates the end of a FORTRAN file and is followed by a hardware EOF when written on tape.
- When the input file is on tape, each hardware EOF encountered designates the end of a FORTRAN file. No hardware EOF is maintained on mass storage.

C

Used to direct FURPUR not to exit through ERR\$ if an error is encountered.

Use of the C option on any FURPUR command will prevent execution of ER ERR\$ on error exit if FURPUR is running in breakpoint or batch. The C option should be used with caution because a bad result of an erroring FURPUR command could be input to a process later on in the runstream.

The C option is valid for all FURPUR commands.

D

This option can be used in conjunction with the G option on an @COPY,G command to write the revised format tape blocks to a noncompressed tape. This format, described in the *Data Structures Programming Reference Manual* (7833 3481) allows increased performance when the tape is read into mass storage with an @COPY,G command. Read performance for both labeled and unlabeled copies is increased by use of the D option when the tape is written.

The @COPY,G to noncompressed tape without the D option continues to write tapes in the obsolete variant (one track per block) format. An @COPY,G command from mass storage to compressed tape continues to write tapes in the revised format.

File-based Copy Options (continued)**E**

Used with @COPY statements that write to unlabeled tapes to improve write performance. At the end of the file, write a single EOF mark instead of writing two EOF marks and backspacing over one of them. Before finishing with the tape (for example, @FREE, @MOVE,B or @REWIND), you must close the output tape with @CLOSE (see Section 7.8) or @MARK (see Section 7.38). The E option is illegal if the output file is not a tape. The E option is ignored if the statement does not write an EOF mark. The E option is ignored for labeled tapes, which always write a single EOF mark.

F

Used to copy the contents of one file into another file. Only system data format (SDF) files can be copied with this option. The input file must be in SDF. Reading of the input file is terminated by the SDF EOF mark. Block sizes for tape files must be 224 words. When the output file is a magnetic tape file, a hardware EOF mark is written following the file.

G

Used to efficiently copy a file from mass storage to tape or from tape to mass storage without regard to file contents. The mass storage file may be cataloged or temporary, sector-addressable or word-addressable, and may reside on fixed or removable mass storage. Multiple tracks are read or written with each I/O request.

- Copying from mass storage to tape

The D, E, M and W options may be included with the G option.

A 28-word logical file descriptor block is the first block written to tape. The logical file descriptor contains information about the source mass storage file, about the date and time of the copy and about the format of the file data blocks that follow.

A two-word track header precedes each 1,792-word data track written to tape, for a total of 1,794 words. Each track header contains the mass storage address of the data track, a checksum of the track data and the sequence number of the track, beginning with zero. Only allocated mass storage tracks up to and including the highest track written are read. The physical tape block size is fixed for each file and is a multiple of 1,794 words, as described for the D option.

The last physical tape block written to the tape may contain fewer tracks than the previous blocks.

File-based Copy Options (continued)

If the mass storage file fills the currently mounted tape volume, the following message is output and the tape is swapped to the next output volume:

```
OUTPUT REEL SWAPPED      BLOCK COUNT n
```

The block count *n* is the number of 1,794-word tracks written to the output volume before the tape swap.

- Copying from tape to mass storage

The H option may be included with the G option.

A 28-word logical file descriptor block is the first block read from tape. Information on the original name of the file and when it was copied from mass storage to tape is formatted into the following message:

```
q*f(c) COPIED ON yyyy-mm-dd hh:mm:ss
```

The track header checksum and the sequence number for each track are validated as the tracks are read from tape. Each track is written to the mass storage address contained in the track header. Exec dynamic allocation algorithms may allocate mass storage tracks that were unallocated when the file was originally written to tape.

If the tape file fills the currently mounted tape volume, the following message is output and the tape is swapped to the next input volume:

```
INPUT REEL SWAPPED      BLOCK COUNT n
```

The block count *n* is the number of 1,794-word tracks read from the input volume before the tape swap.

H

Used with the G option when copying very old multi-volume labeled tapes to mass storage.

Labeled tapes have not been written with swap blocks since FURPUR 27R3A was released in late 1978. To improve tape read performance, labeled @COPY,G format tapes are not checked for swap blocks.

Include the H option to read swap blocks only if this is a multi-volume labeled tape that was or might have been written by a FURPUR level prior to 27R3A.

File-based Copy Options (continued)

M

Used only when the output file is a magnetic tape file.

- Copying from mass storage to tape

Assumed with the G option to copy a sector-formatted mass storage file to magnetic tape. Two hardware EOF marks are written on the tape following the file copied and the tape is positioned between the EOF marks.

- Copying from tape to tape

Used without other options or with the N option for tape-to-tape copying of one or more files (depending on *number-of-files* parameter). If more than one file is being copied, a hardware EOF mark is written on the tape following each file copied and two EOF marks are written after the last file copied to the output tape if the tape is positioned between the two EOF marks.

- Copying an empty file from tape to tape

On labeled tapes, if an empty file set is found on the input tape, the empty file set is duplicated on the output tape and copying continues. For unlabeled tapes, the @COPY,M function terminates at the empty file.

N

Used to copy a magnetic tape file containing an abnormal frame count to another magnetic tape file or to a sector-formatted mass storage file.

P

Used to copy all nondeleted elements from one program file and add them to another. any existing elements that are duplicated as a result of the copy are deleted in the destination file. If the element being deleted is a PROC element, all associated PROC names in the procedure table are deleted. Both program files must be located on sector-formatted mass storage. Can be used in conjunction with the A, O, R, or S options. Used alone, it implies the A, O, R, S combination.

T

Used for tape-to-tape copying of 8-bit labeled tapes.

For the HDR1 label block, the expiration date field is set to zero by FURPUR to force use of the system default or the @ASG statement value for the output tape. The accessibility code field is transferred from the input tape to the output tape.

For the HDR2 label block, the record format, block length, and buffer offset length fields are transferred from the input tape to the output tape.

The block numbering, block count, and data compression fields are not transferred. The block numbering and data compression fields are determined by system defaults or values on the @ASG statement of the output tape.

File-based Copy Options (continued)

The following information for the output tape HDR2 block is reproduced: record format, block length, record length, buffer offset length, block attribute, block numbering, and data compression.

For the HDR1 block on the output tape, the file identifier is the qualifier and file name under which the output tape was assigned, and the expiration date field is set to the appropriate value for the output tape.

If the input tape is a foreign tape that contains HDR3 through HDR9, EOF3 through EOF9, or EOVS3 through EOVS9 label blocks, these label blocks cannot be written by FURPUR to the output tape. The following message is output for header label blocks that cannot be written:

```
Input tape HDR3[, HDR4, ...] label[s] not written to output tape.
```

The following message is output for trailer label blocks that cannot be written:

```
Input tape EOF3[, EOF4, ...] label[s] not written to output tape.
```

The messages lists all of the label blocks read from the input tape.

User header labels following an HDR1-HDR2 group are read and written on the output tape. User trailer labels following an EOF1-EOF2 group are read and written on the output tape. Any user trailer labels following an EOVS1-EOVS2 group on the input tape are ignored. Input reels are swapped when an EOVS1-EOVS2 group is encountered. Reels are swapped on the output tape when a 02 I/O status is received while writing a data block.

The M option is assumed with the T option. The T option is intended for use with 8-bit labeled tapes that are created by ASCII FORTRAN or ASCII COBOL.

U

Causes labels on an output labeled tape to be written in EBCDIC format.

V

Used to copy one file into another file. The input and output file must not both be on magnetic tape or sector-formatted mass storage.

- Copying from mass storage to tape

Variable block size is assumed. The first word of the block (containing the block size) is stripped from the block before it is written into the tape file to be compatible with the block buffering package. Block size is limited to 7,168 words.

File-based Copy Options (continued)

- Copying from tape to mass storage

A word containing the block size is prefixed to the block before it is written on sector-formatted mass storage. The input tape file must be terminated by a hardware EOF mark. Block size is limited to 7,168 words.

A copy to or from sector-formatted mass storage always begins in sector 0, and each block starts in a new sector. If the block size is not divisible by 28, the excess words of the last sector contain random data.

W

Used when the buffered-write mode BUFTAP is enabled on an output tape to assure that all buffered data has successfully been written on tape. After the write of the EOF mark, a synchronize (FSAE\$) I/O function is issued, if necessary. Note that the synchronize increases the time to write this file, as if the buffered-write mode BUFFIL was enabled. To maximize performance, the W option must be used only periodically, to establish error recovery "checkpoints." The W option is illegal if the output file is not a tape. The W option is ignored if the statement does not write an EOF mark. The W option is ignored if the buffered-write mode is BUFOFF, BUFON, or BUFFIL. The W option is ignored for unlabeled tapes when the E option is not specified. In the latter two cases, the tape is implicitly synchronized by the Exec or by the tape unit.

Y

Used in conjunction with the A option to direct FURPUR to copy the executable elements that have a subtype matching one of those listed in the subtype list field, which is specification field 3.

7.14.2. Element-based Copy Options

The following are the options that you can supply to @COPY statements that copy

- A source element into a destination element
- A source element into a destination file
- A source file into a destination element

A, O, R, S

Used to restrict the copy of the specified element from the input file to those matching the type specified. Any combination of A, O, R, and S can be used. Any existing elements that are duplicated as a result of the copy are deleted in the destination file. If the element being deleted is a PROC element, all associated PROC names in the procedure table are deleted.

The options represent the types of elements to be copied (one or more are needed). The element name can be changed by renaming it in *name-2*. Both input and output files must be on sector-formatted mass storage. When a symbolic element is being copied, all element cycles are copied unless an element cycle is specified in *name-1* or if the L option is specified on the call (in which case, the output element produced is designated as cycle zero).

Element cycles cannot be specified in *name-1* for Assembler, FORTRAN, or COBOL procedure elements. Element cycles cannot be specified in *name-2*.

I

Used to copy an SDF file to a program file as a new symbolic element or to copy a symbolic element into an SDF file. The SDF file can be either a tape file or a sector-formatted mass storage file.

- SDF to Element

The SDF file is copied into a new symbolic type element in the destination program file. If the O option is also present, the new element will be created as an omnibus type.

Reading of the input file is terminated by an SDF EOF control record.

The new element retains the SDF records of the file from which it was created. When the new element is referenced, S3, S4, S5, S6 in the control records can be treated as cycle information by many system processors.

- Element to SDF

A symbolic-type element is copied into a new SDF file. If the O option is also present, the new file will be created from a source element of the omnibus type.

All deleted images from the source element are removed.

If the source element is in SIR\$ format, all element cycle information in the image control words is zeroed. Otherwise, the copy is done in blocks with no checking of image control words.

Element-based Copy Options (continued)

L

Used with the S option to copy the latest cycle of the element in *name-1*. The output element is designated as cycle zero. The L option is ignored if the element in *name-1* is an Assembler, FORTRAN, or COBOL procedure element, or if the element in *name-1* is not general symbolic type SDF.

O

Usable only in conjunction with the I option. When present, it indicates use of omnibus elements, rather than symbolic elements, in copies between SDF files and Program File elements.

When copying from a file to an element, the mass storage file cannot contain any unallocated areas (holes).

V

Allows you to broaden your selection of elements by using version masking (the asterisk character) or by omission of either the element or the version name from the *name-1* specification. For details, see Section 7.14.3.

Y

Can only be used in conjunction with the A option. The AY options restrict the copy of executable type elements to the subtypes indicated in the subtype-list in specification 3.

X

For output element specifications (*name-2*), the X option affects the interpretation of an omitted version name. The previous version name can be retained or removed (replaced by blanks). See Section 7.14.3 for a detailed description.

When a procedure element is copied, the procedure name entries are automatically added to the procedure name table of the output file.

If a relocatable element is copied, the relocatable entry-point table of the output file is destroyed and the @PREP statement (see Section 7.46) must be used to recreate it.

Before you do any copy operation from tape, you might need to execute an @MOVE statement (see Section 7.40) to position the tape beyond some EOF mark. No @COPY operation moves backward or forward over an EOF mark before the start of the copy.

7.14.3. Version Masking, Multiple Element Selection, and the V and X Options

There are times when you want to select multiple elements to copy instead of a single individual element. You may want to copy all elements with a particular element or version name. Perhaps you may want to copy all elements with version names starting with the letter A. To allow multiple element selection, FURPUR provides these methods to select a multiple element set on a single FURPUR statement.

The following methods can be used together or separately. These selection methods are activated by using the V option on FURPUR statements that accept an input element specification. Each of these methods is described in detail in the following sections.

1. Omission of *name-1* (input) specification for element name or version name.
2. Using version masking in the version part of *name-1* (input) specification.
3. Omission of *name-2* (output) specification for element name or version name along with use of the X option.

The V option can be used on element-based formats of the following FURPUR statements. File-based formats of these statements have a different meaning, or no meaning, ascribed to the V option.

@CHG,V	input-elt-spec, output-elt-spec
@COPIN,V	tape-name, output-elt-spec
@COPOUT,V	input-elt-spec, tape-name
@COPY,V	input-elt-spec, output-elt-spec
@DELETE,V	input-elt-spec
@PRT,TV	input-elt-spec

The V option can be used in conjunction with any of the element type options (A, O, R, or S). If no element type options are specified to limit the scope of selection, the assumption is that all types are selectable.

The ability to select multiple elements for any of the supported statements is a powerful tool which should be used judiciously. Using the V option allows a repeated action on multiple input elements that may result in unintended deletion of existing elements. For example:

- Renaming multiple input elements to have the same output element and version name.
- Creating a new element into a program file where that element name and version already exists.

In such cases, any existing elements are marked as deleted when new elements with the same element name, version name, and in the case of executable elements, same subtype, are added through renaming of an existing element (@CHG) or creating a new element (@COPY).

Omission of Input Element or Version Names

For input (*name-1*) specifications, the following table shows the effect of omitting either the element or version name in conjunction with the V-option on a statement. An element specification is generally defined as:

[qualifier]file.element[/version].*

For the purposes of the following table, only the components *element* and *version* are addressed. If an input file is omitted, the standard assumption of TPF\$. is enacted.

Omitted Component	Sample Specification	Selection Results
None	File.Elt/Vers	Only the exact input element and version is selected. The presence or absence of the V option has no effect on this command.
Element	File./Vers	All elements with names of any length and composition match, but only elements with the exact version name are selected.
Version	File.Elt	Only the exact element name with a blank version name (no version) is selected.
Both	File.	All elements with names of any length and composition match, but only elements with a blank version name (no version) are selected.

Using a Mask as a Version Name

Another V option method to select multiple elements is to use a *version mask*. A version mask uses the asterisk (*) as a positional wildcard and can consist of 1 to 12 characters. Multiple asterisk characters can be specified in any of the 12 positions available.

The version mask only applies to version names; it does not apply to element names.

The following table shows how asterisks in the version field identify character positions for which any character in the corresponding position of a version field is accepted as a match. Only trailing asterisks can match blanks because blanks cannot be embedded in the body of a version name.

Version Mask	Selection Process	Possible Matches
ELT*****	Selects elements named ELT with any version name, including blank characters (no version).	ELT ELT\A ELT\123456789012 ELT\SYS\$FILE

Version Mask	Selection Process	Possible Matches
DATA\SYS*****	Selects elements named DATA that have version names with characters 1-3 equal to SYS and characters 4-12 equal to anything.	DATA\SYS DATA\SYS\$\$FILE DATA\SYS123456789
LOG****201*	Selects elements named LOG that have version names with characters 5-7 equal to 201 and characters 1-4 and 8 equal to anything.	LOG\01012016 LOG\AAAA2017 LOG\DATA2010 LOG\DATA201
\DATA-****	Selects any element name (omitted element name matches all element names) with version names having characters 1-5 equal to DATA- followed by characters 6-9 equal to anything.	E\DATA-FILE ELT\DATA-2004 ELT\DATA-Z ELT\DATA- ELTNAME\DATA-LOG

Both V option methods, name or version omission and version masking, can be used together.

Element\Version Mask	Selection process	Possible Matches
File.\P*****	The omitted element name selects all elements. However, the use of version masking selects only the elements having a version name beginning with P of any size and any composition.	A\P B\P B\P12345678901 LONGELT\P123

The X Option and Omission of Output Element or Version Names

For output (*name-2*) specifications, the X option affects the manner in which omission of output element and version names are interpreted. The version name in the destination element can be retained or removed, depending on the use of the X option.

The @CHG, @COPY, and @COPIN statements allow an output element specification and may use the X option.

An element specification is generally defined as:

[qualifier]file.element[/version].*

Depending on the X-option, the interpretation of the final destination element and version names will vary. The following table shows the effect of omitting the element and version components in the output specification (or destination) of the copy statement, with or without the X option.

Omitted Component	Output Specification	Interpretation of Output Names	
		Without X Option	With X Option
None	File.Elt/Vers	All selected input elements based on specification 1, will be renamed with the same element and version name, Elt/Vers. As elements are renamed, existing elements with that name will be deleted. Only the last element evaluated will remain undeleted.	
Element	File./Vers	All selected input elements will retain existing element names. The version names will be replaced with the specified version name.	
Version	File.Elt	Change selected element names to the name specified. Remove version names (replace with blanks).	Change selected element names to the name specified. Retain version names.
Both	File.	Retain existing element names. Retain version names.	Retain existing element names. Remove version names (replace with blanks).

7.14.4. @COPY Examples

The following examples illustrate the use of the @COPY statement. The examples are presented in alphabetical order.

```
@COPY, AOV FI./VERS, FO.
```

All nondeleted executable and omnibus elements in program file F1 having a version name of VERS are copied into program file FO. Element and version names remain unchanged.

```
@COPY, ARSV FILE1./V*****, FILE2.
```

All nondeleted executable, relocatable, and symbolic elements in program file FILE1 that have a version name beginning with V are copied into program file FILE2 with element and version names unchanged.

```
@COPY, AY F11.ELT1, F86.ELT2, OM
```

If the element is an executable type and has a subtype of OM, FURPUR copies the element ELT1 from the file F11 to the file F86 and changes its name to ELT2. executable type and has a subtype of OM.

@COPY,AY WILDC.,BEARC.

This statement produces an error message stating that one or more of the subtypes ABS, BOM, OM, SD, and ZM must be specified when you specify the AY option combination.

@COPY,B FORT.,TAP2.,3

The first three FORTRAN files in the file FORT are copied into tape file TAP2. File FORT must contain at least three FORTRAN files. A hardware EOF mark follows each of the three files written on TAP2.

@COPY,F F1.,F2.

The contents of file F1 are copied into file F2. File F1 must be an SDF file.

@COPY FLAP4.,FLAP5.

The contents of sector-formatted mass storage file FLAP4 are copied into sector-formatted mass storage file FLAP5, writing over any previous contents of the FLAP5 file.

@COPY,GM FILLUP.,TANK.

The contents of sector-formatted mass storage file FILLUP are copied into magnetic tape file TANK. If TANK is an unlabeled tape, two EOF marks are written at the end of the output file (TANK), and the tape is positioned between the two EOF marks (M option). Since file TANK is in @COPY,G format, the @FIND and @COPIN statements cannot be used to access the file; however, the @COPY,G format makes more economical use of time and space. When you return the file (@COPY,G) to sector-formatted mass storage, the file is reproduced as it was before, including all tables of contents and deleted elements. Do not attempt to merge two program files, each of which were saved on tape using the @COPY,GM statement, because the second file would overlay the first.

@COPY,I FLIP.,FORK.UPT3/INOUT

The contents of input file FLIP, which is in SDF format, are copied into output file FORK, which is in program file format. Input file FLIP is entered in FORK as an element having UPT3 as its element name and INOUT as its version name. It is set at element cycle 0.

@COPY,M TRAP3.,TRAP6.,9

The nine files that form magnetic tape file TRAP3 are copied into magnetic tape file TRAP6. Each file in output file TRAP6 is separated by an EOF mark, as directed by the M option. The last file copied into the output file is followed by two EOF marks, and the output file is positioned between the two final EOF marks if TRAP6 was an unlabeled tape.

@COPY,NM TAP1.,TAP2.

One file of magnetic tape file TAP1 is copied onto magnetic tape TAP2. Two EOF marks are written on unlabeled TAP2, and the tape is positioned between the two. File TAP1 can contain abnormal frame counts.

@COPY,O F1.,F2.

The nondeleted omnibus elements in program file F1 are copied into program file F2.

@COPY,P FLYBY.,FLIGHT.

The nondeleted elements of program file FLYBY are copied into program file FLIGHT.

@COPY,RS FIRM.C,FIREUP.A

The nondeleted relocatable and symbolic elements with element name C (version name of blanks) in program file FIRM are copied into program file FIREUP as elements with element name A (version name of blanks).

@COPY,RS FOR6.,FOR9.

The nondeleted relocatable and symbolic elements (R and S options) located in program file FOR6 are copied into program file FOR9.

@COPY,SAV FOX./BASE,FPERM.JOB23

The nondeleted executable and symbolic elements with a version name of BASE in program file FOX are copied into program file FPERM. The element name JOB23 and version name of all blanks are assigned to each element copied.

@COPY,SAVX FGX-25./RDSE***,FPERM.MATHPAC

The nondeleted executable and symbolic elements with a 4- to 7-character version name starting with RDSE in program file FGX-25 are copied into file FPERM. The element name MATHPAC is assigned to each element copied and the corresponding version name is retained.

@COPY,SAY F4F.,F8F.,OM/ABS/ZM.

All symbolic elements and all executable elements that have ABS, OM, or ZM subtypes are copied from the file F4F to the file F8F.

@COPY,SRV F.ELT/VERS,FF.ELTNEW/VERSNEW

The nondeleted relocatable and symbolic elements named ELT in program file F that have VERS as a version name are copied into program file FF and given an element name of ELTNEW and version name of VERSNEW.


```
@COPY,SY B17.,B29.,OM/SD/ZM.
```

Because you have not specified the A option with the Y option, the system displays an error message stating that you must do so.

```
@COPY,U TAPE1.,TAPE2.
```

EBCDIC format labels is written to the output tape.

7.15. @@CQUE—Queue Input in a Main Storage Buffer

The @@CQUE statement lets you enter lines of statements and data into a buffer in the computer's main storage. The images in the resulting input queue are processed in the order they were entered. This lets you continue entering input without waiting for the next SOE character.

Format

```
@@CQUE soe-parameter  
input-image  
input-image  
input-image  
.  
.  
.  
@@END CQUE
```

Valid values for the *soe-parameter* field are

N

No SOE characters are issued by the system when in this input queuing mode. The system resumes issuing appropriate SOE characters when the corresponding @@END control statement is received to return the terminal to interactive mode.

S

SOE characters are issued by the system when in this input queuing mode after each input image is transferred to a system main storage buffer. The system continues to issue appropriate SOE characters when the corresponding @@END control statement is received, allowing the terminal to return to interactive mode.

The *soe-parameter* field is a 1-character field. If more than one character or anything other than an N or S is specified, the statement is ignored and the following error is displayed at the terminal:

```
@@ERROR - SYNTAX
```

input-image represents the statements and data you want to enter into the buffer in main storage.

You must provide your own SOE character once you enter CQUE mode if the N is specified or no *soe-parameter* is specified.

7.16.@CYCLE—Alter Cycle Retention Limit

Sets the maximum range of absolute F-cycle numbers to be retained for a cataloged file, or the maximum number of element cycles for a program file symbolic element. When *number* is omitted, cycle information regarding the current F-cycle series is displayed. See Section 3.4 for a detailed discussion of F-cycle sets.

The *filename* or *element-name* parameter of the @CYCLE statement is required.

7.16.1. Set File Cycle Range

Format for Files

```
@CYCLE filename[,number]
```

filename

Specifies the F-cycle set.

number

A decimal value from 0 to 32 that specifies the maximum range of F-cycles for the set. If *number* is omitted, information about the F-cycle set is output.

The specified *filename* must be a member of a cataloged F-cycle set. The *filename* can include an absolute or relative F-cycle number or can be an internal file name. If *number* is specified and if the file has read and write keys, either the file must be assigned with the keys or the keys must be specified on the @CYCLE statement, in any combination. For example, the file can be assigned with the read key and the write key can be specified on @CYCLE.

FURPUR reads the MFD entry for the specified file name, which contains a description of the F-cycle set and all of its members, including their absolute F-cycle numbers and file statuses (currently cataloged, to be cataloged, to be deleted, or not cataloged).

When you omit *number*, information about the entire F-cycle set is displayed in the following format:

```
MAX-RANGE= a           CUR-RANGE= b  
NO. OF F CYCLES= c: d-1, d-2, ... d-b
```

Where *a* is the maximum number of members of the F-cycle set (1 to 32), *b* is the current number of members of the F-cycle set (1 to 32), *c* is the number of members of the F-cycle set that are currently cataloged (1 to 32), and *d-1* through *d-b* are the absolute F-cycle numbers (1 to 999) and file statuses for each of the *b* files in the current range. There are four possible file statuses for members of the F-cycle set.

The following are the displayed formats for absolute F-cycle number *d* for each status:

- d* Absolute F-cycle *d* is currently cataloged.
- **d** Absolute F-cycle *d* is deleted or was not cataloged.
- +*d*+ Absolute F-cycle *d* is to be cataloged. It changes to currently cataloged when freed by the run that created it with @ASG,C or @ASG,U.
- d*- Absolute F-cycle *d* is to be deleted. It changes to deleted when freed by all of the runs that currently have it assigned.

Note: The *b* files in *CUR-RANGE* can have any of the above four file statuses while the *c* files in *NO. OF F CYCLES* are only the F-cycle members that are currently cataloged.

If *number* is nonzero, it specifies the new maximum F-cycle range. Only the F-cycle members in the new range are retained. If there are any F-cycle members outside the new range they are deleted.

If the highest absolute F-cycle number is *highest*, then the new range includes F-cycle numbers from *highest* through *highest* - *number* + 1, in decreasing order. For example, if the highest absolute F-cycle number is 49 and *number* is 16, the new range is absolute F-cycles 49 through 34 (49-16+1). If the formula yields 0 or a negative number, this indicates that the F-cycle set range wraps around from F-cycle 1 back to F-cycle 999. In this case the new range includes F-cycle numbers from *highest* through 1 and 999 through 1000 + *highest* - *number*, in decreasing order. For example, if the highest absolute F-cycle number is 7 and *number* is 12, the new range is absolute F-cycles 7 through 1 and 999 through 995 (1000+7-12). Note that whenever the F-cycle range wraps around, the F-cycle numbers *highest* through 1 are considered to be "higher" than 999 through 1000 + *highest* - *number*. In the second example, F-cycles 7 through 1 are considered to be higher than 999 through 995.

If *number* is nonzero, there must be at least one F-cycle member in the new range that can be assigned to the run. If *filename* is a retained F-cycle member, it is the F-cycle that is used. If *filename* is an F-cycle member that is to be deleted, then beginning with the highest absolute F-cycle the new retained F-cycle range is searched for an F-cycle member that can be assigned. The F-cycle member can be currently cataloged or be marked as cataloged or as deleted, as long as it can be assigned to the run. If no retained F-cycle can be assigned to the run, @CYCLE is terminated in error. See Appendix B.7 for a complete list of error messages associated with @CYCLE.

F-cycle members are deleted when *number* is 0 or when there are F-cycle members outside the new maximum F-cycle range specified by *number*. FURPUR begins by checking that each F-cycle member to be deleted can be assigned to the run (exclusively when *number* is nonzero) and that each is not currently on a symbiont queue. This includes F-cycle members that are currently cataloged, to be cataloged and to be deleted.

@CYCLE—Alter Cycle Retention Limit

If this check fails, no F-cycle members are deleted. Error messages are output to list all of the F-cycle members that cannot be assigned and all of the F-cycle members that are on symbiont queues. @CYCLE is terminated in error.

When all of the F-cycle members to be deleted are assigned to the run and it has been determined that none are on symbiont queues, then FURPUR begins deleting them. The deletion begins with the lowest absolute F-cycle and continues in increasing F-cycle order.

If *number* is 0, then after all F-cycles have been successfully deleted the following message is output and @CYCLE is complete:

```
DROP FLAG SET ON ALL CYCLES
```

If *number* is nonzero, then after the F-cycles outside of the new range have been successfully deleted, the MFD entry for the F-cycle set is updated. The maximum F-cycle range is set to *number*, the following messages are output, and @CYCLE is complete:

	NUMBER-CYCLE	CYCLE-LIMIT
OLD	<i>c</i>	<i>a</i>
NEW	<i>c</i>	<i>a</i>

Where *c* is the current number of members of the F-cycle set and *a* is the maximum number of members for the F-cycle set. The OLD row shows the values at the beginning of @CYCLE and the NEW row shows the values at the end of @CYCLE. The *c* values in the NUMBER-CYCLE column correspond to the NO. OF F CYCLES= *c* value output when *number* is omitted. The *a* values in the CYCLE-LIMIT column correspond to the MAX-RANGE= *a* value output when *number* is omitted.

The following is a summary of @CYCLE file assignment information:

- The specified *filename* can be any member of the F-cycle set that is or can be assigned to the run.
- The specified *filename* can include an absolute F-cycle number, a relative F-cycle number, or can be an internal file name.
- The read and write keys for the F-cycle set must be supplied by assigning *filename* with the keys or by specifying the keys on @CYCLE, in any combination.
- If *number* is nonzero, there must be at least one currently cataloged, to be cataloged or to be deleted F-cycle member in the new range that is or can be assigned to the run.
- No F-cycle members to be deleted can be on a symbiont queue.
- All F-cycle members to be deleted must be assigned to the run by you or by FURPUR.
- Fixed mass storage files and cataloged tape files are assigned with the Y option; removable mass storage files are assigned with the E option.
- When *number* is nonzero, the F-cycle members to be deleted are assigned exclusively (X option).

- Files currently assigned to the run are freed and reassigned under the following conditions:
 - A removable mass storage file assigned with the Y option is freed and reassigned with the E option.
 - A file assigned without required read and write keys is freed and reassigned with the keys, when they are available.

In both cases the file is not restored to its original assignment state when @CYCLE is complete.

- In the following exception cases, files that would otherwise be freed and reassigned are instead marked as not assignable:
 - If the file status is to be cataloged
 - If the file status is to be deleted
 - If the file is assigned with the D option or K option

Notes:

- *@CYCLE operations use the MFD entry initially read, which describes the F-cycle set at the instant that the read is done. If multiple runs are working with the same F-cycle set and particularly if F-cycle members are being cataloged or deleted by other runs, then @CYCLE operations that delete F-cycle members or change the maximum F-cycle range might error or yield unexpected results. Having runs assign files exclusively (X option) might not overcome this problem, since @CYCLE uses the Y or E option on file assignments, which causes exclusive use to be ignored.*
- *Use of the C option on any FURPUR command will prevent execution of ER ERR\$ on error exit if FURPUR is running in breakpoint or batch. The C option should be used with caution because a bad result of an erroring FURPUR command could be input to a process later on in the runstream.*

7.16.2. Set Element Cycle Limit

Format for Elements

@CYCLE *element-name*,*number*

element-name

Specifies a program file symbolic element whose cycle limit is to be changed.

number

Specifies the maximum number of element cycles to be retained.

The A, O, R, S, and Y options are not appropriate for this statement.

When a program file element is named, the @CYCLE statement sets the maximum number of element cycles. If *number* specifies a new maximum less than the current number of element cycles being retained, a new element with the same name is

created and some data image control words can be changed and some images cannot be transferred at all.

Any image deleted in a cycle number less than the lowest cycle number being retained, because of the new lower maximum number of cycles, is not transferred to the new element. Any images added but not deleted in a cycle number less than the lowest cycle number being retained are transferred, but S6 of each data image control word is changed to reflect the new lowest cycle number retained. All other nondeleted and deleted images are transferred to the new element.

Examples

The following examples illustrate the use of the @CYCLE statement:

```
@CYCLE Q*A.B,2
```

Assume that symbolic element B in program file Q*A consists of element cycles 5, 6, 7, and 8. Since the new limit is two cycles, a new element B is created consisting of cycles 7 and 8.

```
@CYCLE Q*D.,2
```

Assume that the master file directory entry for file Q*D indicates that four absolute F-cycles 18, 15, 14, and 12 of the file exist. Since the new limit is two, absolute F-cycles 15, 14, and 12 are deleted. The limit is considered to be the range starting from the highest current absolute F-cycle number.

```
@CYCLE Q*D.
```

Assuming this statement follows the second example, the following F-cycle information is displayed:

```
MAX-RANGE= 2          CUR-RANGE=          1
NO. OF F CYCLES= 1:    18
```

7.17. @@DCT—Change DCT Terminal Characteristics

The @@DCT statement changes characteristics associated with the DCT 500 terminal. The @@DCT and @TTY statements are interchangeable. You can submit the statements from either device (see Section 7.68).

Format

```
@@DCT function-1,value-1[, ...,function-n,value-n]
```

function-n

Indicates which characteristics to change. They are as follows:

C

Changes character delete.

I	Changes input image length.
L	Changes line delete.
P	Changes page length.
S	Changes solicit character.
T	Changes image terminator.
W	Changes page width.

value-n

Is the new value you are setting for the field (character) just specified. Most fields can take a character, octal, or decimal value. However, the I, P, and W functions can take only an octal or decimal value. You can indicate character values by specifying the octal value for ASCII characters. You cannot assign the following as values: T for termination character, L for line delete, or C for character delete.

Examples

```
@@DCT W,96,T,010,S:
```

The page width is changed to 96. The image terminator is octal 10, which is the octal representation for the ASCII backspace character. The solicit character is the colon (:).

```
@@DCT I,40
```

The input image length is 40.

7.18. @DELETE—Delete Files and Elements

The Delete command deletes cataloged files, deletes program file elements (changes elements from current to deleted), or undeletes program file elements (changes elements from deleted to current).

All parameters in the @DELETE statement are optional except *name-1*.

Format

```
@DELETE[,options] name-1 [,name-2,...,name-n,subtype-list]
```

options

When you delete cataloged files, the N option can be used. See Section 7.18.1.

When you delete elements, the A, O, R, S, V, and Y options can be used. See Section 7.18.2.

When you undelete elements, the U option must be specified and the A, O, R, S, V, and Y options can be used. See Section 7.18.3.

Use of the C option on any FURPUR command will prevent execution of ER ERR\$ on error exit if FURPUR is running in breakpoint or batch. The C option should be used with caution because a bad result of an erroring FURPUR command could be input to a process later on in the runstream.

name-n

Specifies the cataloged file to be deleted or the program file element to be deleted or undeleted.

subtype-list

When elements are deleted or undeleted, this field can be used to select executable element subtypes. You must specify the A and Y options in order for this field to be interpreted as a subtype list. When you specify the A and Y options, you must specify a subtype list as the last field.

7.18.1. Deleting Files

When you delete a cataloged file, only the N option applies.

When you specify an external file name, you must specify the F-cycle if it is not the latest F-cycle. If read and write keys exist, you must either specify them or must have assigned the file to the run with the read and write keys.

The file is not actually deleted until all other runs that have the file assigned to them have freed the file. When the file is deleted, the master file directory items are updated. The older F-cycles have their relative F-cycle number increased.

The N option is available for the file DELETE command. The N option deletes the file entry from the master file directory (MFD), but retain any internal file names that were attached to the file before it was deleted.

When a removable pack is in the DN state (brought down by a DN,PACK keyin), the operator can be able to make the pack available so that the delete can be performed. See the *Exec Operations Reference Manual* (7831 0281) for more information on the Down Pack feature.

You can also use the @FREE,D statement to delete files.

To delete all members of an F-cycle set, you can use the @CYCLE command. See Section 7.16 of this manual for more information.

7.18.2. Deleting Elements

One or more elements can be deleted in a program file, based on element name, version name and element type. When a symbolic procedure element is deleted, all procedure table entries associated with the element are deleted. When a relocatable element is deleted, the relocatable entry point table is removed. However, when an object module executable element is deleted, the object module entry point table is not affected.

Until the @PACK command (see Section 7.43) is used, deleted elements can be undeleted (changed from deleted to current) with the @DELETE,U command. See Section 7.18.3 for a detailed description. Once the @PACK command is used, the deleted elements, their associated element text, and their associated procedure table entries, if any, are permanently removed from the program file and can no longer be undeleted.

When the V option is not specified, any combination of the element type options A, O, R, and S can be specified, but at least one must be specified. If the A option is specified the Y option can be included to select only the executable element subtypes specified in the subtype-list field. The element name must be specified. If the version name is not specified, it is assumed to be spaces. If a version name is specified, it cannot include asterisk (*) mask characters. The element cycle cannot be specified. For each element type specified for which an element could not be found and deleted, the following message is output:

```
q*f(c).e/v type NOT FOUND
```

When the V option is specified, any combination of the element type options A, O, R, and S can be specified. If the A option is specified, the Y option can be included to select only the executable element subtypes specified in the subtype-list field. If you do not specify any element type options, then all element types are selected. If the element name is not specified, then elements are selected without regard to element name. If a version name is not specified, it is assumed to be spaces. If version name is specified it can include asterisk (*) mask characters. The element cycle cannot be specified. See Section 7.14.3 for a detailed description of the V option and version name mask characters.

Either with or without the V option, one of following messages summarizes the results of the @DELETE for *name-n*:

```
[a SYM] [b REL] [c ABS] [d OMN]  
NOTHING DELETED
```

In the first message, *a*, *b*, *c*, and *d* are the number of elements of each type that were deleted. If no elements were deleted for an element type, then the corresponding part of the message does not appear. If no elements of any element type were deleted, the second message is output.

Examples

The following examples illustrate the use of the @DELETE statement:

```
@DELETE FLAP.,TARE5.,ZEBRA4.,BAKER.
```

Relative F-cycle 0 of cataloged files FLAP, TARE5, ZEBRA4, and BAKER is dropped from the master file directory (the files are decataloged).

```
@DELETE,S F.ELT1/VERS,F1.ELTY
```

The symbolic elements ELT1/VERS in program file F and ELTY in program file F1 are marked as deleted. Any associated procedure names are also marked as deleted.

```
@DELETE,OV F1./*****
```

All omnibus elements in program file F1 are marked as deleted.

```
@DELETE,UOV F1./*****
```

Undelete the highest relative sequence number for each omnibus element/version/type set in program file F1. If file F1 did not have any deleted omnibus elements before the previous example, then this example "undo" the @DELETE,OV and restore file F1 to its previous state.

```
@DELETE,RSV F2./VERS
```

All relocatable and symbolic elements that have a version name of VERS in program file F2 are marked as deleted. If any relocatable elements are deleted, the relocatable entry point table is removed. If any symbolic procedure elements are deleted, the associated procedure table entries are also deleted.

```
@DELETE,RSVU F2./VERS
```

Undelete the highest relative sequence number for each relocatable and symbolic element/version/type set with a version name of VERS in program file F2. If any relocatable elements are undeleted, the relocatable entry point table is removed. If any symbolic procedure elements are undeleted, the associated procedure table entries are also undeleted. If file F2 did not have any deleted relocatable or symbolic elements before the previous example, then this example "undoes" the @DELETE,RSV and restores file F2 to its previous state, with the exception of the relocatable entry point table. An @PREP (see Section 7.46) is needed to restore the relocatable entry point table.

```
@DELETE,V FAST./V*****
```

All elements having a version name that begins with V in program file FAST are marked as deleted.

```
@DELETE,SAY F101./V1**,OM/SD
```

All symbolic elements and all executable elements in F101 that have a subtype of OM or SD and that have a 2- to 4-character version name starting with the characters V1 are deleted.

```
@DELETE,CUARS F103.MFDUTIL(-1),.ADD/MFDUTIL(-1)
```

Undelete the second-highest relative sequence number for each absolute, relocatable, and symbolic element/version/type set with an element name of either MFDUTIL or ADD/MFDUTIL in program file F103. If any relocatable elements are undeleted, the relocatable entry point table is removed. If any symbolic procedure elements are undeleted, the associated procedure table entries are also undeleted.

```
@DELETE,U F104.(185),.(181)
```

Undelete the elements with sequence numbers 185 and 181 in program file F104. If a relocatable element is undeleted, the relocatable entry point table is removed. If a symbolic procedure element is undeleted, the associated procedure table entries are also undeleted.

```
@DELETE,SY F106./V1**,OM/SD
```

Because you have not specified the A option with the Y option, the system displays an error message stating that you must specify the A option with the Y option.

```
@DELETE,AY F111./V1**
```

This statement is in error because you did not specify one or more of the subtypes ABS, BOM, OM, SD, and ZM when you specified the AY option combination. See Appendix A for the format of the message.

7.18.3. Undeleting Elements

One or more elements can be undeleted (changed from deleted to current) in a program file. The U option must be specified to undelete elements. Undeletion of an element can cause the deletion of an element with the same element name, version name, and element type.

When a symbolic procedure element is undeleted, all procedure table entries associated with the element are undeleted. When a symbolic procedure element is deleted, all procedure table entries associated with the element are deleted. When a relocatable element is undeleted or deleted, the relocatable entry point table is removed. When an object module is undeleted, the object module entry point table is removed.

If a program file initially has no deleted elements, an element @DELETE can be followed by an @DELETE,U with the same options and specifications to restore the program file elements and procedure table entries to their original state. Note that an @PREP (see Section 7.46) might be needed to restore the relocatable entry point table if relocatable elements were undeleted or deleted.

Once the @PACK command (see Section 7.43) is used, the deleted elements, their associated element text, and their associated procedure table entries, if any, are permanently removed from the program file and can no longer be undeleted.

Element/Version/Type Sets

All elements with the same element name, version name, and element type make up an element/version/type set. At most, one of the elements in the set can be undeleted. The rest of the elements in the set are deleted. @DELETE,U allows any deleted member of the element/version/type set to be undeleted. If the set contains an undeleted element, that element is deleted before another member of the set is undeleted.

The members of the element/version/type set can be uniquely identified by either their absolute sequence number or their relative sequence number. This dual identification is unique to element undeletion. The following illustrates the use of @PRT,TL (see Section 7.49.3) to display the complete element/version/type set for element name UTILABS, version name 3R2A, and element type absolute (A option).

```
@PRT,TLA WORKFILE.UTILABS/3R2A
STD#QUAL*WORKFILE(1) PF, 2153/5000 elements, 2765/9999 tracks
ELEMENT TABLE
D NAME VERSION TYPE DATE TIME SEQUENCE PREAMBLE TEXT CYCLE WORD FLAG SECTOR
* UTILABS 3R2A ABSOLUTE 2014-01-18 10:43:44 1712 SIZE 1340 MAX CUR NUM BITS LOCATION
* UTILABS 3R2A ABSOLUTE 2014-02-03 14:09:49 1733 1344 QTR,CLR 153550
* UTILABS 3R2A ABSOLUTE 2014-02-08 09:31:45 1740 1344 QTR,CLR 155600
* UTILABS 3R2A ABSOLUTE 2014-02-16 13:59:24 1748 1347 QTR,CLR 156990
UTILABS 3R2A ABSOLUTE 2014-05-05 12:24:52 1917 1351 QTR,CLR 159119
NEXT AVAILABLE LOCATION- 173687 176916
*File contains 97 deleted elements and 1423 tracks of deleted text
*PLEASE PACK FILE
```

There are five members in this element/version/type set. The first four members are deleted ("*" in D column) and the fifth member is undeleted. The members can be identified by their absolute sequence numbers (SEQ #) 1712, 1733, 1740, 1748, and 1917. The members can also be identified by their relative sequence numbers -4, -3, -2, -1, and 0, respectively. The element with the highest absolute sequence number is always relative sequence number 0. The element with the next highest absolute sequence number is relative sequence number -1, and so on.

Element Selection Methods

There are two methods that can be used to select elements for undeletion.

- An element can be selected by its relative sequence number in an element/version/type set. The A, O, R, or S option selects the element type, element name and version name specified, and the relative sequence number is specified in the element cycle subfield. If the element cycle subfield is not specified, then relative sequence number 0 is assumed. The V and Y options can also be specified.

- An element can be selected by its absolute sequence number in a program file. Element type (A, O, R, and S options), element name, and version name cannot be specified, and the absolute sequence number is specified in the element cycle subfield. The V and Y options also cannot be specified.

Examples

The following examples use the element/version/type set shown with the preceding @PRT,TL.

```
@DELETE,UA WORKFILE.UTILABS/3R2A(-2)
```

This example uses method 1 and specifies relative sequence number -2 of the UTILABS/3R2A absolute element set. It would delete element number 1917 and undelete element number 1740.

```
@DELETE,UA WORKFILE.UTILABS/3R2A
```

This example uses method 1 and specifies relative sequence number 0 (assumed) of the UTILABS/3R2A absolute element set. Following the previous example, it would delete element number 1740 and undelete element number 1917.

```
@DELETE,U WORKFILE.(1733)
```

This example uses method 2 and specifies absolute sequence number 1733. Following the previous example, it would delete element number 1917 and undelete element number 1733.

Undelete Specifications and Operation

The following additional information applies to element selection method 1.

When the V option is not specified, any combination of the element type options A, O, R, and S can be specified, but at least one must be specified. If the A option is specified, the Y option can be included to select only the executable element subtypes specified in the *subtype-list* field. The element name must be specified. If the version name is not specified it is assumed to be spaces. If a version name is specified, it cannot include asterisk (*) mask characters. If the element cycle is not specified, the relative sequence number is assumed to be 0. If the element cycle is specified, it identifies the relative sequence number and must be in the range of -9999 to 0. The same relative sequence number is used for each element/version type set selected by the A, O, R, and S options. For each element type option specified for which an element could not be found and undeleted, the following message is output:

```
q*f(c).e/v type NOT FOUND
```

When the V option is specified, any combination of the element type options A, O, R, and S can be specified. If the A option is specified, the Y option can be included to select only the executable element subtypes specified in the *subtype-list* field. If you do not specify any element type options, then all element types are selected. If the element name is not specified, then elements are selected without regard to element name. If the version name is not specified, it is assumed to be spaces. If the version name is specified, it can include asterisk (*) mask characters. If the element cycle is not specified, the relative sequence number is assumed to be 0. If element cycle is specified, it identifies the relative sequence number and must be in the range of -9999 to 0. The same relative sequence number is used for each element/version type set selected. See Section 7.14.3 for a detailed description of the V option and version name mask characters.

The A and Y options select only the executable element subtypes specified in the *subtype-list* field. Note that it is possible for members of an executable element/version/type set to have different subtypes. If this is the case, then the *subtype-list* must include all of the subtypes in the set for a member to be selected for undeletion.

The following additional information applies to element selection method 2.

No options other than U can be specified. Element name and version name cannot be specified. Element cycle must be specified. It identifies the absolute sequence number and must be in the range of 1 to 26,146. If the specified sequence number does not exist in the file, the following message is output:

```
*ERROR PRA002663: The sequence number specified does not exist in the program file.  
The largest sequence number in the file is n.
```

Since method 1 requires at least one of the A, O, R, S, V, and Y options and since method 2 does not allow any of these options, methods 1 and 2 cannot be specified on the same @DELETE,U command.

For all element selection methods, one of the following messages summarizes the results of the @DELETE,U for this *name-n*:

```
[a SYM] [b REL] [c ABS] [d OMN]  
NOTHING UNDELETED
```

In the first message, *a*, *b*, *c*, and *d* are the number of elements of each type that were undeleted. If no elements were undeleted for an element type, then the corresponding part of the message does not appear. If no elements of any element type were undeleted, the second message is output.

7.19.@ENABLE—Enable Disabled Files

The Enable command removes the disable flag for cataloged files.

All parameters in the @ENABLE statement are optional.

Format

```
@ENABLE[,options] [filename-1,filename-2,...,filename-n]
```

options

C

Used to direct FURPUR not to exit through ERR\$ if an error is encountered.

Use of the C option on any FURPUR command will prevent execution of ER ERR\$ on error exit if FURPUR is running in breakpoint or batch. The C option should be used with caution because a bad result of an erroring FURPUR command could be input to a process later on in the runstream.

The C option is valid for all FURPUR commands.

filename-n

Specifies the cataloged files to be enabled.

The A, O, R, S, and Y options are not appropriate for this statement.

If the file you specified is not disabled, a message to this effect is printed on the listing (normal exit is taken). If there are any keys, you must specify them.

7.20.@END—End of Input for @DATA and @ELT

The @END statement terminates the @DATA and @ELT processors. Refer to the *ClearPath OS 2200 ELT Processor Operations Reference Manual* (8206 2365), UP 13402 for more information.

If encountered in a runstream by anything other than the @DATA or @ELT processors, this statement is ignored. If a READ\$-type request encounters the statement, it is treated in the same manner as any other control statement. In that situation an end-of-file status is given.

7.21.@@END—Terminate Special Modes

The @@END statement terminates special modes. After the special mode is terminated, the terminal is returned to normal input mode.

Format

@@END [*mode*]

mode indicates which special mode to terminate. You can process in more than one mode at a time. If you enter @@END without specifying a mode, all the modes in which you are processing terminate.

The special modes are

CQUE

Input queue in main storage

INQ

Input queue on mass storage

FUL

Full screen mode

CONS

Console mode

ESC

Escape mode

PTI

Paper tape input

CDI

DCT 1000 card read input mode

HOLD/PR/TM

Hold conditions as specified

Note: To initiate a special mode, enter the mode you want preceded by @@.
For example, @@FUL initiates full screen mode.

7.22.@ENDF—Terminate the @FILE Card File

The @ENDF statement marks the end of the images for a file created by the @FILE statement.

Format

@ENDF

The @ENDF statement cannot have a label and cannot be continued.

The @FILE and @ENDF statements can be disabled on your system because they do not have proper security protection. If you use either statement in your runstream when they are disabled, the system removes your run.

These are the security problems:

- Any cataloged file can be deleted by the use of @FILE,AD or @FILE,KD.
- Read/write key validation is bypassed with @FILE.
- On systems with Security Level 1 or higher, the user-ID is not validated for @FILE, and the ACR validation is bypassed. Also, a private file created with @FILE is given the user-ID of EXEC8.

7.23.@ENDX—Terminate CLIST Mode

The @ENDX statement gives a program or processor an end-of-file return and terminates CLIST mode.

It marks the end of the images for a file created by the @FILE statement.

Format

```
@ENDX
```

If you enter @ENDX when your run stream is not in CLIST mode, the Exec returns the message:

```
@ENDX IGNORED - NOT IN CLIST MODE
```

Refer to the *Exec System Software Executive Requests Programming Reference Manual* (7830 7899) for more information about CLIST mode.

7.24.@EOF—Indicate the End of Input

The @EOF statement indicates the end of input from the runstream to the program being executed, which can be a compiler, user application, or any OS 2200 system processor.

Format

```
@EOF [s]
```

Note: *Label is not allowed on this control statement.*

s is an optional, 1-character, user-defined sentinel that, if specified, must be in column 6 of the @EOF statement.

When an @EOF statement is detected on a read request (READ\$, AREAD\$, READA\$, AREADA\$, ATREAD\$, TREAD\$ or SYMB\$), any user-defined sentinel is returned either in the SYMB\$ packet or for non-SYMB\$ requests, right-justified in the activity's A0 register. For non-SYMB\$ requests, control is transferred to the end-of-file return address specified in the packet. Once an @EOF statement is encountered, it is possible to continue reading from the runstream. Attempting to read a control statement other than @EOF inhibits further reading by the program and an abnormal return to the end-of-file return address is given. In this way, the @EOF statement can be used to divide datasets.

Examples:

```
@ACOB, IS FILEZ.SYM, FILEZ.REL
program lines
program lines
program lines
.
.
.
@EOF
```

The @EOF indicates the end of input to the ASCII COBOL compiler and causes the compiler to commence generation of the output relocatable element.

```
@XQT      PROGX
..
@XQT      PROGX
..
data for part 1
..
@EOF A
..
data for part 2
..
@XQT      PROGY
```

All data between the two @XQT statements are to be read by PROGX. The @EOF control statement serves as a separator between the two logical datasets.

7.25.@ERS—Empty (Erase) a File

The @ERS statement releases mass storage from cataloged or temporary, sector-addressable or word-addressable files. The @ERS statement exclusively assigns the files to be processed. Options are available to zero-fill mass storage before it is released and to zero-fill mass storage that is retained.

If the first track of the file is retained and if the first track was previously written, the first 28 words of the file are zeroed. If this is a sector-addressable file, this makes the file available as an initially empty program file (PF). After an @ERS, the @CHG statement T and E options (see Section 7.6) can be used to create an empty large program file (LPF) or an empty large-element program file (LEPF), respectively.

All @ERS parameters are optional.

Format

```
@ERS[,options] [filename-1,filename-2, . . . ,filename-n]
```

options

Options for releasing mass storage

None

For sector-addressable files, release all mass storage past the initial reserve. If the file does not have an initial reserve, release all mass storage. For word-addressable files, release all mass storage, including the initial reserve.

C

Used to direct FURPUR not to exit through ERR\$ if an error is encountered.

Use of the C option on any FURPUR command will prevent execution of ER ERR\$ on error exit if FURPUR is running in breakpoint or batch. The C option should be used with caution because a bad result of an erroring FURPUR command could be input to a process later on in the runstream.

The C option is valid for all FURPUR commands.

I

Release all mass storage, including the initial reserve.

N

Do not release any mass storage.

Options for zeroing mass storage

None

Zero the first sector (28 words) of the file, if the first track (1,792 words) of mass storage is retained and if the first track was previously written.

G

Zero mass storage that is retained.

Z

Zero mass storage before it is released.

filename-i

Specifies the name of the file to process.

If the N option is specified, it overrides both the I and Z options. The following informational message is displayed if the I and/or Z options are included with the N option:

```
N OPTION SPECIFIED - [I][Z] OPTION[S] IGNORED
```

If the I option is specified, it overrides the G option. The following informational message is displayed if the G option is included with the I option:

```
I OPTION SPECIFIED - G OPTION IGNORED
```

When the G option is specified for sector-addressable files, retained mass storage is zeroed from the beginning of the first track to the lesser of the highest track written and the highest track retained. When the G option is specified for word-addressable files, retained mass storage is zeroed from the beginning of the first track to the greater of the highest track written and the initial reserve. Note that the G option can actually cause additional mass storage to be allocated for files that contain unallocated areas (holes).

If the Z option is specified and if file is XPC cached, it must be changed to not cached with an @CHG,F *filename-i*.,DNC (see Section 7.6) before the @ERS. This assures that the mass storage granules are actually zeroed before they are returned to the available pool.

If no *filename-i* are specified, TPF\$ is processed.

7.26.@@ESC—Pass Unaltered Input to a Program or Unaltered Output to a Terminal

The @@ESC statement lets input to a program or output to a terminal pass in the format in which it was entered.

Format

```
@@ESC [function]
```

function can be one of the following:

I

Escape mode for input. The cursor positions are not stripped from the input data while in escape mode.

O

Escape mode for output.

Example

```
@@ESC I
```

7.27.@FILE—Create a Mass Storage Card File

The @FILE statement creates a mass storage file at input time, while the images are being read, and assigns it to the run.

Creating a file with the @FILE statement, rather than by means of the ELT or DATA processors, reduces overhead since it is accomplished at input time rather than at execution time. The data is handled only once when creating the file, rather than being

read again from auxiliary storage with the runstream and then transferred to the file by the DATA or ELT processors at execution time.

The @FILE statement is valid only within a batch run and cannot have been initiated with the @START statement.

Note: *The @FILE statement can be disabled on your system because it does not have proper security protection. See your security officer for information about the availability of this statement.*

Format

```
@FILE[,options] filename[,type/reserve/granule/maximum/  
placement,pack-id-1/.../packid-n,,,ACR-name]
```

The format is identical to the @ASG statement for mass storage devices (except the label field is not allowed). The options are the same as the @ASG statement. If you do not specify an option, the default is U. Use the T option only for removable disk files. Use the X option for previously cataloged mass storage files.

If you do not supply a qualifier on the file name, Q\$Q\$Q\$ is assumed. For private files, the project-ID is picked up from the @RUN statement.

Pack-IDs are used only if the type field specifies removable disk. Sector-formatted files cannot be temporary files, except for removable disks.

An @FILE statement must not be more than 80 columns in length. You cannot continue an @FILE statement to another line.

7.27.1. Security Considerations

The @FILE and @ENDF statements can be disabled on your system because they do not have proper security protection. If you use either statement in your runstream when they are disabled, the system removes your run. These are the security problems:

- Any cataloged file can be deleted by the use of @FILE,AD or @FILE,KD.
- Read/write key validation is bypassed with @FILE.
- On systems with Security Level 1 or higher, the user-ID is not validated for @FILE, and the ACR validation is bypassed. A private file created with @FILE is given the user-ID of EXEC8.

7.27.2. Terminating @FILE

To terminate the @FILE statement, use one of the following statements:

- @ENDF
- @FIN
- Another @FILE statement

Data images and all control statements, except @COL and its accompanying end sentinel, are placed into the created file. The @COL statement and sentinel are processed immediately, and the file is marked when the mode switch is made.

Do not nest @FILE statements for mass storage files. It causes @FILE mode to terminate with an error message.

Example

```
@FILE,T FILE1.
```

7.27.3. Creating a Tape File from Card Images

The @FILE statement creates a tape file at input time, while the images are being read, and assigns it to the run.

Do not nest @FILE statements unless they use the same file name and are for tape files.

Note: The @FILE statement can be disabled on your system because it does not have proper security protection. See your security officer for information about the availability of this statement.

Format

```
@FILE,[options] filename,[type/units/log/noise/processor/tape/  
format/data-converter/block-numbering/data-compression/  
buffered-write,reel-1/reel-2/.../reel-n,expiration-period,  
ring-indicator,ACR-name]
```

The format is identical to the @ASG statement for tape files (label field is not allowed).

The options are the same as the @ASG statement. If you do not specify an option, the default is U. Use the T option for tape files.

An @FILE statement must not be more than 80 columns in length. You cannot continue an @FILE statement to another line.

The @FILE and @ENDF statements can be disabled on your system because they do not have proper security protection. If you use either statement in your runstream when they are disabled, the system removes your run. These are the security problems:

- Any cataloged file can be deleted by the use of @FILE,AD or @FILE,KD.
- Read/write key validation is bypassed with @FILE.
- On systems with Security Level 1 or higher, the user-ID is not validated for @FILE, and the ACR validation is bypassed. Also, a private file created with @FILE is given the user-ID of EXEC8.

Example

```
@FILE,TJ FILE2,U9V/////109A
```

7.28.@FIN—Identify the End of a Run

The @FIN statement identifies the end of a run. The @FIN statement must be the last statement in all runs, whether demand or batch.

The @FIN statement must be coded exactly as shown (entered in the first four columns):

```
@FIN
```

The @FIN statement is never treated as data by any processors, including the DATA and ELT processors. If the Exec reads an @FIN statement in an @ADD stream, it does not terminate the run, and it displays the following message:

```
@FIN IN ADD FILE - IGNORED
```

After you enter the @FIN statement, the run terminates. You can start another run by pressing the XMIT key. If you enter anything other than @@TERM or a \$\$ statement (except for @@SEND), you see this message:

```
Enter userid/password and clearance level:
```

If you want to start another run, respond to this and the subsequent prompts.

If you do not want to start another run, sign off from the system. Refer to @@TERM.

See Section 5.4 for information on executing a run termination task.

See Appendix E for a description of the tail sheet.

7.29.@FIND—Position Tape Within Element Files

Locates an element on a magnetic tape file (file must be in element file format) and positions the tape immediately preceding the element's label block.

All parameters in the @FIND statement are required except *options*.

Format

@FIND[,*options*] *element-name*

options

Can be one of the A, C, O, R, or S options.

Use of the C option on any FURPUR command will prevent execution of ER ERR\$ on error exit if FURPUR is running in breakpoint or batch. The C option should be used with caution because a bad result of an erroring FURPUR command could be input to a process later on in the runstream.

element-name

Specifies the file and the element to be located

The search is made forward until either the element is found or a hardware EOF mark is encountered. When the EOF mark is encountered, the tape is backspaced to the previous EOF mark (or load point, whichever is encountered first) and the search is repeated. If no find is made, the error exit is taken when the EOF mark is encountered.

Normally, the @FIND statement is used just before an @COPIN statement (see Section 7.12) requesting that the element just located (or all elements up to the EOF mark) be inserted into a program file located on sector-formatted mass storage. It is also required before calling a language or systems processor that uses the source input/output routine (SIR\$) to read the element from tape (specified as the SI field).

Note: *Be careful when you do an @FIND operation on other than the first reel of a multireel file. If an EOF is encountered before locating the desired element, the reel is backspaced only to the load point, not to the EOF, which can be located on a previous reel.*

7.30.@FREE and @@FREE—Release Files and Peripheral Devices

The @FREE statement releases files, reels, internal file names, and the exclusive use of files from your run.

When used for mass storage files, the @FREE statement can cause a file to be deleted, cataloged, or just released from your run. What action is taken depends upon what option was used when the file was assigned and what option is used on the @FREE statement.

After you release a file with an @FREE statement, you can no longer reference the file without reassigning it with another @ASG statement. You must free tape drives as soon as possible so that other people can use them.

When you release a temporary file with @FREE, the file is deleted. Temporary files are automatically deleted by run termination. All files are automatically released at run termination. Therefore, it is not necessary to do an @FREE before entering @FIN.

Notes:

- *If a shared mass storage file is application-private, its application-private state is not cleared when the file is freed. Refer to the MHFS Planning, Installation, and Operations Overview (7830 7469).*
- *An @FREE of an alternate print file must not be done before an @BRKPT. An @FREE does not close the print alternate file, and a successive @BRKPT to the free file results in an I/O error.*

Format

```
@FREE[,options] filename
```

options

These are the options:

A

Releases only the specified internal file name given to the file with an @USE statement.

B

Releases only the specified internal file name associated with the file, unless it is the only internal name for the file, in which case the entire file is released from the run.

D

Deletes a cataloged file, in addition to releasing it from the run. If it is a temporary file, it is just released from the run. If the file was assigned (@ASG) with the C or U option, it is not cataloged as it normally would be.

For systems with Fundamental Security, to delete cataloged files assigned with the C or U option, you must have included the correct read and write keys (if any) when you assigned the file to the run.

If the file is owned, you must have delete access to the file in order to delete it.

If your site has Fundamental Security, you must include any read or write keys that are attached to the file in order to delete the file.

If your site has Security Level 1, 2, or 3 and the dynamic configuration parameter `owned_file_read_write_keys` is TRUE, then you must include read or write keys to delete the file.

I

Inhibits the cataloging of a file that was assigned (@ASG) with a C or U option. You can accomplish the same thing with the D option.

R

Releases the specified file from the run, but retains all internal names associated with the file. If the file was assigned (@ASG) with the C or U option, it is also cataloged.

S

Releases the file from the run, but keeps the physical tape unit assigned to the run.

When an unlabeled tape is being mounted, the following prompt is displayed on the system console to ask for the tape's reel number:

```
n ANSWER WITH REEL NO ON unit E,REEL
```

When this tape is freed (@FREE) and the next one assigned, the same scenario is repeated.

However, a special case occurs when an autoloader is installed and the S option is used on the @FREE statement: For the first tape, the operator is prompted for the reel number as described above. If this tape is released using @FREE,S, the next unlabeled tape in the autoloader is not prompted for the reel number.

This is a useful feature that can significantly help improve efficiency, especially when a large number of unlabeled tapes are to be processed.

You can reassign the unit to your run by using a normal @ASG statement. Just be sure the type you specified corresponds to the unit reserved for your run. Since a pooled unit can be used only by your run, your request for a tape file can be satisfied immediately.

X

Releases exclusive use of the file when the file was assigned with the X option. The file is not released from the run. This lets other runs use the file.

If you do not specify the A or X option, the file or device is released from your run. If the file was assigned (@ASG) with the C or U option, it is also cataloged. If it is a temporary file, the file no longer exists.

When you are releasing tape files from your run, the S option lets you release just the file and not the physical unit. The physical unit is reserved exclusively for your run. Other users cannot use the unit until your run terminates, or until you reassign the unit, and then release it completely.

The Exec issues I/O to write two tape marks. If tape-mark buffering is enabled, the Exec then issues an FSAFE\$ request. The TMB feature increases the importance of checking for an FSAFE\$ failure, because the feature introduces the potential of leaving a tape in an incomplete state, with missing file marks or data. Currently no status is returned to the user for any closing tape operation, including @FREE.

Table 7-3 shows options allowed for releasing different types of files and devices. An asterisk (*) indicates the option is allowed but has no effect.

Table 7-3. FREE Options for Releasing Files and Devices

Item Being Released	Valid Options
Mass storage files to be cataloged	Blank, A, B, D, I, R, and X*
Already cataloged mass storage files	Blank, A, B, D, R, and X
Temporary mass storage files	Blank, A, B, D, R, and X*
Tape files to be cataloged	Blank, A, B, D, I, R, and S
Already cataloged tape files	Blank, A, B, D, R, and S
Temporary tape files	Blank, A, B, D, R, and S
Tape devices	Blank, A, B, D, R, and X*
Internal file name	Blank, A, B, R, S, and X*

If you assigned the cataloged file with the D or K option on the @ASG statement, any @FREE statement that releases the file from the run deletes the file.

filename

Identifies the file or device to be released from your run. This can be either the name given to the file when it was created or an internal file name given to the file with an @USE statement.

Example

```
@FREE,D FREMONT1.
```

This statement frees file FREMONT1 from the run and also deletes the file.

7.31.@@FUL—Enter Input a Full Screen at a Time

The @@FUL transparent statement initiates full-screen mode, which lets you enter data and statements a full screen at a time. The screens of information are stored on mass storage until you terminate full-screen mode with an @@END FUL statement (See Section 7.21).

You must provide your own SOE character for the first line of your input; the other lines do not require an SOE character. You must also provide your own SOE character on the @@END FUL statement.

It is a good idea to leave the first line on the screen blank, in case the system gives you a timeout warning. Otherwise, the first line of your input is lost.

If the input lines entered produce any output, it is held until full-screen mode is terminated.

Format

@@FUL

Example

```
@@FUL (xmit)*FULL SCREEN INPUT MODE*  
line of input (cr)  
line of input (cr)  
line of input (cr)  
line of input (cr)  
.  
.  
.(xmit)  
@@END FUL (xmit)
```

line of input

Represents the lines of data and statements you want to enter. Each line on the screen is considered a separate input image.

(cr)

Indicates that you must press the carriage return key after each line of input.

(xmit)

Indicates that you must press the XMIT key after each screen of input, before the @@END FUL statement. If you do not press the XMIT key before you enter the @@END FUL statement, you lose your last screen of data.

Note: The carriage return key only moves the cursor to the next line, but the XMIT key transmits the data on the screen to the system.

7.32.@HDG and @@HDG—Provide a Heading for Printer Output

The @HDG statement lets you specify a heading to be printed on every page of the printout, as well as page numbers and the date when the file was created. Use @HDG with the @BRKPT statement when you are directing output into a user-defined file.

Any number of @HDG statements can appear in the runstream.

Note: *If your site has Security Level 2 or higher, printouts can be labeled with the clearance level and compartment set associated with your print file. See Section 7.3 for more information.*

Format

```
@HDG[,options heading.printcontrol]
```

options

N

Suppresses printing of heading, date, and page number.

P

Starts page numbering with page 1 instead of current page number.

X

Suppresses printing of date and page number.

heading

Contains the heading that is to appear within the top margin of each page (two print lines before the main body of text printed on the page). This heading can be any length up to a maximum of 96 characters (including embedded blanks); any additional characters are ignored.

The heading starts with the second character after the options field. Leading blanks are permitted in the heading. The end of the heading text is signified by the end of the statement, a period, or the 96-character limit, whichever occurs first. The text can be followed by print control fields.

If the top margin is less than two lines (including the heading line), the heading is not printed. Unless suppressed (N or X options), the date and page number are printed to the right of the heading. Page numbering begins with the page count current to the print file unless the P option is specified, in which case page numbering begins with page 1.

The continuation character (;) and the period character are not permitted as part of the heading text.

printcontrol

This field consists of a period immediately followed by a print control function. This field can be repeated up to a maximum of 96 characters including the heading.

Some types of printers do not support all print control functions.

These are the print control function formats:

A

Allows error recovery. The I and A functions control the automatic recovery of errors for individual output files. These functions can be used in pairs more than once in an individual file to control automatic recovery over specific areas of output. If autorecovery is inhibited, a console message asks the onsite operator whether recovery must be attempted on the file.

B,*lpi ctid,eltname*

Specifies print cartridge (band) *ctid* or change lines-per-inch to *lpi*, or both. Nonstandard cartridge-ID code buffer can be found in *eltname* in the file SYS\$*PRINTER\$. Except for *lpi*, this is for 0776 printers only.

lpi

Lines-per-inch; *lpi* must be 8, 6, or * (an asterisk indicates device default) as desired; no leading spaces between the comma and *lpi* are allowed. *lpi* can be the only field in this print control function; however, a change in *lpi* implicitly changes the page size. Therefore, unless special forms are desired, use an M function on the same statement as the B function to keep the page size the same.

ctid

Cartridge identification code. This is the hexadecimal cartridge code that appears on the cartridge case. Print data following this image is printed after the operator has replaced the print cartridge currently loaded with the one specified here (if necessary). This field is ignored at print time if the printer is not a type 0776 printer.

eltname

If a special cartridge to be used is not known to the Exec, *eltname* is the name of the element in the SYS\$*PRINTER file, in which this load code buffer can be found. Coding *eltname* with no *ctid* is an error. *ctid* can be the only field supplied if the cartridge is known to the Exec and *lpi* is not needed.

H,options,page,text

Provides a page heading for the printout, where:

options

- N Do not print heading.
- X Suppress printing page number and date.

page

Is the page number to be used for the first page on which this heading appears. If no page number is given and there is no X or N option, the page numbering is not affected; it continues in its normal sequence.

Note: *The Exec symbiont handlers do not print page numbers greater than 262,144.*

text

Is the text of the heading you want to appear within the top margin of each page (up to 96 characters long). This text cannot contain periods. The heading is printed at the top of each succeeding page, where the page size is determined by the margin function (M), described later. If the top margin is less than two lines (including the heading line), the heading is not printed.

I

Inhibits error recovery.

L,nn

Spaces the printer to logical line number *nn*-1. The first logical line number is 0, (1 = 0); that is, both L,0 and L,1 position the printer to position 1 before the first printable line. This statement causes the printer to be logically positioned so that a print statement with a space count of 1 is printed on printable line *nn*.

M,length,top,bottom,lpi

Provides margin setting information for readjusting page length, top and bottom margins, and the number of lines per inch. Hardware limitations require that the page length for a 0776 printer be no more than 192 lines. Other printers can also have limitations on page length.

length

Is the number of lines per page. This field can be a decimal value (maximum of three digits), an asterisk (*), blank, or X.

An asterisk specifies the system standard value; a blank specifies no change from the value currently in effect. An X specifies absolute line spacing; it changes the method of calculating spacing, and all subsequent fields (*top*, *bottom*, and *lpi*) are invalid. After an M,X function, spacing is not truncated by an end-of-page condition, and top and bottom margins are zero. Any *lpi* change must have been set up by a previous print control function. Any M or B statement causes absolute line spacing mode to be exited.

top

Is the number of blank lines used for the top margin (must be less than 63). This field can be a decimal value (maximum of two digits), an asterisk (*), or blank.

An asterisk specifies the system standard value; a blank specifies no change from the value currently in effect.

bottom

Is the number of blank lines used for the bottom margin (must be less than 63). This field can be a decimal value (maximum of two digits), an asterisk (*), or blank.

An asterisk specifies the system standard value; a blank specifies no change from the value currently in effect.

lpi

Is the number of lines per inch. This field can be a decimal value, an asterisk (*), or blank.

A decimal value can be only one digit (either 6 or 8). An asterisk specifies the system standard value; a blank specifies no change from the value currently in effect.

R

Inserts a logical break into print output files; used by the print symbionts when skipping forward through a print file. When the symbiont encounters the R function, it stops skipping. It then asks the operator to indicate whether a reprint is desired.

S,text

Requests special forms for processing the print file. The text can be up to 48 characters long. When this control function is encountered for onsite printers, the text is displayed on the operator's console. For remote batch devices, the text is displayed on a remote printer. The text cannot contain any periods.

U

Ignores any following print control functions. This option is reserved for users. It is ignored by the device handlers and not scanned for syntax.

W,line width

Changes the maximum print width allowed. The standard line width value is 22, which is equivalent to 132 characters. The line width value is one-sixth of the number of characters you want the line width changed to. (This is necessary to use the full width of onsite printers that allow more than 132 characters per line.) Some terminals also require other statements.

Notes:

- *The H function does not take effect until spacing statements have crossed a page boundary.*
- *The S function takes effect immediately and causes a page eject.*
- *The M function causes top of page action and takes effect immediately.*
- *The B,lpf function causes top of page action and takes effect immediately, but the B ctid function does not.*
- *After a W function, the maximum allowable character count for an ASCII print request through SYMB\$ is the smallest multiple of 4 that is greater than or equal to six times the value specified on the W function.*

The @HDG statement does not cause the printer to space to the top of the next page, unless a print control that causes a page eject is used.

The maximum total length from the end of the options field to the last character is 361 characters. A period followed by a space ends the @HDG statement.

Terminating @HDG

The following statements terminate any current @HDG statement in effect:

```
@BRKPT PRINT$
```

or

```
@BRKPT PRINT$/part-name
```

Examples

```
@HDG,P PYREP COMPANY - ANNUAL REPORT
```

Heading PYREP COMPANY - ANNUAL REPORT is printed at the top of each page, along with the date and the page number. Page numbering begins with page number 1 (P option).

```
@HDG,X PAYROLL REPORT - RUN-1
```

The specified heading appears at the top of each printed page, but numbers and date are suppressed (X option).

@HDG, N

The N option suppresses further printing of the heading, date, and page numbers.

@HDG, M, 66, 3, 3

The M option gives a new margin setting.

7.33.@@HOLD—Hold Terminal Messages and Print Output

The @@HOLD statement places a hold on terminal messages and print output.

Format

@@HOLD [*type/type*]

type can be the following values:

TM

Holds all terminal messages. Terminal messages are generated by @@TM, TM keyins, and TB keyins. Immediate TB keyins (TB /I) are displayed.

PR

Holds print output.

The system standard (default) is both TM and PR.

Examples

@@HOLD TM

Holds all terminal messages.

@@HOLD

Holds all terminal messages and print output. Same as @@HOLD TM/PR.

7.34.@@INQ—Queue Input in a Mass Storage Buffer

The @@INQ statement lets you enter lines of statements and data into a buffer on mass storage. This statement is similar to the @@CQUE statement. The major difference is that statements entered in @@CQUE mode are processed immediately. Statements entered in @@INQ mode are not processed until you enter the @@END INQ or @@END statement.

Format

```
@@INQ soe-parameter



.
.
.
@@END INQ or @@END
```

Valid values for the *soe-parameter* field are

N

No SOE characters are issued by the system when in this input queuing mode. The system resumes issuing appropriate SOE characters when the corresponding @@END control statement is received to return the terminal to interactive mode.

S

SOE characters are issued by the system when in this input queuing mode after each input image is transferred to a system mass storage buffer. The system continues to issue appropriate SOE characters when the corresponding @@END control statement is received allowing the terminal to return to interactive mode.

The *soe-parameter* field is a 1-character field. If more than 1 character or anything other than an N or S is specified, the statement is ignored and the following error is displayed at the terminal:

```
@@ERROR - SYNTAX
```

input-image represents the statements and data you want to enter into the buffer on mass storage.

You must provide your own SOE character once you enter INQ mode if the *N* is specified or no *soe-parameter* is specified.

If you use @@INQ mode outside of an active run, the input images are discarded until you begin a run with an @RUN statement.

If you enter @RUN statements while in @@INQ mode without a demand run active, the resulting runs are considered remote batch runs. If a demand run is already active, the @RUN statements in the INQ are ignored.

7.35.@@INS—Change the Insert Point on the Display

The @@INS statement lets you change the insert point on the screen display. The insert point is the line on the display where new lines first appear.

Format

```
@@INS [n]
```

n indicates the line on the display to be used as the insert point. The lines are counted down from the top.

For standard 24-line displays, the normal insert point is at line 23 and the @@INS statement with no fields gives the default of line 23.

Examples

```
@@INS 22
```

The insert point is the 22nd line down from the top. This saves any text on the 23rd line.

```
@@INS
```

This gives the default insert point at line 23. This is the same as @@INS 23.

7.36.@JUMP—Branch from Within the Runstream to Another Point

The @JUMP statement advances control to the specified labeled statement in the runstream.

Format

```
@JUMP label
```

label specifies the label or name attached to a subsequent statement to which control is passed. This field can be a decimal value (from 1 to 999999), indicating how many statements are to be skipped. Those statements that cannot have labels must not be included in the count. A numeric value of 0 as a field is not permitted.

The @JUMP statement must refer to a statement not yet processed.

Examples

```
@JUMP TAG
```

Control is advanced to the control statement containing label TAG.

```
@JUMP 4
```

After execution of the @JUMP statement, control is advanced to the fourth control statement. The fourth control statement can have a label field.

7.37.@LOG and @@LOG—Place Information in the System Log

The @LOG and @@LOG statements are used to insert user-specified information in the system log. The system log is a file maintained by the Exec as a historical record of the activity in the system.

Format

```
@LOG message
```

message is a user-specified message to be placed in the system log.

The message can be up to 132 characters long, including embedded blanks. The first nonblank character is considered the beginning of the message. The end of the message is the last character before the end of a line, before a comment (.), or within the 132-character limit. The message cannot begin with a comma (,) or a slash (/), and cannot include the continuation character (;). The blank-period-blank (.), which introduces a comment, also cannot be used.

When the Exec encounters the @LOG statement, it prefixes the message with your run-ID and the date and time, and enters it in the system log.

Examples

```
@LOG TRANSPORT PROBLEM NO. 12 . Revised 5-1
```

In this example, the message consists of 26 characters (the blank preceding the period is considered part of the message) and is terminated by blank-period-blank (.) before the comment. The following additional text is not transmitted to the log

```
Revised 5-1
```

7.38.@MARK—Mark an EOF on Tape

The @MARK statement writes an EOF mark on a tape. If the tape is unlabeled, FURPUR writes a second EOF mark and positions the tape between the two EOF marks.

All parameters except *filename-1* are optional.

Format

```
@MARK[,options] filename-1[,filename-2,...,filename-n]
```

options

C

Used to direct FURPUR not to exit through ERR\$ if an error is encountered.

Use of the C option on any FURPUR command will prevent execution of ER ERR\$ on error exit if FURPUR is running in breakpoint or batch. The C option should be used with caution because a bad result of an erroring FURPUR command could be input to a process later on in the runstream.

The C option is valid for all FURPUR commands.

E

Used for unlabeled tapes to improve write performance. Write a single EOF mark instead of writing two EOF marks and backspacing over one of them. Before finishing with the tape (for example @FREE, @MOVE,B or @REWIND), you must close the output tape with @CLOSE (see Section 7.8) or with another @MARK. The E option is ignored for labeled tapes, which always write a single EOF mark.

W

Used when the buffered-write mode BUFTAP is enabled on the tape to assure that all buffered data has successfully been written to tape. After the write of the EOF mark, a synchronize (F\$SAFE) I/O function is used, if necessary. Note that the synchronize increases the time to write this file, as if the buffered-write mode BUFFIL was enabled. To maximize performance, the W option must be used only periodically, to establish error recovery "checkpoints." The W option is ignored for unlabeled tapes when the E option is not specified. In these cases the tape is implicitly synchronized by the Exec or by the tape unit.

filename-i

Specifies the tape files on which to write the EOF marks.

@MARK is typically used under the following conditions:

- After output tape functions that do not write EOF marks, such as @COPOUT with any of the AORSVX options (see Section 7.13) and @COPY with neither the G nor the M options (see Section 7.14.1)
- To close an unlabeled output tape when the E option has been specified on @COPOUT or @COPY to improve the tape write performance. In this case an @MARK,E can be used to write a second EOF mark to close the unlabeled output tape.

Notes:

- If you use the @MARK statement on an unlabeled tape immediately after @COPOUT, @COPY, or @MARK have written an EOF mark, you are creating a logical end-of-file on the tape. Even though additional data can be written to the tape, the data cannot be accessible once the tape is rewound or released.
- If you use the @MARK statement on a labeled tape immediately after @COPOUT, @COPY, or @MARK have written an EOF mark, you are creating a null file on the tape. A null file is a file that has Exec tape labeling system label blocks and EOF marks but does not have any user data blocks. A null file does not usually serve a useful purpose on a labeled tape.

7.39.@MODE and @@MODE—Change the Mode Settings for a Tape File

The @MODE statement lets you change the hardware mode settings that were initially specified when a tape file was assigned. To change the mode settings, the file must be currently assigned to your run.

You must be aware of the following two terms that are most important in this subsection:

- The Half-Inch Cartridge (HIC) tape that can be used to record 18-track or 36-track formatted data.
- The Digital Linear Tape (DLT)
- Linear Tape Open (LTO): This device is released as different generations of the Ultrium technology. To denote the different generations, the generation number is appended to LTO. Therefore, a sixth generation LTO device is referred to as LTO6.

The only other way to change the mode settings for a file is to release the file from your run (@FREE) and then reassign it (@ASG).

Format

```
@MODE,options filename [,noise/processor/tape/format/  
data-converter/block-numbering/data-compression/  
buffered-write]
```

All fields on the @MODE statement are optional except *filename*.

options

The system does not default to a particular option if none is specified. The specified options are not placed in the master file directory because they apply only to the current run.

If you enter an invalid option, the statement is rejected. These are the options:

E

Indicates even parity. This option is only allowed for 7-track tapes.

H

This option is available on 9-track and U9840D tape subsystems. If the density option "H" is specified with the general assign mnemonic "T", then one of the devices supporting the highest density that allows "H" density option is assumed by the Exec. This option is not available for other tape subsystems.

With the "H" density option now allowed for two different tape types, such as 9-track and U9840D; and these tapes selectable through the general assign mnemonic "T", the Exec's selection sequence criteria remains basically the same except that either 9-track or U9840D would get assigned.

The selection sequence is dependent on what is configured, available, premounted, and what is in the CONV default list for tape assignment mnemonics and how it is ordered.

Note: *The potential for a run stream compatibility issue, for files assigned using the general assign mnemonic T option and the H density option, exists if the CONV default has both U9840D and U9H devices on the system.*

The following variable error message is returned if an invalid density option is used:

```
Assign mnemonic [asg_mnemonic] does not support density option [n].
```

L

This option is available on all 18-track half-inch cartridge, U9840, and UT10KA tape subsystems. If the density option "L" is specified with the general assign mnemonic "T", then one of the devices supporting the highest density that allows "L" density option is assumed by the Exec. This option is not available for other tape subsystems.

With the "L" density option now allowed for three different tape types, such as 18-track half-inch cartridge, U9840, and UT10KA; and these tapes selectable through the general assign mnemonic "T", the Exec's selection sequence criteria remains basically the same except that 18-track half-inch cartridge, U9840, or UT10KA, would get assigned. The selection sequence is dependent on what is configured, available, premounted, and what is in the CONV default list for tape assignment mnemonics and how it is ordered.

Note: *The potential for a run stream compatibility issue, for files assigned using the general assign mnemonic T option and the L density option, exists if the CONV default has both UT10KA and HICL devices on the system.*

The following variable error message is returned if an invalid density option is used:

```
Assign mnemonic [asg_mnemonic] does not support density option [n].
```


M

This option is a density option for the U9840C cartridge subsystems. This M option is also currently used for 36-track half-inch cartridge and CTS9940B tape subsystems. All other density options (for example, L and H) are not allowed for the U9840C subsystem.

The M option is available on all 36-track half-inch cartridge, U9940B, and U9840C tape subsystems. If the M option is specified with the general assign mnemonic T option, then one of the devices supporting the highest density that allows the M density option is assumed by the Exec. This option is not available for other tape subsystems.

With the M option now allows three different tape types (36-track half-inch cartridge, U9940B, and U9840C), and these tapes selectable through the general assign mnemonic T, the Exec's selection sequence criteria remains basically the same, except that 36-track half-inch cartridge, U9940B or U9840C, would get assigned.

The selection sequence is dependent on what is configured, available, premounted, and what is in the CONV default list for tape assignment mnemonics and how it is ordered.

Note: *The potential for a run stream compatability issue, for files assigned using the general assign mnemonic T option and the M option, exists if the CONV default has both U9840C and HICM devices on the system.*

The following error message is returned if an invalid density option is used:

```
E:207133 Assign mnemonic [asg_mnemonic] does not support density option [n].
```

O

Indicates odd parity (assumed for 9-track equipment). This option is allowed for 9-track and 18-track tapes.

S

Indicates density mode of 6250 bpi. This option is available only for UNISERVO 26, 28, 32, 34, and 36 subsystems.

V

Indicates density mode of 1600 bpi on 9-track tape units. This option is available on all 9-track tape units. Indicates a density of 85,937 bpi for DLT subsystems. If the density option is not specified for DLT subsystems, then the highest density supported by the DLT tape device is assumed. This option is not available for other tape subsystems.

If you specify more than one density, the @MODE statement is rejected.

Options specified on the @MODE statement must be compatible with the hardware currently assigned. An incompatible option causes the statement to be rejected. For example, specification of the H option when the tape drive is a UNISERVO 28 causes the entire @MODE statement to be rejected.

filename

Is the file on which you are changing the hardware mode settings.

noise

Is the noise constant (1 to 63) to be used instead of the one currently in use. For more information about the noise subfield, refer to the explanation of the @CAT statement.

processor/tape

Indicates the type of translation required, where processor defines the format of the data in the processor, and the tape subfield defines the format of the data on tape. For more information about the processor/tape subfield, refer to the explanation of the @CAT statement.

format

Specifies the data transfer format for the word-to-byte conversion. This applies to all tapes.

The format subfield is also used in determining the data compression mode for write operations for the 5073 tape subsystem.

Permissible format entries are as follows:

Q

Quarter word

8

8-bit packed (default on 9-track, 1/2-inch cartridge tape, and 1/4-inch cartridge tape)

If no value is specified in the format subfield when using a 5073 tape subsystem, 9-bit data compression is enabled.

data-converter

The allowable values are

DC

Three 8-bit bytes are converted from four 6-bit tape characters.

OFF

Turns off the data-converter. If the data-converter is not on, the request is accepted but no action occurs.

block-numbering

Determines whether block numbering is on. Block numbering is available only on 5061, HIC, DLT and LTO tape subsystems. For the HIC, DLT and LTO cartridge subsystems, you cannot turn on block numbering with this subfield because it is performed by the hardware and is transparent to you.

Specify one of the following values:

BLKON

Turns block numbering on (invalid for HIC, DLT and LTO tape subsystems).

BLKOFF

Indicates that blocks must not be numbered.

data-compression

Reduces the amount of physical tape required to record data by removing repetitive data before recording it (available for 5061, HIC, DLT and LTO subsystems only). Data-compression is generally performed on a file basis.

For the half-inch cartridge tape subsystem, these are the *data-compression* values:

CMPON

Enables 8-bit data compression when Q is specified on the *format* subfield; enables 9-bit data compression when either 8 or nothing is specified on the *format* subfield. Enables LTO-DC for an LTO cartridge tape devices.

CMPOFF

Disable data compression.

CMP8ON

Enables 8-bit mode of data compression.

CMP9ON

Enables 9-bit mode of data compression.

CMPEON

Enables enhanced data recording capability (EDRC) mode of data compression.

CMPLON

Valid for a HIS98 drive. Enables LZ1 data compression.

For DLT tape subsystems, these are the valid data-compression values and their meanings:

CMPLON

Enables LZ1 data-compression. Future Unisys tape products can use this compression value and not be compatible with DLT cartridge tapes.

CMPOFF

Disable data compression.

Notes:

- Although you can specify 8- or 9-bit data compression with CMP8ON and CMP9ON, you can allow the system to select the best compression ratio by specifying the format of the data.
- You can experience a loss in performance if you use the read backward function (the RB\$ function of the ER IOW\$) of the 8- or 9-bit data compression feature on a 5073 tape subsystem.
- Tapes created using 8- or 9-bit data compression cannot be used with Exec levels lower than 41R6/41R6M.
- When you are using EDRC data compression, user data block sizes are limited to 128K bytes or smaller (on 5073 devices).
- When you are reading a tape with EDRC compressed data, EDRC mode need not be enabled, as long as an EDRC-capable device has been assigned (for example, CMEOPT was used on the assignment).
- Tapes created using EDRC data compression cannot be used with Exec levels lower than SB4R7.
- Tape devices that support EDRC data compression do not support data translation (ASCII/EBCDIC) or 8- or 9-bit data compression.
- The value CMPEON can be used for 1/2-inch cartridge tape subsystems that support EDRC data compression.

buffered-write

Specifies whether buffered writes are to be used when a tape is mounted on a device controlled by a control unit that has buffering capability.

These are the allowable values:

BUFTAP

Enables buffered write mode for user data blocks and file marks for the entire tape.

BUFFIL

Enables buffering of write data including label blocks and tape marks for a file written to labeled tapes. The system ensures that writing of the data and file marks occurs at the close of a file.

BUFON

Enables buffered write mode for user data blocks.

BUFOFF

Disables any buffered write mode.

The @MODE statement is rejected if BUFTAP or BUFFIL is used and the device does not support tape mark buffering (TMB). Even though TMB is supported on the CTS5236, the BUFTAP value is not. The rejection message is

```
E:256233 unit is not compatible with request.
```

The following rules apply when using BUFTAP, BUFFIL, BUFOFF, and BUFON:

- When the user switches modes from BUFFIL to BUFON or BUFOFF, writing the buffer to tape is postponed until the next I/O to the tape subsystem. This also applies when switching modes from BUFTAP to other modes.
- If BUFTAP, BUFFIL, BUFOFF, or BUFON is not specified, and the configuration tag TAPBUF (dynamic name_tape_buffering) is zero, the system uses the default value BUFON.
- If BUFFIL is specified for an unlabeled tape, the data blocks for writes are buffered and one synchronization per file is done by a WEF\$ function, just as it is without the BUFFIL specification.
- When using TMB, it is the user's responsibility to retain the original data for possibly performing error recovery, if that is the user's requirement. The Exec provides no means for the user to recover this data. With the use of the BUFTAP option, it is possible for the user to have received good status on previous writes of thousands of files, when in fact none of the files has made it to tape, and if an error occurs while they are being written to tape, the remaining blocks in the tape subsystem's buffer can never get written to tape. The Block ID SAFE (BSAFE\$) I/O function provides a means for the user to determine what block has been actually written to tape by the tape subsystem.
- When using BUFTAP, an FSAFE\$ can be performed at any point if a user wants to ensure that buffered data/tape marks have been written to tape. It is highly recommended that an FSAFE\$ be performed and its status checked after all user data/tape marks have been written to the tape device, but before releasing the tape. This ensures that the user is notified of any errors in writing any remaining buffered data/tape marks to the tape. This also allows the user to know that an error on an @FREE, TSWAP\$, TINTL\$, or TLBL\$ was not the result of writing any user buffered data/tape marks to the tape.

Example 1

```
@MODE TAPE.,////////CMPEON
```

The tape, assigned by a previous @ASG ECL statement as the file TAPE, explicitly requests EDRC data compression.

Example 2

```
@MODE,EL FILEY,30
```

The E and L options override the mode settings initially assigned to FILEY. A new noise level of 30 is also specified for the file.

Support for 4780 18- and 36-track tapes is provided by standard Exec. Compression on the 4780 subsystems is called improved cartridge recording capability (ICRC) and is totally compatible with the 5073 enhanced data recording capability (EDRC).

Device characteristics for compressed tape readability are

- Unisys proprietary compressed cartridge tapes cannot be read on 4780 drives, but EDRC compressed cartridges from 5073 drives can be successfully read.
- A cartridge drive on a 4780 that does not have compression hardware can read only uncompressed cartridge tapes.
- The 4780 drives that have compression hardware can read EDRC compressed tapes without having to explicitly request CMPEON.

Also, 36-track tapes have more capacity on current cartridges because data is written in both directions on the tape, and more capacity with longer tapes (1,100 feet [E-carts]) can hold four times more data than current cartridges.

The following tape assign and catalog modes are available for 4780 drives:

```
@MODE,T.,////////CMPOFF/BUFON
```

The CTS9840, DLT7000, CTS9940B, U9840D, and UT10KA tapes always have expanded buffering so the EBCOPT is not needed to get a CTS9840, DLT7000, or CTS9940B.

The following tape assign and catalog mode is available for 5136 tape subsystems:

```
@MODE T.,////////CMPEON/BUFON
```

7.40.@MOVE—Position Tape Files

This command moves a magnetic tape file forward or backward over a specified number of hardware EOF marks.

The *filename* parameter is required for the @MOVE statement. All other parameters in the @MOVE statement are optional.

Format

```
@MOVE[,options] filename[,number]
```

options

Specifying B moves the tape backward. Omitting B moves the tape forward.

Use of the C option on any FURPUR command will prevent execution of ER ERR\$ on error exit if FURPUR is running in breakpoint or batch. The C option should be used with caution because a bad result of an erroring FURPUR command could be input to a process later on in the runstream.

filename

Specifies the name of the tape file. This parameter is required.

number

Specifies the number of EOF marks to be skipped. If you do not specify *number*, FURPUR specifies 1.

When you direct FURPUR to move a tape forward, the move stops either at the beginning of a file or between the two EOF marks comprising an end-of-recorded area sentinel. FURPUR never moves the tape beyond the first EOF mark of a pair of EOF marks that constitute an ERA sentinel. If you attempt to move beyond an ERA sentinel, FURPUR positions the tape to this sentinel and informs you that the tape is positioned at the ERA sentinel.

If FURPUR encounters a swap indicator during a forward move, it swaps to the next reel and continues the move forward on the next reel.

Exercise care when moving tape in the backward direction. Assume that tape file BOB is positioned at file 6:

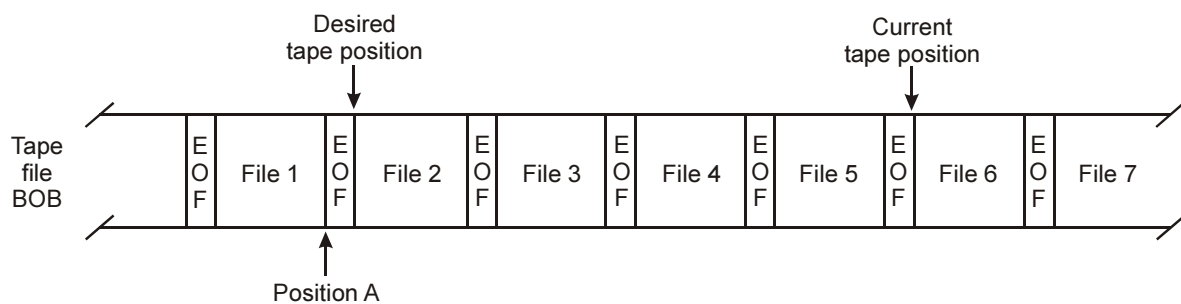


Figure 7-2. Positioning Tape Files

To position the tape to the start of file 2, execute the following sequence:

```
@MOVE,B BOB,5  
@MOVE BOB,1
```

The first statement moves the tape to position A, and the second statement moves the tape to the start of file 2.

If an @MOVE,B statement for a multireel tape file encounters the load point of the reel it is currently on, a diagnostic message is given and the ERR\$ exit is taken.

When you use a 1/4-inch cartridge tape subsystem, you cannot use an @MOVE,B statement. If you do, the operation is not performed and an appropriate error message is displayed.

7.41. @MSG and @@MSG—Send a Message to Various Console Types

The @MSG statement sends a message to an I/O, communications, hardware-confidence, or system console from a run.

Format

@MSG[,*options*] *message*

options

C

Sends a message to a communications console.

H

Sends a message to a hardware-confidence console.

I

Sends a message to an I/O console.

N

Suppresses message display. The message appears only in the user's print file. N option overrides W option and is used to place a comment in the print file.

S

Sends a message to the operator's (system) console.

W

Holds your run until the operator responds to your message. Your message remains displayed on the operator's console until the operator responds. For all demand runs where the print file has not been breakpointed (BRKPT), the response appears on your terminal. If the print file has been breakpointed, the response appears in the breakpointed (BRKPT) file. If the @MSG,W is from a batch run and the operator replies with an 'X', then the run is aborted.

The message can be directed to one console category only. The order of precedence is I, C, H, and S. If none of these options is present, the message is displayed on the operator's console.

message

Contains the message to be sent. The message can be up to 60 characters long, including embedded blanks. The first nonblank character is considered the beginning of the message. The end of the message is the last character before the end of a line, before a comment (.), or within the 60-character limit.

The message cannot begin with a comma (,) or a slash (/). The @MSG can include the continuation character (;) or blank-period-blank (.) which introduces a comment, but the @@MSG cannot include either character. Keep your messages as concise as possible.

When the message is displayed, it is prefixed with your run-ID and is entirely in uppercase letters, regardless of whether you used uppercase or lowercase letters when you entered it.

Examples

```
@MSG,I EXPECT 2 REELS OF OUTPUT FOR FILE XYZ
```

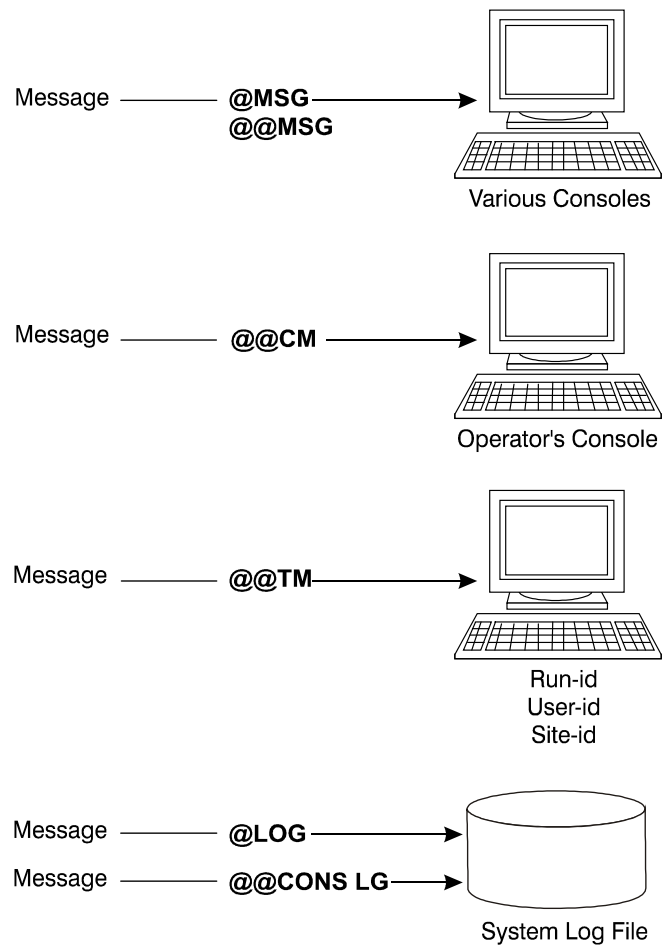
Illustrates a 37-character message to be displayed at the I/O console (I option). The message is strictly informational, and no operator response is expected.

```
@MSG,W IS REMOTE HOOKUP READY? . Answer required
```

This 23-character message requires a reply from the operator (W option). Since no particular console has been specified, the message is displayed on the operator's console. Run processing does not continue until the operator replies to the message. The comment, Answer required, is not displayed.

@MSG and @@MSG—Send a Message to Various Console Types

Figure 7–3 shows which message to use for a specific destination.



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Figure 7–3. Sending Messages

7.42.@@NOPR—Terminate a Local Printer

The @@NOPR statement terminates output to a local printer.

Use this statement after an @@PRNT statement. The @@NOPR statement causes subsequent output to be sent only to the terminal.

Format

```
@@NOPR
```

Example

```
@@PRNT
This list of filenames ...
.
.
.
@@NOPR
```

An @@PRNT statement is terminated with the @@NOPR statement. The text following the @@NOPR statement is sent only to the terminal.

7.43.@PACK—Remove Deleted Elements

The Pack command rewrites an entire program file, removing specified types of elements (depending on the options specified) and all elements marked as deleted. The @COPOUT and @COPY,P statements (see Sections 7.13 and 7.14.1) have the same effect on output files since they do not copy deleted elements. Mass storage space that is no longer needed is returned to the system.

The FURPUR @PACK command is used with large program files to remove elements in the file which have previously been marked as deleted. The D, L, and M options on the @PACK statement are not supported for large program files. If entered, these options are allowed, but are ignored.

FURPUR calls the PACK function in Program-Callable FURPUR (PCFP) to perform the pack operation on large program files. For this reason, updated levels of both products must be installed concurrently to pack large program files.

The FURPUR @PACK command can also be used with large element program files (LEPFs). See Section 3.7 for more information on LEPFs.

All parameters in the @PACK statement are optional.

Format

```
@PACK[,options] [filename-1,filename-2,...,filename-n, subtype-list]
```

filename-n

Specifies the program files to be packed.

subtype-list

Use the last specification to specify the subtype mnemonics of executable elements when you specify the AY option combination.

options

None

Used to remove all deleted elements and release all unused granules except those granules initially reserved. If granule zero remains allocated (see N option) and there are no nondeleted elements, the first sector is zero-filled.

A

Used to remove all elements except nondeleted executable elements.

C

Used to direct FURPUR not to exit through ERR\$ if an error is encountered.

Use of the C option on any FURPUR command will prevent execution of ER ERR\$ on error exit if FURPUR is running in breakpoint or batch. The C option should be used with caution because a bad result of an erroring FURPUR command could be input to a process later on in the runstream.

The C option is valid for all FURPUR commands.

D

Used in conjunction with the M option to inhibit creation of entry-point tables. This option is ignored if no M option is specified

I

Used to release all unused granules including initial reserve.

L

Used to print diagnostic messages if any duplicate relocatable entry points are found. (Used only when a prep is done as part of the @PACK command. Also see D, M, and P options.)

M

Used to create a minimum-size table of contents by starting each program file table in the sector following the previous nonempty program file table. The element text is started in the first track not used for the table of contents. Both a relocatable and an object module entry-point table are created unless the D option is specified.

After an @PACK,M statement, the room to add new elements to the program file is highly restricted. If any procedure tables exist or an entry point table was created, you can be able to add one or two elements, if any at all, because the next table starts at the next available sector. If no procedures or relocatable entry points exist, the space available to add new elements varies up to a track, which allows up to 179 entries in the element table. This is due to the starting element text address being forced to a track boundary.

If you want to expand the table of contents, you must transfer the elements to another file. You can use @COPY,P or @COPOUT statements for this transfer. Alternatively, you can delete existing elements followed by an @PACK statement without the M option.

N

Used to prevent the release of any granules.

O

Used to remove all elements except nondeleted omnibus elements.

P

Used to create both entry-point tables. The prep is done before FURPUR releases unused space. The P option is assumed if the M option is specified and the D option is not specified.

R

Used to remove all elements except nondeleted relocatable elements.

S

Used to remove all elements except nondeleted symbolic elements.

Y

Used in conjunction with the A option to cause FURPUR to save all executable elements with a subtype that matches one of the subtypes specified in the subtype list.

Z

Used to zero out granules released during the @PACK process.

Note: The D, L, and M options are not supported for large program files. If entered, these options are allowed, but are ignored.

You can specify any combination of A, O, R, and S options; or, if you want to select executable subtypes, you can select the AY option combination. If you specify no options, FURPUR removes only all deleted elements.

Thus, FURPUR stops the pack process to prevent destroying your file.

Normal Termination

If the file has position granularity, the term "FILE SIZE" in the completion message is the number of positions required to contain the program file.

If the file has track granularity, the term "TEXT" in the completion message is the number of tracks required to contain the element text, and TOC is the number of tracks required to contain the table of contents.

The number of elements of each element type is printed at termination.

Abnormal Termination

The @@X T or @@TERM operator commands or system stops cause abnormal termination resulting in possible loss of data. You must examine the file to verify that all elements are complete.

Packing a Large Program File

When the file that you specify on the @PACK statement is a large program file (LPF), FURPUR calls PCFP to pack the file. During this operation, PCFP utilizes a temporary file, whose size is equal to the length of the file being packed, as a temporary storage for all of the elements to be retained. B.7 describes the error messages for all of the possible errors that can be encountered during the pack of an LPF.

7.44.@PASSWD and @@PASSWD—Use and Change Passwords

In batch mode, you supply your user-ID on the @RUN statement, and your password and executing clearance level on the @PASSWD statement. You can place the @PASSWD statement anywhere in the runstream between the @RUN statement and the @FIN statement. For batch runstreams initiated with a @START statement, the @PASSWD statement is not required and if it is found, it is ignored.

Long passwords, both case-sensitive and case-insensitive, are accepted on @PASSWD and @@PASSWD statements.

The current state of password case sensitivity is maintained when an @@PASSWD is performed. For example, if the user's password is case-sensitive at the time of @@PASSWD, the new password is also case-sensitive.

You can include several @PASSWD statements in the same runstream. However, if any of the passwords are illegal, the system terminates your run. If more than one @PASSWD statement is in error, the error associated with the first incorrect @PASSWD statement is placed in your run's print file.

Do not place the @PASSWD statement within an @ADD file. This compromises the secrecy of your password, and the system ignores the @PASSWD statement anyway.

When you use the @START statement to execute a batch run, do not include an @PASSWD statement in the created runstream. If you do, security is compromised because the password is made visible. In addition, the @PASSWD statement is always ignored when this situation occurs.

Notes:

- *You cannot use the @PASSWD statement to change passwords if the machine-generated passwords feature is implemented at your site. The @PASSWD statement is still valid for password validation in batch mode.*
- *If your system has Fundamental Security, this subsection can or cannot be applicable, depending on how your system is configured.*

- If your site has Security Level 1 or higher, this subsection is applicable. Check with your security officer if you are not sure of your security level.

Format

The format of the @PASSWD statement is

```
@PASSWD[,N] old-password[/new-password] [clearance level,comp-1/comp2.../comp-n]
```

N

Indicates the *old-password/new-password* field is not present. The N option can only be specified in @RUN,/B or @@INQ runstreams.

old password

Is the current password for your user-ID.

new password

Is your new password, if you are changing your password. Do not include any blank character spaces between the slash (/) and the old and new passwords (old password/new password).

clearance-level

If your site has Security Level 1 or higher, you can specify an executing clearance level. The clearance level you specify must be in your clearance level range. If you do not specify a clearance level, it is set to the minimum clearance level in your range.

If you enter more than one @PASSWD statement, the system uses the clearance level specified or assumed on the last @PASSWD statement in the runstream as the clearance level for the entire run.

comp-n

If your site has Security Level 2 or higher, you can use this field to supply compartment information on the @PASSWD statement. If you cannot fit all the compartment names on one line, use a semicolon (;) as the continuation character. The continuation character cannot appear before the first compartment name.

If you enter more than one @PASSWD statement in the batch runstream, the compartment set associated with the last @PASSWD statement processed is the set associated with your run.

You can specify the null compartment set by entering "NULL," or you can specify your maximum compartment set by entering "MAX." If you enter the @PASSWD statement with no compartments listed, your default compartment set is given to your run. The maximum or default compartment sets are given to your run at the time it is put into backlog.

If your system has the compartment feature, and your user compartment set changes while your run is in backlog, the run terminates in error when it comes out of backlog. This happens because the READ\$ file cannot be assigned by the run. You receive this message:

```
READ$FILE REJECT STATUS 264733
```

For example, you are executing with compartments A, B, and C. While a run you entered is in backlog, the security officer removes compartment C from your user compartment set. The run comes out of backlog and terminates in error because the READ\$ file has compartments A, B, and C associated with it and you are no longer executing with compartment C.

If you make any errors in the password or clearance level fields of the @PASSWD statement, you do not receive any messages on the compartments field, whether or not you enter the compartment information correctly.

The clearance level and compartment set security attributes are displayed in the print file following the @RUN statement. Messages regarding password replacement or compartment set validation follow the display of the mandatory attributes.

Examples

Here is an example of using an @PASSWD statement with a remote batch run:

```
@RUN, /B BCD, 802910, CAPRICORN
.
.
.
.
@PASSWD OMNI/RUXMEN 10, PAYROLL
.
.
```

These are examples of using compartments with the @PASSWD statement:

```
@PASSWD RQZ , PAYROLL/ACCOUNTING/INVENTORY

@PASSWD RQZ 63, PAYROLL/;
ACCOUNTING/;
INVENTORY

@PASSWD RQZ HIGHCL, PAYROLL/TOPSECRET
```

Changing Your Password Using @@PASSWD

The @@PASSWD statement lets you change your password in demand on systems where it is allowed. The system administrator may disallow it because of the risk that an unauthorized person might see the password on the screen as it is being entered. Unlike the response to the initial user-id/password solicitation, which can be blanked out to protect its confidentiality, the system is unable to prevent the display of the characters you enter in the @@PASSWD statement. The administrator might also disallow the use of @@PASSWD because it conflicts with the user of Configured Password Profiles. @@PASSWD is never allowed in Security Level 3. Attempts to change your password on a system where it is not allowed returns the error

```
System configuration disallows password changes by @@PASSWD.
```

Format

```
@@PASSWD old-password/new-password
```


For example, the following statement changes your password from QMKE to PAX01.

```
@@PASSWD QMKE/PAX01
```

Caution

The password specified in an @@PASSWD statement is visible to observers. Security can be compromised if you execute this statement. The most secure way to change your password is during sign-on.

A demand terminal is user authenticated before the user is allowed to enter a batch runstream by using the @RUN,/B statement. The same user-ID is required on the @RUN,/B image or the run is removed. To allow a user-ID with a nonzero authentication type access to the @RUN,/B interface, an N option exists for the @PASSWD statement to indicate that the *password/newpassword* field is not present. An @PASSWD,N statement with no parameters indicates the defaults for clearance level and compartment sets must be used. The user can optionally specify other than default clearance level or compartment sets.

The N option is available to all @RUN,/B users regardless of authentication type. If the @RUN,/B runstream contains an @PASSWD image without the N option, the verification for password still takes place. If the user-ID is not an Exec authentication type (AT=0), the run is removed with the following error message:

```
@PASSWD ERROR: Userid's authentication type is not valid for batch.
```

Table 7–4 summarizes @PASSWD processing for batch runs.

Table 7–4. @RUN,/B Following a Demand Sign-on

Authentication Type = 0		Authentication Type > 0	
Fail Cases	Pass Cases	Fail Cases	Pass Cases
No @PASSWD image and no @PASSWD,N image in runstream	- Only @PASSWD,N images in runstream - Only @PASSWD images in runstream - Multiple and mixed @PASSWD images and @PASSWD,N images	- At least one @PASSWD image (without ",N") in runstream - No @PASSWD,N image in runstream	Only @PASSWD,N images in runstream

Note: An @RUN statement in @@INQ mode, with no demand run open, is an implied @RUN,/B.

7.45. @PCH—Punch Program File Elements

The @PCH command creates 80-column card images from program file elements.

All parameters in the @PCH statement are optional except *element-name*.

Format

```
@PCH[,options] element-name[,sequence-character]
```

options

Can be one or more of the following options:

A,O,R,S

Specify the type of element to be processed. Any combination of A, O, R, or S can be used, but at least one must be given. See Section 6.4 for additional information on these options.

C

Used to direct FURPUR not to exit through ERR\$ if an error is encountered.

Use of the C option on any FURPUR command will prevent execution of ER ERR\$ on error exit if FURPUR is running in breakpoint or batch. The C option should be used with caution because a bad result of an erroring FURPUR command could be input to a process later on in the runstream.

Q

Set punch control to create ASCII card images.

You can use the following options only in conjunction with the S option:

D

Set data mode to allow statement images to be read by the ELT processor. FURPUR generates an @ELT,ID statement with F\$\$ in the parameter field 3 as the end sentinel. The last card image created contains the corresponding end sentinel (@END F\$S).

G

Produce output containing compressed symbolic images. Compressed symbolic output is produced only in Fieldata code. ASCII code images are first converted to Fieldata code.

H

Place sequence number into columns 73 to 80 of each image. The *sequence-character* must contain an alphabetic sequence from 1 to 3 characters. The characters are left-justified and overlay columns 73 to 80.

J

Compress input images and sequence output card images in columns 73 to 80.

Note: Do not specify the G and J options in the same statement.

element-name

Specifies the element to be processed.

sequence-character

Specifies the alphabetic sequencing characters when you choose the H option.

Description

Card readers and card punches are no longer supported. @PCH is supported for compatibility with existing runstreams. @PCH output can be directed to a file by @BRKPT PUNCH\$/filename ... @PCH command ... @BRKPT PUNCH\$, but its use is discouraged. The FURPUR processor ensures that the input elements contain the control statements needed to reinsert the elements a program file with the same name in a later run. The first image of the procedure element is an @PDP,I control statement. For all other elements, it is an @ELT,I statement. The file name on the @PDP or @ELT control statement is the name of the file from which the element was created.

Relocatable and executable elements are automatically (without special option) sequenced in columns 79 to 80. Sequencing starts with AA and ends with ZZ, starting over with AA if necessary. If you specify the H option, FURPUR sequences symbolic images in columns 76 to 80. The card sequence is 100 apart and is preceded by the designated alphabetic string given in the *sequence-character* field on the @PCH statement.

The output is either preceded by a properly formatted @ELT,I statement or an @PDP,I statement. The command statement images produced by @PCH have the same element name as the @PCH statement. Thus, if you assign the file containing the element that was created by @PCH to a subsequent run, all that is necessary to reintroduce the element is to include the images produced by the @PCH statement in the runstream.

Examples

The following examples illustrate the use of the @PCH statement:

```
@PCH,S UPDATE.RUNPROG
```

Symbolic element RUNPROG in program file UPDATE is copied and transformed onto 80-column card images.

```
@PCH,SRJH A.B,XYZ
```

Symbolic element B of program file A is transformed into 80-column card images. The input images are sequenced in columns 76 to 80. The identification

sequence, XYZ, is placed into columns 73 to 75. The input images are also compressed.

Relocatable element B of program file A is also transformed and output. The text has been previously sequenced, and the FURPUR processor sequences the preamble.

7.46.@@PMOD—Control Output Sent to a Printer Connected to Your Terminal

By defining the peripheral mode operations, the @@PMOD statement lets you control the way in which data is sent from a UTS 400 terminal in demand mode to a printer. To return to print transparent mode (normal mode), enter @@NOPR.

Format

```
@@PMOD [function]
```

function

Can be one or more of the following:

F

Prints form; the terminal handler uses the print form function to send data to the printer.

P

Prints all; the terminal handler uses the print function to send data to the printer.

X

Prints transparent; the terminal handler uses the print transparent function to send data to the printer (default).

Example

```
@@PMOD P
```

Sends all characters between the SOE (or home position) and the cursor to the printer (except FCC sequences).

7.47.@PREP—Create Entry-Point Tables

Using the PREP command creates relocatable and object module entry-point tables from the preambles of relocatable elements and from the control information in object modules.

All parameters in the @PREP statement are optional.

Format

```
@PREP[,options] [filename-1,filename-2,...,filename-n]
```

*options***C**

Used to direct FURPUR not to exit through ERR\$ if an error is encountered.

Use of the C option on any FURPUR command will prevent execution of ER ERR\$ on error exit if FURPUR is running in breakpoint or batch. The C option should be used with caution because a bad result of an erroring FURPUR command could be input to a process later on in the runstream.

The C option is valid for all FURPUR commands.

F

Forces a prep of a file. Without the F option, FURPUR does not perform a prep if an entry-point table already exists.

L

Prints duplicate relocatable entry points found with the sequence numbers of the corresponding elements.

filename-n specifies the program files for which entry-point tables are to be created.

The addition of a relocatable element into a program file or the deletion of a relocatable element from a program file destroys any existing relocatable entry-point table in that file. The object module entry-point table is unaffected by the addition of an object module to a program file, and the deletion of an object module only causes the entry points of that element to be lost from the object module entry-point table.

If the @PREP operation is unsuccessful because there are too many relocatable entry points for the space allocated to the relocatable entry-point table, use the @PACK,M statement on the file (see Section 7.43). Note that because the object module entry-point table exists in the text area of a program file, it is essentially open-ended and is bounded only by the "MAX" size of the file. Because of this, the object module entry-point table is never too small for object module entry points on which an @PREP statement has been used.

The visibility of entry points not contained in an entry point table depends on the type of entry point. A relocatable entry point is not visible unless it exists within the relocatable entry-point table. In contrast, an object module entry point need not be in the object module entry-point table for it to be visible to the Linking System. The only advantage in performing a prep (@PREP statement) of object module entry points is that the Linking System can resolve link faults and external references faster when the associated entry points are contained in the object module entry-point table.

7.48.@@PRNT—Start a Local Printer

The @@PRNT statement directs output to both the terminal and a local printer (a printer connected directly to a terminal). All output following the @@PRNT statement is sent to both the terminal and the local printer.

Terminate @@PRNT with an @@NOPR statement.

Format

```
@@PRNT
```

Example

```
@@PRNT
output from statements here goes to both the printer and the terminal
```

The text following the @@PRNT statement is sent to the printer and to the terminal.

7.49.@PRT—List Files, Elements, and Master File Directory

Obtains a listing of the text of a symbolic element, the table of contents of a program file, information regarding temporary files, or the master file directory items of cataloged files. The statement does not list executable, omnibus, or relocatable elements. You can list these elements with the LIST processor.

All parameters in the @PRT statement are optional.

Format

```
@PRT[,options] [name-1,name-2,...,name-n]
```

options

Are specified and described in the subsections that follow.

name-n

Specifies any of the following, depending on the options indicated:

- The name of a cataloged file in any format
- The name of a program file
- The name of an element
- The name of a temporary file
- An account number
- A project-ID
- A removable disk pack-ID and disk equipment type
- The number of elements to print
- A subtype list

When *name-n* is a subtype list or the number of elements to print, it must be the last specification on the statement.

Description of Options

None

Used to display the master file directory.

When no name fields are specified and the requesting run is privileged, the entire master file directory is displayed alphanumerically by project-ID. When the requesting run is nonprivileged, all public and ACR-controlled files in the master file directory are listed along with all files cataloged with the project-ID of the requesting run. (Read/write keys are displayed if present in the MFD.) FURPUR displays files sorted first by the directory-ID (only applicable when file sharing is configured), then by the project-ID and the account number, and finally by the qualifier and file name.

When file sharing is configured, the first line of the display contains the DIRECTORY keyword and the name of the directory for local files. Following from this line, FURPUR displays all the applicable files in the local directory. After this display, a line containing the DIRECTORY keyword and the name of the directory for shared files is displayed. Following from this line, FURPUR displays all the applicable files in the shared directory.

If the shared subsystem is not available when you perform the @PRT statement, FURPUR displays the following message:

```
FILES ON SHARED SUBSYSTEM ARE NOT AVAILABLE
```

A, O, R, S

Used in conjunction with the T option to list the table of contents entry for the type(s) of elements in a program file specified by the options. The A,O,R, and S options and the AY option combination can be used with the B, L, and V options but must be accompanied by the T option.

B

Used in conjunction with the I or T options to list the files assigned to the run or the table of contents in descending sequence order (for example, to list the most recently inserted items first). The B option can be used with any combination of A, O, R, S, and V options and the AY option combination when accompanied by the T option.

FURPUR allows only three names if you use the B option with the AY option combination. It ignores all other names. If you do not specify the AY option combination, FURPUR allows only two names and ignores all others. The number of items printed is limited to the number given as *name-2*. There is no number limit if *name-2* is not given or is left blank when the AY option combination and a subtype list are given.

C

Used to direct FURPUR not to exit through ERR\$ if an error is encountered.

Use of the C option on any FURPUR command will prevent execution of ER ERR\$ on error exit if FURPUR is running in breakpoint or batch. The C option should be used with caution because a bad result of an erroring FURPUR command could be input to a process later on in the runstream.

The C option is valid for all FURPUR commands.

D

Used to display the names of all cataloged files currently residing wholly or partially on the named removable disk packs. When the requesting run is nonprivileged, all read/write keys and project-IDs not matching the project-ID of the run are replaced by slashes. The files are sorted alphanumerically by project-ID, unless the N option is also specified, in which case the files are sorted alphanumerically by account number. Completion of an @PRT,D statement requires the packs to be mounted.

The display for the @PRT,D statement includes a one- or two-line header, followed by a line item for each file on the pack.

If file sharing is configured, the system displays a second line. This line contains the DIRECTORY keyword and the name of the directory associated with the pack.

The specified *name-i* have the following format:

```
pack-id[/type]
```

If *type* is specified it must be an assign mnemonic for a sector-addressable mass storage file (see Table D-1). If *type* is not specified it defaults to F.

E

Used in conjunction with the T and L options to suppress the display of duplicate object module entry points. For program files containing many object module entry points, the E option can provide a very large performance increase by avoiding the search process for duplicates.

F

Used to output information about cataloged and temporary mass storage and tape files. Different information is presented for each of the following file types:

- Cataloged mass storage files
- Cataloged tape files
- Temporary sector-addressable mass storage files
- Temporary word-addressable mass storage files
- Temporary tape files

See Section 7.49.1 for a detailed description of the output format for each file type.

I

Used to output information about files that are currently assigned to the run. The format for each file is

```
q*f(c),equipment,@asg-options [internal-name-1, ...] [{STD|SHARED}]
```

where *q*f(c)* is the external file name, *equipment* is the file equipment mnemonic, *@asg-options* are the options used to assign the file to the run, *internal-name-1* are all of the internal file names attached to the file, if any, and STD or SHARED is the file directory if file sharing is configured on the system.

If an internal name is attached to a file that is not assigned to the run, then NAME ITEM is output instead of *equipment* and *@asg-options*. Internal file names are listed in the order in which they were attached to the file, with the most recently attached name listed first.

The files SYSS\$*DATA\$, SYSS\$*LIB\$, SYSS\$*RLIB\$, SYS\$LIB\$**product* and DIAG\$ are not output, unless the L option is also specified.

The files are output in the order in which they were assigned to the run, with the first file assigned output first. The B option can be included to reverse the order in which the files are output, with the most recently assigned file output first. The B option does not affect the order in which the internal file names are listed. The L option can be specified with the B option.

All files assigned to the run are output, unless *name-2* specifies a number of files to output. For example:

```
@PRT,IBL ,20
```

This would output the 20 most recently assigned files, with the most recently assigned files listed first. Without the B option, the first 20 files assigned to the run would be output. Note that *name-1* is always ignored when the I option is specified.

L

Used in conjunction with D, N, P, T, or no options on an @PRT statement (in a demand run) to obtain a complete rather than the abbreviated listing normally produced for demand runs. Used with the I option on the @PRT,I statement to print all files assigned to the run, including those otherwise excluded (see I option).

M

Used in conjunction with the L option on the @PRT statement to allow file caching information to appear as part of the @PRT,L display. Retrieving file caching information for each file causes a major increase in system overhead and in the time to process @PRT,L, so include the M option only if file caching information for all files is needed.

N

Used in conjunction with the T option to specify that only summary information should be produced for the named program files. Information for individual elements is not produced, even when element names are included with the

program file names. When the N option is specified, the A, B, E, O, R, S, V and Y options are ignored. If the L option is also specified, the N option is ignored.

When the T option is not specified, the master file directory is displayed by account number. When the requesting run is privileged, the lack of a name field causes the entire master file directory to be displayed and sorted alphanumerically by account number. When name fields are specified, only the files cataloged with the specified account numbers are displayed. They are displayed in the order specified in the name fields.

In nonprivileged mode, a name field is unnecessary, and if specified, is ignored. In this mode, you see all unowned cataloged files with the account number of the requesting run displayed.

You also see displayed all owned cataloged files with the account number of the requesting run, if the user-ID of the requesting run is the owner of these files. They are sorted alphanumerically by project-ID. (Read/write keys are not displayed.) If the M option is specified and if this is either a batch run or if the L option is specified, then the output format is the same as for @PRT,F.

If the shared subsystem is not available when you perform the @PRT,N statement, FURPUR displays the following message:

```
FILES WITH SPECIFIED PROJECTS ON SHARED SUBSYSTEM ARE NOT AVAILABLE.
```

P

Used to display by project-ID. When the requesting run is privileged, the lack of a name field causes the entire master file directory to be displayed and sorted alphanumerically by project-ID. When name fields are specified, only the files cataloged with the specified project-ID are displayed. They are sorted in the order specified in the name fields.

When you are in nonprivileged mode, you do not need to specify a name field. If you do, FURPUR ignores it. In this mode, you see all unowned cataloged files with the project-ID of the requesting run displayed. You also see all owned cataloged files with the project-ID of the requesting run, if the user-ID of the requesting run is the owner of these files. (Read/write keys are not displayed.) If the M option is specified and if this is a batch run or if the L option is specified, then the output format is the same as for @PRT,F.

If the shared subsystem is not available when you perform the @PRT,P statement, FURPUR displays the following message:

```
FILES WITH SPECIFIED PROJECTS ON SHARED SUBSYSTEM ARE NOT AVAILABLE.
```

T

Used to display the table of contents for each specified program file or program file element.

U

Used to display the current usage of the removable disk pack specified. The name fields are the same as the @PRT,D statement. The output consists of the first line of the @PRT,D output, which displays pack-ID, tracks available, and current number of assigns. If file sharing is configured, the output consists of the first two lines of the @PRT,D output, the second line displaying the DIRECTORY keyword, and the name of the directory associated with the pack.

V

Used in conjunction with the T option to list the table of contents for elements with the same version name as specified in the name field. When the version name is omitted from the name field, only those elements having a blank version name are listed. The V option can be used with any combination of A, B, O, R, and S options, but must be accompanied by the T option.

Y

When you specify this option, you must also specify the A option and, at the end of the statement, the subtype list. You use the AY option combination to limit the display of elements in a program file to executable elements that have a subtype matching one of the subtypes specified in the subtype list.

7.49.1. @PRT,F Output Format

You can use @PRT,F to output information about cataloged and temporary mass storage and tape files. FURPUR uses a number of information sources to gather the information presented. Different information is presented for each file type, as described in detail under the topics below. Each bullet describes the information on one output line.

Date and Time reporting

@PRT,F reports date and time information related to files using the Unisys standard time display format *yyyy-mm-dd hh:mm:ss [zone]*. The *zone* mnemonic is left blank when the time zone mnemonic is not available. This occurs when the source timestamp is in TDATE\$ format.

Cataloged Mass Storage Files

For cataloged mass storage files most of the information comes from the MFD entry for the file. Some file assignment and file caching information is gathered from other sources. The following information is output:

- STD or SHARED, the directory for the file if file sharing is configured.
- The project-ID and account number for the file.
- The complete name of the file, including qualifier, filename, F-cycle, optional read key and optional write key. The assign mnemonic, initial reserve, granularity and maximum size. If this is a removable disk file, the pack-IDs for the file.

- The 'MODES:' line containing a list of the following file modes, separated by commas:
 - PUBLIC, PRIVATE or SEMI-PRIVATE, the file privacy,
 - WRITE if the file is write-only.
 - READ if the file is read-only.
 - X-USE if the file is assigned exclusively.
 - *a*-OPT for each of the options C, D, G, K, S, U, and V with which the file is cataloged or assigned.
 - DROP if the file is marked as to-be-deleted.
 - UNLOD if the mass storage has been unloaded to tape.
 - DATA-DSABLD if the file is data disabled because of a cache destage failure on an XPC.
 - DSABL if the file is hardware disabled, software disabled, or FAS disabled.
 - ASG-D if the file is assigned to any run.
 - SYM if the file is on a symbiont queue.
 - ENCRYPT if the file is encrypted.
- The file owner user-ID if the file is owned. The file ACR if the file is semi-private. The ACR owner user-ID or UNOWNED, if the file is semi-private and if either the file is unowned or the ACR is unowned.
- One of the following file caching statuses if XPC file caching is configured:

```
CACHED-ON: xpc-name, SPEC=spec
NOT-CACHED DUE TO reason, SPEC=spec
CACHING STATE AND DEVICE NAME NOT AVAILABLE, SPEC=D
FILECACHING INFORMATION NOT AVAILABLE-FILECACHE$ ERROR: status OCCURRED.
```
- The number of granules assigned. The number of granules assigned to each of the eight groups (GP1 through GP8) if the file is loaded. The number of granules backed up to tape by FAS if the file is unloaded.
- The highest granule assigned and the total number of file assignments.
- The highest track written.
- The file catalog date and time and the last reference date and time.
- The backup date and time if the file has been backed up by FAS. NOT-CURRENT if the file has been updated since it was backed up.

- A descriptive line for each file disable if the file is software disabled, hardware disabled, FAS disabled or data disabled.

Example

```
* * DIRECTORY:  STD      * *
* * PROJ:  DEV          ACCNT: 298631      * *
DEV*TESTFILE(1),FSCSI/100/TRK/999,SPAR02/SPAR03
MODES: SEMI-PRIVATE,READ,DSABL,ASG-D
OWNER: DEV027 ACR: D27A2
NOT-CACHED DUE TO XPC NOT AVAILABLE,SPEC=D
NO. OF GRANULES ASG-D:      240      GP2=240
HIGHEST GRANULE ASG-D:      239      TOTAL ASSIGNMENTS:    931
HIGHEST TRACK WRITTEN:      233
CAT:      2005-12-14 08:42:39      LAST REF: 2016-08-25 10:48:59 CDT
BACKUP: 2016-08-20 13:38:18 CDT
WARNING: FILE ASSIGNED AT TIME OF SYSTEM CRASH
```

- A descriptive line with encrypted KEY NAME and KEY VERSION, if file is encrypted.

Example

```
* * DIRECTORY:  STD      * *
* * PROJ:  ////////// ACCNT: ////////// * *
DT*DT(3),F/0/TRK/99
MODES: PRIVATE,ENCRYPT
OWNER: DARETEST2
KEY NAME:  THISKEYCANBEALPHANUMERIC      KEY VERSION: 1
NO. OF GRANULES ASG-D:      0
HIGHEST GRANULE ASG-D:      0      TOTAL ASSIGNMENTS:      4
HIGHEST TRACK WRITTEN:      0
CAT:      2025-09-02 12:52:52 CDT      LAST REF: 2025-09-18 10:23:30 CDT
```

Cataloged Tape Files

For cataloged tape files many of the tape specifications in the MFD entry for the file can be dynamically changed with the @ASG and @MODE statements. FURPUR gathers the information that currently applies when a tape file is assigned to the run. The following information is output:

- STD or SHARED, the directory for the file if file sharing is configured.
- The project-ID and account number for the file.
- The complete name of the file, including qualifier, filename, F-cycle, optional read key and optional write key. The assign mnemonic. The volume-IDs for the file. 'LTO' immediately follows a volume-ID if this is an LTO tape file and if the volume is currently mounted. It identifies the type of LTO cartridge that is mounted.
- The 'MODES:' line containing the same list of file modes as shown for cataloged mass storage files. The UNLOD and DATA-DSABLD modes do not apply to cataloged tape files.

- The file owner user-ID if the file is owned. The file ACR if the file is semi-private. The ACR owner user-ID or UNOWNED, if the file is semi-private and if either the file is unowned or the ACR is unowned.
- The 'TAPEMODES:' line containing a list of the following tape modes:
 - The density and parity if this is an open reel tape, VTH virtual tape, or DVD.
 - The buffered-write mode if this is a half-inch cartridge tape. In addition to the standard BUFOFF, BUFON, BUFFIL, and BUFTAP values, BUFDEF indicates the cataloged tape file is not assigned to the run and the buffered-write mode is determined by the Exec configuration parameter tape_buffering.
 - Q-WORD or 8-BIT, the data transfer format.
 - The processor/tape translation, if specified.
 - DC if data converter is specified.
 - BLKON if block numbering is specified.
 - CMPON, CMPEON, or CMPLON if data compression is specified.
 - EBCOPT if expanded buffer is specified.
 - CTL=*name* if a cartridge tape library pool name is specified.
- The file catalog date and time and the last reference date and time.
- The backup date and time if the file has been backed up by FAS.
- A descriptive line for each file disable if the file is software disabled, hardware disabled, FAS disabled, or data disabled.

Example

```
* * DIRECTORY: SHARED * *
* * PROJ: DEV          ACCNT: 298631      * *
DEV*CATTAPE4(1),LTO4,L60026(LTO6)/L60027/L60028
MODES: PUBLIC,X-USE,ASG-D
OWNER: DEV016
TAPEMODES: BUFTAP,8-BIT /CMPON
CAT:      2013-10-25 10:17:52          LAST REF: 2016-08-25 10:48:59 CDT
```

Temporary Sector-Addressable Mass Storage Files

For temporary sector-addressable mass storage files, the following information is output:

- The complete name of the file, including qualifier, filename, and F-cycle. The equipment mnemonic, initial reserve, granularity, and maximum size.
- The highest track written.

- The highest granule assigned.
- The status of caching for the file if XPC file caching is configured. The statuses are the same as those shown for cataloged mass storage files.

Example

```
DEV*TPF$(0),MEMFL/0/TRK/256
HIGHEST TRACK WRITTEN: 28
HIGHEST GRANULE ASG-D: 31
NOT-CACHED DUE TO XPC NOT AVAILABLE,SPEC=D
```

Temporary Word-Addressable Mass Storage Files

For temporary word-addressable mass storage files, the following information is output:

- The complete name of the file, including qualifier, filename, and F-cycle. The equipment mnemonic and maximum size in number of words.
- The highest track assigned.

Example

```
DEV*WAD(0),SCDISK/1792
HIGHEST TRACK ASSIGNED: 1
```

Temporary Tape Files

For temporary tape files, the following information is output:

- The complete name of the file, including qualifier, filename, and F-cycle. The equipment mnemonic. One or two logical device names. The current volume-ID. 'LTO*n*' immediately follows if this is an LTO tape file and if the volume is mounted. It identifies the type of LTO cartridge that is mounted. The next volume-ID to mount if it has been identified. The 'TOTAL REELS / REEL INDEX *j*' volume summary if there is more than one volume for the tape file.
- The 'TAPEMODES:' line containing the same list of tape modes as shown for cataloged tape files.

Example

```
DEV*TAPE(0),U47M,T51A3/T51A4 20362H/20363H TOTAL REELS 4 REEL INDEX 2
TAPEMODES: BUFON,8-BIT
```

7.49.2. Listing the Master File Directory

The @PRT statement reports the files from the local MFD and from the shared MFD, if a shared directory is available. For @PRT, the @QUAL,D statement does not affect the selection of directories.

You can select a subset of MFD files by using the N or P options. You can also limit the selected files to specified removable pack(s) by using the D option. See Section 7.48 “Description of Options” topic.

The long report (using the L option or in batch mode) formats each selected file in a format similar to @PRT,F. Otherwise, the short format provides a one line summary for each selected file.

If the requesting run is nonprivileged, slashes replace all read and write keys, as well as all project-IDs and account numbers not matching the run to protect exposure of sensitive data.

7.49.3.Listing the Table of Contents

The @PRT,T statement is used to output information from the table of contents of program files. Refer to Sections 3.8.4 through 3.8.7 for additional information about program files. The @PRT,T statement provides the following three levels of information detail.

- Long format (@PRT,TL)
- Short format (@PRT,T)
- Summary format (@PRT,TN)

If both the L option and the N option are specified, long format is selected and the N option is ignored.

The following describes, in order, the @PRT,T output and indicates for which formats the information is produced:

- File and element summary information—all formats
- Element Table information—long and short formats; the elements selected for output depends on the format, the presence of an element name, and the presence of the A, B, O, R, S, V, and Y options.
- Procedure Table information—long format; when element name and the A, B, O, R, S, V, and Y options are not included.
- Relocatable Entry Point Table information—long format; when element name and the A, B, O, R, S, V, and Y options are not included.
- Object Module Entry Point Table information—long format; when the table is present, when element name, and the A, B, O, R, S, V, and Y options are not included.
- Invalid Element Table Entry Count –long format only; this is only produced if the long format element table validation finds invalid values in element table entry fields.
- Deleted element summary information—all formats when the file contains at least one deleted element.

File and Element Summary Information

The summary information output for all formats is as follows:

filename type, a/b elements, c/d tracks

where:

filename

The name of the file in format: *directory-id#qualifier*filename(F-cycle)*

type

The program file type that can have a value of "PF", "LPF", "Standard LEPF", or "Large LEPF". Refer to Section 3.8.4 for additional information about program file types.

a

The number of elements in the program file, including deleted elements.

b

The maximum number of elements that could possibly be contained in the program file based on the current allocation.

c

The number of tracks occupied by program file tables and text.

d

The maximum number of tracks for the file.

You can use the @PACK statement when the current number of elements or tracks approach their respective maximum values. Refer to the Section 7.43 for a detailed description of @PACK.

Element Table information (Long Format)

For long format, a column header line is output to identify the element information. One element is output for each line. The following is a description of the column headers and contents:

D

The deleted element flag. An asterisk (*) appears in this column if the element has been replaced by another element with the same element name, version name, and element type. Refer to the Section 7.18.3 for information on undeleting elements.

NAME and VERSION

The element name and version name, respectively. Version name can be spaces.

TYPE

The element type. The basic element types are executable, omnibus, relocatable, and symbolic. Executable, omnibus, and symbolic types can also have subtypes. The element type and, if available, subtype appear in this column. For symbolic elements, "-Q" is appended to the type and subtype, if the element is marked as being in the ASCII character set.

DATE and TIME

The date and time in the format *yyyy-mm-dd hh:mm:ss [zone]*. The *zone* is blank when the time zone mnemonic is not available. This occurs when the source timestamp is in TDATE\$ format. Usually denotes the date and time when the element was originally created.

SEQUENCE NUMBER

The sequence number of the element in the file. Elements are numbered sequentially beginning with 1.

PREAMBLE SIZE

This field applies only to relocatable elements. The number of sectors of relocatable preamble text in the element.

TEXT SIZE

The number of sectors of element text.

CYCLE WORD

The element cycle information is present only for symbolic elements. Three columnar fields for the cycle information are shown.

MAX	Maximum number of element cycles
CUR	Current absolute cycle number
NUM	Number of element cycles present

Refer to Section 3.5 for a more detailed description of element cycles.

FLAG BITS

This information is present only for executable and relocatable elements.

The partial-word sensitivity values can be:

QTR	If the element is quarter-word sensitive.
THR	If the element is third-word sensitive.
BOTH	If the element is both quarter-word and third-word sensitive.

The arithmetic fault mode values can be:

SET	If the element has arithmetic fault compatibility mode set.
CLR	If the element has arithmetic noninterrupt mode set.
INS	If the element is insensitive to arithmetic fault handling.

SECTOR LOCATION

The sector location of the beginning of the element text, relative to the beginning of the file.

After all the selected elements have been output, the sector location of the element text for the next element that will be added to the file is shown on the following line.

NEXT AVAILABLE LOCATION

The next available sector location value appears to the far right, aligned with the SECTOR LOCATION column.

Element Table Information (Short Format)

Short format includes the element type and element name only. No column header is output. One element is output for each line. Deleted elements are not output. The element information is output in the following format:

type name

where:

type

The element type. If the element has a subtype, only the subtype is included. For an omnibus element, the subtype is preceded by "O-" so that it can be differentiated from a symbolic element subtype.

name

The element name in the format:

name[/version][(s-cycle)]

The (s-cycle) field is present only for symbolic elements and is the current absolute cycle number (see Section 3.5.2).

Procedure Table Information

Procedure Table information is produced only for long format when element name and the A, B, O, R, S, V, and Y options are not included.

A program file can have, in order, an Assembler Procedure Table, a COBOL Procedure Table, and a FORTRAN/PLUS Procedure Table. If a table does not have any entries, the following is output:

```
table-name EMPTY
```

If a table has entries, a column header line is output to identify the procedure information. The entries are sorted by procedure name before they are output. FURPUR sorts blocks of 1,680 Assembler Procedure Table entries, 839 COBOL Procedure Table entries, and 1,680 FORTRAN/PLUS Procedure Table entries. If a table exceeds this number of entries, then a separator line with 12 asterisks (*) is output, followed by the next block of procedure table entries. Each block is sorted separately. For the Assembler Procedure Table and the FORTRAN/PLUS Procedure Table, three procedure entries are output per line. For the COBOL Procedure Table, two procedure entries are output per line.

Procedure entries in these tables are associated with symbolic procedure elements. Assembler Procedure Table entries are associated with element type ASM PROC. COBOL Procedure Table entries are associated with element type COB PROC. FORTRAN/PLUS Procedure Table entries are associated with element types FOR PROC and PSP SYMB.

The following is description of the column headers and contents:

D

The deleted entry flag. An asterisk (*) appears in this column if the procedure entry and the element associated with it are marked as deleted.

NAME

The procedure name. Assembler and FORTRAN/PLUS procedures names can be up to 12 characters and COBOL procedure names can be up to 30 characters.

LOCATION

The word location of the start of the symbolic procedure text, relative to the start of the file. This location is always within the element text of the associated procedure element.

LINK

The sequence number of the associated procedure element.

Relocatable Entry Point Table Information

Relocatable Entry Point Table information is produced only for long format when element name and the A, B, O, R, S, V, and Y options are not included.

If the table does not have any entries, the following is output:

```
RELOCATABLE ENTRY POINT TABLE EMPTY
```

If the table has entries, a column header line is output to identify the entry point information. The entries are sorted by entry point name before they are output. FURPUR sorts blocks of 1,680 Relocatable Entry Point Table entries. If the table exceeds this number of entries, then a separator line with 12 asterisks (*) is output, followed by the next block of Relocatable Entry Point Table entries. Each block is sorted separately. Five entry points are output for each line.

Entry points are associated with relocatable elements, with element type RELOCATABLE.

The following is a description of the column headers and contents.

D

The deleted entry flag. This should not be set for the Relocatable Entry Point Table.

NAME

The entry point name, which can be up to 12 characters.

LINK

The sequence number of the associated relocatable element.

Object Module Entry Point Table Information

Object Module Entry Point Table information is produced only for long format when the table is present and when element name and the A, B, O, R, S, V, and Y options are not included. The following separate object module information tables are output:

- The prepped Object Module Entry Point Table lists entry points processed by @PREP (see Section 7.46) or @PACK,P (see Section 7.43).
- The non-prepped Object Module Entry Point Table lists entry points for object modules added to the file after the last @PREP or @PACK,P or all entry points if neither @PREP nor @PACK,P is performed.
- The duplicate Object Module Entry Point Table lists entry points that are included in more than one object module.

Object module entry points are associated with executable elements, with element subtype OBJ-MODULE, BOUND-OM, and ZOOM.

Prepped Object Module Entry Point Table

If there are no prepped object module entry points, the following is output:

```
PREP'D OBJECT MODULE ENTRY POINT TABLE EMPTY
```

If there are prepped object module entry points, a column header line is output to identify the entry point information. The entry points are not sorted and appear in element sequence number order. Two entry points are output for each line.

The following is a description of the column headers and contents.

T

The truncated entry point name flag. An asterisk (*) appears in this column if the entry point name is over 32 characters in length.

NAME

The first 32 characters of the entry point name.

SEQ

The sequence number of the associated object module element.

POS

The position of the entry point within the associated object module element, relative to the other entry points in the element.

TYPE

The entry point type can have one of the following values:

CB	Common bank
CNST	Constant
CODE	
DATA	
FG	Fixed gate subsystem
UNKN	Unknown

Non-Prepped Object Module Entry Point Table

If there are no non-prepped object module entry points, the following is output:

```
NON-PREP'D OBJECT MODULE ENTRY POINTS EMPTY
```

If there are non-prepped object module entry points, they are output in the same format described for prepped object module entry points.

Duplicate Object Module Entry Point Table

The duplicate object module entry point table is not produced if the E option is specified.

If there are no duplicate object module entry points, the following is the output:

```
NO DUPLICATE OM ENTRY POINTS FOUND
```

If there are duplicate prepped and non-prepped object module entry points, the following is output:

```
LIST OF DUPLICATE OM ENTRY POINT NAMES AND ASSOCIATED ELT SEQ #:
```

The duplicate entry points are sorted by entry point name before they are output. One entry point is output per line. Depending on the number of elements that contain the entry point, the entire list of element sequence numbers may require more than one line.

The duplicate entry point information is output in the following format:

```
ENTRY POINT: name FOUND IN ELT SEQ NUM: seq-num-1 seq-num-2 ...
```

where:

name

The first 64 characters of the entry point name.

seq-num-i

The sequence numbers of object module elements that contain the entry point name.

Invalid Element Table Entry Count

Additional verification of each selected element table entry is performed when the long format is requested. Errors are reported for each element and are summarized when the program file has been completely processed.

Fields that are verified are listed in the following table.

Validated Field	How error conditions are indicated
TYPE	Illegal element types are reported as type *UNDEFINED*
DATE and TIME	The DATE and TIME fields contain the text: *ERR* e1, e2 Where: e1 is the AEDIT\$ error code. See the <i>SYSLIB Programming Reference Manual</i> (7833 1733). e2 is the timestamp that is in error. Both errors are reported in octal.
TEXT SIZE	If the element size is illegal, then asterisks indicate the condition.

If invalid field(s) are encountered for any elements, one of these messages is output:

```
*File contains 1 invalid element table entry
*File contains n invalid element table entries
```

where:

n

The count of element table entries with one or more invalid fields.

Deleted Element Summary Information

For all formats the deleted element summary information is output when the file contains at least one deleted element. The information is output in the following format:

```
*File contains e deleted element[s] and f track[s] of deleted text
```

where:

e

The number of deleted elements in the program file.

f

The number of tracks of deleted text associated with the deleted elements. If there is less than one track (one to 63 sectors), then "<1" is included.

If the file contains 100 or more tracks of deleted text, the following is output:

```
*Please @PACK file
```

This message is provided as a reminder that the @PACK statement can be used to release deleted text to the available mass storage pool, making it available for allocation to other files and users.

7.49.4.Examples

The following examples illustrate some of many possible uses of the @PRT statement. This is not a comprehensive set of all possible combinations of options, but this section does present a variety of frequently used combinations.

For reporting of date and time attributes, the time zone mnemonic is blank when the time zone mnemonic is not available. This occurs when the source timestamp is in TDATE\$ format.

```
@PRT PROGFILE.SAM/XYZ
```

The most recent cycle of symbolic element SAM, version XYZ, in program file PROGFILE is listed.

```
@PRT,D PAC1/F14
```

All cataloged files residing on removable disk pack PAC1 are displayed. PAC1 must be an F14 disk pack.

```
@PRT,F BASE.
```

Information describing the file BASE is displayed. BASE can be either temporary or cataloged.

```
@PRT,F TPF$.
```

The output for this temporary file shows that the file is not cached due to invalid equipment type:

```
WCB*TPF$(0),MEMFL/0/TRK/256
HIGHEST TRACK WRITTEN:    0
HIGHEST GRANULE ASG-D:    0
NOT-CACHED DUE TO INVALID EQUIPMENT TYPE,SPEC=D
```

@PRT,FC SYS\$LIB\$*FURPUR.

The output for this cataloged file shows that the file is cached:

```
* * PROJ: ////////// ACCNT: ////////// * *
SYS$LIB$*FURPUR(16),F/0/TRK/9999
MODES: SEMI-PRIVATE,READ,V-OPT,ASG-D
OWNER: HAMANN ACR: RDONLY
CACHED-ON: XPC1,SPEC=D
NO. OF GRANULES ASG-D:      117      GP2=117
HIGHEST GRANULE ASG-D:      143      TOTAL ASSIGNMENTS: 16544
HIGHEST TRACK WRITTEN:      143
CAT:      1995-10-27 04:14:08      LAST REF: 2016-01-16 12:57:14 CST
BACKUP: 2016-02-28 13:19:12 CST
```

@PRT,F J.

The output for this cataloged file shows that 100 tracks are unloaded to tape:

```
* * PROJ: SECURE ACCNT: BBD88C * *
JCB*J(1),F/0/POS/100
MODES: PUBLIC,UNLOD
OWNER: 9JCB
NO. OF GRANULES ASG-D:      100      UNLOADED=100
HIGHEST GRANULE ASG-D:      0      TOTAL ASSIGNMENTS:      3
HIGHEST TRACK WRITTEN:      0
CAT:      2013-08-27 04:14:08      LAST REF: 2016-09-16 12:57:14 CDT
BACKUP: 2016-09-17 13:19:12 CDT
```

@PRT,F S.

The output for this cataloged file shows that it has been written to since the last backup:

```
* * PROJ: SECURE ACCNT: BBD88C * *
JCB*S(1),F/0/TRK/1000
MODES: PUBLIC
OWNER: 9JCB
NO. OF GRANULES ASG-D:      25      GP2=25
HIGHEST GRANULE ASG-D:      51      TOTAL ASSIGNMENTS: 51
HIGHEST TRACK WRITTEN:      51
CAT:      2012-02-27 04:30:18      LAST REF: 2016-08-27 17:09:11 CDT
BACKUP: 2016-08-27 17:45:49 CDT      NOT CURRENT
```

@PRT,F W.

The output for this cataloged file shows that it is disabled; the reason is given in the last line:

```
* * PROJ: SECURE  ACCNT: BBD88C      * *
JCB*W(1),F/0/TRK/1000
MODES: PUBLIC,DSABL
OWNER: 9JCB
NO. OF GRANULES ASG-D:      25      GP2=25
HIGHEST GRANULE ASG-D:      51      TOTAL ASSIGNMENTS: 51
HIGHEST TRACK WRITTEN:      51
CAT:      2010-04-05 12:18:20      LAST REF: 2016-02-07 18:10:09 CST
BACKUP: 2016-02-07 19:42:31 CST
DISABLE: FILE DISABLED DUE TO DIRECTORY ERROR
```

@PRT,F TAPE.

This display shows the typical TAPEMODES line for a reel tape with a density value of 6250 fpi (fpi=frames/bits per inch).

```
WCB*TAPE(0),U36N, T36N2 SCRTCH
TAPEMODES:6250-FPI,ODD-PARITY,8-BIT
```

@PRT,F TAPE.

This display is for a cataloged cartridge tape file with several attributes displayed on the TAPEMODES line.

```
* * DIRECTORY:  STD      * *
* * PROJ: BUILD      ACCNT: BBD88C      * *
TEST1R1*TAPE(1),HIS98,R00059
MODES: PUBLIC
OWNER: ADMIN
TAPEMODES: BUFFIL,8-BIT /CMPLON
CAT:      2010-06-25 12:38:54      LAST REF: 2016-01-24 12:38:54 CST
```

@PRT,LM

File caching information is included for each file, in the same format as for @PRT,F (see the example for @PRT,F SYS\$LIB\$*FURPUR). Retrieving file caching information for each file causes a major increase in system overhead and in the time to process @PRT, so include the M option only if file caching information for all files is needed. The only difference in the display for @PRT,LM is that an error in FILECACHE\$ is encountered during @PRT,LM. The NOT AVAILABLE message appears once. No further attempts is made to access FILECACHE\$.

@PRT,P MERCURY

Information from the master file directory items for all cataloged files whose project-ID is MERCURY is listed subject to current system security restrictions. The project-ID MERCURY must be the same as the project-ID of the run. If not, no listing is generated, unless the run is privileged.

@PRT,T PROGFILE.

The table of contents for program file PROGFILE is listed following the format given in the description. The period must follow the file name; otherwise, the specified name is considered to be an element name in TPF\$.

@PRT,TB F1.

The table of contents for program file F1 is listed, beginning with the most recently inserted elements.

@PRT,TL ELEY

A complete table of contents is listed for all elements in TPF\$ with the name of ELEY and a version name of blanks.

@PRT,TLAY F6F.,F7F.,F8F.

This statement is in error because it did not specify one or more of the subtypes ABS, BOM, OM, SD, and ZM when it specified the AY option combination.

@PRT,TLBAY F4F.,25,OM/SD/ZM.

The table of contents is displayed for up to 25 executable elements that have the subtype OM, SD, or ZM in the file F4F. The most recently inserted elements are displayed first.

@PRT,TLSY F11.OM/BOM.

Because the A option is not specified with the Y option, FURPUR displays an error message stating that the A option must be specified with the Y option. See Appendix A for the message format.

@PRT,TO F2.

All omnibus elements in the table of contents of program file F2 are listed.

@PRT,TS F1.

All symbolic elements in the table of contents of program file F1 are listed.

@PRT,TV FILE./VERS

All elements having a version name of VERS in the table of contents of program file FILE are listed.

@PRT,TSV FAST./V*****

All symbolic elements with a version name beginning with a V in the table of contents of program file FAST are listed.

```
@PRT,TN F5A.,F5B.
```

List only the file and element summary and the deleted element summary for program files F5A and F5B.

7.50.@QUAL and @@QUAL—Specify a Qualifier or Directory for File Names

The @QUAL statement lets you use implied directory-IDs or qualifier names for subsequent control statements involving references to external file names.

You can change the following with the @QUAL statement:

- The default directory-ID
- The implied directory-ID
- The default qualifier
- The implied qualifier

The default directory-ID and default qualifier are the directory-ID and qualifier given on the last @QUAL,D statement performed in a run. The implied directory-ID and implied qualifier are the directory-ID and qualifier given on the last @QUAL (with no options) performed in a run.

Any number of @QUAL statements can appear throughout the runstream. Each time an @QUAL (with no options) statement is encountered, the new directory-ID or qualifier overrides the implied directory-ID or qualifier specified on a previous @QUAL (with no options) statement. Each time an @QUAL,D statement is encountered, the new directory-ID or qualifier overrides the default directory-ID or qualifier specified on a previous @QUAL,D statement.

Format

```
@QUAL[,options] [directory-id#][qualifier]
```

options

D

Specifies the default directory-ID or qualifier. When you specify the directory-ID, your run's default directory is set to it. When you specify the qualifier, your run's default qualifier is set to it.

R

Reverts your run's default and implied directory-ID and qualifier back to the values they had when your run started. When you use this option, you cannot specify a directory-ID or qualifier.

You can use only one option.

If you do not specify an option, you must include a directory-ID, qualifier, or both. If you give the directory-ID, your run's implied directory-ID is set to it. If you give the qualifier, your run's implied qualifier is set to it.

directory-id

Specifies the master file directory used on subsequent file names that do not specify a directory-ID.

Caution

If you change the default directory-ID to SHARED, it affects all file name references (except for the references that explicitly specify a directory-ID), including assign requests issued by the Exec. This can cause errors on subsequent control statements, processor calls, and alternate library references.

qualifier

Specifies the name extension used to qualify subsequent file names.

The qualifier must follow the rules given in the explanation of naming files.

```
@ASG,A STD#PROJECTID*FILEA(2) .
```

7.50.1. Initial and Default Values

Before a run opens, the implied and default qualifiers are initialized to a run's project-ID. The initial values for the default and implied directory-IDs depend on the system's configuration. On systems without the Multi-Host File Sharing (MHFS) feature, before a run begins, the default and implied directory-IDs are initialized to STD. For MHFS systems, the initial values depend on whether the system is a single master file directory search or a double master file directory search.

Whenever you specify a file name without the qualifier or directory-ID or both, the Exec assumes default values for them. Whenever you omit the qualifier, but you specify an asterisk, the implied qualifier is used. If you omit both the asterisk and the qualifier, then the default qualifier is used. The selection of the default directory-ID depends on the system configuration.

7.50.2. Single MFD Search

For single search, the default and implied directory-IDs are initialized to STD.

Systems with MHFS that have single master file directory search and systems without MHFS both respond as follows:

- If you omit the directory-ID, but specify the #, the system uses the implied directory-ID.
- If you omit both the # and the directory-ID, then the default directory-ID is used.

When you specify the file name as **filename*, the implied qualifier is assumed.

When you specify the file name as *#qual*filename*, the implied directory-ID is assumed.

When you specify the file name as *filename*, the default directory-ID and qualifier are assumed.

As an example of a single MFD search, assume the following statements are in the same run and the system has the file sharing feature with single master file directory search:

```
@RUN RUNID,ACCT,PROJECTID
@QUAL STD#1STQUAL
@USE M,FILEX
@FTN FILEA.DO/ABC
@QUAL 2NDQUAL
@ASG,AX FILEJ(5).
@FREE #*FILEA
@QUAL,D SHARED#DEFAULTQUAL
@ASG,A FILEB(-4).
```

The first @QUAL statement specifies that all subsequent file name references having a preceding asterisk are interpreted as having qualifier 1STQUAL. File name references that have a preceding pound sign are interpreted as having directory-ID STD. The @USE statement is interpreted as

```
@USE M, STD#PROJECTID*FILEX (-0)
```

but the file name specified on the @FTN statement shown in the example is interpreted as

```
STD#PROJECTID*FILEA (-0) .DO/ABC
```

The qualifier on the second @QUAL statement supersedes the implied qualifier given on the first @QUAL statement. Therefore, subsequent file name references that have a preceding asterisk have implied qualifier 2NDQUAL. The @ASG statement is interpreted as

```
@ASG, AX STD#PROJECTID*FILEJ (5) .
```

but the file name specified on the @FREE statement is interpreted as

```
STD#2NDQUAL*FILEA (-0) .
```

With the third @QUAL statement, the default directory-ID is changed to SHARED and the default qualifier is changed to DEFAULTQUAL. Therefore, subsequent file name references that do not specify a directory-ID or a pound sign and do not specify a qualifier or an asterisk have a directory-ID of SHARED and a qualifier of DEFAULTQUAL. The file name specified on the @ASG,A statement is interpreted as

```
SHARED#DEFAULTQUAL*FILEB (-4) .
```

7.50.3. Double MFD Search

On systems with MHFS with the double MFD search, the default and implied directory-IDs are not initialized to anything before a run opens.

Whenever you omit the directory-ID, these rules apply:

- When you perform an @ASG or @FREE, the section for the name of the user is searched first to see if the qualifier*filename(cycle) is known to the run. This section is found either by the USE name or by being assigned to the run.
- If the file is assigned to the run, then the directory-ID of the known file is assumed. If the file exists in both master file directories, and both files are known to the run, then the standard master file directory is assumed.
- If the file is not known to the run, then the following apply:
 - ASG,C; @ASG,U; or @CAT), the standard master file directory is searched for the qualifier*filename. If any cycle of the file is found, then STD is the assumed directory-ID. If no cycle is found, then the shared master directory is searched for the qualifier*filename. If any cycle of the file is found, then SHARED is used as a directory-ID.

- Cataloging requests that do not specify a cycle (or use relative cycle -0) are rejected if any cycles of the file exist (both master file directories are searched); otherwise, STD is the assumed directory-ID.
- When a cataloged file is assigned (for example, @ASG,A), the standard master file directory is searched for the file. If any cycle of the file is found, then STD is assumed as the directory-ID. If no cycle is found, then the shared master file directory is searched. If the file is found, then SHARED is assumed and that file is assigned; otherwise, the request is rejected.

Note: *The same qualifier*filename can appear in both the STD and SHARED master file directories. Be careful, when assigning a file without a directory-ID, to make sure that the correct file is assigned when duplicate qualifier*filename cycle sets exist.*

- Once you enter @QUAL,D, the default directory has a value and the assumed defaults are the same as on a system that does not have the double MFD search configured. Similarly, once an @QUAL (with no options) is entered with a directory-ID specified, the implied directory-ID has a value and is used as on a system that does not have the double MFD search configured.

The following examples show double MFD search.

Assume the following statements are in the same run, and the system has file sharing with double master file directory search:

```
@RUN RUNID,ACCT,PROJECTID
@CAT FILEA(2) .
@CAT SHARED#FILEA(2) .
@ASG,A FILEA(2) .
@USE XX,FILEB
@ASG,C XX.
@ASG,A #FILEA(2) .
@QUAL,D STD#DEFAULTQUAL
@QUAL STD#
@ASG,A FILEC(-3) .
```

The three statements in sequence following the @RUN statement show that an assign statement assigns the file in the standard master file directory if the file exists in both the master file directories. The images are interpreted as

```
@CAT STD#PROJECTID*FILEA(2) .
@CAT SHARED#PROJECTID*FILEA(2) .
@ASG,A STD#PROJECTID*FILEA(2) .
```

The next two statements show that cataloging assumes STD. In addition, the @USE statement does not select a default directory-ID. This is not determined until the file is assigned. The images are interpreted as

```
@USE XX,PROJECTID*FILEB
@ASG,C STD#PROJECTID*FILEB.
```

The @ASG,A #FILEA(2). statement shows that the assignment is for a file that is already assigned.

After the two @QUAL statements are done, the default and implied directory-ID s both have valid values. Hence, any file name that does not specify a directory-ID or that has only a # specified always has a default directory. The Exec now supplies defaults that are exactly the same as on a system that does not have the double MFD search configured.

The @ASG is interpreted as

```
@ASG,A STD#DEFAULTQUAL*FILEC(-3) .
```

7.51.@REWIND—Rewind Tape Files

Rewinds magnetic tape files back to the load point of the first reel.

All parameters in the @REWIND statement are optional.

Format

```
@REWIND[,options] [filename-1,filename-2,...,filename-n]
```

options

Are the C and I options.

If the I option is specified, the tape file is rewound with interlock; if omitted, the tape file is rewound without interlock.

Use of the C option on any FURPUR command will prevent execution of ER ERR\$ on error exit if FURPUR is running in breakpoint or batch. The C option should be used with caution because a bad result of an erroring FURPUR command could be input to a process later on in the runstream.

The C option is valid for all FURPUR commands

filename-n

Specifies the tape files to be rewound.

7.52.@@RLD—Cause Screen Images to Scroll Down

The @@RLD statement causes the lines of the display to scroll down instead of up, as they normally do.

This statement could be used, for example, when there is something important in the top part of the screen and you do not want it to scroll off the screen.

Format

```
@@RLD n
```

n indicates the line on the display from which output is to scroll down. The lines are counted down from the top.

You can restore the display to its normal setting with an @@RLU 23 statement if your terminal has a standard 24-line display.

Examples

```
@@RLD 1
```

All the images on the screen scroll down starting at line 1.

```
@@RLD 5
```

All the images on the screen scroll down starting at line 5. The images preceding line 5 are saved.

7.53.@@RLU—Cause Screen Images to Scroll Up

The @@RLU statement causes the images displayed to scroll up from a specified line on the screen. This statement can be used to reverse the action taken by the @@RLD statement.

Format

```
@@RLU n
```

n indicates the line on the display from which output is to scroll up. The lines are counted down from the top.

Examples

```
@@RLU 22
```

The images scroll up starting at line 22. The images after line 22 are saved.

```
@@RLU 23
```

This is the normal scrolling.

7.54.@@RQUE—Stop an Output File and Place It in a Queue

The @@RQUE statement stops printing the present batch output file, requeues it for later printing (@@SEND statement), and returns the terminal to inactive mode. You can issue this statement from any terminal in remote batch mode.

Format

```
@@RQUE
```

7.55.@RSTRT—Restart a Run

The @RSTRT statement lets you restore information from the checkpointed data and resume processing. See the *CKRS Programming Reference Manual* (7830 7360) for more information.

Format

```
@RSTRT[,scheduling-priority/options/processor-dispatching-priority]  
run-id,[acct-id/user-id,project-id, filename.element,ckpt-number,  
eqpmnt-type,reel-1/... reel-n]
```

scheduling-priority

Indicates what preference the run must be given in relation to all other runs that are available for execution. The scheduling priority can be any letter from A through Z, where A is the highest priority and Z is the lowest. If you specify a scheduling priority that is higher than is permitted for the account number (acct-ID) you specify, the Exec changes the priority to the highest priority allowed for that account.

processor-dispatching-priority

Indicates the processor dispatching priority that is used to place the run's activities into a switching group once the run is opened. The processor dispatching priority can be any letter from A through Z, where Z is the lowest.

options

A

Does not terminate the run for certain nonfatal errors.

B

Uses the saved (checkpointed) absolute element if the absolute element cannot be found in the original program file. Restart always restores the current state of the program.

J

Assigns an unlabeled checkpoint tape. This must agree with the @CKPT statement.

M

Prints full diagnostic messages without diagnostic snapshots while processing the full checkpoint data.

N

Does not print error message or diagnostics in the run's print file.

P

Displays error message on the console.

Q

Prints diagnostic snapshots while processing full restart data.

R

Restores the saved mass storage symbiont files. (Tape symbiont files are always restored.)

S

Performs only a partial restart.

U

Prevents verification of files. Use this option to bypass file verification if the CKRS_FILE_VERIFY parameter is set to true.

V

Forces verification of files. Use this option to force file verification if the CKRS_FILE_VERIFY parameter is set to false.

Note: If both the U and V options are specified on a restart request, the U option overrides the V option.

Y

Lets the restart initialization processor print diagnostic snapshots in the run's print file.

Z

Prints diagnostic snapshots while processing partial restart data.

run-id

Identifies the run-ID for the restarted run. The original run-ID or the system generated run-ID of the restarted run must match the generated run-ID of the run that took the checkpoint.

acct-id

Specifies the account number. This is optional. However, the account number must match the account number of the checkpointed run. This ensures that all private files assigned to the checkpointed run is assigned to the restarted run.

user-id

Identifies the user-ID of the checkpointed run. This ensures that the restarted run is able to assign all of the owned files that were assigned to the checkpointed run.

project-id

Specifies the project name for the restarted run. If you do not specify one, the project-ID is Q\$Q\$Q\$.

filename

Identifies the external file name of the checkpoint data. The file name is optional if you specify reel-number-n and the tape is unlabeled. A file name without a period is treated as an element in TPF\$. A file name can include the directory-ID.

element

Specifies the element from which the checkpoint data can be retrieved. This element cannot reside in a Large Element Program File (LEPF).

ckpt-number

Specifies the checkpoint number; you must include the ckpt-number if the checkpoint is on tape. The ckpt-number is a sequence number that specifies which checkpoint on the tape to use.

eqpmnt-type

Specifies the equipment type.

reel-n

Identifies the reel number of the tape containing the checkpoint.

Examples

See the *CKRS Programming Reference Manual* (7830 7360) for examples of @RSTRT.

7.56.@RUN—Identify a Run to the Exec

A run is a sequence of tasks linked to form a self-contained unit of work. A run is controlled by a series of control statements, which tell the system what to do.

Start a run with an @RUN statement and end it with an @FIN statement. Between the two, you can include the following:

- Other control statements to create or manipulate files or run a program
- Lines of a source program to be compiled and executed
- Instructions to a system processor
- Data for an executing program to process

If your user-ID is configured to generate the @RUN statement for you, you do not have to enter an @RUN statement; the system does it for you, based on the information you provide when you sign on. However, if your user-ID is configured for the @RUN statement to be user-entered, you must enter an @RUN statement.

There are two basic types of runs: batch and demand.

7.56.1. Batch Run

A batch run is a run you submit as a complete job. It is designed to run independently, from start to finish, without further instructions from you. Use batch runs for routine and repetitive applications that process large volumes of data or require a lot of processing time.

When you submit a batch run for processing, the Exec stores the runstream in a file (READ\$) until the system has enough resources to open the run. Even after the Exec opens a run, the run does not have exclusive use of the system resources; it must share them with other active runs.

As the run is processed, the system reads the images from READ\$ and processes them in order. Any output produced during the run is stored in a file (PRINT\$). When the run terminates, this output is printed.

There are two kinds of batch runs: canned runs and remote batch runs.

Canned Runs

Canned runs are batch runstreams that you prepare in advance, by using a text editor such as EDIT 1100 (the IPF 1100 editor). You store the canned runstream in an SDF file or symbolic element for later use.

Using canned runs is a timesaving technique you can use for repetitive functions and for runstreams that you want to use more than once.

There are two kinds of canned runstreams: full and partial. The main difference is that full canned runstreams include @RUN and @FIN statements, while partial canned runstreams do not.

Full canned runstreams are complete batch runs that you start by using the @START statement. Partial canned runstreams are a sequence of control statements (and other input, if desired) that you add to an active run by using the @ADD statement.

The following are examples of canned runstreams that can be initiated with @START:

```
@RUN R231,03412,CAPER,10/100 . last run
```

Run R231 of project CAPER is assigned standard system scheduling and processor-dispatching priorities by the Exec. Expenses incurred by the run are changed to account 03412. Run time is estimated at 10 minutes; results are required an hour after the run is submitted. The run cannot terminate automatically when run time or page estimates are exceeded unless automatic termination was provided for when the system was configured. The Exec uses the system's standard page-output estimate because that field is omitted. The run is scheduled by the Exec for execution according to its priority (start-time is omitted). This run statement includes a comment (.last run).

```
@RUN,C/P 716,03412,CAPER,,300
```

Run 716 of project CAPER has a scheduling priority of C. The P option indicates that the run must be terminated automatically if more than 300 pages of output are produced. Expenses incurred by the run are charged to account 03412. The Exec uses the system's standard value for estimated run time, since none is specified in the statement, and assumes that there is no deadline, since none is provided. The Exec considers the run immediately (because start-time is omitted) and schedules the run according to its priority.

```
@RUN,A 201,90431010,EXODUS1,S50,50,D830
```

Run 201 of project EXODUS1 has a scheduling priority of A. Run expenses are charged to account 90431010. Run-time is estimated at 50 seconds with an expected output of less than 50 pages. The start-time field specifies that the run not be considered for execution before 8:30 a.m. Unless provided for at system generation, automatic run termination does not occur if running time or output estimates are exceeded, since these options are not specified.

Remote Batch Run

You enter a remote batch run at a terminal without saving the runstream in a file or element. You enter the entire runstream at a terminal, much as you would for a demand run. To indicate that you want to start a remote batch run and not a demand run from a demand terminal, use the B option on the @RUN statement (see Section 7.56). As you enter the runstream, it is stored in a READ\$ file and then queued for processing like other batch runs after the @FIN statement is entered. Responses to statements do not appear on your terminal, as they would if you were entering a demand run; responses and all other output are stored in a PRINT\$ file and printed when your run terminates.

Starting a Batch Run

You must include the @PASSWD statement in a batch run. For more information on the @PASSWD statement (see Section 7.44).

Notes:

- *You cannot start a remote batch job if your user-ID is configured for the @RUN statement to be always system generated.*
- *Do not initiate long-running, computing-intensive programs. Use the B option with the @RUN statement to improve system scheduling and to free the terminal for further use.*
- *When you enter the @RUN statement from a remote terminal, use the N option. This inhibits postmortem dumping and dynamic diagnostic dumping.*

Format

```
@RUN[,scheduling-priority/options/processor-dispatching-priority run-id,acct-id/user-id,project-id,run-time/deadline,pages/cards,start-time]
```

scheduling-priority

Indicates what preference the run must be given in relation to all other runs that are available for execution. The scheduling priority can be any letter from A through Z, where A is the highest priority and Z is the lowest. If you specify a scheduling priority that is higher than is permitted for the account number (acct-id) you specify, the Exec changes the priority to the highest priority allowed for that account.

Examples:

```
@run,B//A runid.... (scheduling priority B, processor dispatching priority A)
```

```
@run,//A runid .... (scheduling priority from default, processor dispatching priority A)
```

Notes:

- *Normally, when the Exec is holding a run in backlog because resources are not available (such as a file), other lower-priority runs are not allowed to start. To avoid holding up the start of runs, start as many runs as possible at the same priority. However, if the dynamic configuration tag `bypass_facility_held_runs` is set, lower priority runs can open.*
- *When a batch run is initiated and its required resources are unavailable, the run is placed on the hold queue. The status of such runs can be viewed in the backlog queue using the BL D console command, which provides detailed information about held runs.*

The hold queue is monitored to evaluate the currently queued actives. Periodically all runs held on the hold queue are removed and reevaluated. The run is then either allowed to proceed, or it is requeued to the hold queue.

For runs waiting on the exclusive use of a local file, the Exec issues a console message indicating how long the run has been waiting. This message is generated approximately every 20 minutes, based on the time the run was queued to the hold queue.

If a run is waiting for local resources, the run is removed from (and potentially requeued to) the hold queue every 60 minutes. If a run is waiting on shared resources, the run is removed from (and potentially requeued to) the hold queues every 5 seconds. Hence these console messages never appear for runs held for shared files.

Maximum and Default Processor Dispatching Priority vs. Configuration

The following table shows the maximum and default priority information used for a run based on the values of USERONCONF (user-ID enforcement if TRUE) and RUNLVLENF (run level enforcement of quota sets if TRUE).

USERONCONF	RUNLVLENF	Maximum Processor Dispatching Priority	Default Priority Dispatching Priority
False	False	"A"	"M"
False	True	From quota set	From basic accounting block
True	True	From quota set	From basic accounting block
True	False	From basic accounting block	From basic accounting block

Notes:

- *These values follow the same pattern set by the maximum and default scheduling priority used for a run.*
- *If the processor dispatching priority is specified on the @RUN but a maximum is not present in the accounting file, the maximum run scheduling priority is used as a maximum for the processor dispatching priority.*
- *If the processor dispatching priority is not specified on the @RUN and a default is not present in the accounting file, the run scheduling priority (from @RUN statement or default) is used for the processor dispatching priority.*

options

Your site can be configured to use the terminations specified by the P and T options, whether or not the options are specified.

processor-dispatching-priority

Indicates the processor dispatching priority that is used to place the run's activities into a switching group once the run is opened. The processor dispatching priority can be any letter from A through Z, where Z is the lowest.

General Options**B**

Treats the run input from a demand terminal as batch input and schedules the run as a batch run.

D

Allows more files to be assigned to the run. For each file assigned to a run, several data structures are needed to manage that assignment. This option causes the device area descriptor (DAD) tables to be moved to a separate run-level bank, thus freeing up PCT space for additional file assignments. See the *Exec Administration Reference Manual* (7831 0323) for further information on this option.

This option is ignored if the configuration parameter User_DAD_Bank is zero.

O

Allocates DAD tables for assigned files in the run's PCT when the configuration parameter User_DAD_Bank is set to 2. This option is provided to maintain compatibility regarding the location of the DAD tables.

If both the D option and O option are used, then the D option is ignored.

Note: Options E through M, which control the minimum size of the PCT, are not meaningful on XPA hardware because PCTs are given the maximum configured PCT size allowable by your site (see configuration tag PCTMAX).

P

Terminates the run if the page estimate is exceeded.

R

Reschedules the run if a recoverable system failure occurs. If the R option is specified and the associated run is open at the time of system failure, the run is reinstated in the schedule queue during system recovery. When the run is selected for reopening, bit 4 is set in the run's condition word.

The following actions and restrictions govern the use of the R option:

- All queued print and punch files generated by the open run being rescheduled are removed from the print and punch queues and decataloged during the recovery, with the exception of user-defined files that have been sent to a printer by using @SYM with the U option. These files are not decataloged; however, they are removed from the appropriate print or punch queue.
- The R option is ignored on all demand runs and these runs are never rescheduled.
- The user is responsible for handling facility assignments in the run to resolve potential conflicts during the restart of the run. The condition word can be examined by the run to determine whether the run has been restarted.

S

Processes this run in sequence with the previous run submitted from the same site-ID; holds this run until the previous run completes. The S option is used for runs entered with @RUN,/B or from RSI\$ remote batch terminals. It is ignored for demand runs and for batch runs started with the ST keyin or the @START control statement.

T

Terminates the run if the standard unit of processing (SUP) usage estimate is exceeded.

X

Disregards system holds when opening run (requires a security privilege). The name of the new dynamic configuration parameter is batch_run_x_option_modes. On Security Level 1 or higher systems, the security privilege SSRUNXOPT is required for runs specifying the X option on @RUN or @START control statements and the ST keyin.

Options for Postmortem and Dynamic Dumping

N

Does not write the program or processor to the diagnostic file. The PMD statements are not honored, but are printed along with the message PMD NOT ALLOWED. Diagnostic dumps are not printed. This option overrides the W and Y options.

Y

Always writes the program or processor to the diagnostic file, regardless of the source of the program (even for system processors from SYS\$*LIB\$ or SYS\$LIB\$ files). Appropriate PMD statements are honored, and diagnostic dumps are printed. Use this option only when a program in the SYS\$*LIB\$ or SYS\$LIB\$ files is being debugged.

Note: System processors are not currently allowed to be zero overhead object modules (ZOOM) or object modules.

W

Specifies that only user programs that error are to be written to the diagnostic file. In addition, for batch programs, the appropriate PMD processor is loaded to dump the program in error. An automatic @PMD call assumes the L and P options; an automatic @PADS call assumes the DIAG\$ standard procedure; and an automatic @DAP call assumes the NPE\$PMD function. Diagnostic dumps are automatically printed only if the program taking a diagnostic dump terminates in error. The W option saves overhead.

The time at which dumps are taken is different for basic mode and extended mode programs. For basic mode programs, dumps are performed at program termination time. For extended mode programs (object modules or ZOOMs), dumps are performed for each activity when a contingency occurs (contingencies include both error conditions and normal activity termination).

Note: *The diagnostic file formats for basic mode programs and extended programs are completely different.*

When you specify both the W and Y options, all user programs and LIB\$ processors are written to DIAG\$. In addition, an automatic postmortem dump of error user programs and LIB\$ processors is received.

When you omit the N, Y, and W options, the program, unless it is a processor loaded from the system library (LIB\$) or an alternate library, is dumped to the diagnostic file (DIAG\$), and normal postmortem dump action occurs. See the *PMD Programming Reference Manual* (7862 7288) for information concerning normal PMD action. PMD statements encountered after a LIB\$ processor call statement are not honored and appear on the printout, along with the message

PMD NOT ALLOWED

Absolute programs collected with the TYPE NPEDIAG directive only has activities that terminate in error written to the diagnostic file, regardless of the W option or no-option specification. An automatic PMD is only performed according to the W and Y options.

The rules governing postmortem dumping also apply to the printout for diagnostic dumping. The no-option mode must be used whenever one or more programs other than those in library LIB\$ are being debugged. This is the normal mode for debugging runs.

If you use an @@CONS E,N *run-id* command, the N option (no postmortem dump) on the @@CONS E command overrides the Y and W options.

run-id

Identifies the run to the Exec. The run-ID consists of 1 to 6 characters in any combination of letters and numbers. If you omit run-ID, the Exec uses RUN000. If you specify a run-ID that duplicates a run-ID already in the system, the Exec modifies your newly submitted run-ID to make it unique. Both the original and the modified run-IDs appear on the operator's console, in the system log, and in the printer listing.

acct-id

Specifies an account number used for billing purposes. The account number consists of 1 to 12 characters in any combination of letters, numbers, periods (.), and hyphens (-).

If your site has Fundamental Security, a file can be private by account number (depending on how security is configured at your site). This means files can be assigned only by runs whose account number is the same as the account number of the run that created the file.

If your system has Security Level 1 or higher and user-IDs in your security management product are set up to create owned files (this is the default), files are private by the user-ID that created the file.

If you omit *acct-ID*, the Exec uses 000000.

A system is configured as either restrictive or nonrestrictive.

If the system is restrictive, all runs using an account number that is not registered with the Exec are rejected.

If the system is nonrestrictive and the account number submitted is not registered with the Exec, the system operator is notified and can accept or reject the run.

user-id

Identifies the user of the account specified (*acct-ID*). The user-ID consists of 1 to 12 characters in any combination of letters, numbers, periods (.), and hyphens (-). If your site requires a user-ID and you do not enter one, the system uses the user-ID you entered at sign-on time. If you do enter a user-ID on the @RUN statement, it must match the one you entered when you signed on or the system rejects the run.

project-id

Classifies the run for file access and accounting purposes. The project-ID is also used as the qualifier for file names when you do not provide one. Project-IDs can be up to 12 characters and can include any combination of letters, numbers, hyphens (-), and dollar signs (\$).

Unowned files can be private by project-ID (depending on how security is configured at your site). This means files can be assigned only by runs whose project-ID is the same as the project-ID of the run that created the file.

Owned files are private by the user-ID that created the file.

If you omit project-ID, the Exec uses Q\$Q\$Q\$.

run-time

Estimates run time in standard units of processing (SUP). This is roughly the same as minutes, but you can use a leading S to specify seconds. When your run exceeds the estimated run time, the Exec does the following:

- Notifies the operator, allowing the operator to manually terminate the run
- Terminates the run if the T option is specified or if automatic run termination was specified when the system was configured

The Exec can be configured to maintain a quota limit on run time for each account. If the Exec is maintaining a quota limit and you specify a run time that exceeds the quota limit, the quota limit is used. In addition, if you do not specify a run time at all, the Exec uses the quota limit.

When the Exec is not maintaining a quota limit and you do not specify a run time, the Exec uses the system standard value of 2,184 minutes. If you specify a run time greater than 2,184 minutes, the Exec replaces it with 2,184.

In all cases, when you specify a run time of 0, the Exec uses a run time of 1 minute.

deadline

Is a request that a batch run be completed by a specific deadline. The deadline is ignored if no estimated run time and the run is started as a normal batch run. As the deadline approaches and the batch run has not been started yet, the system raises the run's priority to complete processing by the requested deadline and at that time the run becomes a deadline batch run.

Deadline time can be specified as either elapsed time since run submission or time of day. Time of day is based on a 24-hour clock using the local time base. The format is hhmm for elapsed time and Dhhmm for time of day. Both are limited to maximum of 2400, that is 24 hours. If the specified time of day is earlier than the current time, the system considers it as the time of the next day.

Caution

Be careful if you decide to use deadline runs. System performance can be severely hampered as the deadline time approaches and your run priority increases. Also, depending on how your site has set Quota limits, deadline processing can require privileges that you do not have. If you want to use deadline runs, check with your site administrator first.

Indicate deadline as either elapsed time since run submission or the time of day. Enter the deadline in the following format:

[D] *hhmm*

D

Indicates that the deadline is a time of day (within the next 24 hours). If you omit the D prefix, the deadline is the elapsed time since the start time (the deadline is then the length of time during which the run must be completed).

hh

Specifies hours.

mm

Specifies minutes.

You do not need to use leading zeros for *hhmm*.

Here are some examples of deadlines:

D910

Run must be done by 9:10 a.m. (on the day of run submission if submitted at or before 9:10 a.m.; on the next day if submitted after 9:10 a.m.)

D2110

Run must be done by 9:10 p.m. (on the day of run submission if submitted at or before 9:10 p.m.; on the next day if submitted after 9:10 p.m.)

100

Run must be done 1 hour after submission.

230

Run must be done 2.5 hours after submission.

If the run is not completed by the specified deadline, the Exec raises the priority of the run so that it completes as soon after the specified deadline as possible. This action, however, can degrade system operation since other runs, including demand runs, can be given relatively fewer resources.

A system can be configured to prevent the use of deadlines. If you specify a deadline when it is not allowed, the deadline is ignored.

pages

Estimates the number of printed pages expected as output from the run.

The Exec can be configured to maintain a quota limit on pages for each account. If the Exec is maintaining a quota limit and you specify a value that exceeds the quota limit, the quota limit is used. If you do not specify pages, the Exec uses the quota limit.

When the Exec is not maintaining quota limits and you do not specify pages, the Exec uses a system standard value. If the page estimate is exceeded, the Exec does one of the following:

- Notifies the operator and gives the operator the option of terminating the run
- Terminates the run if the P option is specified or if automatic run termination was specified when the system was configured

When the Exec is not maintaining quota limits and you specify pages, the maximum number allowed is 131,071. If you specify more than 131,071, the Exec replaces it with 131,071.

cards

Estimates the number of cards expected as output from the run

The Exec can be configured to maintain a quota limit on cards for each account. If the Exec is maintaining a quota limit and you specify a value that exceeds the quota limit, the quota limit is used. If you do not specify *cards*, the Exec uses the quota limit.

When the Exec is not maintaining quota limits and you do not specify cards, the Exec uses a system standard value. If the cards estimate is exceeded, the Exec does one of the following:

- Notifies the operator and gives the operator the option of terminating the run
- Terminates the run if the C option or automatic run termination was specified at system generation

When the Exec is not maintaining quota limits and you specify cards, the maximum number allowed is 131,071. If the cards estimate exceeds 131,071, the Exec replaces it with 131,071.

start-time

Specifies the earliest time when a run can be processed.

Start-time is a request to delay the start of a batch run until a specified time. Before that time is reached, the Exec places the run in a hold state. Once the start-time has been reached, the Exec releases the run from the hold state and schedules it under normal priority rules. The Exec does not interpret other @RUN statement fields such as deadline and priority until the start-time has been reached. The start-time is considered the time of run submission when considering the deadline.

Start-time time can be specified as either elapsed time since run submission or time of day. Time of day is based on a 24-hour clock using the local time base. The format is hhmm for elapsed time and Dhhmm for time of day. Both are limited to maximum of 2400, that is 24 hours. If the specified time of day is earlier than the current time, the system considers it as the time of the next day.

[D] *hhmm*

D

Indicates that the start-time is a time of day (within the next 24 hours). If you omit the D prefix, the start-time is elapsed time from when the Exec reads this @RUN statement to when the Exec starts the run.

hh

Specifies hours.

mm

Specifies minutes.

You do not need to use leading zeros for *hhmm*, but if you specify a *start-time* greater than 24 hours, your run is flagged for error termination.

Remote Batch Run Examples

```
@RUN, /B BCD, 802910, CAPRICORN
```

BCD is the run-ID, 802910 is the account number, and CAPRICORN is the project-ID.

```
@RUN, A/BR PLB, 729610/RMH, MASTER
```

This remote batch run has a scheduling priority of A. Because of the R option, the run is rescheduled if a recoverable system failure occurs. The account number is 729610 with user-ID RMH, and the project-ID is MASTER. RMH must also be the user-ID entered at sign-on time.

7.56.2. Demand Run

A demand run is an interactive run. You enter ECL and other types of statements one at a time at a terminal and generally wait for each statement to be processed before entering another. Demand runs let you correct errors right away and get answers more quickly than you could by using batch runs.

To start a demand run after signing on the system, you can have to enter an @RUN statement, or the system can automatically generate one for you, depending on how your user-ID is configured. The system can prompt you for certain kinds of information after you sign on.

Demand runs do not use READ\$ and PRINT\$ files for input and output. All input is taken directly from your keyboard. All output is displayed directly on the screen of your terminal.

To terminate a demand run, use the @FIN statement.

You can still add statements and data from files and send output to printers, but you must use special statements to do so. @ADD lets you add data files or symbolic elements to your run (see Section 7.1). @BRKPT and @SYM let you save and direct output to a file or a printer (see Section 7.3 and Section 0).

Note: *Do not execute long-running, computing-intensive programs while in demand mode; use batch runs instead. This improves system scheduling and frees the terminal for other uses.*

Starting a Demand Run

Format

```
@RUN[, scheduling-priority/options processor-dispatching-priority  
      run-id,acct-id/user-id,project-id]
```

scheduling-priority

Specifies printing priority of A through Z in the print queue.

processor-dispatching-priority

Indicates the processor dispatching priority that is used to place the run's activities into a switching group once the run is opened. The processor dispatching priority can be any letter from A through Z, where Z is the lowest.

options

Your site can be configured to use the terminations specified by the P and T options whether or not the options are specified.

General Options

B

Treats the run input from a demand terminal as batch input and schedules the run as a batch run.

D

Allows more files to be assigned to the run. For each file assigned to a run, several data structures are needed to manage that assignment. This option causes the device area descriptor (DAD) tables to be moved to a separate run-level bank, thus freeing up PCT space for additional file assignments. See the *Exec Administration Reference Manual* (7831 0323) for further information on this option.

This option is ignored if the configuration parameter User_DAD_Bank is zero.

O

Allocates DAD tables for assigned files in the run's PCT when the configuration parameter User_DAD_Bank is set to 2. This option is provided to maintain compatibility regarding the location of the DAD tables.

If both the D option and O option are used, then the D option is ignored.

Note: Options E through M, which control the minimum size of the PCT, are not meaningful on XPA hardware because PCTs are given the maximum configured PCT size allowable by your site (see configuration tag PCTMAX).

Options for Postmortem and Dynamic Dumping

N

Does not write the program or processor to the diagnostic file upon run completion. PMD statements are not honored, but are printed along with the message 'PMD NOT ALLOWED'. Diagnostic dumps are not printed. Specify the N option whenever the run is being executed for production purposes because this eliminates the overhead of saving the program for a possible PMD statement. The N option provides minimum overhead. This option overrides the Y option.

Y

Writes all programs and processors in the run to the diagnostic file. All PMD statements are honored, and diagnostic dumps are printed. Use this option only when a program in SYS\$*LIB\$ or the SYS\$LIB\$ files is being debugged.

When you omit the N and Y options, all programs, except processors loaded from the system library (LIB\$) or an installable library, are dumped to the diagnostic file (DIAG\$), and normal postmortem dump action occurs. See the *PMD Programming Reference Manual* (7862 7288) for details.

PMD statements encountered after a processor call statement are not honored and are printed out along with the following message:

PMD NOT ALLOWED

The rules governing postmortem dumping also apply to the printout for diagnostic dumping. Use the no-option mode whenever one or more programs, other than those in the library LIB\$ or an installable library, are being debugged. This is the normal mode for debugging runs.

If you use an @@CONS E,N *run-id* command, the N option (no postmortem dump) on the @@CONS E command overrides the Y option.

run-id

The rules governing postmortem dumping also apply to the printout for diagnostic dumping. Use the no-option mode whenever one or more programs, other than those in the library LIB\$ or an installable library, are being debugged. This is the normal mode for debugging runs.

The run-ID identifies the run to the Exec. It consists of 1 to 6 characters in any combination of letters and numbers.

If you omit *run-id*, the Exec uses RUN000.

If you specify a run-ID that duplicates a run-ID already in the system, the Exec modifies your newly submitted run-ID to make it unique. Both the original and the modified run-IDs appear on the operator's console, in the system log, and in the printer listing.

acct-id

Specifies an account number. The account number consists of 1 to 12 characters in any combination of letters, numbers, periods (.), and hyphens (-).

Unowned files can be private by account number (depending on how security is configured at your site). This means files can be assigned only by runs whose account number is the same as the account number of the run that created the file.

If your system has Security Level 1 or higher and user-IDs in your security management product are set up to create owned files (this is the default), files are private by the user-ID that created the file.

If you omit *acct-id*, the Exec displays

MISSING ACCOUNT

A system is configured as either restrictive or nonrestrictive.

If the system is restrictive, any run using an account number that is not registered with the Exec is rejected.

If the system is nonrestrictive and the account number submitted is not registered with the Exec, the operator is notified and can accept or reject the run.

user-id

The user-ID subfield is optional. If your site requires a user-ID and you do not enter one, the system uses the user-ID you entered at sign-on time. If you do enter a user-ID, it must be the same as the one you used to sign on, or the system rejects the run. If user-IDs are required by your site, a user-ID must be specified on batch jobs.

project-id

Classifies the run for file access and accounting purposes. The project-ID is also used as the qualifier for file names when you have not provided one. The project-ID can be up to 12 characters and can include any combination of letters, numbers, hyphens (-), and dollar signs (\$).

Unowned files can be private by project-ID (depending on how security is configured at your site). This means files can be assigned only by runs whose project-ID is the same as the project-ID of the run that created the file.

Owned files are private by the user-ID that created the file.

If you omit project-ID, the Exec uses Q\$Q\$Q\$ as the project-ID.

Example

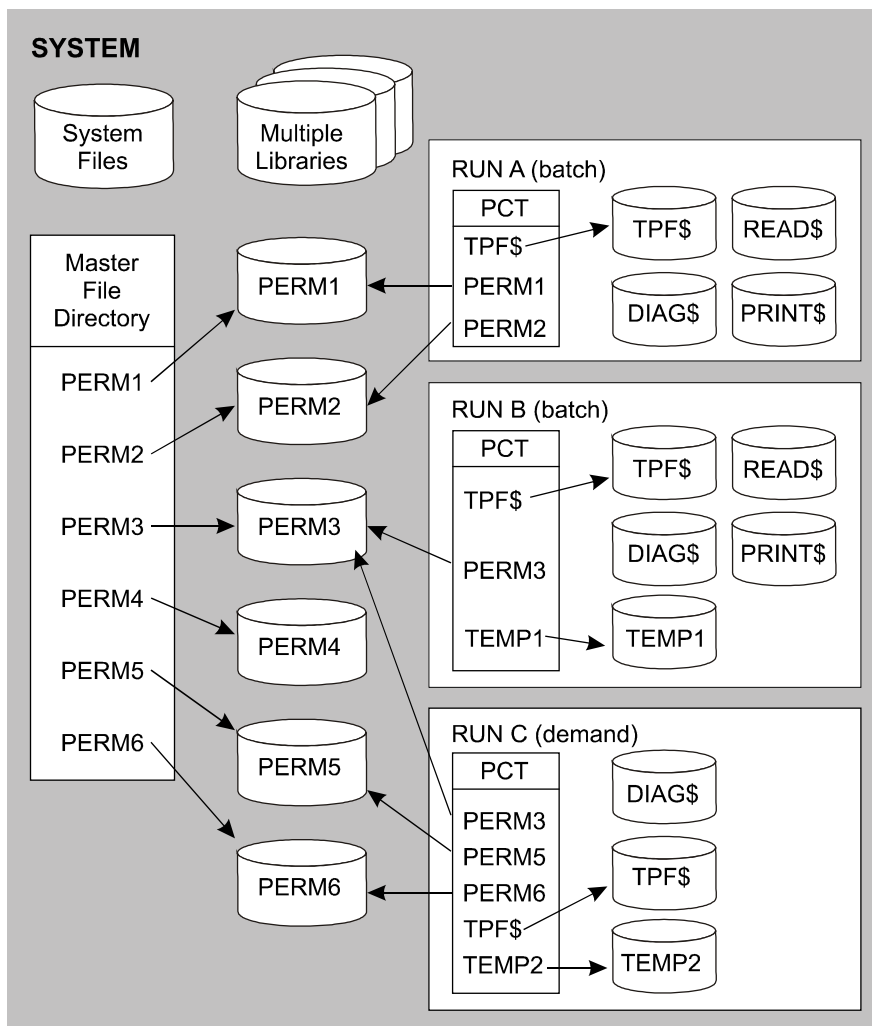
```
@RUN LMN,766120,PROD-SCHED
```

This is an example of how your @RUN statement might look if your user-ID is configured to either require you or allow you to enter your own @RUN statement. LMN is the run-ID, 766120 is the account number, and PROD-SCHED is the project-ID.

7.56.3. How Runs and Files Work Together

Figure 7-4 shows how the run and file concepts fit together. In this example

- All permanent files (PERM1 through PERM6) are listed in the master file directory; the temporary files (TEMP1 and TEMP2) are not.
- All files assigned to a run are listed in the run's program control table (PCT).
- Runs A and B are batch runs, since they include READ\$ and PRINT\$ files. Run C is a demand run, since it does not include READ\$ and PRINT\$ files.



000792

Figure 7-4. Run and File Concepts

7.57.@@SEND—Receive a Print File Queued to Your User ID

The @@SEND statement tells the system to send you the print files queued to your user-ID or your terminal's site-ID.

You cannot use the @@SEND statement during an active run. The best time to use @@SEND is immediately after you have terminated a run (@FIN), but before you have logged off the system (@@TERM) or the network (\$\$SOFF).

Normally, the output is just displayed at your terminal. If a local printer is connected to your terminal, you can also receive a printout of the queued output by entering the @@PRNT statement before you use the @@SEND statement.

Note: You cannot be allowed to use @@SEND, depending on your system's security configuration. Check with your site administrator or security officer.

Format

```
@@SEND[,U] [filename]
```

U

Sends you the files queued to your user-ID. If you do not specify this option, the files queued to your terminal's site-ID are sent. You can queue files to a site-ID or user-ID with the @SYM statement.

filename

Identifies a specific file you want sent to your terminal. If you do not specify a file name, you receive all files queued to your user-ID or site-ID.

The following are examples of the @SEND statement that lets you print files queued to your user-ID or your terminal's site-ID:

```
@@SEND
```

This statement sends you all print files queued to your terminal's site-ID.

```
@@SEND,U
```

This statement sends you all print files queued to your user-ID.

```
@@SEND URSA*TEST1.
```

This statement sends you print file URSA*TEST1, which is queued to your terminal's site-ID.

```
@@SEND,U URSA*TEST2.
```

This statement sends you print file URSA*TEST2, which is queued to your user-ID.

7.58.@@SETC—Store a Value in a Condition Word

The @SETC statement stores values in the condition word.

Format

```
@SETC[,options] [f/]value[/j]
```

options

A

Clears bit 5 of the condition word. This allows a normal termination of this run.

I

Sets bit 5 of the condition word. This inhibits run termination after a program error terminates. The I option permits runs with independent tasks or tasks with nonfatal errors to continue with normal processing after any error terminations occur. It also permits tests, during which error conditions are expected to be encountered, to continue. When an error condition is encountered, console messages and log entries inform the run. See the *ER Programming Reference Manual* (7830 7899) for information on error termination contingencies.

This option is overridden by the E keyin. Refer to the *Exec Operations Reference Manual* (7831 0281) for more information on the E keyin.

If you specify the I option, standard units of processing (SUP) accumulate beyond the time an error is detected.

N

Sets bit 3 of the condition word. This inhibits page ejects between processor calls.

P

Clears bit 3 of the condition word. This enables page ejects between processor calls.

If the configuration parameter control-image-spacing is nonzero, there is a line spacing equal to the value of control-image-spacing instead of a page eject.

f

Specifies the type of operation used to alter the part of the condition word designated by j. The operation can be AND, OR, or XOR. This field is optional. If the f field is specified, the result of the logical operation between the designated part of the condition word and the value specified in the control word statement is stored in the designated part of the condition word. If you do not specify the f field, the designated bits are replaced in the condition word.

value

Specifies a positive octal value not exceeding four digits in length. The *value* is used in the logical operation specified by *f* or is entered into the designated portion of the condition word. The value is right-justified, zero-filled prior to a partial word store into the designated portion. If the magnitude of value exceeds the capacity of the designated portion, truncation occurs.

j

Designates the portion of the condition word to change. The portion can be T1, T2, S2, S3, or S4. If you omit *j*, the default is T2. Within T1 and S2, only bits 7 and 8 are modified. Entries T2, S3, and S4 designate the corresponding word parts. *Tn* and *Sn* are partial word designators. They refer to a third or sixth of a word, respectively. The following diagram shows the partial word designators in relation to a word of storage:

T1		T2		T3	
S1	S2	S3	S4	S5	S6
0	6	12	18	24	30 35

You can examine the condition word in either a runstream or an executing task. You can alter only bits 7 and 8, T2, or T3. You can modify bits 7 and 8 with either a control statement or an Executive request (ER SETC\$). You can modify T2 with a control statement and T3 with an Executive request (ER SETC\$).

The condition word is set to all zeros at the start of a run, unless one of the following occurred:

- The run was started by an @START statement with a set field, in which case T2 initially contains the value of that field.
- The run has been rescheduled across a recovery boot through use of the R option on the @RUN statement, in which case bit 4 is set.

Notes:

- @SETC,A and @SETC,I do not affect the ABORT\$ and EABT\$ Executive requests or E-keyin unsolicited keyin.
- @SETC,I alters the ER ERR\$ processing and other error processing, such as guard mode and illegal operations.

Examples

```
@SETC 6
```

Loads octal 6 into T2 of the condition word. (T2 is assumed since the *j* field is omitted.)

@@SKIP—Skip Lines of Output

```
@SETC 10/S3
```

Loads octal 10 into S3 of the condition word. The value is treated as octal even when the leading zero is omitted.

```
@SETC,I 4/S4
```

Loads octal 4 into S4 of the condition word. Sets bit 5 of the condition word.

```
@SETC 7777/T1
```

Sets bit 7 and 8 of the condition word.

7.59.@@SKIP—Skip Lines of Output

The @@SKIP statement is used to skip a specified number of lines of output. It is usually used after an output interrupt (MSG WAIT key).

Format

```
@@SKIP n
```

n

Indicates the number of lines of output that must be skipped. You can skip up to 63 lines.

If you specify that a certain number of lines must be skipped and fewer lines than that are output by a program, the system continues to skip lines of output, even if they are responses to statements you enter.

If you make a mistake with the @@SKIP statement, you can reset it by entering @@SKIP 0 after an output interrupt (MSG WAIT key).

Examples

```
@@SKIP 5
```

This skips five lines of output.

```
@@SKIP 0
```

No lines are skipped. This is the default.

7.60.@@SOLI—Solicit for User-ID/Password

The @@SOLI statement generates a traditional user-ID/password solicit for demand users while the Authentication and Session Initiation Subsystem (ASIS) is temporarily not available or draining.

Normally, if the ASIS is not available or draining and demand terminal users try to sign on, they receive an error message:

```
The Authentication Subsystem is not available. This sign-on is
rejected. Contact the Site Administrator.
```

The user can then enter an @@SOLI command to force a traditional Exec solicit for a user-ID/password.

Format

```
@@SOLI
```

7.61.@START and @@START—Start Canned Runstreams

The @START statement starts canned runstreams that were previously entered into the system. The statements and data that make up such a runstream must be stored in either a data file or a symbolic element within a program file. The symbolic element cannot reside within a large element program file (LEPF). Such a file or element can be created by

- An editor
- A user program that writes system data format (SDF) files (data files) or symbolic elements to a program file

The first statement in the stored runstream must be an @RUN statement that complies with the rules detailed in 7.56. At the end of the file or element, the run is automatically terminated (the system provides an implied @FIN statement).

If you currently have an active run on a demand terminal, and you are not in control mode (such as during program execution), you can start a canned run from that terminal by using the @@START statement.

You can use the @START statement to

- Submit runs for execution at a time or time interval of your choosing (for example, on a daily basis).
- Start a run that generates data for input to another run.
- Start a run that builds a cataloged file containing an entire runstream and then calls for the runstream to be scheduled. (Thus, you can use the @START statement to initiate parallel processing, since tasks from different runs can be executed concurrently.)
- Start batch runs from demand terminals. The run-control statements for the batch run must have been previously stored in files or elements. The run is scheduled as a batch run, and its output is sent to onsite peripherals.

The @START statement has two formats.

Format 1

Use this format when you want to use all fields from the stored @RUN statement except for the account number (acct-ID), which is the same as the run in which the @START statement is used:

```
@START runstream-name
```

where *runstream-name* identifies the file or element that contains the stored runstream. The runstream-name field must include the name of a file. If the runstream name is a file name only, the file must be a data file.

If the runstream name is an element name without a version name, the system uses the symbolic element with a blank version name, if one exists. If one does not exist, the system uses the first (not necessarily the oldest) nondeleted symbolic element in the named file, regardless of its version name. The specified element cannot reside in a Large Element Program File (LEPF).

Format 2

Use this format when you want to change all or part of the @RUN statement in the stored run:

```
@START[,scheduling-priority/options/processor-dispatching-priority]  
runstream-name[,setc,run-id,acct-id/user-id,project-id,runtime/  
deadline,pages/cards,start-time]
```

scheduling-priority

Indicates what preference the run must be given in relation to all other runs that are available for execution. The scheduling priority can be any letter from A through Z, where A is the highest priority and Z is the lowest. If you specify a scheduling priority that is higher than is permitted for the account number (acct-ID) you specify, the Exec changes the priority to the highest priority allowed for that account.

processor-dispatching-priority

Indicates the processor dispatching priority that is used to place the run's activities into a switching group once the run is opened. The processor dispatching priority can be any letter from A through Z, where Z is the lowest.

Any fields included on the @START statement replace the corresponding fields in the stored @RUN statement. The fields on the @RUN statement are replaced for a particular run but not deleted.

Note: *At the time a runstream is submitted to the Exec for execution, the Exec makes a copy of the runstream in an Exec file to execute. Therefore, if you make any changes to a runstream after the @START statement is finished and before the run is opened, these changes cannot be reflected in the copy of the runstream that the Exec executes.*

priority

Indicates what preference the run must be given in relation to all other runs that are available for execution. The priority can be any letter from A through Z, where A is the highest priority and Z is the lowest. When the Exec is holding a run in backlog because resources are not available (such as a file), other lower-priority runs are not allowed to start. To avoid holding up the start of runs, start as many runs as possible at the same priority.

Note: Normally, when the Exec is holding a run in backlog because resources are not available (such as a file), other lower-priority runs are not allowed to start. To avoid holding up the start of runs, start as many runs as possible at the same priority. However, if the dynamic configuration tag *bypass_facility_held_runs* is set, lower priority runs can open.

options

If specified, options replace those on the @RUN statement for the stored runstream. Following is a list and abbreviated description of the @RUN options and the two special @START options, U and Z.

B

Treats the run input from a demand terminal as batch input and schedules the run as a batch run.

Note: Options E through M, which control the minimum size of the PCT, are not meaningful on XPA hardware because PCTs are given the maximum configured PCT size allowable by your site (see configuration tag PCTMAX).

N

Does not write the program or processor to the diagnostic file. PMD statements are not honored, but are printed along with the message PMD NOT ALLOWED. Diagnostic dumps are not printed. This option overrides the W and Y options.

P

Terminates the run if the page estimate is exceeded. Your site can be configured to use this termination, whether or not the option is specified.

R

Reschedules the run if a recoverable system failure occurs.

T

Terminates the run if the standard unit of processing (SUP) usage estimate is exceeded. Your site can be configured to use this termination, whether or not the option is specified.

U

Lets you start a runstream under a user-ID other than your own. With this option, the run executes under the user-ID specified on the @RUN statement in the runstream. The MAC and DAC attributes associated with the user-ID on the @RUN statement must be the same or dominate those of the runstream's starter. See Section 4.3 for more information on MAC and DAC. Also, see the *ClearPath OS 2200 Apex Help* (8207 4154).

V

The V option can be used on an @START statement to start a run at the minimum clearance level and default compartment set of the user-ID under which the started run executes (the target user-ID). Without the V or Z option, the started run assumes the starter's executing mandatory attributes. That user-ID depends on how the V option is used, as follows:

- @START with V option used alone—The starting run's user-ID (the run from which @START was executed).
- @START with both the U and V options—The user-ID specified by the @RUN statement in the runstream being started.

The V option allows trusted user-IDs to start runstreams that have more privilege. Without the V option, the started run assumes the starter's executing mandatory attributes (clearance level 0 and default compartment set of NULL). The V option is ignored for systems with fundamental security.

See Table 7-5 for references to the various @START statement formats when using the V option.

W

Specifies that only user programs and processors in the system files that encounter an error are written to the diagnostic file. Diagnostic dumps are printed only if the program producing the diagnostic dumps terminates in error.

X

Disregards system holds when opening a run; requires a security privilege.

Y

Always writes the program or processor to the diagnostic file, regardless of the source of the program. All PMD statements are honored, and diagnostic dumps are printed.

Z

Lets you start a runstream with mandatory security attributes at system high. Without the Z option or the U option, the started run assumes your executing mandatory attributes. If you specify the U option but not the Z option, the started run assumes the mandatory attributes of the user-ID on the @RUN statement in the runstream.

The Z option exists so that a started run can execute with system high attributes. Executing at system high means that the run has all compartments and the highest clearance level. See Section 5.2 for more information on compartment sets and clearance levels.

To use the Z option, the following conditions apply:

- You must have the privilege to start runs at system high.
- You must have system-high capability, or if you specify the U option, the user-ID on the @RUN statement in the started runstream must be able to run at system high.

The Z option is not useful unless your system has Security Level 1 or higher.

Table 7–5 summarizes the security attributes of a started run when you use different combinations of the U and Z options, and also shows the validations required to use each of the option combinations.

Table 7-5. Security Validations for @START Statements

START Image	Validations	Executing Clearance Level of Started Run	Executing Compartment Set of Started Run	User-ID of Started Run	Privilege Set of Started Run
@START	Starter must have read access to the file containing runstream.	Starting run's executing clearance level	Starting run's executing compartment set	Starting run's user-ID	Starting run's privilege set
@START, / U	Starter must have read access to the file containing the runstream and execute access to started run's user-ID. Started run's user-ID must own the file unless the starter has SSSTUBYPASSO or SSBYPASSOWNR privilege.	Starting run's executing clearance level	Starting run's executing compartment set	Started @RUN image user-ID	Started @RUN image user-ID privilege set
@START, / V	Starter must have read access to the file containing the runstream. The start file CL and CS must equal the starter's minimum CL and default CS unless the starter is sufficiently privileged to change the start copy file to the minimum CL and default CS of the starter's user-ID.	Minimum clearance level of starting run's user-ID	Default compartment set of starting run's user-ID	Starting run's user-ID	Starting run's privilege set

Table 7-5. Security Validations for @START Statements

START Image	Validations	Executing Clearance Level of Started Run	Executing Compartment Set of Started Run	User-ID of Started Run	Privilege Set of Started Run
@START, /UV	Starter must have read access to the file containing the runstream and execute access to the started run's user-ID. The started run's user-ID must own the file unless the starter has SSSTUBYPASSO or SSBYPASSOWNR privilege. The start file CL and CS must equal the minimum CL and default CS of the started @RUN image user-ID, unless the starter is sufficiently privileged to change the start copy file to the minimum CL and default CS of the started @RUN image user-ID.	Minimum clearance level of started @RUN image user-ID	Default compartment set of started @RUN image user-ID	Started @RUN image user-ID	Started @RUN image user-ID privilege set
@START, /Z	Starter must have read access to the file containing runstream, have privilege to start runs at system- high, and have system-high capability.	63	ALL compartments	Starting run's user-ID	Starting run's privilege set

Table 7-5. Security Validations for @START Statements

START Image	Validations	Executing Clearance Level of Started Run	Executing Compartment Set of Started Run	User-ID of Started Run	Privilege Set of Started Run
@START, /UZ	Starter must have read access to the file containing the runstream, and execute access to started run's user-ID, plus privilege to start runs at system- high. Started run's user-ID must own file and must have system-high capability unless the starter has SSSTUBYPASSO or SSBYPASSOWNR privilege.	63	ALL compartments	Started @RUN image user-ID	Started @RUN image user-ID privilege set

runstream-name

Identifies the file or element that contains the stored runstream. The runstream name must include the name of a file. If the runstream name is a file name only, the file must be a data file.

If the runstream name is an element name without a version name, the system uses the symbolic element with a blank version name, if one exists. If one does not exist, the system uses the first nondeleted symbolic element in the named file, regardless of its version name. The specified element cannot reside in a Large Element Program File (LEPF).

setc

Specifies a positive octal number not exceeding 4 digits in length to be placed in T2 of the condition word of the run being scheduled.

run-id

Specifies the new run-ID; can be 1 to 6 characters in any combination of letters and numbers.

acct-id/user-id,project-id

If runstream-name is a private file and the file is unowned, the @START statement must include the project-ID or account number of the run that created the file. (Whether you must specify a project-ID or an account number depends on how your system is configured.)

If you do not specify acct-ID, the system uses the account number of the run that contained the @START statement.

If you do not specify the U option, the system always uses the user-ID of the run containing the @START statement. If the @RUN statement for the stored run includes a user-ID, it is ignored.

If runstream-name is a private file and the file is owned, the user-ID that is starting the run must be the same as the one that owns the file.

An attempt to start a runstream using the @START statement from the SYS run is handled as follows:

- If a valid user-ID is specified on the @START image, the run is allowed to execute.
- If the user-ID on the @START image does not exist, an error is returned and the run is terminated.
- If no user-ID is specified on the image, the overhead user-ID is used and the run is allowed to execute.

runtime/deadline,pages/cards,start-time

Here is an example of this option:

If you have an inventory system that includes the following two data files

- File INV*MASTER (current inventory information)
- File INV*DTRANS (the day's transactions)

Then each day, you execute the same three absolute elements, as follows:

- INV*PROG.D-UPDATE (updates the inventory file using the day transactions)
- INV*PROG.STATUS-RPT (generates a stock status report based on the inventory file)
- INV*PROG.TRANS-LIST (generates a list of the transactions posted against the inventory file)

Instead of executing these three programs individually, you could create a canned run to execute all of them at the same time. You then use the @START statement to initiate the canned run whenever you need to.

To illustrate this, suppose you put the following statements into an element called INV*RUNS.DAILY:

```
@RUN R231,03412,INV84,10,150
@ASG,A INV*PROG.
@ASG,A INV*MASTER.
@ASG,A INV*DTRANS.
@XQT INV*PROG.D-UPDATE
@XQT INV*PROG.STATUS-RPT
@XQT INV*PROG.TRANS-LIST
@FIN
```

To execute this entire sequence of statements, just enter the following statement:

```
@START INV*RUNS.DAILY
```

Or, here is another example for this option:

```
@START FILEC.ELTB,,R231,03414,,,50
```

The run in element ELTB of file FILEC is scheduled for execution. The system uses the run-ID (R231), account number (03414), and page estimate (50) on the @START statement instead of those on the @RUN statement in the stored runstream.

7.62.@SYM and @@SYM—Queue Print Files and @SYMCN Control Statement

The @SYM statement queues previously created print files to a printer, group of printers, or user-ID, for printing. It also directs the PRINT\$ file (the primary print file) to an alternate device.

Request output with the @SYM statement rather than using SITE from a text editor, which ties up your terminal. To protect classified data, your security officer cannot let you use @SYM to queue print files to another terminal or user-ID. For information on using @SYM and @BRKPT together, see Section 7.3.

Format

Use this format to queue any data file (you cannot specify program files with the @SYM statement) to a printer; you can also use this format to queue a file to any terminal that is configured in the system generation:

```
@SYM, options filename, [number, device, pname-1/pname-2/pname-3  
/.../pname-n, banner]
```

Use this format to queue print files to a user-ID instead of to a printer:

```
@SYM, options filename, [number, user-id/U, pname-1/pname/pname-3  
/.../pname-n, banner]
```

options

A

Specifies that all files on the tape are to be printed in the order that they appear on the tape; therefore, the file name given on the @SYM statement must be a tape file.

D

Deletes the current part of the file when the file part is closed, and produces no output. The file name must be PRINT\$.

The device field is ignored when this option is used.

F

Directs this part and all future parts of the PRINT\$ file to the device specified, until the Exec finds another @SYM statement, with or without the F option, for a primary PRINT\$ part. When the Exec processes the second @SYM statement, the device specification from the original @SYM,F statement is lost. You can reinstate it with another @SYM,F statement.

J

Print the top and bottom separator page regardless of the three system generation parameter values: *demand_batch_top_separator*, *rsi_remote_top_separator*, and *rsi_bottom_separator*.

K

Do not print the top and bottom separator page.

L

If security labeling is configured, prints the label on all pages of the printout.

N

If security labeling is configured, only the banner and trailer pages are labeled.

U

Inhibits deletion of the file when processing is completed. This option is applicable only to user-defined files. Do not use this option unless it is necessary. Overuse of the U option loads the system with unnecessary files. Also, modifying the file can result in unpredictable output.

Note: If both the J and K options are specified, the J option is used. The J and K options are meaningful only for print files. The options are ignored on an @SYM of a punch file.

filename

Identifies the file to be queued. If the file name is a user-defined file, it must be a cataloged file. It cannot be in a to-be-cataloged state. The file name must be an internal or external file name or PRINT\$ (the primary print file). If an internal name is specified, the associated file must be assigned to the run.

When file name is PRINT\$, the @SYM statement applies only to the current part of the PRINT\$ file. In this case, you must not have breakpointed the PRINT\$ file to a user-defined file with the @BRKPT statement.

number

Indicates how many copies of the file are to be printed. If omitted, one copy is assumed. You can specify up to 63 copies.

device

Identifies the device on which the file is to be printed. This can identify a specific onsite device, a specific remote site, or a group of onsite devices (that is, the group might be all onsite printers). Device group and remote site identifiers are defined when the system is configured. If omitted, the output file is printed at the default printer for the terminal used for the run.

For RSI\$ terminals with configured site-IDs, the default is the site-if of the terminal. For RSI\$ terminals with nonconfigured nongeneric site-IDs, the default is the system default print queue. For RSI\$ terminals with generic site-IDs, the default is the user-ID of that user.

You can queue a file to any terminal that is configured in the system generation (instead of to a printer) by using the terminal's site-ID in the device field. To receive a file that has been queued to your terminal, use the @@SEND statement.

user-id/U

Is any valid user-ID within the system. The /U indicates that you want the print file queued to a user-ID instead of a device. A slash (/) must precede the U.

If the given user-ID is invalid, the Exec issues the message:

```
ILLEGAL USERID SPECIFIED.
```

You can queue the print file to your own user-ID or to some other valid user-id. The print file is then either printed at the printer connected to the specified user's terminal (if there is one) or displayed at the user's terminal (the @@PRNT statement directs output to the printer). The file is printed or displayed when you enter @@SEND,U.

pname-n

Are the labels of the symbiont file parts of a multifile tape to be printed. If omitted, only the first part on the tape is processed. This field is not applicable to mass storage files.

The order in which the part names are specified on the @SYM statement must correspond to the order in which the parts are located on tape. However, not all parts on the tape need be processed. Any parts not named on the @SYM statement are bypassed.

banner

Specifies the banner to appear on the first page of the printout. The banner must be a maximum of 12 characters. If this option is omitted, the run-ID is used.

Note: *The banner field is ignored for a nonstandard file if the file is saved and restored with the SV and SR operator keyins. The run-ID of the @SYM statement is used for the banner in this case.*

Once the print file has been printed, it is deleted, unless you include the U option. If you intend to queue the same print file with more than one @SYM statement, you must use the U option. If you do not, the print file ceases to exist when all copies of the file are printed.

After you have queued a print file to a printer, you can check where your file is in the print queue by using the @@CONS SQ statement.

You cannot delete a file that is listed in a print queue unless you first remove it from the queue with the @@CONS SX statement.

After queuing a file to a printer, avoid using the print file until after the file has been printed.

@SYMCN Control Statement

A user program can dynamically specify line advancement before control statement images are printed. The options on this control statement override dynamic configuration parameter *control-image-spacing*. The @SYMCN control statement is available in the runstream as well as through ER CSF\$ and ER ACSF\$.

Options on @SYMCN statements are

L

The standard line spacing for processor call statements is used.

If configuration parameter *page-eject-on-processor calls* is TRUE, or if the run has done @SETC,P, a page eject occurs before processor control statements are printed. Line spacing is 3 lines for all other control statements.

If dynamic configuration parameter *page-eject-on-processor-calls* is FALSE, or if the run has done @SETC,N, line spacing is 3 lines for all control statements.

N

Advance *n* lines before any control statement image is printed where *n* is the number specified for dynamic configuration parameter *control-image-spacing*. The value *n* is used for all control statements including processor statements. If *n* is zero, a value of 3 is used.

Continuation images of a control statement advance *n* lines or 4 lines, whichever is less.

Security Labels on Printouts

If your site has Security Level 2 or higher, printouts can be labeled with the clearance level and compartment set associated with your file and with a sequence number. Check with your security officer or site administrator to determine whether you have this feature.

If you have print labeling configured, the security label always appears on the first (banner) and last (trailer) pages of your printout. To print the label on all the pages, specify the L option on the @SYM statement. Use the N option to print the label on only the banner and trailer pages. If you do not specify either option, the label can or cannot appear on all pages, depending on how your system is configured.

If your site labels every page, remember that a printed page holds fewer lines of text than if your site did not label printouts. Check with your security officer to find out the allowable maximum width of your printouts. Printout width must be the same as or larger than the configured width for labels.

The security label appears on the top and bottom of each page, except the banner page, where it appears below the banner. Labels are one to four lines long (up to 225 characters) and they are preceded and followed by a single blank line.

The symbolic clearance level of the file appears first in the label, followed by the compartment set. A space separates each word in the label. For example, Figure 7-5 shows a security label for a user with executing clearance level SECRET and executing compartment set PAYROLL, PERSONNEL, and PLANNING.

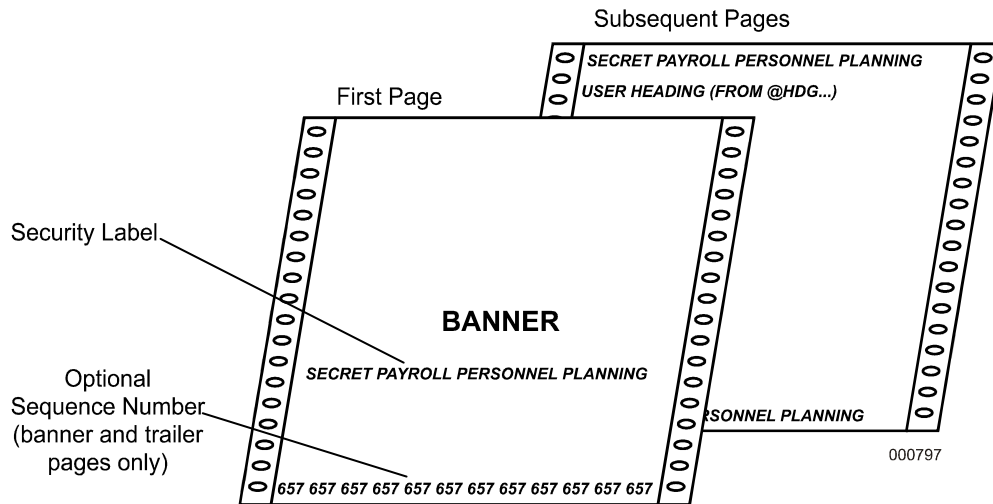


Figure 7-5. Sample Printout Label

The following four special cases change the label appearance:

- If your file has the system compartment set associated with it, the label says
clearance level ALL
- If compartments are not installed on your system, the label contains only the clearance level.
- If no compartments are associated with the file, the label says
clearance level NULL
- If the label is longer than 225 characters, it says
clearance level ALL (The security label is too long to be printed.)

Examples

```
@SYM LPRNT,2,PR6
```

This @SYM statement queues user-defined print file LPRNT to print queue PR6. Two copies of the printout are generated. After the file is printed, it is deleted.

```
@SYM,L LPRNT,,REMPR1
```

This @SYM statement queues user-defined print file LPRNT to printer REMPR1. If security labeling is configured, the security label appears on all pages of the printout.

7.63.@@TERM—Sign Off from the System

The @@TERM statement terminates your terminal at the end of a run. Enter this statement as follows after an @FIN statement when you do not want to start a new run:

```
@@TERM
```

Although you could enter an @@TERM statement before an @FIN statement to terminate the run and the terminal simultaneously, you must use it only as a last resort. If you enter the @@TERM statement without first entering an @FIN, the following happens:

- Files that were supposed to be deleted at the end of the run due to an @ASG,D statement are not deleted.
- Files that were supposed to be cataloged at the end of the run due to an @ASG,C statement are not cataloged.

7.64.@TEST—Test a Condition Word to Control Execution

The @TEST statement tests the value of the condition word to select the particular control statements to be executed or skipped.

When more than one test is to be made, the control statement is scanned until a test is met or all parameters are exhausted. When a test is met, the control statement immediately following the @TEST statement is skipped.

Format

```
@TEST f/value[/j], [f/value/j...]
```

f

Specifies the type of comparison to make. If the test is met, the next control statement is skipped. If the test is not met, the next control statement is executed.

Permissible test types are:

TE	Test for equal
TNE	Test for not equal
TG	Test for greater than
TLE	Test for less than or equal
TEP	Test for even parity
TOP	Test for odd parity

value

Specifies a positive, octal value not exceeding 12 digits to be compared with that portion of the condition word specified by the *j* field.

j

Specifies the portion of condition word to be tested. Permissible entries are W, H1, H2, T1 through T3, and S1 through S6. If you omit this field, the default is T2. Tests with *j* field of T1, T2, or T3 result in sign extension of that portion of the condition word for the specified test. The following diagram illustrates this:

W					
H1			H2		
T1		T2		T3	
S1	S2	S3	S4	S5	S6
0	6	12	18	24	30 35

Examples

@TEST TE/6/S3

S3 of the condition word is tested to see whether it is equal to a value of octal 6; if equal, the next control statement in the stream is skipped. If unequal, the next control statement is executed.

@TEST TG/6,TE/4

T2 of the condition word is tested for two conditions: greater than octal 6 or equal to octal 4. If either condition is met, the next control statement in the stream is skipped. If neither condition is met, the next control statement in the stream is executed.

@TEST TG/10/H2

H2 of the condition word is tested for a greater-than octal 10 condition. If the condition is met, the next control statement in the stream is skipped; otherwise, the next control statement is executed. The value is treated as an octal value even if the leading zero is omitted.

@TEST TLE/4/T2

T2 portion of the condition word is tested to see whether it is less than or equal to octal 4. If so, the next control statement in the stream is skipped; otherwise, it is executed.

```
@TEST TE/4000/T2
```

Due to sign extension of T2, this test always results in the next control statement being executed.

```
@TEST TOP/20/T3
```

T3 of the condition word is tested to see whether the bit represented by 020 is set. If it is, the next control statement in the stream is skipped; otherwise, it is executed.

7.65.@@TM—Send a Message to Another User

The @@TM statement sends a message (in demand runs only) to a specific run-ID, user-ID, or site-ID. The ID must be active to receive the message.

If you have the limited keyin demand console mode or a higher console mode, you can also send terminal messages by using the @@CONS TM statement. See the *Exec Operations Reference Manual* (7831 0281) for the syntax and a description of the TM keyin.

Note: The @@TM statement might not be available on your system. Check with your site administrator.

The format of the @@TM statement depends on whether it is being sent to a run-ID, user-ID, or a site-ID.

Formats

```
@@TM run-id/R message
```

Sends the message to the specified run-ID. No space is needed after the /R.

```
@@TM user-id/U message
```

Sends the message to all active runs with the specified user-ID. No space is needed after the /U.

```
@@TM site-id message
```

Sends the message to the specified site-ID. A space must follow the site-ID.

Examples

```
@@TM WPS/R DO YOU HAVE TAPE 3556A?
```

Sends message DO YOU HAVE TAPE 3556A? to run-ID WPS.

```
@@TM JAM/U DO YOU HAVE TAPE 3556A?
```

Sends message DO YOU HAVE TAPE 3556A? to user-ID JAM.

```
@@TM RE2211 DO YOU HAVE TAPE 3556A?
```

Sends message DO YOU HAVE TAPE 3556A? to site-ID RE2211.

7.66.@@TOUT—Change Your Timeout Period

If your terminal is idle for a certain length of time, the system assumes that you are not using the terminal, so it terminates the run and signs the terminal off (timeout). The @@TOUT statement lets you override the standard value that is normally used for this timeout period (typically 30 minutes).

A timeout warning is sent after a session is idle for the period of time defined as the timeout period. The session is closed if it continues to remain idle after the warning for the same period of time defined as the timeout period.

Not all users are allowed to use this statement. The new timeout period you establish with the @@TOUT statement remains in effect until you enter another @@TOUT statement or your run terminates (@FIN). When your run terminates, the timeout period is reset to the standard value.

Format

```
@@TOUT t
```

t

Indicates how long (in minutes) you want the timeout period to be. It can be from 1 to 2047 minutes (about 34 hours). The actual limit that you work with depends on your site. If you do not provide a time or enter a value of 0, the system defaults to the standard value.

To find out what your current timeout period is, enter an @@TOUT ? statement. This does not change the timeout period.

To disable the timeout mechanism, enter an @@TOUT D statement.

Examples

```
@@TOUT 30
```

The timeout period is 30 minutes.

```
@@TOUT ?
```

A message tells you what the timeout period is.

7.67.@@TRNS—Ignore Escape/EOT Characters

The @@TRNS statement causes the escape and end-of-transmission characters to be ignored. (In other words, this allows you to send stop code characters without functioning as the stop code.) The function of this statement is similar to @@ESC and is for users sending file images from a host to a demand terminal.

Format

```
@@TRNS
```

7.68.@@TTY—Change TTY Terminal Characteristics

The @@TTY statement changes characteristics associated with the TTY terminal.

Format

```
@@TTY function-1,value-1[,...function-n,value-n]
```

The *function-n* and *value-n* fields are the same.

Example

```
@@TTY W,30
```

The page width is changed to 30.

7.69.@USE and @@USE—Reference a File by Another Name

The @USE statement lets you reference a file by more than one file name. The alternate file names created with @USE are called internal (use) file names. The original name of a file is called the external file name. (See Section 3.3 for more information on naming files.) If you create more than one internal file name for a file, you can reference that file by any one of those internal names.

You can want to create internal file names for the following reasons:

- To simplify file references by using a shorter internal file name instead of a long external file name
- To differentiate between files that have the same basic file name, but different qualifiers or file cycles
- To connect internally programmed file names to external file names
- To specify a directory-ID on an external file name (Most control statements that have a *filename* field do not allow a directory-ID to be specified with the file name.)

Format

```
@USE[,I] fname-1,fname-2
```

|

Automatically releases the specified file from the run, as well as all internal file names associated with the file, the next time a task, such as a user program, terminates. (A task is a discrete processing step involving the execution of an absolute element.)

The file and internal file names are released even if one of the following applies:

- Errors occur when the task terminates.
- The specified file is not assigned to your run.

fname-1

Identifies the new, alternate name for the file (an internal file name). This name can be up to 12 characters long. You cannot specify a directory-ID, qualifier, file cycle, or read/write keys for an internal file name.

fname-2

Identifies the file for which you are providing a new internal name. This can be either a previously specified internal (use) name or an external file name.

If your system has Multi-Host File Sharing (MHFS), you can specify the directory-ID (STD or SHARED) on the *fname-2* field. If your system has MHFS and you do not specify a directory-ID, the Exec uses the implied directory-ID (if you include the #) or the default directory-ID (if you do not include the #).

Note: A directory-ID cannot be used when *fname-2* is a previously specified internal (use) name.

All internal file names equated to an external file name are maintained for the duration of the run, or until you issue an @FREE,A statement.

Once the names are equated, you can reference the file by its internal or external file name from within a program or the runstream. If a conflict of file names exists, you must attach an internal name to the file with the conflicting external name before any references to that file are attempted.

When a file is referenced, the list of internal file names is searched first. If the file cannot be found, the list of external file names is searched. Generally, when you use @FREE or @DELETE to release a file, all of the attached internal names disappear. However, if you try to use @DELETE or @DELETE,C on a noncataloged file in batch mode, the internal names remain attached to the file.

Use @PRT,I to display the internal names of your currently assigned tapes and files.

Examples

```
@USE FILEB,COMPANY*PAYROLL
@USE COST,FILEB
```

In the first statement, internal file name FILEB is equated to external file name COMPANY*PAYROLL. In the second statement, internal file name COST is equated to internal file name FILEB. The file can now be referenced by the original name (COMPANY*PAYROLL) or one of the internal file names (COST or FILEB).

```
@USE NEWCYC,ID*FILESET(0)
@USE OLDCYC,ID*FILESET(-1)
@USE OTHER,QUAL*FILESET
```

This example shows how you can use internal file names to differentiate between files that have the same basic file names, but have different F-cycle numbers or qualifiers. The internal file name NEWCYC identifies the current cycle (relative F-cycle 0) of file ID*FILESET, where OLDCYC identifies the previous cycle (relative F-cycle -1) of the same file. Internal file name OTHER identifies the file QUAL*FILESET, which has the same basic file name as the other file (ID*FILESET), but a different qualifier.

```
@USE Q,BACKUP
@USE BACKUP,EZ
@USE EZ,STD#REG*FILE
```

For the remainder of this run, file Q can be referenced by any of these names: BACKUP, EZ, and STD#REG*FILE.

```
@USE MS,SHARED#MISC*FILE
@ADD MS.
```

File SHARED#MISC*FILE is given internal name MS, and then used on the @ADD statement. (The @ADD statement does not allow a directory-ID to be specified.)

```
@USE SHAREDFILE.,SHARED#ANY*FILE
@COPY SHAREDFILE.,BACKUP*FILE.
```

This example shows you how to use an internal name to specify a directory-ID for a file on an @COPY statement. The @COPY statement does not allow you to specify a directory-ID with the file name. By associating the internal file name with the correct directory-ID, you can circumvent this limitation.

7.70.@@X—Halt Processors

At times you can want to stop your current run processing to do something else. You can stop a processor at any time with the @@X statement.

Format

```
@@X [functions] [, data]
```

functions

Let you specify what kind of stop you want. Specifying just @@X (with no *function* characters) defaults to @@X TIOCR.

C

Jumps to an address previously specified in the program (remote break contingency). You can use either CEND\$ or CRTN\$.

Although most requests can terminate contingency mode, the CEND\$ request is designed for this purpose. It is the most efficient method unless another request is needed.

The CRTN\$ request removes the interrupted activity from contingency mode and returns control at the address (not address+1) specified in H2 of word 0 of the contingency packet. This lets the user activity process a contingency and return to the original execution path without destroying a register on the return. This avoids the risk of another contingency interrupt overlaying the packet between a CEND\$, jump-indirect-on-the-packet sequence. If user activity is not in contingency mode, the return is to the ER+1. When an activity not in contingency mode executes this request, it has no effect.

If the activity is returning from a contingency that registered the RBASE\$ function, CRTN can rebase the banks at the time the contingency occurs. The CRTN\$ request reestablishes the environment when the contingency occurred, rebases the banks previously unbased, and returns you to the return address specified in the contingency packet.

The *data* field is only available when the C function is present, either alone (@@X), followed by a comma (@@X C,*data*), or with other *function* characters followed by a comma (@@X CT,*data*). See the description of the *data* field that follows.

Discards all backed-up input. This includes all images from @ADD files and images entered in @@CQUE and @@INQ mode (includes FUL, PTI, and TCI).

I

If the corresponding @@END INQ has not been received, the terminal remains in @@INQ mode but the file is purged of all images. The next input becomes the first image in the @@INQ file.

O

Discards all backed-up output.

R

Terminates the active processor or program and all future actions of the program or processor, including removal of all entries from the QECL stack.

T

Terminates the program currently being executed.

data

Lets you pass along some unique information along with the contingency.

The *data* field is only available when the C function is present, either alone (@@X), followed by a comma (@@X C,*data*), or with other *function* characters followed by a comma (@@X CT,*data*). Special field characters such as a slash (/), semicolon (;), and a stand-alone period (.) can never be used. The *data* field is case sensitive.

The *data* field can be a maximum of 6 characters in length. Passing more than 6 characters results in the error message

@@X ERROR - TOO MANY DATA CHARACTERS

This keyin can penetrate wait conditions. It can be directed at a specific activity or activities. Distribution of this and other externally generated contingencies (those generated at the keyboard or the system console) are best controlled by the program. It is unlikely that a user sitting at the keyboard would know enough about where a program is executing to control which activity must field contingency conditions. Also, the distribution is viewed as an attribute of the run rather than of any particular subsystem. To help control of external contingency distribution, the EXTCGY\$RECV is provided.

ER II\$ activities can be activated by @@X C as described in the *ER Programming Reference Manual* (7830 7899). Input can now be received in register A0 by either the @@X C or II keyins. Also, instead of generating the II activity with data, a contingency can be generated instead. Depending on when the ER II\$, II keyin, or @@X C(T) operations are performed, activating the II activity or generating a contingency for it is purely an asynchronous event by OS 2200.

Normal operation is as follows:

- An II keyin activates an II\$ activity if one is present or if an II\$ activity does not exist, external contingencies are generated for all activities that have chosen to accept them.
- An @@X keyin both activates an II\$ activity (if one is present) and generates external contingencies for all activities that have chosen to accept them.

Table 7–6 describes the conditions that affect an II activity and their respective results. Translation of any ASCII characters to Fielddata characters conform to those in the conversion table shown in the *ER Programming Reference Manual* (7830 7899).

Table 7–6. Actions Taken with Respect to an II Activity

Keyin	ER II\$ Activated	Contingency Generated
@@X C[T]	A0 = 6 Fielddata characters or Fielddata spaces if no data was passed.	Cgy = 020010 or 020110 Aux = 4 ASCII characters or zero if no data was passed.
II Keyin	A0 = 6 Fielddata characters or Fielddata spaces if no data was passed.	Cgy = 010010 Aux = 6 Fielddata characters or zero if no data was passed.

Examples

@@X CO

This example shows how to stop output without terminating your program.

@@X

This default means that all options are used.

@@X ,

This default also means that all options are used, and the auxiliary data consists of ASCII spaces.

@@X ,INTR

This default also means that all options are used.

@@X CT,QUit

The auxiliary data is "QUit."

@@X CT, QUIT

After the comma @@X follows the ECL command format rules: spaces are allowed before the actual function fields.

@@X CT,QU IT

The space after "QU" terminates the data field scan.

@@X C,

Jumps to an address previously specified in the program; no value is in the data field, so auxiliary data consists of spaces.

```
@@X T,TEST
```

Forces abnormal program termination. No data is passed because the data field is only available when the C function is used.

```
@@X IO,DATA
```

Again, no data is passed because the data field is only available when the C function is used.

```
@@X C,TESTING
```

Keyin error. The data field has exceeded the maximum number (6) of characters allowed.

Note: Specifying program termination with all the options (as in the second and third examples) can corrupt common banks, database files, and user files. In general, you must avoid this type of program termination.

7.71. @XQT—Execute Absolute Elements or Object Modules

The @XQT statement initiates the execution of an absolute element or object module. This element cannot reside in a Large Element Program File (LEPF).

Format

```
@XQT, [options] [filename.eltname]
```

options

Are user-specified and are initially presented to a program in a register. You can retrieve them with ER OPT\$. (These options are also used by OS 2200 system processors.)

The registers and their contents are as follows:

A2 and A3

Hardware feature bit mask:

Bits 0-35	Unused
Bits 36-41	Numerical Host-ID, (1 = A, 2 = B ...)
Bits 42-63	Unused
Bit 64	VERSION3E type hardware
Bit 65	Unused
Bit 66	Queuing Architecture supported

Bit 67	Unused
Bit 68	Unused
Bit 69	Basic mode
Bit 70	Extended mode
Bit 71	Unused

Mnemonics and values for these fields in SYS\$RLIB\$.ERU\$ are

VERSION3E\$	0200
BTPBTU\$	0100
QARCH\$	040
EIS\$	020
BYTE\$	010
BMODE\$	04
XMODE\$	02
FPD\$	01

A4

Program type:

4	Demand
5	Deadline batch
6	Batch

A5

Bit 0	Processor was reloaded from QECL stack.
Bit 1	Print file is breakpointed (@BRKPT).
Bit 2	R11–R15 have been established.
Bit 3	Repeat option specified on entry of QECL stack caused reload.

@XQT—Execute Absolute Elements or Object Modules

Bit 4	Processor was searched for in the alternate library file and the file was assigned on the user's behalf.
Bit 5	If set, the program was called as a processor. If cleared, an @XQT started the program.
Bit 6 - 9	Unused
Bits 10-35	Contain the options from the @XQT or processor call (A = bit 10, Z = bit 35)

R1

Date (in Fieldata DATE\$ format in local time):

Bits 0-11	Month
Bits 12-23	Day of the month
Bits 24-35	Year (last two digits)

R2

Current time (TDATE\$ format in local time):

Bits 0-5	Month
Bits 6-11	Day
Bits 12-17	Year (modulo 1964)
Bits 18-35	Time in seconds from midnight

R3

Accumulated processor time for the run (in 200-microsecond increments)

R11/12

System type (two words, left justified)

R13/14

Exec master level (two words, left justified)

R15

System site-ID (left justified)

filename.eltname

Specifies the absolute element or object module to be executed. This element cannot reside in a Large Element Program File (LEPF).

If you do not provide a file name with an element name (*eltname*), TPF\$ is assumed. If you do not provide an element name, the system executes the absolute element or object module that was most recently placed in TPF\$, or in the file name you specified on the @XQT call.

If you specify an element name without a version name, the system executes the absolute element or object module with that name that has a blank version name, if one exists. If one does not exist, the system executes the first (not necessarily the oldest) nondeleted absolute element or object module in the program file that has the name *eltname*, regardless of its version name.

For basic mode programs, when there are no absolute elements in TPF\$, the system collects all of the relocatable elements in TPF\$ (implied collection) and executes the resulting absolute element. The newly created absolute element is given the name NAME\$ and placed in TPF\$.

Examples

```
@XQT PAR.12C
```

For an extended mode program, executes object module 12C in file PAR.

```
@XQT FILEA.XYZ
```

For a basic mode program, executes absolute element XYZ in file FILEA.

```
@XQT
```

For a basic mode program, executes the most recent absolute element placed in TPF\$. If no absolute element is present, the Collector creates an absolute element (NAME\$) from all relocatable elements in TPF\$ and executes that absolute element.

Program Load Error

Refer to the *ER Programming Reference Manual* (7830 7899) for more information.

@XQT—Execute Absolute Elements or Object Modules.

Appendix A

ECL and Terminal Sign-on Messages

This appendix contains messages you might receive at your terminal or in a printout.

Notes:

- *Some messages start with a symbol (such as * or -@@), or a variable (such as X). These messages are listed alphabetically by the first word in the message.*
- *Terminal sign-on messages are listed with a 1, 2, or 3 endnote—see the last page of this Appendix for explanations.*

A

ABNORMAL RETURN FROM READ\$

A program that expects to be called as a processor has been called with @XQT.

ACCOUNT SUA QUOTA EXCEEDED

No system units of accounting (SUA) remain for the account specified on the @RUN statement at initial scheduling time.

A COMPARTMENT NAME YOU ENTERED IS NO LONGER IN USE. PLEASE START OVER.³

You entered a compartment name and it was validated, but the name changed while you were entering other compartment names. The system displays the prompt for you to enter your compartment set again, and you must start over by entering compartment names.

ACR Name ►

You encounter this prompt while using the TLABEL processor to attach an ACR to a tape. Enter the name of the ACR you want attached to the tape.

Assign of file-id failed - inconsistent security information.

The security information in the master file directory (MFD) is inconsistent with the security label.

AT LEAST 1 DATA CHECK OCCURRED WHILE OUTPUTTING FILE

If a data check occurs, the file is not terminated.

AWAIT/DACT AMBIGUITY

The Exec determined that all activities of the run were either in an AWAIT state (ER AWAIT\$) or a DACT state (ER DACT\$) and could not be activated by the run.

B

BAD EXEC INTERNAL STATUS LOAD NOT PERFORMED

An internal Exec error occurred when trying to build bank load packets, bank control packets, or stacks for ZOOMs.

BAD INPUT SEQUENCE

The input control message was rejected because it conflicts with the current status of the run.

x BADLY CODED FIELD

Fill from previous field was requested and the filled field is illegal or ambiguous.

BAD RUN STATEMENT

The @RUN statement is improperly formatted. This message is displayed only for demand runs; for batch runs, the device is placed in the run search mode. If the statement was submitted from a demand terminal, it can be retyped immediately in the proper format.

BAD ZOOM ZCI - LOAD NOT PERFORMED Bank Group=*gggggg* L,BDI=*llllll*

The Exec is unable to load and execute the ZOOM. Either a corrupted program file or corrupted software is responsible for this error condition, where:

gggggg

Is the number of the bank group being processed

llllll

Is the L,BDI of the bank being processed

BATCH RUN MODE - ENTER RUN STREAM

An @RUN,/B has been received and this message announces that input can now be entered.

BATCH START TIME OF 24 HOURS EXCEEDED

User's batch start time was too large. The activity is terminated. Message RUNSTREAM ANALYSIS TERMINATED also appears on the run's output.

BRKPT

User's breakpoint setting matched and no contingency was registered. The activity is terminated.

BRKPT PRINT\$ IS ILLEGAL FOR DEMAND

The @BRKPT PRINT\$ statement was used from a demand terminal before the @BRKPT PRINT\$/file.

C

CATALOGUED FILE MASS STORAGE QUOTA ERROR RECEIVED

The Exec encountered an error while attempting to retrieve the quota limits for a cataloged mass storage file.

CATALOGUED FILE MASS STORAGE QUOTA LIMIT EXCEEDED

The allowable cataloged mass storage tracks in the quota set for this account number have been depleted.

Choose your account index(1-5):¹

Pick the account index you wish.

Choose your project-identifier index (1-5; <XMIT>=Q\$Q\$Q\$):¹

Pick the project-id index you wish, or transmit to use Q\$Q\$Q\$.

Choose your project-identifier index (1-5; <XMIT>=firstproj):¹

Pick the project-id index you wish, or transmit to use the first project-id.

COMPARTMENT SET VALIDATION FAILED
RESUBMIT RUN

One of the specified compartments was decontrolled while your run was in backlog. The run is removed from backlog.

@@COMPLETE

Processing of a transparent ECL statement has completed normally.

CONNECT TIME QUOTA EXCEEDED

The demand run has been open longer than the time allowed in its quota. The system immediately terminates the site-id and the offending run.

CONS ERR - SYNTAX*

You receive this message for one of these reasons:

- The host-id you specified on the @@CONS statement is not valid.
- You entered @@CONS[,*hostid*] without accompanying text.
- You entered HK *keyin* or HM *message* as the accompanying text.

CONS ERR - HOSTID NOT AVAILABLE*

The host-id you specified on the @@CONS statement is valid, but is not currently active in the file sharing environment.

@@CONS WARNING - THE KEYIN TEXT WAS TRUNCATED. VERIFY THE RESULT

You entered, through @@CONS, a keyin that had too many characters. The limit, not including the keyin prefix or read-and-reply number, is 66 characters for unsolicited keyins or 68 for replies to read-and-reply messages. The Exec truncated and executed the keyin. Check to make sure the keyin did what you intended.

CONTACT SITE ADMINISTRATOR AND REPORT COMMON BANK ERROR NUMBER *nn*.

This message is printed with other common bank error messages. It is meant to direct a user to the administrator for resolution of the problem. It is assumed that the user of a common bank is not able to correct what is quite possibly a system problem. The administrator can retrieve further information regarding the problem from the system log. See the *Exec Operations Reference Manual* (7831 0281) for an explanation of the common bank error numbers.

Continue entering compartment set (or XMIT):³

You entered a line of compartment names by using the continuation character (;) at the end of the line, and the system validated the names. Continue to enter more compartment names, or XMIT if you are finished entering compartment names.

COPY I/O ERROR *xx*

An I/O error was encountered while copying the stored runstream from the start file to an Exec file (*xx* is the I/O error code).

COPY FILE UNOBTAINABLE *xxxxxxxxxxxx*

The file specified on the @START statement cannot be copied to an Exec file (refer to Appendix C for the facility status bits (*xxxxxxxxxxxx*) returned).

CTL SYSTEM ANSWERED MESSAGE FOR devnam WITH: *nnnnnn*

This is a cartridge tape library message (CTL) that appears in the @FIN tailsheet. The message indicates how the CTL responded to a previous message for the run.

x CYCLE SPECIFICATION IGNORED

The cycle you specified in the processor-output or symbolic-output field is meaningless and is ignored. Does not cause error return.

D

DATA CHECK ERROR - FILE TERMINATED

The remainder of the file is lost.

DATA IGNORED - IN CONTROL MODE

Data statements were encountered when the Exec was attempting to read control statements; that is, a program or processor was not in control of the run at the time these statements were encountered.

DEVICE NAME SPECIFIED IS NOT CONFIGURED

You tried to queue (@SYM) a file to an illegal or nonexistent device or group.

DEV. NOT CONFIGD FOR OUTPUT FILES

An @@SEND was received from a demand terminal that is not configured for receiving batch output files.

*DIAG CANNOT HOLD ALL BANKS TO BE WRITTEN - PMD INCOMPLETE

There are more banks to be written to DIAG\$ than the file can hold. Only those banks that will fit into the file have been written out. There is no guarantee that the control bank is included in the banks written, hence a PMD may not be possible.

Do you wish to change your password? <Y>,N¹

Your password will expire soon. Answer 'Y' to change it now, or 'N' to ignore.

E

ELEMENT UNOBTAINABLE xx

The element specified on the @START statement cannot be found (xx is a program file search code).

END ERROR

You specified an improper mode on an @@END statement.

@END IGNORED - IN CONTROL MODE

An @END statement was encountered when the coarse scheduler was attempting to read control statements; that is, the DATA OR ELT,D processor was not in control of the run at the time this statement was encountered.

@ENDX IGNORED - NOT IN CLIST MODE

An @ENDX statement was encountered when the coarse scheduler was attempting to read control statements; that is, a program or processor was not in CLIST mode at the time this statement was encountered.

ENTER COMPARTMENT SET (OR NULL, MAX, XMIT=DEFAULT):³

This message solicits you for compartment set information during a demand sign-on sequence (see Section 5).

ENTER DISCRETIONARY SECURITY ATTRIBUTE (PUBLIC,PRIVATE,ACR)

You encounter this prompt while using the TLABEL processor to change the access list of a tape. Enter PUBLIC, PRIVATE, or ACR (abbreviations of two characters allowed).

ENTER SESSION ESTABLISHMENT REQUEST¹

Enter the \$\$OPEN statement in response to this message.

ENTER YOUR CLEARANCE LEVEL (0-63; <XMIT>=DEFAULT):²

You may receive this prompt after changing your password.

ENTER YOUR NEW PASSWORD:

You entered a new password that is not valid. Please enter a new password.

Enter your project identifier:¹

Enter a valid project-id.

Enter your user-id/password:¹

This is the first prompt after opening a Demand or TIP session.

Enter your user-id/password and clearance level:²

This is the first prompt after opening a Demand or TIP session.

Enter your new password (or transmit to request another password)¹

Your password has expired since you last signed on. The system has generated a new password for you. Enter your new password to verify the change or transmit to generate another password.

Enter your new password (or Transmit (XMIT) to not change your password): ¹

Enter your account number:¹

Enter a valid account number.

@EOF IGNORED - IN CONTROL MODE

An @EOF statement was encountered when the Exec was attempting to read control statements; that is, a program or processor was not in control of the run at the time this statement was encountered.

ERROR 26*ACR DOES NOT EXIST

The ACR you specified when attempting to update a security tape label with @TLABEL does not exist. You must create the ACR before you can use it.

ERROR 27*ACR IS MARKED AS DELETED

The ACR you specified when attempting to update a tape security label with @TLABEL has been marked for deletion.

ERROR 28*ACR CANNOT BE ACCESSED

You cannot access the ACR you specified when updating a security tape label with @TLABEL. You do not own the ACR and do not have the privilege to bypass the owner of the ACR.

ERROR 32*TAPE ACCESS NOT ALLOWED

When attempting to update a security tape label with @TLABEL, you failed security validation for the tape.

ERROR ASSIGNING COPYFILE FOR STARTED *runstream*

A system error prevented assignment of an Exec file for copying the started *runstream*.

ERROR - DYNAMIC DUMPS NOT CLOSED

I/O error encountered when writing the system diagnostic file (DIAG\$).

```
-@@ERROR (error codes)  
-@@ERROR - ILLEGAL TYPE  
-@@ERROR - SYNTAX
```

These three messages indicate that an error was encountered while processing a transparent statement. The *error codes* are the same error codes returned on a dynamic request of the control statement.

ERROR ENCOUNTERED WHILE OUTPUTTING FILE

An I/O error occurred with a code of 48, 118, or 128, and the inhibit recovery flag had not been set by an ER PRTCN\$. The message is displayed if the recovery process is successful or if the operator did not terminate the file.

ERROR ENCOUNTERED WHILE OUTPUTTING FILE

When there is successful symbiont output file error recovery, this message is displayed at the end of the current output file.

Recovery is unsuccessful if either the I/O error is nonrecoverable or the operator terminated the file by responding to the I/O error message with a response that does not allow recovery (such as G); or if recovery is inhibited, the operator responds NO to the message that asks whether recovery should be attempted.

ERROR LOADING PROGRAM

An error was detected in the absolute element. Bad element or the element is possibly destroyed.

ERROR OCCURRED WHILE THE linking *system* WAS ACTIVE.

The linking *system*, during program load or link fault processing, has terminated abnormally. Contact your systems analyst for resolution.

ESTIMATED RUN TIME EXCEEDED

The run either exceeded the system standard or the time specified on the @RUN statement, and the T option was set.

EVERY PAGE SECURITY LABELING OVERRIDDEN FOR FILE: [*directory qualifier*filename*] USER-ID: [*user-id*]

You specified the N option on the @SYM statement. Security labels do not appear on pages between the banner and trailer pages on the printout of the specified file for the specified *user-id*.

EXEC WAS UNABLE TO COMPLETE SYM OF FILE WITH CHANGED NAME

Someone changed the name of the file you specified on the @SYM statement.

*EXECUTION TERMINATED *

The execution of the demand run has been terminated by the demand symbiont @@X T statement.

F

FILE ACCESS DENIED xxxxxxxxxxxx

The file name specified on the @START statement cannot be assigned because the starter does not have access rights to it. Refer to Table C-2 for the facility inventory status bits (xxxxxxxxxx) returned.

@FILE AND @ENDF ARE NOT ALLOWED ON THE SYSTEM

You are not allowed to use the @FILE and @ENDF statements because of the security configuration on your system.

x FILE ASSIGN ERROR - nnnnnn

A fatal error occurred during the opening of the run when the Exec attempted to assign files TPF\$, DIAG\$, or SYS\$*LIB\$. The value represented by *x* is the name of the file; the value represented by *nnnnnn* is a 6-digit octal message number that explains why the assignment failed (refer to Appendix C).

FILE BEING BREAKPOINTED TO IS IN USE

An @BRKPT statement is issued for a file currently being used as a print or punch file.

x FILE CANNOT BE READ

Input file is in read-inhibited mode due to absence of read key, setting of write-only mode for file, or use of the Y option on the file assignment.

FILE ERROR

- A FACILITY REJECT received on @ASG of a program file.
- A FACILITY WARNING received on @ASG of a program file and status indicates the required read key is not supplied.
- Facility status on @USE was nonzero.
- Status returned from indicated file was not a program file.
- The file requested on an @XQT or processor call statement could not be assigned. If the run is not in demand mode, it is terminated.

FILE FORMAT ERROR. FILE TERMINATED

An image length error occurred and the image function was set.

FILE FORMAT ERROR. TERMINATED BY OPERATOR

An image length error occurred and the operator did not try recovery.

FILE FREED WHILE PROCESSING START

The file name specified on the @START statement was freed before the @START statement completed.

x FILE IS IN USE

Either you are requesting the Exec to open a print or punch file that is already open (it has an alternate symbiont control table), or the internal name you specified on an alternate symbiont request is not the same as the one you used when you first opened the file.

x FILE IS TAPE

You cannot specify tape files in the processor-output or symbolic-output fields.

FILE NOT ASSIGNED: x

You tried to free a file that was not assigned.

x FILE NOT AVAILABLE - STATUS: *n*

An attempt to assign a file failed. *n* is the facility request status code (see Appendix C).

FILE NOT CATALOGED ON MASS STORAGE

The file name specified on the @START statement is not cataloged on mass storage.

FIRST FILENAME IS IN ERROR

First file name was not given for an @BRKPT or @SYM statement, or an @BRKPT statement was given for an inactive alternate file. This message is also printed if the D option is specified on an @SYM statement and the file name specified is not generic name PRINT\$.

H

HOLD ERR - NOT DEMAND

@@HOLD is allowed only for demand runs.

HOLD ERR - SYNTAX

Syntax error on @@HOLD.

I

I/O ERROR ffffffff

ffffffff = LIB\$ or CSINTNAME\$. CSINTNAME\$ refers to the file specified on the preceding @XQT or processor call statement.

I/O ERROR ENCOUNTERED

An error occurred when the Exec tried to read the file specified on the @START statement.

I/O ERROR *nn* while reading from ADD file: *directory#qualifier*filename (fff)*

I/O ERROR *nn* was received while trying to read input images from an @ADD file. *directory#qualifier*filename (fff)* shows the external file name and F-cycle of the file on which the error occurred. Processing of the run cannot continue and this message is immediately followed by the message RUNSTREAM ANALYSIS TERMINATED.

I/O ERROR *xx*. FILE TERMINATED

An I/O error occurred that is not one of the recoverable types (*xx* is the error code).

I/O ERROR *xx*. TERMINATED BY OPERATOR

An I/O error occurred and the operator terminated the file (*xx* is the error code).

I/O ERROR *xx* WRITING DIAG\$ - PMD NOT COMPLETED

An I/O error occurred while writing the program to DIAG\$. PMD is incorrect (*xx* is the error code).

I/U OPTION CONFLICT

Both I and U options were given on the processor call statement. This situation is ambiguous.

ID NOT ACCEPTED

A user-id or password entered by the user is illegal.

ILLEGAL ACCOUNT¹

You entered a syntactically correct account number but you are not allowed to use that number. Depending on system configuration, you must start the sign-on process over again, or else the operator determines whether you are allowed to access the system.

ILLEGAL ACCOUNT FILE REFERENCE

A nonprivileged run attempted to perform a privileged function on the summary account file.

nn ILLEGAL CHARACTER *x*

The Exec encountered an illegal character *x* at column *nn* of the control statement.

ILLEGAL COMMON BANK BDI *iii*

A common bank BDI referenced by the program is invalid.

ILLEGAL CONTINUATION

Continuation of the control statement is not allowed or the next control statement has the control character (@) in the first column. The control statement is rejected, and if it is a batch run, the run is terminated.

x ILLEGAL DEVICE

Output file is not sector formatted, or input file is neither tape nor sector formatted.

ILLEGAL ENTRY POINT TO A GUARANTEED ENTRY COMMON BANK

The entry point specified by the user program is not that of the guaranteed entry common bank that is initially based; or more than one guaranteed entry common bank is initially based.

ILLEGAL EQUIP ON @FILE*

A file of the wrong device type was assigned. The user's read file is closed with an @FIN statement, the run is marked as removed, and the file from the first @FILE statement is not cataloged.

ILLEGAL EQUIP TYPE ON FIRST FILE

The device type on the first @FILE statement did not specify tape equipment. This may be encountered when processing multiple @FILE statements, or when a T option is found on the @FILE statement for a mass storage file.

x ILLEGAL FIELD

Field is ambiguous with option given (for example, l option specified and source output field coded).

nn ILLEGAL OPTION *x*

An illegal option *x* was encountered at column *nn*. The control statement is rejected, and if it is a batch run, the run is terminated.

ILLEGAL USERID

The user-id you specified on the @RUN statement is not installed under the account you specified, and either the operator has rejected it or the Exec is configured to reject all runs with undefined user-ids. If you submitted the run from a demand terminal, you can enter another @RUN statement. Batch runs are removed.

@@IMAGE IGNORED - TRANSP CTL IN PROGRESS

A transparent statement has been rejected because another transparent statement from this terminal is currently being processed.

IMPROPER RUNSTREAM IN FILE

The first image in the file or element specified on the @START statement is an invalid @RUN statement.

INITI LOAD ERROR

A type 04 program load error has occurred. See the *ER Programming Reference Manual* (7830 7899) for more information.

INLBL* reel# LABELED date time

This message confirms that you have updated the security label of tape *reel#* at date time during a batch run.

INPUT WAS IGNORED. AN ILLEGAL COMPARTMENT WAS ENTERED.³

Either you entered a compartment name incorrectly or the compartment name is not in your security record.

INTERVENING STATEMENTS SKIPPED

A conditional statement (@TEST, @JUMP) has been encountered and has caused one or more control statements to be bypassed.

INVALID BANK NAME TABLE IN ABSOLUTE ELEMENT

The bank name table in an absolute element contains duplicate bank names. This message is given during the initial load of a program.

INVALID EQUIPMENT TYPE FOR ADD FILE

INVALID SDF LABEL WORD

The input element is not in system data format or has been corrupted.

INVALID USERID FOR BATCH

The user-id specified on an @START statement could not be found in the system file.

L

L IS LEGAL FOR TAPE ONLY

The L option was specified on the @BRKPT statement and the file does not reside on tape.

LABEL FORMAT ERROR

You tried to illegally access a labeled tape, or a hardware error occurred when you try to validate a tape/disk pack label.

linking system INTERNAL ERROR

An auxiliary status word is printed with this message.

M

MASS STORAGE OVERFLOW

Mass storage request cannot be satisfied because mass storage is not currently available or pack is in down state.

MASTER OR SUBMASTER KEY IN ERROR

You attempted to use a privileged instruction.

Maximum number of user-created log entries has been exceeded

Your runstream has reached the maximum log count. The Exec passes control to an error routine and does not accept any more @LOG statements from the run.

nn MAX NUMBER OF CHARACTERS EXCEEDED

The character at column *nn* of the control statement is not a field/subfield terminator (, or /), and the maximum number of characters for this subfield has been reached. The control statement is rejected, and if it is a batch run, the run is terminated.

nn MAX NUMBER OF FIELDS OR SUBFIELD EXCEEDED

The character at column *nn* of the control statement is the field or subfield terminator (, or /), and no more fields or subfields are allowed. The control statement is rejected, and if it is a batch run, the run is rejected.

MAX CARDS

The estimated card output has been exceeded. A system generation parameter or the C option on the @RUN statement specified that the run be aborted if this occurred.

MAX PAGES

The estimated page output has been exceeded. A system generation parameter or the P option on the @RUN statement specified that the run be aborted if this occurred.

MAX TIME

The estimated running time has been exceeded. A system generation parameter or the T option on the @RUN statement specified that the run be aborted if this occurred.

MAX VOLUNTARY DELAY TIME

This message appears on the tail sheet of any batch run that exceeds its quota of voluntary delay time. This message appears only if the run has a quota abort.

MISSING ACCOUNT

The account is required, but it is not specified on the @RUN statement. If you submitted the run from a demand terminal, you can enter another @RUN statement. Batch runs are terminated.

MISSING USER-ID

The user-id is required, but it is not specified on the @RUN statement. If the run originated from a demand device, another @RUN statement can be entered. Batch runs are terminated.

MORE INFO? (A, X, R, J, E, B)

This message is displayed when a demand program terminates after encountering an error or aborts after an error identification line has been printed. This facility lets the demand user obtain more information about the error. If no additional information is desired, you can enter another control statement. The responses available to this message are any combination of the following letters:

- A Printout of the A-registers
- X Printout of the X-registers
- R Printout of the R-registers
- J Jump history stack saved at the time of the error
- E Message explaining the error code
- B Interpretation of bank descriptor indexes (BDI) and D-bits

MSG QUEUE FULL

The addressee on an @@TM statement already has seven messages queued, the maximum allowed.

N

NESTING OF @FILE STMT IS ILLEGAL

The @FILE statement was encountered with a file name different from the preceding @FILE statement. This occurred before encountering an @ENDF statement associated with the first @FILE statement. The user's read file is closed with an @FIN statement, the run is marked as removed, and the file from the first @FILE statement is not cataloged.

NO FILE SPECIFIED

You did not include a file name on an @START statement.

NO FILENAME AND/OR ELEMENT NAME SPECIFIED

You issued a control statement without giving the required file name.

NO PCT BUFFER AVAILABLE - PMD NOT POSSIBLE

A postmortem dump (PMD) is not possible at this time because there is not enough program control table (PCT) space available for the Exec. This applies only to @RUN statements using the W option (@RUN,W).

NO QUOTA FOR THIS TIME/TYPE

Quota limits are required but do not exist for this run type under the account specified on the @RUN statement, or quota limits do not exist for the current time under the account specified on the @RUN statement.

NO RUN ACTIVE

While in terminal inactive mode, an input image was received other than an @RUN or a transparent symbiont control statement.

NO SPACE FOR MAJOR SAVE ON ABORT\$ CONTINGENCY

Space is not available in the PCT for register save, when processing an ABORT\$ contingency. The run is terminated.

NO SUAs REMAINING FOR ACCOUNT

No standard units of accounting (SUA) remain for the account specified on the @RUN statement at run selection time.

NO SUAs REMAINING FOR USER-ID

No standard units of accounting (SUA) remain for the user-id specified on the @RUN statement at run selection time.

NONZERO I/O STATUS FROM USER FILE *ss*

After an I/O request on the file, an I/O status was returned that indicates either the file has been destroyed or there is incorrect data in the absolute element.

x NOT A PROGRAM FILE

The file you specified on the processor call statement cannot be used as a program file.

O

OPERATOR REMOVED RUN

The operator removed the run from the backlog by using the RM keyin. A possible solution to eliminate receiving this error is to have valid user-id, account number, and characters in your run-id.

OPERATOR TERMINATED RUN BY AN E-KEYIN

The operator replied with an E to terminate the run, or a demand user entered an @@X T.

OPERATOR TERMINATED RUN BY AN X-KEYIN

The operator entered an X keyin to terminate the run.

OPERATOR TERMINATED RUN KILLED VIA AN E-KEYIN

The operator entered an E keyin for this run. When used for a demand run, an E keyin terminates the task that is currently executing.

OPERATOR TERMINATED RUN KILLED VIA AN X-KEYIN

The operator entered an X keyin for this run. When used for a demand run, an X keyin terminates the task that is currently executing.

P

PARITY ERROR

A parity error has been detected in at least one character of the input image. The entire image is discarded (for teletypewriter only).

@@PASSWD ERROR: ILLEGAL COMPARTMENT

The compartment name is not valid, or you are not allowed to access it. Your run terminates because of the error.

@@PASSWD ERROR: INVALID CLEARANCE LEVEL

One of the @PASSWD statements has a clearance level specified that failed validation.

@@PASSWD ERROR: INVALID USER-ID/PASSWORD

The password you entered as your current password is not legal.

@PASSWD ERROR: THE N-OPTION IS NOT ALLOWED FOR REMOTE BATCH RUNS.

@PASSWD ERROR: USERID'S AUTHENTICATION TYPE IS NOT VALID FOR BATCH.

@PASSWD ERROR: YOU CANNOT CHANGE YOUR PASSWORD IN BATCH MODE

You attempted to change a password using the @PASSWD control statement in a batch run. You cannot use the @PASSWD statement because the machine-generated passwords feature has been configured at your site.

@@PASSWD ERROR: YOU MAY NOT CHANGE YOUR PASSWORD YET.

You tried to change your password too soon after the last time you changed your password.

@@PASSWD ERROR: YOUR CURRENT PASSWORD WILL EXPIRE TOMORROW.

Your password is valid, but it is due to expire tomorrow.

@@PASSWD ERROR: YOUR NEW PASSWORD CONTAINS INVALID CHARACTERS.

You tried to include invalid characters in your new password (for example, comma, slash (/), blank).

@@PASSWD ERROR: YOUR NEW PASSWORD IS TOO LONG.

You tried to replace your password with one that is more than the maximum characters allowed.

@@PASSWD ERROR: YOUR NEW PASSWORD IS TOO SHORT.

You tried to replace your password with one that is shorter than required for your system.

@@PASSWD ERROR: YOUR NEW PASSWORD MUST BE DIFFERENT THAN YOUR OLD PASSWORD.

Your new password must not be identical to your current password.

@@PASSWD ERROR: YOUR NEW PASSWORD MUST BE DIFFERENT THAN YOUR USER-ID.

You tried to change your password to the same characters as your user-id.

@@PASSWD ERROR: YOUR PASSWORD HAS EXPIRED.

Your current password has expired and you have not replaced it. Your run is terminated.

** @PASSWD STATEMENT IGNORED **

You have placed an @PASSWD statement in an @ADD file, or have entered @PASSWD in demand. The correct demand statement is @@PASSWD.

PCT EXPANDED BEYOND SYSTEM LIMITS

One of the following has occurred:

- The number of main storage blocks required for expansion of this run's PCT exceeds system generation parameter PCTMAX. When a run aborts with this message, a postmortem dump of the PCT (obtained by using @PMD,P) may show one of the following to be the cause:
 - Excessive number of DAD tables (consider using initial reserves when creating files)
 - Excessive number of activities (check for ER FORK\$ loop)
 - Excessive number of files assigned (check for ER CSF\$ loop)
- The size of this run's PCT does not exceed PCTMAX, but the PCT could not be expanded because the PCT is temporarily locked in main storage while an initial bank load for the program is occurring. Using the E through L options on the @RUN statement, try assigning a larger initial value for the PCT.

PCT AT MAX SIZE - UNABLE TO COMPLETE ADD

A PCT buffer is unavailable. This stops you from assigning any more files to your run. To get more PCT space, perform an @FREE on the files you are currently not using that are assigned to your run.

PCT FULL. CANNOT COPY STARTED RUNSTREAM

A PCT buffer is unavailable; this prevents copying the started runstream to an Exec file.

PCT OVERFLOW ON INITIAL LOAD

The total number of PCT blocks requested is greater than system generation parameter PCTMAX. This message is specifically given during an initial load.

PCT/PROGRAM SIZE EXCEEDS USER CORE

This program is too large to fit the available main storage.

PMD NOT ALLOWED

Postmortem dump is not allowed for a system processor (called from the system library) unless a Y option appeared on the @RUN statement. If an N option appears on the @RUN statement, postmortem dumps of any programs are not allowed. If a W option appears on the @RUN statement in batch mode, postmortem dumps can be done only if the preceding task terminates because of an error.

PROCESSOR CALL ERROR

A program that expects to be called as a processor has been called with @XQT.

PROGRAM COULD NOT BE LOADED

Additional information regarding this message is found in accompanying messages.

PROGRAM EXCEEDED MEMORY QUOTA

The storage requested by the run exceeds its quota limit.

PROGRAM NOT FOUND

The requested program or processor is not in the given file (the system library or TPF\$, depending on the statement). If the run is batch, it is terminated.

PROGRAM TOO LARGE PROGRAM SIZE *nnnnn* BLK SCORE SIZE *xxxxxx* BLOCKS

The program is too large to fit in the space available to user programs. The program requires *nnnnn* main storage blocks. *nnnnn* and *xxxxxx* are octal numbers.

PUNCH FILE CANNOT BE PUNCHED: NO PUNCH DEVICES CONFIGURED

Punch device is not configured.

Q

QECL IMAGE WAS NOT A PROCESSOR REQUEST

Only processor calls and @XQT statements are allowed in an ER QECL\$ request packet; all other control statements are ignored.

QUOTA ABORT

The run has exceeded one of several quota values. The exact quota parameter exceeded appears either on the tail sheet or immediately preceding this message.

R

READ\$FILE REJECT STATUS xxxxxx

The READ\$ file cannot be assigned.

x READ ONLY OUTPUT FILE

Output file is in write-inhibited mode, due to absence of write key, setting of read-only mode for file, or use of the Y option on the file assignments.

REAL TIME-LOAD PARAMETER IGNORED

A run requesting execution of a real time-load absolute element has an account number for which real time is not permitted. In this case, the element is executed without real-time load privileges.

REAL TIME NOT ALLOWED FOR THIS ACCOUNT/USER-ID

Your account/user-id is not configured for real-time processing.

REAL TIME PROGRAM ATTEMPTED PCT EXPANSION

A PCT expansion attempt for a real-time program could not be done. If the program expects PCT expansion, it should use an @RUN statement option to initialize PCT size so expansion is not attempted.

RELOAD ERROR

It was not possible to install the new bank on the system. Unless an I/O error occurred while loading the bank's template into storage, the old bank remains installed on the system. Otherwise, the bank is no longer installed. Additional information regarding this message is found in accompanying messages.

REMARK**CATALOGED TAPE FILE ENCOUNTERED

REMARK**MFD UPDATED TO MATCH SECURITY LABEL

When labeling or updating a tape label, TLABEL encountered a cataloged tape file. The MFD lead item of the file was updated so the security attributes are consistent with the tape label.

REMARK**reel# tape*name UPDATED date time

The security label for *tape*name* on *reel#* has been updated at given date and time.

nn REQUIRED FIELD OR SUBFIELD MISSING

A field or subfield that is required on the control statement has not been specified. The omission was detected when the Exec reached a field or subfield terminator (, or /) or the end of the control statement, at column *nn*. The control statement is rejected, and if it is a batch run, the run is terminated.

RUN ALREADY ACTIVE

An @RUN statement was received while in demand run mode.

RUN-ID NOT ACTIVE

The run-id on an @@TM is not currently active.

RUN REMOVED DUE TO INVALID USER-ID/PASSWORD

Either the user-id or the password on the @PASSWORD statement is illegal.

RUN REMOVED DUE TO SECURITY ERROR

The user's run has been removed due to an illegal password, or a password was not found in the runstream.

RUNSTREAM ANALYSIS TERMINATED

The run has been terminated because of an error condition, and the remaining control statements are not processed.

S

SACCNT ERROR x

An internal Exec error was encountered while referencing the summary account file.

SECOND FILENAME IS IN ERROR

The second file name on your @BRKPT statement is not assigned to your run.

SECOND FILENAME NOT ASSIGNED, PUNCH\$ NOT BREAKPOINTED

SECOND @FILE STATEMENT FORMAT ERROR

A syntax error was encountered on the second @FILE statement. SECOND implies that one or more @FILE statements have already been processed. The user's read file is closed with an @FIN statement, the run is marked as removed, and the file from the first @FILE statement is not cataloged.

SECOND NAME IS ILLEGAL

A second file cannot be given on an @BRKPT of a read alternate or print alternate file.

SECURITY LABELING IS OFF, L OR N OPTIONS IGNORED

You included the L or N option on the @SYM statement, but security labeling is not configured on your system.

The symbolic-input field on a processor call specifies an S-cycle that does not exist, or cycle field is in the wrong format.

The system cannot find the element you specified in the symbolic-input field of a processor call statement.

SI: IMPROPER LABEL BLOCK

SI IS NOT SIR TYPE SDF

SI: MISSING FIELD

SI: TAPE I/O ERROR - STATUS: *n*

SIR EDIT ERR cause range correction-image

SITE-ID NOT ACTIVE

SPECIFIED ADD ELEMENT DOES NOT EXIST.

START ADDRESS NOT WITHIN LIMITS OF INITIALLY BASED BANKS

A-23

STARTER DENIED ACCESS TO TARGET USERID

The starter that issued the @START statement with the U option specified does not have execute access to the user-id or the stored RUN statement.

ST KEY ERROR - V AND Z ARE CONFLICTING OPTIONS. RUN NOT STARTED.

The V and Z options are incompatible and cannot be used together in @START statements or ST keyins.

ST KEY ERROR - V OPTION REQUIRES PRIVILEGES. ER SUMOD\$ ERR 0xx.

An error occurs if none of the following conditions are met when using an @START,/V or @START,/VU statement:

- The start file clearance level and compartment set do not equal the target user-id's minimum clearance level and default compartment set.
- The start file is not system low (clearance level of 0 and default compartment set of NULL).
- The requestor is not sufficiently privileged to change the start copy file to the minimum clearance level and default compartment set of the target user-id.

@SYM OF FILE MARKED TO BE CATALOGED ILLEGAL

You cannot queue (@SYM) a file unless it is cataloged. If you assigned your print file with a C or U option, you must release the file (@FREE) before you can queue it (@SYM).

@SYM OF SHARED FILE NOT DONE--FILE SHARING ENVIRONMENT DOWN

You have used @SYM on a print file that is cataloged as a shared disk file and the system file sharing environment has been brought down by an MH keyin. You must wait for file sharing to be restored, and then resubmit the @SYM statement.

@SYM OF TEMPORARY FILE IS ILLEGAL

Only cataloged data files can be used as print files.

*SYSTEM HOLD ON DEMAND RUNS*¹

The operator has placed a hold on the opening of demand runs. Demand symbiont statements are accepted, and the remote batch input mode (RUN,/B) can be entered to place runs in the batch backlog.

SYSTEM WARNING - MAX CARDS

The card estimate on the @RUN statement has been exceeded. It is doubled, and the program/run continues.

SYSTEM WARNING - MAX PAGES

The page estimate on the @RUN statement has been exceeded. It is doubled, and the program/run continues.

SYSTEM WARNING - MAX TIME

The estimated time on the @RUN statement has been exceeded. It is doubled, and the program/run continues normally.

T

TAPE IMAGE LOST*REREAD

Applies only to demand terminals; indicates that images were lost while inputting images in paper tape mode.

TAPE LABEL FORMAT ERROR

Tape labeling errors receive the following diagnostics:

devnam ACCESS NOT ALLOWED ON *aaa*

where *devnam* is the device name of the unit and *aaa* is the reel number. If read-and-write inhibit is set beyond the first label group, the following message is logged:

*qualifier*filename(fff)* ACCESS NOT ALLOWED ON FILE *n*

where *qualifier*filename(fff)* is the external file name and F-cycle of the tape file. File *n* is the tape file in which the error occurred.

Errors that occur beyond the first label group are logged to the user. The following messages are used:

*qualifier*filename(fff)* {EOF1|EOF2|HDR1|HDR2} LABEL NOT FOUND FOR FILE *n*

*qualifier*filename(fff)* FILE *n* EXPED *x* BLOCKS REC *y*

*qualifier*filename(fff)* I/O ERROR *z* ON {R|W} ON {TRL|HDR} LABEL ON FILE *n*

<i>x</i>	The number of blocks in the EOF1, EOF2, or HDR1 labels (HDR1 in reverse processing).
<i>y</i>	The number of blocks actually read when the tape mark was received.
<i>z</i>	The actual I/O error status received (other than 0).
R	Read
W	Write

TRL	Indicates trailer label group; and HDR indicates header label group.
EXPED	Expected
REC	Received

The first two error log messages indicate that something is missing on the tape. In both cases, recovery is attempted. In the case where a label is missing, the system tries to determine whether any type of labels exist. If so, it tries to position accordingly. If not, it repositions after the last valid label group or at the tape mark read by the user.

In the case where file *n* expected *x* blocks and received *y*, the system has picked the block count in the EOF1 or EOVI label, compared it to the number of blocks extended, and found them not equal.

In the case where the number of blocks read by the user does not correspond to the block count in the EOF1, EOF2, or HDR2, the error message is logged to the user, and processing continues.

If an I/O error is detected beyond the first label group and the I/O is a read, the input buffer is checked to see whether there was a partial read of a label. If there was, processing is attempted. If there was not, the file is set to read- and write-inhibited, and the user is informed that access to that file is not allowed.

TARGET USERID DOES NOT OWN START FILE

The @START statement with the U option specified requires that the user-id on the stored RUN statement own the start file.

TEMPORARY FILE MASS STORAGE QUOTA LIMIT EXCEEDED

The allowable temporary mass storage tracks in the quota set for this account number have been depleted.

*TERMINAL INACTIVE*¹

The terminal is now inactive. The next ECL statement expected is an @RUN (or @RUN,/B). (Your user-id may be configured so that the @RUN statement is automatically generated.)

TERMINAL INACTIVE - OUTPUT FILE AVAILABLE

The terminal is inactive, and a batch output file is available. An @@SEND causes the file to be transmitted.

*TERMINATED DUE TO SYSTEM HOLD ON TERMINALS*¹

The onsite operator has placed a hold on the recognition of demand sites.

*TERMINATED DUE TO UNAVAILABLE SYSTEM RESOURCES*¹

You cannot log onto the system at this time because of a lack of system resources or because the maximum number of terminals allowed are already active. For example, if the Exec ASA becomes tight, then new batch and demand runs are held by the system to prevent further strain on system resources.

THE AUTHENTICATION SUBSYSTEM IS NOT AVAILABLE. THIS SIGN-ON IS REJECTED.
CONTACT YOUR SITE ADMINISTRATOR.¹

The system cannot process passwords at this time.

The Exec cannot assign *qualifier*filename* -- will try to assign *filename*.¹

The Exec was processing an @BRKPT PRINT\$ statement from a batch run, but could not assign the next primary print part. The Exec skipped that part and tried to assign the next one.

THE NEW PASSWORD YOU ENTERED CONTAINS INVALID CHARACTERS.¹

You cannot use a space, comma (,), or slash (/) in your password.

THE NEW PASSWORD YOU ENTERED IS TOO LONG.¹

Your new password is longer than the maximum length currently allowed by the system.

THE NEW PASSWORD YOU ENTERED IS TOO SHORT.¹

Your new password is shorter than the minimum length currently required.

THE TRANSMISSION OF DATA FROM A CLASSIFIED USER IS NOT ALLOWED.

For security reasons, you are not allowed to use the @@TM, @@CONS TM, or @@SYM statements.

THIS TERMINAL HAS BEEN DISABLED.¹

You have made too many attempts to sign on to the system from this terminal. The system disables the terminal until the security officer frees it.

TIMEOUT CHANGE NOT ALLOWED

The user-id is not allowed to change terminal timeouts by using the @@TOUT t statement.

TIMEOUT DISABLED

The timeout mechanism is disabled because of an @@TOUT D statement. The terminal will not time out.

TIMEOUT DISABLE NOT ALLOWED

The user-id cannot disable the timeout mechanism by using @@TOUT D.

*TIMEOUT PERIOD IS *t* MINUTES*

The current terminal timeout period in minutes is indicated by *t*. When the timeout value indicated on the @@TOUT is greater than the timeout length allowed for the user-id, the new timeout period is the maximum value allowed for the user-id, as indicated by *t*.

*TIMEOUT WARNING*¹

No activity has occurred on the line for a predefined interval. If another time interval elapses without activity, the terminal is terminated.

TM KEY ERROR

Syntax error on @@TM.

TOO MANY COMMON BANKS DEFINED

The number of common banks referenced by this program is greater than the system limit.

TOO MANY SPECIFICATIONS

A processor call statement is too complex.

TOO MANY USER BANKS DEFINED

The number of user banks defined for this program is larger (250) than the maximum allowed.

U

UNABLE TO ASSIGN ADD FILE

The file you specified on an @ADD statement does not exist. Correct the statement and try again.

UNABLE TO GENERATE UNIQUE RUNID FOR *runid*

The system is unable to generate a unique run-id after going through the possible combinations for the last two characters of the submitted run-id.

UNABLE TO INSTALL COMMON BANKS

It was not possible to install a common bank that the executing program uses. This message is printed with the following message:

CONTACT SITE ADMINISTRATOR AND REPORT COMMON BANK ERROR NUMBER *nn*.

UNRECOVERABLE I/O ERROR WHEN READING FILE *filename*

The Exec encountered an unrecoverable I/O error when searching the *filename* for a program or processor during a run. If it is a batch run, the run is terminated.

UNISYS 1100 OPERATING SYSTEM LEV. xxRx xx.xx (RSI)

UNISYS 1100 OPERATING SYSTEM LEV. DS1R2 (RSI)

Indicates you have successfully logged onto the system.

UNISYS 1100 OPERATING SYSTEM LEV. xxRx xx.xx (RSI) HOST x

Indicates you have successfully logged onto a system with MHFS, on host-id x.

U OPTION REQUIRES SENTRY CONFIGURED

Specification of the U option on the @START statement requires that SENTRY CONTROL be configured in the system.

USERID AND ACCNT NEEDED ON RUN STATEMENT

The @START statement was entered with the U option specified, but either the user-id or account-id was not present on the stored @RUN statement. (The account-id on the stored @RUN statement can be absent if it is specified on the @START statement.)

USER-ID DIFFERENT FROM LOG-ON

The user-id supplied on the demand @RUN statement is different from the user-id verified at log-on.

USER-ID NOT ACTIVE

The user-id on an @@TM is not currently active.

USER-ID SUA QUOTA EXCEEDED

No standard units of accounting (SUA) remain for the user-id specified on the @RUN statement at initial scheduling time.

USER JUMP HISTORY REL ADD-BDI

The last jumps and their corresponding bank descriptor indexes (BDI) are printed as a result of standard action taken by the Exec on a hardware fault interrupt caused by the user. If any absolute jump address falls outside the user limits, the address - BDI is printed as 000000 - 000000.

V

VOLUNTARY DELAY TIME QUOTA EXCEEDED

A demand run has exceeded its quota of voluntary delay time. The Exec immediately terminates the site-id and the run.

W

*WAIT-LAST INPUT IGNORED¹

This applies only to demand terminals. The last input statement was rejected, probably because another statement was still being processed.

WARNING - CONSOLE OUTPUT LOST

Some output specified on an @@CONS statement may be lost.

WARNING IMPROPER ACTION C-OPTION ILLEGAL FOR NON-PUNCH TYPE DEVICE

This is caused by an @SYM,C statement with printer symbiont name (warning only).

WARNING: MAX CONNECT TIME LIMIT IN 5 MINUTES

The demand run has only 5 minutes of its connect time quota remaining.

WARNING: MAX VOLUNTARY DELAY TIME LIMIT IN 2 MINUTES

The demand run has only 2 minutes of its voluntary delay time quota remaining.

WARNING: YOU CANNOT CHANGE YOUR PASSWORD USING @@PASSWD

You attempted to change your password using the @@PASSWD control statement. You cannot use this statement because the machine-generated passwords feature is configured at your site.

devnam WRONG REEL, LOAD INITIALIZED VOLUME AE *reel*

You see this message if a tape with a security label is mounted and you request a prelabeled blank tape. You need a tape with an initialized volume to be mounted.

X

X OPTION REQUIRES USERID

A run control statement from a card reader on a start control statement has specified the X option but contains no user-id.

X OPTION SECURITY VIOLATION

A run control statement from a card reader on a start control statement has specified the X option and failed a security system check. Probably, the user-id does not have the X option privilege.

@@X ERROR - TOO MANY DATA CHARACTERS

The *data* field cannot be more than 6 characters.

Y**YOU HAVE ENTERED AN INVALID ACCOUNT INDEX.¹**

You did not respond correctly to the account index solicitation (see Section 5.2).

YOU HAVE ENTERED AN INVALID ACCOUNT NUMBER.¹

You entered a syntactically incorrect account number when responding to the account number solicitation.

YOU HAVE ENTERED AN INVALID CLEARANCE LEVEL.²

You entered a clearance level that you are not allowed to use.

YOU HAVE ENTERED AN INVALID PASSWORD.¹

You entered the wrong password as your current password.

YOU HAVE ENTERED AN INVALID USER-ID.¹

You entered a user-id that is not registered on the system.

YOU MAY NOT CHANGE YOUR PASSWORD YET.¹

You tried to change your password before you are allowed.

YOUR CURRENT PASSWORD WILL EXPIRE TOMORROW.¹

Your password expires tomorrow.

YOUR NEW PASSWORD MUST BE DIFFERENT THAN YOUR OLD PASSWORD.¹

You cannot change your password to itself.

YOUR NEW PASSWORD MUST BE DIFFERENT THAN YOUR USER-ID.¹

You cannot change your password to the same characters as your user-id.

YOUR PASSWORD HAS BEEN REPLACED.¹

Your new password was accepted by the system.

YOUR PASSWORD HAS NOT BEEN REPLACED.¹

Your new password was not accepted by the system.

YOUR PASSWORD HAS EXPIRED. ENTER YOUR NEW PASSWORD.¹

Your password has expired since you last signed on. Supply a new password.

YOUR PASSWORD HAS EXPIRED.¹

ENTER YOUR NEW PASSWORD (OR TRANSMIT (XMIT) TO NOT CHANGE YOUR PASSWORD)

You may either enter a new password or skip changing your password.

YOUR PASSWORD HAS EXPIRED.¹

YOUR NEW PASSWORD IS: xxyyzzzz (xx-yy-zzzz)

ENTER YOUR NEW PASSWORD (OR TRANSMIT TO REQUEST ANOTHER PASSWORD)

Your password has expired since you last signed on. The system has generated a new password for you. Enter your new password to verify the change or transmit to generate another password.

YOUR PASSWORD HAS EXPIRED.¹

YOUR USERID WILL BE DISABLED IF YOU DO NOT CHANGE YOUR PASSWORD

YOUR NEW PASSWORD IS: xxyyzzzz (xx-yy-zzzz)

ENTER YOUR NEW PASSWORD (OR TRANSMIT TO REQUEST ANOTHER PASSWORD)

Your password has expired since you last signed on. The system has generated a new password for you. Enter your new password to verify the change or transmit to generate another password. Your user-id will be disabled if you do not change your password.

YOUR USERID CANNOT SIGN-ON; IT HAS NO_PASSWORD_ACCESS SET.¹

Your user-id is not allowed to use demand or TIP access.

YOUR USER-ID DOES NOT HAVE ACCESS TO THE REQUESTED APPLICATION.¹

The TIP application you are accessing is not allowed for your user-id.

YOUR USER-ID DOES NOT HAVE ACCESS TO THE REQUESTED PROJECT-ID.¹

The project-id validation failed for a batch @RUN image or a user-entered demand @RUN image.

YOUR USER-ID HAS NO ACCESS TO BATCH.¹

Your user-id is not configured for batch access. Your run is terminated because of an error.

YOUR USER-ID HAS NO ACCESS TO DEMAND.¹

Your user-id is not configured for demand access.

YOUR USER-ID HAS NO ACCESS TO TIP.¹

Your user-id is not configured for TIP access.

YOUR USER-ID IS DISABLED. CONTACT YOUR SECURITY OFFICER¹

Your user-id was disabled because you did not change your password or because too much time has elapsed since your last sign-on. Contact the security officer for instructions.

YOUR USER-ID WILL BE DISABLED IF YOU DO NOT CHANGE YOUR PASSWORD¹

Z

Z OPTION REQUIRES STARTED RUN USERID HAVE SYSTEM HIGH CAPABILITY

You specified the Z option on the @START statement, but your user-id or the user-id you included on the @START statement, along with the U option, cannot execute at system high.

Z OPTION REQUIRES STARTER TO HAVE Z OPTION PRIVILEGE

You specified the Z option on the @START statement, but do not have the SSSTRZOPT privilege required to use the Z option.

Endnotes

- 1 Applies to all system security levels (fundamental, security level 1 and higher).
- 2 Security Level 1 or higher.
- 3 Security Level 2 or higher.

Appendix B

FURPUR Messages

B.1. Functional and Error Messages

The functional and error messages are divided into the following categories:

- Messages prefixed with I/O error codes
- Messages without prefixed I/O codes

For all commands that complete successfully with an END message (such as END DELETE, END REWIND, or END ERS) and can accept more than one file name on the command line, only one END COMMAND message will be issued upon successful completion even if several files were processed by the function.

Use of the C option on any FURPUR command will prevent execution of ER ERR\$ on error exit if FURPUR is running in breakpoint or batch. The C option should be used with caution because a bad result of an erroring FURPUR command could be input to a process later on in the runstream.

B.1.1. Messages Prefixed with I/O Error Codes

13 THE PERIPHERAL UNIT WAS DECLARED DOWN BY AN UNSOLICITED OPERATOR
KEYIN.

Call your system operator.

14 TRANSLATE ERROR - NO TRANSLATE FEATURE OR TRANSLATE TABLE NOT LOADED.

Call your hardware maintenance representative.

16 A PORTION OF THE I/O PACKET IS NOT WITHIN PROGRAM LIMITS.

Call your software maintenance personnel.

17 I/O NOT COMPLETED - LACK OF EXPOOL SPACE.

Call your software maintenance personnel.

21 AN ATTEMPT WAS MADE TO REFERENCE A FILENAME FOR WHICH NO ASSIGNMENT
HAS BEEN MADE.

Call your software maintenance personnel.

23 PACKET ADDRESS SPECIFIED IN A0 IS NOT WITHIN PROGRAM LIMITS, OR
DEFINES A PACKET SPLIT BETWEEN THE INSTRUCTION AND DATA BANKS OF
A PROGRAM.

Call your software maintenance personnel.

24 THE FUNCTION CODE IS NOT DEFINED FOR THE ASSIGNED EQUIPMENT TYPE.

Call your software maintenance personnel.

25 THE I/O ACCESS WORD REFERS TO A BUFFER WHICH IS WHOLLY OR
PARTIALLY OUTSIDE OF THE PROGRAM AREA.

Call your software maintenance personnel.

27 AN I/O REQUEST WAS MADE WITH THE STATUS WORD OF THE REQUEST
PACKET SET TO 040 INDICATING THE PACKET IS ALREADY IN USE.

Call your software maintenance personnel.

B.1.2. Messages Without Prefixed I/O Error Codes

EM CALL ERROR - SEVERITY: *a*, PRODUCT_ID: 0**bbbbbb**, MSG_ID: 0*c*

On an Extended Mode (EM) CALL to an Exec system service an error code was returned. Refer to the *OS 2200 System Services Programming Reference Manual* (7833 4455) to interpret the octal values in this error message.

END COPY

Copy, I has been completed successfully.

END DELETE

The deletion of a cataloged file has been completed successfully.

END REWIND

Rewind has been completed successfully.

EOF ENCOUNTERED, MOVE 1 OCCURRED.

This message is displayed when you attempt to copy and the tape is at the file mark position. Thus, when you performed the @COPY statement, FURPUR moved the tape to the correct position but did not complete the operation.

n EOFs ENCOUNTERED.

This message is a normal response to an @MOVE statement. It is accompanied by other messages if the position you requested in the @MOVE statement cannot be attained.

THE END OF THE RECORDED AREA HAS BEEN ENCOUNTERED.

The @MOVE statement you specified moves the access pointer on a diskette beyond the last data set recorded on the diskette. The associated message, *n* EOFs ENCOUNTERED, indicates the number of data sets actually encountered before the end of the diskette.

WARNING - BUFTAP PERFORMANCE ADVANTAGE NEGATED WITHOUT THE E OPTION

A @COPOUT, @COPY, or @MARK statement without the E option has been done to an unlabeled output tape that has BUFTAP buffered-write mode enabled to improve tape write performance. Without the E option these statements write two EOF marks and move backward between them. This backward tape movement causes the tape unit to write the data from the tape unit buffer to the physical tape. This negates the performance advantages of BUFTAP, which require constant forward tape motion and no implicit or explicit tape synchronization requests. Include the E option with these statements to gain the full performance advantage of BUFTAP buffered-write mode.

WARNING - SOLAR reported a software license compliance exception with auxiliary status code *n*

Where *n* is the decimal auxiliary status code returned by SOLAR and is documented in Appendix F, "Conditioned File Copy Auxiliary Status Codes" in the *Software Library Administrator (SOLAR) User Reference Manual* (7831 0604).

B.2. Quarter-Inch Tape Subsystem Operations

The following error messages are displayed when a syntax error involving quarter-inch tape subtype operations is encountered on one of the FURPUR commands:

ATTEMPT TO WRITE BEYOND PHYSICAL END-OF-TAPE ON THIS TAPE TYPE.

THIS TAPE TYPE CANNOT MOVE BACKWARD.

THIS TAPE TYPE CANNOT WRITE BLOCK SIZES LARGER THAN 65,536 BYTES.

END-OF-RECORDED AREA ENCOUNTERED, APPENDING POSSIBLE.

B.3. Error Messages Associated with Element Subtype Selection

The following error messages are displayed when a syntax error involving element subtype selection is encountered on one of the FURPUR commands that allow element subtype selection:

ABS, OM, SD, ZM, OR BOM SUBTYPE MNEMONIC REQUIRED WITH Y OPTION.

This error message is displayed when the AY combination has been specified, but one or more of the items in the subtype list is not one of the valid mnemonics: ABS, BOM, OM, SD, ZM.

ILLEGAL SEPARATOR IN ELEMENT SUBTYPE FIELD

This error message is displayed when an incorrect item separator (that is, something other than a slash) is found in a subtype list.

Y OPTION WITHOUT THE A OPTION IS ILLEGAL.

This error message is displayed when the Y option is not accompanied by the A option.

B.4. Error Messages Associated with the Prep of Object Modules

The following error messages are associated with the prep operations.

The fifth message in this set occurs when, in the process of building the object module entry-point table, the maximum size of the file is reached.

This message

OBJECT MODULE PREP HALTED

follows the display of each of the following messages:

ERROR OCCURRED WHILE READING FTI DURING OBJECT MODULE PREP.

ERROR OCCURRED WHILE WRITING FTI DURING OBJECT MODULE PREP.

ERROR OCCURRED WHILE WRITING OBJECT MODULE ENTRY POINT TABLE.

FILE OVERFLOW OCCURRED DURING OBJECT MODULE PREP.

NOT A PROGRAM FILE.

UNABLE TO CREATE OBJECT MODULE ENTRY POINT TABLE.

UNABLE TO READ EXISTING OBJECT MODULE ENTRY POINT TABLE.

B.5. Error Messages Returned by PCFP

FURPUR calls Program-Callable FURPUR (PCFP) to perform all or part of the processing for the following statements:

- @COPY *mass-storage-file-1,mass-storage-2.* (see Section 7.13 for all @COPY statement formats)
- @COPY,AORS *program-file-1,program-file-2.*
- @COPY,G *mass-storage-file.,tape-file.*
- @COPY,G *tape-file.,mass-storage-file.*
- @COPY,P *program-file-1,program-file-2.*
- @DELETE,UAORS *program-file-1,...* (See Section 7.17.3)
- @ERS *mass-storage-file-1,...* (See Section 7.24)
- @PACK *large-program-file-1,...* (See Section 7.42)

If PCFP detects errors, it returns error information to FURPUR. The error information includes an Extended Language Message System (ELMS) condition identifier, an auxiliary error code, and a file identifier.

If ELMS message delivery is available, FURPUR calls ELMS to display the error message. The following is a typical PCFP error message displayed by ELMS:

```
*ERROR PRA002001: Unexpected I/O status 11 was returned by a read (R$)
request to the file with index 1.
```

If ELMS message delivery is not available or if ELMS is unable to display the error message, FURPUR displays the error information returned by PCFP in the following messages:

```
ELMS MESSAGE DELIVERY IS NOT AVAILABLE
PCFP ERROR CODE: nnnn, AUX_ERROR_CODE: xxxxxxxxxxxx, ERROR_FILE: y[, SUB_ERROR_CODE: zzzz]
REFER TO PCFP PROGRAMMING REFERENCE MANUAL (7830 9796) FOR MESSAGE TEXT
```

Where *nnnn* is the decimal PCFP error code (part of the returned ELMS condition identifier), *xxxxxxxxxxxx* is the octal auxiliary error code, *y* is the file identifier, and *zzzz* is the decimal sub error code, if it is nonzero.

The PCFP errors listed in this section are the ones that will be most commonly encountered for the FURPUR statements that are performed by PCFP. This is neither a complete list of PCFP error messages nor a complete list of possible PCFP errors that can occur for the FURPUR statements. Refer to the *Program-Callable FURPUR (PCFP) Programming Reference Manual (7830 9796)* for a complete description of PCFP error messages.

For the error messages listed in this section, if *xx* is shown, it is the auxiliary error code, usually displayed in octal. If *y* is shown, it is the decimal file index of the file in error. FURPUR adjusts the file index returned by PCFP so that it corresponds to the FURPUR statement field number for the file. If the file in error was a temporary file assigned by PCFP, the file index is set to 99.

0021 The performance advantage of buffered-write mode BUFTAP is negated because `Omit_Log_End_Mark` is `FALSE`.

A `@COPY,G` statement without the `E` option has been done to an unlabeled output tape that has BUFTAP buffered-write mode enabled to improve tape write performance. Without the `E` option PCFP writes two EOF marks and moves backward between them. This backward tape movement causes the tape unit to write the data from the tape unit buffer to the physical tape. This negates the performance advantages of BUFTAP, which requires constant forward tape motion and no implicit or explicit tape synchronization requests. Use `@COPY,GE` to gain the full performance advantage of BUFTAP buffered-write mode.

2001 Unexpected I/O status *xx* was returned by a read (`R$`) request to the file with index *y*.

2002 Unexpected I/O status *xx* was returned by a scatter-read (`SCR$`) request to the file with index *y*.

2004 Unexpected I/O status *xx* was returned by a multiple-request (`MR$`) read (`R$`) request to the file with index *y*.

2005 Unexpected I/O status *xx* was returned by a move-forward (`MF$`) request to the file with index *y*.

2006 Unexpected I/O status *xx* was returned by a move-backward (`MB$`) request to the file with index *y*.

2011 Unexpected I/O status *xx* was returned by a write (`W$`) request to the file with index *y*.

2012 Unexpected I/O status *xx* was returned by a gather-write (`GW$`) request to the file with index *y*.

2014 Unexpected I/O status *xx* was returned by a multiple-request (`MR$`) write (`W$`) request to the file with index *y*.

2015 Unexpected I/O status *xx* was returned by a write-end-of-file (`WEF$`) request to the file with index *y*.

2021 Unexpected I/O status *xx* was returned by a file update wait/synchronize (`FSAFE$`) request to the file with index *y*.

2022 The Words Transferred value returned by a scatter-read (`SCR$`) I/O request was different than PCFP expected for the file with index *y*.

- 2023 The Words Transferred value returned by a gather-write (GW\$) I/O request was different than PCFP expected for the file with index y.
- 2024 The Words Transferred value returned by a multiple-request (MR\$) I/O request was different than PCFP expected for the file with index y.
- 2101 The file is not or cannot be assigned as readable.
- 2102 The file is not or cannot be assigned as writable.
- 2201 A track sequence number error was detected while reading the source file.
- 2202 A checksum error was detected while reading the source file.
- 2205 The copy operation could not be performed, because the source file is written in a format not compatible with the requested format for the destination file.
- 2212 The source tape file was written in SDF format, which does not match the format specified in the function packet.
- 2215 The format of the logical file on the source tape file could not be recognized by PCFP.
- 2216 The source tape is a conditioned file. The site is not authorized to access the file.

The source tape file is a conditioned file created by SOLAR. The conditioned file cannot be copied from the source tape because SOLAR has determined that the site does not have the required keys registered. The file is not copied. Refer to Appendix F, "Conditioned File Copy Auxiliary Status Codes" in the *Software Library Administrator (SOLAR) User Reference Manual* (7831 0604) for further information on circumstances under which this error can be reported by SOLAR. Use the @MOVE statement to move past this file on the tape.

- 2217 The source tape file is a null file.

A labeled tape null file consists of Exec tape labeling system header and trailer label blocks and associated EOFs but no user data blocks. An unlabeled tape null file consists of a single EOF at load point followed by user data blocks. Use the @MOVE statement to move past a null file.

- 2222 The descriptor for the logical file on the source tape file is badly formed.

- 2233 The source tape file cannot be read because it is positioned at its logical end.
- There are no further logical files on the tape. You must move the tape backward before you can read another logical file from the tape.
- 2235 The destination tape file is not positioned immediately following an EOF mark.
- The tape must be positioned immediately following an EOF mark before you can copy another logical file to it.
- 2261 SOLAR returned auxiliary status code *n* when processing the conditioned source tape file. The site is not authorized to access the file.
- SOLAR detected an abnormal condition when attempting to authenticate the keys contained in the conditioned file. Refer to Appendix F, "Conditioned File Copy Auxiliary Status Codes" in the *Software Library Administrator (SOLAR) User Reference Manual (7831 0604)* for a detailed description of the decimal auxiliary status code contained in the message.
- 2301 The source tape file could not be read because it has inconsistent block sizes. The number of words transferred (xxxxxx) is not a multiple of the tape block size.
- Where xxxxxx is the decimal number of words. This error frequently indicates invalid or corrupted data on the tape file. Note that the tape block size for FURPUR format tape files should be a multiple of 1,794 words (2 header words plus 1,792 data words).
- 2302 The source tape file could not be read because it has a badly formed track header.
- 2340 Unexpected status 11 with I/O status xx was returned by ER TLBL\$ against the tape file with index y.
- 2341 Unexpected status xx was returned by ER TLBL\$ against the tape file with index y.
- 2342 Unexpected status xx was returned by ER TINTL\$ against the tape file with index y.
- 2343 Unexpected status xx was returned by ER TSWAP\$ against the tape file with index y.
- 2344 The function cannot be performed because of a previous Locate Block (LBLK\$) I/O function against the tape file with index y.

2501 The maximum size of the destination file is too small.

The function was terminated when the destination file was filled to its maximum size.

2520 Device Area Descriptor (DAD) table sector number xxxxxx is too large for the DREAD\$ function of ER MCON\$.

The cataloged mass storage source file on this @COPY command has too many DAD sectors and cannot be completely copied to the destination file.

2603 The function could not be completed because the program file table of contents for table x does not have enough open entries.

Where x is a decimal table number. Completion of the function would require more table entries than the program file table of contents can contain.

The following lists the table numbers:

Table	Description
1	Element table
2	MASM procedure table
3	COBOL procedure table
4	FORTTRAN/PLUS procedure table
5	Relocatable entry point table

2605 The operation could not be performed because the program file with index y is missing all or part of its table of contents.

2606 The operation could not be performed because the program file with index y is missing all or part of its element text.

2609 The LEPF program file element with sequence number xx has an illegal element text length granularity value.

2610 The LEPF program file element with sequence number xx has a text size that exceeds the maximum.

2611 Symbolic procedure, relocatable and absolute elements cannot be copied to an LEPF.

2612 The element with the sequence number specified is not correctly linked by element name, element version, or element type.

The element sequence number links contained in the program file pointer table and the element entries are incorrect or inconsistent. The program file table of contents should be rebuilt with @PACK (see Section 7.42) to correct the links.

2650 The Basic Service Package (BSP) detected a program file format error with the error code xx for the file with index y.

2654 An unexpected error xx was returned by the Basic Service Package (BSP)

2655 An unexpected I/O error xx was returned by the Basic Service Package (BSP)

2656 System Error: An undefined error code xx was returned by the Basic Service Package (BSP).

2659 System Error: Installation of a more recent SYSLIB level is required. The referenced SYSLIB Basic Service Package (BSP) does not support all of the functionality required by PCFP.

Information returned by the SYSLIB BSP\$ Read File Table Index (RFTI\$\$) function indicates that the level of SYSLIB referenced does not support functionality required by the PCFP program file functions.

2663 The sequence number specified does not exist in the program file. The largest sequence number in the file is xxxxx.

The largest sequence number xxxxx is a decimal number.

2701 The SDF file or element being operated on does not contain legal SDF text. The text has no Label Control Record.

2702 The symbolic element being operated on contains a type of SDF text that the PCFP function cannot handle.

2706 PCFP could not copy the SDF file or symbolic element because it contains no SDF end-of-text control record.

3001 The file with index y could not be assigned and the error returned by the attempted assign was xxxxxx.

3002 The file with index y could not be exclusively assigned and the error returned by the attempted exclusive assign was xxxxxx.

3004 The file with index y does not exist.

3031 Error message number xxxxxx was returned while trying to assign a temporary file that PCFP needs to complete the processing of the requested function.

The following error codes are returned by PCFP during the @PACK of an LPF or a large LEPF. In each case, the display of the error message is followed by the display of a second error message describing the error that was detected:

- 3502 During a program file pack, the copy of the program file elements to the temporary intermediate file resulted in the following error:
- 3503 During a program file pack, the copy of the temporary intermediate file back to the program file resulted in the following error:
- 3504 During a program file pack, the clearing of the temporary intermediate file resulted in the following error:
- 3506 During a program file pack, the prepping of the program file resulted in the following error:

B.6. Error Messages Associated with the @CHG Statement

The following messages are for the @CHG[*options*] *filename-1,name-1* statement used to change cataloged file attributes. Refer to Section 7.5.1 for a description of the @CHG statement.

Normal Termination Messages

END CHG.

Normal completion of @CHG when no specific change message is output.

REQUESTED KEY CHANGE COMPLETED

Normal completion of read key and write key changes.

REQUESTED MODE CHANGE COMPLETED

Normal completion of privacy change specified by the K, P, or Q option and/or read-only and write-only mode change specified by the V, W, or Z option.

Messages Associated with the I, K, M, P, Q, V, W, and Z Options and with Key Changes

abc OPTIONS CONFLICT

More than one of the options from the set K, P, and Q is specified or more than one of the options from the set V, W, and Z is specified. The conflicting options are specified by *abc*.

ACR NAME MUST BE SPECIFIED WITH K OPTION

The K option is specified to change the privacy state to ACR-controlled, but an ACR is not specified in *name-1*.

@csf\$-image-in-error

CSF\$ STATUS nnnnnnnnnnnn

The facilities request on the first line (for example, @USE, @ASG, @FREE) was submitted by ER CSF\$, resulting in the octal error status nnnnnnnnnnnn shown on the second line. This is followed by additional explanatory error messages.

CANNOT CHANGE OBSOLETE FORMAT ACRS

This error occurs only for Security Level 1 and higher systems when the number of security words in the file's Lead Item 0 is not seven, indicating an obsolete ACR format.

q*f(c) CANNOT BE ASSIGNED FOR REQUESTED CHANGE

Follows ER CSF\$ error message and indicates the F-cycle member that cannot be assigned.

q*f(c) CANNOT BE REASSIGNED FOR REQUESTED CHANGE

Follows ER CSF\$ error message and indicates the F-cycle member that cannot be freed and reassigned.

q*f(c) KEY(S) NEEDED

A read key or a write key or both are required for *filename-1* and were not provided.

q*f(c) KEY(S), READ/WRITE/DELETE/V-OPT ACCESS OR PRIVILEGE NEEDED

An ER MCON\$ DBITS\$ or DKEY\$ function returned a status 037 indicating that the run does not have the required privileges to change the file.

q*f(c) MCON\$ - DBITS\$ ERROR - nnnnnnnnnnnn

The octal status nnnnnnnnnnnn was returned by the ER MCON\$ DBITS\$ function when attempting to change *filename-1* as specified by the K, P, Q, V, W, and Z options.

q*f(c) MCON\$ - DKEY\$ ERROR - nnnnnnnnnnnn

The octal status nnnnnnnnnnnn was returned by the ER MCON\$ DKEY\$ function when attempting to change the read and write keys specified in *filename-1* to the keys specified in *name-1*.

*q*f(c)* MCON\$ - DREAD\$ ERROR - *nnnnnnnnnnnn*

The octal status *nnnnnnnnnnnn* was returned by the ER MCON\$ DREAD\$ function when initially attempting to retrieve information about the *filename-1* F-cycle set.

REQUESTED CHANGE NOT COMPLETED FOR F-CYCLE MEMBERS: *c-1*, *c-2*, ...

One or more members of the *filename-1* F-cycle set cannot be assigned to perform the requested changes. The absolute F-cycle numbers are identified by the *c-i*. The absolute F-cycle number format can be *c* (currently cataloged), *+c+* (to be cataloged) and *-c-* (to be deleted).

'FUNDAMENTAL SECURITY DOES NOT SUPPORT ACRS'

The K option cannot be specified on a Fundamental Security system.

SPECIFICATION 2. FILE NAME MUST BE SAME AS PREVIOUS SPEC

To perform a read key and write key change, the qualifier, filename, and cycle specified in *filename-1* must be the same as the qualifier, filename, and cycle specified in *name-1*.

SPECIFICATION 2. IS REQUIRED TO CHANGE KEYS

Because the K, P, Q, V, W, and Z options are not specified, a read and write key change is being requested; however, the filename and new read and write keys are not specified in *name-1*.

For other than Fundamental Security systems, ER SUMOD\$ is used to change the privacy state, as specified by the K, P, or Q options. Table B-1 shows ER SUMOD\$ octal error types and octal error codes and the associated error message output by @CHG.

Table B-1. ER SUMOD\$ Errors

Error Type	Error Code	Error Message
05	040	USER-ID SPECIFIED DOES NOT EXIST IN SYS\$*SEC@ACR\$
012	041	ATTEMPT TO PERFORM MODIFICATION WITHOUT PROPER PRIVILEGE
05	042	USER-ID SPECIFIED HAS BEEN DELETED
012	043	TO-BE-ATTACHED ACR DOES NOT EXIST
012	044	INVALID DATA SPECIFIED IN PACKET
012	045	MFDF CONTAINS BAD LINKS(ITEMS CANNOT BE LOCATED)
012	046	MFDF I/O ERROR OCCURRED ON ACCESS OF FILE MAIN OR LEAD ITEM
012	047	SPECIFIED ACR DOES NOT EXIST
012	052	INVALID PACKET LENGTH
012	057	MODIFIER FAILED WRITE FILE SECURITY RECORD ACCESS
012	062	UNOWNED SHARED FILE CANNOT HAVE AN ACR
012	064	OWNER OF FILE AND OWNER OF ACR MUST BE THE SAME
012	071	ER SUMOD\$ CAN ONLY BE PERFORMED BY HOME SUBSYSTEM
0mm	0nn	ERROR, TYPE 0mm CODE 0nn, RETURNED FROM CALL TO SUMOD\$

Messages Associated with the F Option

File caching designator must be included with the F option.

Name-1 must specify one of the file caching attribute values D, DC, DNC, or XRP.

FILECACHE\$ file error. Code: 017

explanatory-message

There are seven explanatory messages for file error subcodes 00 through 06. For all other subcodes, the explanatory message 'Sub code is 0nn' is output.

FILECACHE\$ interface error. Code: 0nn

[*explanatory-message*]

There are nine explanatory messages for error codes 01, 02, 012, 013, 014, 015, 016, 020, and 021.

The FILECACHE\$ call is not configured on this system.

File caching is not available.

Messages Associated with the N Option

FILE KEY AND FILE CYCLE VALUES ARE IGNORED WHEN INCLUDED AS PART OF THE NEW FILENAME.

Name-1 includes file cycle, read key and/or write key specifications that are ignored. The file name change proceeds with the qualifier and/or filename specifications included in *name-1*.

M\$CON\$ NAMCHG\$ ERROR - 022 - THE REFERENCED FILE IS NOT ASSIGNED TO THE RUN OR IS NOT THE REQUIRED TYPE OF FILE.

M\$CON\$ NAMCHG\$ ERROR - 037 - RUN DOES NOT HAVE SUFFICIENT PRIVILEGES TO PERFORM THIS FUNCTION.

M\$CON\$ NAMCHG\$ ERROR - 044 - THE QUALIFIER OR FILENAME CONTAINS ONE OR MORE CHARACTERS NOT ALLOWED FOR THAT FIELD.

M\$CON\$ NAMCHG\$ ERROR - 051 - THE M\$CON\$ NAMCHG\$ FEATURE IS NOT PRESENT ON ALL HOSTS.

M\$CON\$ NAMCHG\$ ERROR - 052 - THE FILENAME ALREADY EXISTS.

M\$CON\$ NAMCHG\$ ERROR - 053 - AT LEAST ONE CYCLE OF THE FILE SET IS UNLOADED OR ON SYMBIONT QUEUE

M\$CON\$ NAMCHG\$ ERROR -nnnnnnnnnnnn- OCCURRED.

The octal status *nnnnnnnnnnnn* was returned by the ER M\$CON\$ NAMCHG\$ function when attempting to change the name of *filename-1*. The error code is other than 03, 022, 037, 044, 051, 052, or 053.

M\$CON\$ NAMCHG\$ WARNING- 03 - A MEMBER OF THE FILE SET IS ON THE SYMBIONT QUEUE AND MAY NOT BE PRINTED.

SECOND FIELD OF EITHER QUALIFIER OR FILENAME OR BOTH REQUIRED

Name-1 includes some specifications, but does include either qualifier or filename.

SPECIFICATION 2. ELEMENT NAME ILLEGAL

Name-1 should not include element name with qualifier and/or filename.

SPECIFICATION 2. IS REQUIRED WITH THE N OPTION

Name-1 must specify qualifier and/or filename.

WARNING - NAMCHG\$ APPLIED TO USER ASSIGNED FILE -
FILE MAY NOT BE REFERENCED BY NEW NAME UNTIL IT IS FREED

Messages Associated with the E and T Options

FILE MUST BE EMPTY BEFORE USING THIS FUNCTION.

The first word of the file specified by *filename-1* is not zero. The file must be newly cataloged with @ASG or @CAT or must be erased with the @ERS statement.

B.7. Error Messages Associated with the @CYCLE Statement

The following error messages are for the @CYCLE *filename,number* statement that operates on cataloged file F-cycle sets. Some of the messages use the absolute F-cycle number and file status formats *c* (currently cataloged), *+c+* (to be cataloged) and *-c-* (to be deleted). See Section 7.15.1 for a description of the @CYCLE statement and of the F-cycle file status formats.

ALL SPECIFIED F-CYCLE MEMBERS CANNOT BE DELETED
x OF y F-CYCLE MEMBERS DELETED
[F-CYCLE MEMBERS DELETED: *c-1*, *c-2*, ...]

The deletion of F-cycle members could not be completed. A previous error message should have reported the error that was detected. The *x* F-cycle members with the lowest absolute F-cycle members were deleted out of a total of *y* F-cycles in the deleted F-cycle range. The third line lists the absolute F-cycle numbers and original file statuses of the deleted F-cycle members.

```
@csf$-image-in-error (e.g. @USE, @ASG, @FREE)
CSF$ STATUS nnnnnnnnnnnn
```

The facilities request on the first line was submitted by ER CSF\$, resulting in the octal error status *nnnnnnnnnnnn* shown on the second line. This is followed by additional explanatory error messages.

```
MAXIMUM F-CYCLE RANGE CANNOT BE CHANGED
[F-CYCLE MEMBERS DELETED: c-1, c-2, ...]
q*f(c) MCON$ - DCYC$ ERROR - nnnnnnnnnnnn
```

The octal status *nnnnnnnnnnnn* was returned by the ER MCON\$ DCYC\$ function when attempting to change the maximum F-cycle set range to number. If any members of the F-cycle set were deleted before the error, the second line lists the absolute F-cycle numbers and original file statuses.

```
MAXIMUM F-CYCLE RANGE MUST BE 32 OR LESS
```

The *number* specified on @CYCLE *filename,number* was not a decimal number, 0 through 32, inclusive.

```
NEW F-CYCLE INFORMATION NOT AVAILABLE
[F-CYCLE MEMBERS DELETED: c-1, c-2, ...]
q*f(c) MCON$ - DREAD$ ERROR - nnnnnnnnnnnn
```

The octal status *nnnnnnnnnnnn* was returned by the ER MCON\$ DREAD\$ function when attempting to retrieve information about the changed F-cycle set. The ER MCON\$ DCYC\$ function was successfully completed before this error was encountered. If any members of the F-cycle set were deleted before the error, the second line lists the absolute F-cycle number and file status.

```
NO F-CYCLE MEMBER IN THE SPECIFIED RANGE CAN BE ASSIGNED
[F-CYCLE MEMBERS THAT CANNOT BE ASSIGNED: c-1, c-2, ...]
```

In the new maximum F-cycle range *number* there were no F-cycle members that could be assigned to the run. The absolute F-cycle numbers and file statuses of any F-cycles that exist in the retained range are listed on the second line. No F-cycle members are deleted and the maximum F-cycle range is not changed.

```
NO F-CYCLE MEMBERS DELETED
[F-CYCLE MEMBERS THAT CANNOT BE ASSIGNED: c-1, c-2, ...]
[F-CYCLE MEMBERS THAT ARE ON A SYMBIONT QUEUE: c-1, c-2, ...]
```

During assignment of files to be deleted not all of the members in the deleted F-cycle range could be assigned to the run or one or more of the members was on a symbiont queue. The absolute F-cycle numbers and file statuses of the F-cycles in error are listed on the second or third line, as appropriate.

```
q*f(c) KEY(S) NEEDED
```

The status returned on an ER FITEM\$ or on an ER CSF\$ @ASG image indicated that a read key or a write key or both are required for this file and were not provided.

*q*f(c)* MCON\$ - DREAD\$ ERROR - *nnnnnnnnnnnn*

The octal status *nnnnnnnnnnnn* was returned by the ER MCON\$ DREAD\$ function when initially attempting to retrieve information about *filename*.

*q*f(c)* MCON\$ - EXIST\$ ERROR - *nnnnnnnnnnnn*

The octal status *nnnnnnnnnnnn* was returned by the ER MCON\$ EXIST\$ function when attempting to determine if an F-cycle set member is on a symbiont queue. The NO F-CYCLES DELETED messages previously described precede this message.

RUN DOES NOT HAVE DELETE ACCESS TO F-CYCLE SET

When attempting to delete the lowest absolute F-cycle with a @FREE,D CSF\$ image, octal status 40000000010 was returned, indicating that this run does not have delete access to the F-cycle set.

B.8. Error Messages Associated with the @PRT Statement

Messages Associated with @PRT,D and @PRT,U

@csi\$-image-in-error

E:*nnnnnn facility-inventory-error-message*

The first error message is the @USE, @CAT or @ASG facilities statement used to assign the \$DGETaa\$ file. The second error message is contained in C.3. Note that when FURPUR displays the variable fatal error messages with numbers 200033 - 240033 the variable information (shown in *italics* in C.3) is not available to FURPUR and is replaced by the 'insert indicator' ^1, ^2, etc. For example, message number 203233 is shown in C.3 as 'E:203233 *device-name* is not up' and would be displayed by FURPUR as 'E:203233 ^1 is not up'.

FILES \$DGETAA\$ THROUGH \$DGETZZ\$ ARE ALREADY CATALOGED

FURPUR cannot catalog the required file on the specified *pack-id* because the 676 filenames that can be used for each qualifier are all currently cataloged in the MFD. Any \$DGETaa\$ file that is not currently assigned to a run can safely be deleted. Deleting these unused files may also improve the efficiency of @PRT,D and @PRT,U.

INCORRECT EQUIPMENT TYPE SPECIFIED

The specified *type* is not an assign mnemonic for a sector-addressable mass storage file.

PACK-ID IS NOT A REMOVABLE PACK

The specified *pack-id* is a fixed pack instead of a removable pack.

PACK-ID NOT FOUND

The specified pack-id does not exist.

Messages Associated with @PRT,T

SORT SUBROUTINE ERROR *xx* OCCURRED
LIST OF DUPLICATE OBJECT MODULE ENTRY POINT NAMES TERMINATED

The SORT subroutine error messages can be listed during the execution of the @PRT,TL command. An explanation of the error code *xx* can be found in the *OS 2200 SORT/MERGE Programming Guide* (7831 0687).

*File contains 1 invalid element table entry
*File contains *n* invalid element table entries

For the long format report only, additional verification of element table entries is performed. If any invalid fields are found, the count of entries containing errors is reported in the summary messages.

The individual fields with errors are reported in the main @PRT,T output as shown in this example.

ELEMENT TABLE												
D	NAME	VERSION	TYPE	DATE	TIME	SEQUENCE NUMBER	PREAMBLE SIZE	TEXT SIZE	CYCLE MAX	WORD CUR	FLAG NUM	SECTOR BITS LOCATION
	ELEMENT-1	VERSION-1	SYMBOLIC	2014-11-12	12:04:28	1		*****	5	0	1	1792
	ELEMENT-2	VERSION-2	*UNDEFINED*	2014-11-12	12:04:28	2		67				1859
	ELEMENT-3	VERSION-3	SYMBOLIC -Q	*ERR* 022,	0300137012345	3		67	5	0	1	1926
	NEXT AVAILABLE LOCATION-											1993
	ASSEMBLER PROCEDURE TABLE EMPTY											
	COBOL PROCEDURE TABLE EMPTY											
	FORTRAN PROCEDURE TABLE EMPTY											
	RELOCATABLE ENTRY POINT TABLE EMPTY											
	*File contains 3 invalid element table entries.											

In the example:

ELEMENT-1/VERSION-1 has an invalid text size. Asterisks are displayed in the TEXT SIZE column.

This occurs for Large Element Program Files (LEPF) if the size calculation based on the granularity setting results in a text size larger than allowed.

It also occurs if conflicting granularity bits are set.

ELEMENT-2/VERSION-2 has an invalid element type. *UNDEFINED* is displayed in the TYPE column.

This occurs if the element type is outside of the range of 1-7.

ELEMENT-3/VERSION-3 has an invalid timestamp. *ERR* 0yy, 0xxxxxxxxxxxx is displayed in the DATE and TIME columns.

This occurs if an invalid timestamp is encountered. The specific error code associated with the processing and the invalid timestamp are displayed.

Oyy

is the octal AEDIT\$ error code which was returned when submitting the timestamp to AEDIT\$ for conversion to the Unisys standard display format. In this case, 022 indicates an illegal month in the TDATE\$ timestamp. These error codes can be found in the *SYSLIB Programming Reference Manual* (7833 1733) in the section documenting the AEDIT\$ routines.

Oxxxxxxxxxxx

is the one-word TDATE\$ timestamp, in octal, which failed timestamp validation. In this case, 0300137012345 indicates a month of 24, which is invalid.

Appendix C

Facility Request Status Codes

The system displays facility request messages and status codes in response to the ECL statements for facility requests: @ASG, @CAT, @FREE, @MODE, @USE, and @QUAL.

The prefixed messages are divided into three categories:

- Informational messages are prefixed with the letter I and an error number. For example:

```
I:001433 run run-id held for unloaded file rollback for nn min.
```

where *run-id* identifies the run being held, and *nn* is the number of minutes the run has been held so far.

- Warning messages are prefixed with the letter W and a message number. For example:

```
W:120133 file is already assigned.
```

- Error messages are prefixed with the letter E and an error number. For example:

```
E:200233 devname is an invalid device name.
```

where *devname* is the character string that was supposed to be a device name.

Table C-1 shows the message number ranges.

Table C-1. Facility Inventory Message Ranges

Message Number Range	Type of Message
000133 – 002333	Variable informational
020133 – 077733	Fixed informational
100033 – 12003	Variable warning
120133 – 177733	Fixed warning
200033 -- 240033	Variable fatal error
240133 -- 377733	Fixed fatal error

C.1. Facility Inventory Informational Messages

Variable Informational Messages

I:000133 Run xxxxxx held for unit for abs device/CU unit assign
for *nn* min.

I:000233 Run xxxxxx held for unit for abs rem disk for *nn* min.

I:000333 Run xxxxxx held for pack availability for *nn* min.

I:000433 Run xxxxxx held for reel availability for *nn* min.

I:000533 Run xxxxxx held for com line availability for *nn* min.

I:000633 Run xxxxxx held for com group availability for *nn* min.

I:000733 Run xxxxxx held for mass storage space for *nn* min.

I:001033 Run xxxxxx held for tape unit availability for *nn* min.

I:001133 Run xxxxxx held for exclusive file use release for *nn* min.

I:001233 Run xxxxxx held for need of exclusive use for *nn* min.

I:001333 Run xxxxxx held for disk unit availability for *nn* min.

I:001433 Run xxxxxx held for rollback of unloaded file for *nn* min.

I:001533 Run xxxxxx held for file cycle conflict for *nn* min.

I:001633 Run *nn* held for disk pack to be mounted for *nn* min.

I:001733 *device-name* is selected *reel-id filename-index run-id*

I:002033 Run xxxxxx held for control of caching for *nn* min.

I:002133 Run xxxxxx held for diskette unit availability for *nn* min.

I:002233 Run xxxxxx held for diskette to be mounted for *nn* min.

I:002333 {ASG|CAT|FREE|MODE|QUAL|USE} complete

C.2. Facility Inventory Warning Messages

Variable Warning Messages

- W:100033 Assign mnemonic *asg-mnem* does not allow block numbering to be turned off, BLKOFF subfield is ignored.
- W:100133 Assign mnemonic *asg-mnem* does not allow data compression to be turned off, CMPOFF subfield is ignored.

Fixed Warning Messages

- W:120133 File is already assigned.
- W:120233 File is already assigned, this image ignored.
- W:120333 File is assigned to another run.
- W:120433 Filename not known to this run.
- W:120533 Filename not unique.
- W:120633 I/O error encountered on MFD during FREE.
- W:120733 Line is assigned in simulation mode.
- W:121033 Not all warning messages printed.
- W:121133 Placement field ignored.
- W:121233 File cannot be deleted because it is on the print queue.
- W:121333 A read key exists on the file.
- W:121433 File is cataloged as a read-only file.
- W:121533 Security ACR does not allow read access.
- W:121633 Security ACR does not allow write access.
- W:121733 Security ACR does not allow execute access.
- W:122033 Security clearance level allows read only access.
- W:122133 File was assigned during system failure.
- W:122233 File is unloaded.
- W:122333 A write key exists on the file.
- W:122433 File is cataloged write-only.

Facility Request Status Codes

W:122533	Option conflict with previous assign options, all options ignored except i, q, x, y, or z.
W:122633	Option(s) conflict with previous assign options-option conflict ignored.
W:122733	Expiration exceeds the configured maximum. Configured maximum used.
W:123033	Read/write keys are ignored in an ownership system for owned files.
W:123133	G option not enforced.
W:123233	X option ignored, file already exclusively assigned by this run.
W:123333	System is configured to ignore Z option for batch in control mode.
W:123433	System is configured to release unused mass storage space.
W:123533	Security compartments allow read only access.
W:123633	The file cycle set is private and owned - therefore this cycle is private.
W:123733	The file cycle set is semi-private and owned - therefore this cycle is semi-private.
W:124033	The file cycle set is public and owned-therefore this cycle is public
W:124133	Security does not allow the use of unlabeled tapes. If written to, this tape will become labeled.
W:124233	The file is security disabled.
W:124333	Media Manager is not available. The Media Manager spec was BYPASS, but Media Manager is not available.
W:124433	File cannot be deleted because an XPC contains data for this file.
W:124733	The memory assign mnemonic specification is ignored for an XPC cached file. An attempt was made to change an XPC cached file's assign mnemonic to DMEM or FMEM.

W:125133 The file created is semiprivate because the caller has a to-be-attached ACR.

Files with ACR-controlled access are semiprivate.

C.3. Facility Inventory Error Messages

Variable Fatal Error Messages

E:200033 Unit *device-name* not compatible with assign mnemonic.

E:200133 *device-name* is an invalid device/control unit name.

E:200233 *device-name* is an invalid device name.

E:200333 Device *device-name* already in use by this run.

E:200433 Device *device-name* is fixed.

E:200533 Device not available on control unit *unit-name*.

E:200633 Pack *pack-id* cannot be mounted-equipment type does not permit interchanging of packs.

E:200733 Insufficient number of units available on control unit *unit-name*.

E:201033 *asg-mnem* is not a configured assign mnemonic.

E:201133 *device-name* is not a configured name.

E:201233 *name* is not a line name or group name.

E:201333 The operator does not allow absolute assignment of pack *pack-id*.

E:201433 Illegal option combination *fj*.

E:201533 Illegal option *x*.

E:201633 Addition of option(s) causes illegal combination *xx*.

E:201733 Pack *pack-id* already in use by this run.

E:202033 Pack *pack-id* failed label checking.

E:202133 Pack *pack-id* is assigned on a different equipment type.

E:202233 Pack *pack-id* is not a removable pack.

E:202333 Placement device *device* is not fixed mass storage.

Facility Request Status Codes

E:202433 Quota does not allow use of x option.

E:202533 reel *nnnnnn* already in use by this run.

E:202633 reel *nnnnnn* is not in the master file directory.

E:202733 Security does not allow use of x option.

E:203033 Available unit not found on CU unit-name because space is not available or quota limit exceeded.

E:203133 *unit-name* is not in the reserved state.

E:203233 *device-name* is not up.

E:203333 *name* is not in the up or reserved state.

E:203433 Device *device-name* is already actively caching.

E:203533 Device *device-name* is not capable of caching.

E:203733 Directory of device *device-name* is not equal to directory of file.

E:204033 Directory of pack *pack-id* is not equal to directory of file.

E:204133 Directory of placement device *device-name* is not equal to directory of file.

E:204233 Assignment of pack *pack-id* could not be completed due to insufficient system resources.

E:204333 Assignments not allowed on pack *pack-id* because the operator has placed assignment restrictions on the pack.

E:204433 Assign mnemonic *asg-mnem* does not support 6-bit packed format.

E:204533 Assign mnemonic *asg-mnem* requires the J option.

E:204633 Assign mnemonic *asg-mnem* does not allow a noise field.

E:204733 Assign mnemonic *asg-mnem* does not support M-L-H-V density options.

E:205033 Assign mnemonic *asg-mnem* does not allow block numbering subfield.

E:205133 Assign mnemonic *asg-mnem* does not allow data compression subfield.

E:205233 Security does not allow the use of xxxxx equipment type.

E:206233 Assign mnemonic *asg-mnem* does not allow buffering to be turned off.

E:206333 Assign mnemonic *asg-mnem* does not support M-L-V-S density options.

E:206433 [*mmspec*] is an invalid Media Manager specification.

E:206533 [*cartridge|non-cartridge*] equipment type specified for [*non-cartridge|cartridge*] tape

The supplied equipment type is inconsistent with the media type for the supplied reel; either the reel is a cartridge and the equipment type on the @ASG is not, or the equipment type on the @ASG is a cartridge and the reel is not.

E:206633 Volume [*volume*] unknown to Media Manager; FJ option combination is not allowed.

The FJ option combination was specified when assigning a tape that is not controlled by the Media Manager. The only time FJ is allowed is when the tape is controlled by the Media Manager.

E:206733 Volume [*volume*] unknown to Media Manager, and BYPASS of Media Manager is not allowed.

E:207033 ACS name *acs-name* is not a legal name

When CSC uses the CARTTAPELIB\$ interface, CSC will determine the mappings between ACS names and libraries. In addition, CSC will determine the Blank-Preference to use, free-standing or ACS drive.

E:207233 Assign mnemonic *aaaaaa* does not support the media type *xxxxxx* for volume *vvvvvv*.

The assign mnemonic conflicts with the media type for the given volume.

E:207333 Packs cannot be added due to the lack of removable directory space on pack *pack-id*.

E:207433 File cannot be created due to lack of removable directory space on pack *pack-id*.

Fixed Fatal Error Messages

E:240133 Absolute device specification is not allowed on cataloged file.

E:240233 Absolute assignment of tape not allowed with CY or UY options.

Facility Request Status Codes

E:240333 Illegal options on absolute assignment.

E:240433 Account could not be found.

E:240533 CY and UY options are not allowed with removable disk.

E:240633 Assign mnemonic does not allow pack-ids.

E:240733 Assign mnemonic cannot be longer than 6 characters.

E:241033 Assign mnemonic must be specified with a packid.

E:241133 Assign mnemonic must be word addressable.

E:241233 File is being dropped.

E:241333 Blank reel-id must be alone.

E:241433 Attempt to change assign mnemonic.

E:241533 Illegal attempt to change assignment type.

E:241633 Attempt to change generic type of the file.

E:241733 Attempt to change granularity.

E:242033 Attempt to change initial reserve of write inhibited file.

E:242133 Attempt to change maximum granules of a file cataloged under a different account.

E:242233 Attempt to change maximum granules on a write inhibited file.

E:242333 Assignment of units of the requested equipment type is not allowed.

E:242433 Illegal value specified for F-cycle.

E:242533 File cycle out of range.

E:242633 Cannot catalog file because read or write access not allowed.

E:242733 Illegal value specified for data converter.

E:243033 Data converter and translator cannot be used together.

E:243133 Delete access is not allowed.

This error may be the result of an attempt to create a new cycle without the security attributes required to access an old cycle.

E:243233 Creation of file would require illegal dropping of private file.

E:243333 Illegal value specified for data transfer format.

E:243433 Duplicate media-ids are not allowed.

E:243533 Syntax error in image submitted to ER CSI\$.

E:243633 Length of image submitted to ER CSI\$ is bad.

E:243733 Illegal control statement type submitted to ER CSI\$.

E:244033 Expiration not allowed on CAT image.

E:244133 Expiration exceeds the configured maximum.

E:244233 Illegal value specified for expiration period.

E:244333 A filename is required on the image.

E:244433 File is already catalogued.

E:244533 File is not catalogued.

E:244633 File can not be recovered because master file directory (MFD) information has been corrupted.

E:244733 File is not tape.

E:245033 Facility inventory internal error 1, undefined generic equipment type.

E:245133 Attempt to delete via @FREE,D but file was not assigned

E:245233 Illegal value specified for granularity.

E:245333 Illegal character(s) in placement field.

E:245433 Insufficient number of units available.

E:245533 Internal name is required.

E:245633 I/O error encountered on MFD assign count not decremented.

E:245733 I/O error encountered on the master file directory.

E:246033 I/O error encountered on MFD mediaids not read successfully.

E:246133 I/O error encountered on MFD--(+1) cycle cannot be deleted.

E:246233 File initial reserve granule limits exceeded.

Facility Request Status Codes

E:246333 Illegal value specified for initial reserve.

E:246433 Read and/or write keys are needed.

E:246533 Line already assigned to this run.

E:246633 Line or group is not available.

E:246733 Logical channel not allowed with absolute device assign.

E:247033 Illegal value specified for logical channel.

E:247133 Maximum granules less than highest granule allocated.

E:247233 File maximum granule limits exceeded.

E:247333 Illegal value specified for maximum.

E:247433 Maximum is less than the initial reserve.

E:247533 Maximum number of mediaids exceeded.

E:247633 Maximum number of packids exceeded.

E:247733 Maximum number of reelids exceeded.

E:250033 File named on the mode is not assigned.

E:250133 Mass storage has overflowed.

E:250233 Number of units not allowed with absolute device assign.

E:250333 Illegal value for number of units.

E:250433 Illegal noise constant value.

E:250533 Noring cannot be specified with SCRTCH or BLANK reelid.

E:250633 Illegal use of I option.

E:250733 I option is the only legal option on USE.

E:251033 Options M and L are not allowed with 9 track tape units.

E:251133 Options V and S are not allowed with 7 track tape units.

E:251233 Number of packids on image not equal to number of packs assigned to file.

E:251333 Number of packids on the image not equal to number of packids in master file directory.

E:251433 Packids on image are not equal to packids assigned to file.

E:251533 Packids on image are not in same order as file's packids.

E:251633 Packid is required.

E:251733 Data bank overflow attempting to acquire file-related buffer space.

E:252033 F-cycle of +1 is illegal with A option.

E:252133 Placement field is not allowed with CAT.

E:252233 Placement requested on a non--mass storage device.

E:252333 Placement is not allowed with a removable disk file.

E:252433 Illegal syntax in placement subfield.

E:252533 Incorrect privacy key for private file.

E:252633 Quota set does not allow absolute device assignment.

E:252733 Loading of file causes quota mass storage limits to be exceeded.

E:253033 Quota mass storage limits exceeded.

E:253133 Quota number of units exceeded.

E:253233 Quota set does not allow communications assign.

E:253333 Incorrect read key.

E:253433 File is not cataloged with read key.

E:253533 Number of reelids on image greater than number of reelids in master file directory.

E:253633 None of the packs are currently registered.

E:253733 Relative F-cycle conflict.

E:254033 Ring specification not allowed on CAT image.

E:254133 Illegal value for ring specification.

E:254233 Scrtch is not allowed on a cataloged file.

E:254333 File backup is not available.

E:254433 Security does not allow absolute device assignment.

Facility Request Status Codes

E:254533 Security access list does not allow access.

This covers the case in which the object has private or ACR-controlled access.

E:254633 Security clearance level does not allow access.

This occurs when your clearance level is less than that of the object and when the object's clearance level is lower than the user's.

E:254733 Security does not allow delete access.

This message is received when the requestor attempts to delete a shared file without having the privilege to do so.

E:255033 Security group name validation failed.

E:255133 Facility inventory internal error - Security packet error.

E:233233 Simulation is not allowed for a C/SP line.

E:255333 Simulation is not configured.

E:255433 S option not allowed with a group mnemonic.

E:255533 Error in translation specification.

E:255633 Requested translator is not configured.

E:255733 Image contains an undefined field or subfield.

E:256033 Requested unit(s) not available.

E:256133 A unit is not assigned to the file.

E:256233 Unit not compatible with request.

E:256333 Units and logical subfields not allowed on CAT image.

E:256433 User modified image or MFD for the file during facility processing.

E:256533 Attempt to change to word addressable not allowed.

E:256633 Incorrect write key.

E:256733 File is not cataloged with write key.

E:257033 Hold for reel rejected because of Z option.

E:257133 Hold for pack rejected because of Z option.

E:257233 Hold for tape unit rejected because of Z option.

E:257333 Hold for disk unit rejected because of Z option.

E:257433 Hold for x-use rejected because of Z option.

E:257533 Hold for release of x--use rejected because of Z option.

E:257633 Hold for rollback of unloaded file rejected because of Z option.

E:257733 Hold for com line rejected because of Z option.

E:260033 Hold for com group rejected because of Z option.

E:260133 Hold for *rem disk rejected because of Z option.

E:260233 Hold for *DEV/CU rejected because of Z option.

E:260333 Hold for F-cycle conflict rejected because of Z option.

E:260433 Hold for mass storage space rejected because of Z option.

E:260533 Maximum number of use names and/or filenames allowed to a run exceeded.

E:260633 File may contain corrupted data. file resided in cache disk memory at time of unrecoverable hardware error.

E:260733 Run has been aborted.

E:261033 The ER CSI\$ request packet has non zero values in a reserved field.

E:261133 The file index passed in the ER CSI\$ request packet is invalid.

E:261233 The ER CSI\$ request causes the maximum number of filenames using file indices to be exceeded.

E:261333 The exec is in control of caching, request cannot be held.

E:261433 Hold for control of caching rejected because of Z option.

E:261533 A diskette id is required on the image.

E:261633 CAT of diskette file is not allowed.

E:261733 Hold for diskette unit rejected because of Z option.

E:262033 Too many diskette ids on the image.

Facility Request Status Codes

E:262133 Illegal ACR name.

E:262233 All cycles of a file must be same access mode.

E:262333 Acr specification illegal with A or T options.

E:262433 Quota does not allow creation of shared files.

E:262533 Quota does not allow deletion of shared files.

E:262633 Directory id and qualifier may not appear on image when R option is used.

E:262733 Absolute control unit assign of a diskette unit is not allowed.

E:263033 Directory id or qualifier must appear on image.

E:263133 Block numbering can only be changed at load point.

E:263233 Data compression requires block numbering.

E:263333 Data compression and translator cannot be used together.

E:263433 Illegal value for block numbering.

E:263533 Illegal value for data compression.

E:263633 Directory id of file is invalid.

E:263733 File sharing is down, Use of shared files is not allowed.

E:264033 Shared files cannot be temporary.

E:264133 No tape units are configured on this system. Tape unit cannot be allocated.

E:264233 File sharing is reserved. Creation of additional shared files is not allowed. Additional shared files cannot be assigned unless at least one shared file is already assigned.

E:264333 The file sharing feature is not configured. Use of shared files is not allowed.

E:264433 Data compression requires GCR recording mode.

E:264533 Absolute assignment of unit which is duplexing not allowed.

E:264633 The specified ACR does not match the ACR already controlling access to the other file cycles.

E:264733 Security compartment validation failed.

E:265033 A cartridge allows only L density option.

E:265133 Illegal value in buffered write subfield.

E:265233 File is privately assigned to an application. Access is not allowed.

See the *MHFS Planning, Installation, and Operations Overview* for information on the application-private feature.

E:265333 Security does not allow the use of unlabeled tapes, the J option is not allowed.

E:266033 The file is security disabled, no access allowed.

E:266133 The file name section you were pointed to has been deleted.

E:266233 Free not allowed. File is in use by the exec.

E:266333 Hold for CTL availability rejected due to `Z' option.

This message is displayed when you specify @ASG,Z with a CTL-pool name and the CTL is unavailable.

E:266433 Cartridge-id is not available in the CTL.

This message is displayed when you specify @ASG with a CTL-pool name and either the cartridge (or one of the cartridges in a two-unit assign) is not found in the CTL database, or the BLANK pool has been depleted.

E:266533 All cartridges be located in the same CLU.

This message is for two-unit assignment and does not occur when the cartridges are located in different cartridge library units (CLU).

E:266633 Cartridge tape library system is not available.

This message is displayed with you specify @ASG with a CTL-pool name and either the CTL is not currently registered or it is in a REJECT state.

E:266733 Ctl-pool name is not recognized.

This message is displayed when you specify an unknown CTL-pool name on an @ASG statement for a BLANK tape.

Facility Request Status Codes

E:267033 Memory files must be in the local directory.

This message is returned with any attempt to allocate a shared file into memory. This error can occur on a subsequent assignment of an existing shared file requesting a switch to a memory file assign mnemonic, as well as to the normal initial assignment.

E:267133 Media manager is not loaded.

A Media Manager specification was supplied on the @ASG statement, but the Media Manager is not loaded.

E:267233 Bypass of Media Manager is not allowed.

The Media Manager spec is not configured to allow bypass because you specified BYPASS in the mmspec field of the @ASG statement.

E:267333 File contains data for a TIP duplexed file. Access is allowed only via TIP.

E:267433 Illegal value for the expanded buffer capability option.

E:267533 A file cannot be assigned with both the old and new names within the same run.

E:267633 CTL-pool name cannot be used with a non-library mnemonic.

You receive this message when you use a CTL pool name other than NONCTL on mnemonics, U47NL, U47M, or NLCART.

E:267733 NONCTL cannot be used with a library mnemonic.

The Exec gives you U47L, U47LM, or LCART when using NONCTL for mnemonics.

E:270033 Hold for network device availability because of the Z option.

E:270133 Facility Inventory internal error: network device could not be marked assigned.

E:270333 The B option is not allowed when assigning a virtual tape.

E:270433 The DLT cartridge only allows the V density option.

The DLT cartridge does not allow S, M, L, or H options.

E:270533 A file is cataloged in a different directory.

E:270633 Security does not allow creation of Shared Files.

This message is received when the requestor attempts to create a shared file without having the privilege to do so.

E:270733 Security does not allow assignment of Shared Files.

This message is received when the requestor attempts to assign a shared file without having at least one privilege to do so.

E:271033 Read-only equipment is requested in a way that implies writing to it.

E:271133 Impersonated user-ids are not allowed to assign tapes.

E:271233 Impersonated user-ids are not allowed to do absolute assigns.

E:271333 Packs can only be added to removable files using an A option assignment.

E:271433 Packs cannot be added to removable files using a Y option assignment.

E:271533 Packs cannot be added to removable files that are currently assigned.

E:271633 Packs can only be added to removable files with a single file cycle.

E:271733 Adding packs requires delete access.

E:272033 Insufficient resources for File Encryption Information.

E:272133 System does not support file encryption type.

E:272233 No file owner or system file encryption information.

E:272333 File owner userid record is marked as deleted.

E:272433 Unable to access file owner userid record.

E:272533 Find of active key matching key name and type failed.

E:272633 Failed to retrieve file encryption information.

E:272733 Illegal Encryption Specification on ASG/CAT statement.

E:273033 Illegal Encryption Specification on ASG statement.

C.4. PCN Fixed Messages

E:000140 No message has the given number.

E:000240 No message lookup could be performed because the program control table (pct) overflowed.

E:000340 No message lookup could be performed because the e\$ormsg element is not in sys\$*lib\$ file.

E:000440 No message was displayed because an error was detected in the e\$ormsg element or an I/O error occurred while reading it.

E:000540 User submitted too few inserts for insertion type message.

E:000740 The request list for insertion type messages is too long.

E:001040 Packet identifier is invalid.

E:001140 Error packet has a non 0 value in field which must be 0.

E:001240 An insert could not be split across lines.

E:001340 The returned message buffer is not big enough to hold the message.

E:001440 Use of output via rsi is not allowed by this requestor.

E:001540 system default profile updated.

I:001640 operator profile updated.

E:110040 Invalid runid specified for er termrun\$.

E:110140 Runid specified for er termrun\$ is no longer active.

E:110240 Caller of er termrun\$ cannot terminate own runid.

E:110340 Bad packet level specified for er termrun\$.

E:110440 Runid specified for er termrun\$ is not batch.

E:110550 Wait time specified for er termrun\$ has expired.

E:110640 Reserved field in er termrun\$ packet is not zero.

C.5. User Profile Error Messages

- E:000143 Unable to assign profile file q*f: frsc-number. User will be issued System Default Profile (if one exists).

- E:000243 Profile file q*f is program file but does not contain PROFILE\$ element. User will be issued System Default Profile (if one exists).

- E:000343 Error in Profile file qual*fn at line number *n*. Processing will continue at next line in file.

- E:000443 User lacks read access to Profile file qual*fn. User will be issued System Default Profile (if one exists).

- E:000643 Passive Profile for User-id *nnnnnn* is not available. User will be issued System Default Profile (if one exists).

- E:000743 Multiple errors occurred while reading User Profile file qual*fn. Profile file processing has been terminated.

- E:001043 User Profile file processing terminated at line *n* in file qual*fn due to internal error number: *nn*. Notify Site Administrator.

- E:001143 System Defined Environment Variable VALUE at line number *n* in Profile file qual*fn. exceeds maximum length. VALUE will be truncated.

- E:002135 System Defined Environment Variable VALUE at line number *n* in System Default Profile file qual*fn. exceeds maximum length. VALUE will be truncated.

- E:002235 System Defined Environment Variable VALUE at line number *n* in Operator Profile file qual*fn. exceeds maximum length. VALUE will be truncated.

C.6. Facility Inventory Status Bits

If a facility request made by one of the facilities control statements (@ASG, @MODE, @CAT, @FREE, @QUAL, and @USE) is in error, a status word is generated in which the various bits define the error. See Table C-2 for descriptions of the bit sets.

Table C-2. Facility Inventory Status Bits

Bit Set	Description
0*	Request not accepted; check other bits for reason.
1*	Field error in control statement other than syntax. Option conflict (for example, MHL, OE, or IB) or noise constant specification error. Requested hardware not currently part of the system. CSF\$ request returns status of 60000000000 when mass storage has overflowed. (This description gave a few examples only. There are other possible errors for this bit.)
2	File specified is already assigned or cataloged (@ASG and @CAT), already released (@FREE), or not assigned (@MODE). The request is rejected for @CAT and @MODE statements.
3	The file was previously cataloged.
4*	Equipment type specified on the @ASG statement is not compatible with the cataloged type, or file specified on the @MODE statement is not magnetic tape.
5	Data bank overflow attempting to acquire file-related buffer space. Down pack environment full. (Either the local or shared pack status table is full.)
6	That portion of the file name used as the internal name for I/O packets is not unique.
7	X option specified; file already in exclusive use.
8*,**	Incorrect read key for cataloged file.
9*,**	Incorrect write key for cataloged file.
10*	Write key that exists in the master file directory is not specified in the @ASG statement (file assigned in the read-only mode).
11	Read key that exists in the master file directory is not specified in the @ASG statement (file assigned in the write-only mode).
12*	Read key specified in the @ASG statement; none exists in the master file directory.
13*	Write key specified in the @ASG statement; none exists in the master file directory.
14*	An A option was specified in the @ASG statement, and the file name cannot be found in the master file directory.

Table C-2. Facility Inventory Status Bits

Bit Set	Description
15*	Invalid or duplicate reel number or <i>pack-id</i> specified on an @ASG statement for a cataloged tape or fixed or removable pack file; or duplicate reel numbers or pack-ids specified on an @ASG statement for a temporary tape or removable pack file. The requested tape reel is already assigned by this run. Assignments on <i>pack-id</i> inhibited.
16*	Mass storage file has been rolled out (only if the Z option is used; otherwise, the run is held until the file is rolled in).
17*	Request on wait status for facilities. For a tape file, this usually means a tape unit is not currently available. For a disk file, this usually is caused by an exclusive use conflict with another concurrent run (only if the Z option is used; otherwise, the run is held until the file is rolled in).
18*	Option conflict on facilities statement: the D and K options were specified; or C or U (or P, R, or W in combination with C or U) was specified for a file that already exists in the directory; or C was specified on an @CAT statement; or an illegal option was specified on an @USE statement (I is the only valid option); or conflicting density options were specified on a tape assignment.
19*	File assigned exclusively to another run.
20	Find was made on a cataloged file request, and the file was already assigned to another run.
21*	File to be decataloged when no run has the file assigned.
22*	Project-id or account number incorrect for cataloged private file.
23	Reserved.
24	Read-only file cataloged with an R option.
25	Write-only file cataloged with a W option.
26	Equipment requested is down.
27*	File specified in an @ASG statement is disabled because the links pertinent to its master file directory items have been destroyed.
28	File specified on the @ASG statement has been disabled because the file was assigned during a system failure.
29*	File specified on the @ASG statement has been disabled because the file has been rolled out and the backup copy is unrecoverable, unless an @ENABLE statement, followed by an @ASG,A statement, is used to retry the loading operation.
30*	Cataloging of the requested F-cycle forces deletion of a currently assigned F-cycle. F-cycle generation is inhibited due to existence of +1 F-cycle. F-cycle requested is not within the currently acceptable range.

Table C-2. Facility Inventory Status Bits

Bit Set	Description
31*	The quota limits do not allow this request. A, V, or G option was used and is not permitted for this account.
32	Security violation.
33-35	Reserved for future use.

* Request was rejected. For dynamic requests through a CSF\$ request, bit 0 is set in the status word returned in register A0.

** If the statement was submitted by a CSF\$ request, the run is aborted and no status word is returned in register A0.

Appendix D

Equipment Mnemonics

This appendix lists and describes the mass storage and tape assign mnemonics that you can specify on the type field of the @CAT and @ASG statements, and lists the selection sequences supplied by Unisys.

D.1. Mass Storage Assign Mnemonics

Table D-1 lists the valid mass storage assign mnemonics and shows the primary device associated with each mnemonic.

Table D-1. Mass Storage Assign Mnemonics

Assign Mnemonic	Primary Device
D	Word-addressable mass storage, independent of equipment type
F	Sector-addressable mass storage, independent of equipment type
D2 or F2	Device assigned according to site configuration*
D3 or F3	Device assigned according to site configuration*
D4 or F4	Device assigned according to site configuration*
D14 or F14	Device assigned according to site configuration*
D14C or F14C	Mass storage on M9614 disk on a cache subsystem
D14D or F14C	Mass storage on M9614 disk on a noncache subsystem
D17 or F17	Device assigned according to site configuration*
D24 or F24	Device assigned according to site configuration*
D25 or F25	Device assigned according to site configuration*
D30 or F30	Mass storage on 8430 disk
D33 or F33	Mass storage on 8433 disk
D34 or F34	Device assigned according to site configuration*
D40 or F40	Device assigned according to site configuration*
D50 or F50	Device assigned according to site configuration*

Table D-1. Mass Storage Assign Mnemonics

Assign Mnemonic	Primary Device
D53 or F53	Mass storage on 7053 storage configured as solid-state disk
D54 or F54	Device assigned according to site configuration*
D60 or F60	Device assigned according to site configuration*
D70C or F70C	Mass storage on 8470 disk on a cache subsystem
D70M or F70M	Mass storage on 8470 disk (movable head portion)
D80C or F80C	Mass storage on 8480 disk on a cache subsystem
D80M or F80M	Mass storage on 8480 disk (movable head portion)
D81 or F81	Mass storage on 8481 disk
D81C or F81C	Mass storage on 8481 disk on a cache subsystem
D94 or F94	Mass storage on 9494 disk
D94C or F94C	Mass storage on 9494 disk on a cache subsystem
DCS or FCS	Mass storage on SSD1000 (SSDISK) disk, 9494, M9720, or on a Zitel 1122/PS/1124 subsystem.
DMD or FMD	Mass storage on MDISK, 9494, M9720, 9760, or on a Zitel 22/PS/1124 subsystem.
DMEM or FMEM	2200/XPA main storage file. Mass storage allocated in memory (MEMFL).
DSCSI or FSCSI	Mass storage small computer system interface (SCSI) disk.
DSSD or FSSD	Mass storage on SSD1000 solid-state disk.
MSDFT	Mass storage on SSDISK default equipment type.
SRCITM	Mass storage on SSDISK fastest equipment type.
TAPITM	Mass storage on slowest equipment type.

* This mnemonic represents a device that is no longer supported. See Table D-2 for selection sequences supplied by Unisys.

Note: The selection sequences in Table D-2 pertain only to fixed mass storage.

Table D–2 shows the mass storage selection sequences supplied by Unisys.

Table D–2. Default Mass Storage Assign Mnemonic Selection Sequences

Type	Order of Satisfying Request
D or F	SSDISK, D7053, D9614, D9494, MDISK, D8481, D8480, D8470, SCDISK, D8430, D8433
D2 or F2	D8433, D8430, SCDISK, D8470, D8480, D8481, MDISK, D9494, D9614
D3 or F3	D8433, D8430, SCDISK, D8470, D8480, D8481, MDISK, D9494, D9614
D4 or F4	D8470F, D8450F, D9614, D9494, MDISK, D8481, D8480, D8470, SCDISK, D8430, D8433
D14 or F14	D8433, D8430, SCDISK, D8470, D8480, D8481, MDISK, D9494, D9614
D14C or F14C	D9614, D9494, MDISK, D8481, D8480, D8470, D8450, SCDISK, D8451, D8430, D8433
D14D or F14D	D9614, D9494, MDISK, D8481, D8480, D8470, SCDISK, D8430, D8433
D17 or F17	D9614, D9494, MDISK, D8481, D8480, D8470, SCDISK, D8430, D8433
D24 or F24	D8433, D8430, D8451, SCDISK, D8450, D8470, D8480, D8481, MDISK, D9494, D9614
D25 or F25	D8433, D8430, SCDISK, D8470, D8480, MDISK, D9494, D9614
D30 or F30	D8430, D8433, D8451, SCDISK, D8450, D8470, D8480, D8481, MDISK, D9494, D9614
D33 or F33	D8433, D8430, SCDISK, D8470, D8480, D8481, MDISK, D9494, D9614
D34 or F34	D8433, D8430, SCDISK, D8470, D8480, D8481, MDISK, D9494, D9614
D36 or F36	D8451, D8430, D8433, SCDISK, D8450, D8470, D8480, D8481, MDISK, D9494, D9614
D40 or F40	D8433, D8430, SCDISK, D8470, D8480, D8481, MDISK, D9494, D9614
D50 or F50	D9614, D9494, MDISK, D8481, D8480, D8470, SCDISK, D8430, D8433
D50C or F50C	D8450, SCDISK, D8451, D8430, D8433, D8470, D8480, D8481, MDISK, D9494, D9614
D50F or F50F	D8450F, D9614, D9494, MDISK, D8481, D8480, D8470, D8450, SCDISK, D8451, D8430, D8433
D50M or F50M	D8450, SCDISK, D8451, D8430, D8433, D8470, D8480, D8481, MDISK, D9494, D9614
D51 or F51	D8451, D8430, D8433, SCDISK, D8450, D8470, D8480, D8481, MDISK, D9494, D9614
D53 or F53	D7053, D9614, D9494, MDISK, D8481, D8480, D8470, SCDISK, D8430, D8433, SSDISK
D54 or F54	D9614, D9494, MDISK, D8481, D8480, D8470, SCDISK, D8430, D8433

Table D-2. Default Mass Storage Assign Mnemonic Selection Sequences

Type	Order of Satisfying Request
D59F or F59F	D9614, D9494, MDISK, D8481, D8480, D8470, D8450, SCDISK, D8451, D8430, D8433
D59M or F59M	SCDISK, D8451, D8430, D8433, D8450, D8470, D8480, D8481, MDISK, D9494, D9614
D60 or F60	D8433, D8430, SCDISK, D8470, D8480, D8481, MDISK, D9494, D9614
D63 or F63	D8433, D8430, D8451, SCDISK, D8450, D8470, D8480, D8481, MDISK, D9494, D9614
D70C or F70C	D8470, SCDISK, D8430, D8433, D8480, D8481, MDISK, D9494, D9614
D70F or F70F	D8470F, D8450F, D9614, D9494, MDISK, D8481, D8480, D8470, D8450, SCDISK, D8451, D8430, D8433
D70M or F70M	D8470, SCDISK, D8430, D8433, D8480, D8481, MDISK, D9494, D9614
D80C or F80C	D8480, D8470, D8450, SCDISK, D8451, D8430, D8433, D8481, MDISK, D9494, D9614
D80M or F80M	D8480, D8470, SCDISK, D8430, D8433, D8481, MDISK, D9494, D9614
D81 or F81	D8481, D8480, D8470, SCDISK, D8430, D8433, MDISK, D9494, D9614
D81C or F81C	D8481, D8480, D8470, SCDISK, D8430, D8433, MDISK, D9494, D9614
D94 or F94	D9494, MDISK, D8481, D8480, D8470, SCDISK, D8430, D8433, D9614
D94C or F94C	D9494, MDISK, D8481, D8480, D8470, SCDISK, D8430, D8433, D9614
DCS or FCS	SSDISK, D7053, D9614, D9494, MDISK, D8481, D8480, D8470, SCDISK, D8430, D8433
DMD or FMD	MDISK, D8481, D8480, D8470, SCDISK, D8430, D8433, D9494, D9614
DMEM or FMEM	MEMFL, SSDISK, D7053, D9614, D9494, MDISK, D8481, D8480, D8470, SCDISK, D8430, D8433
DSCSI or FCSI	SCDISK, D8430, D8433, D8470, D8480, D8481, MDISK, D9494, D9614
DSSD or FSSD	SSDISK, D7053, D9614, D9494, MDISK, D8481, D8480, D8470, SCDISK, D8430, D8433

Table D-2. Default Mass Storage Assign Mnemonic Selection Sequences

Type	Order of Satisfying Request
MSDFT	MEMFL, SSDISK, D7053, D9614, D9494, MDISK, D8481, D8480, D8470, SCDISK, D8430, D8433
SRCITM	SSDISK, D7053, D9614, D9494, MDISK, D8481, D8480, D8470, SCDISK, D8430, D8433
TAPITM	D8433, D8430, SCDISK, D8470, D8480, D8481, MDISK, D9494, D9614, D7053, SSDISK

D.2. Tape Assign Mnemonics

Table D-3 lists the valid tape assign mnemonics and shows what device is associated with each mnemonic.

Table D-3. Tape Assign Mnemonics

Mnemonic	Device
T	Tape unit, type independent (any available on system, regardless of format)
U9	9-track tape unit, density independent (800, 1600, or 6250 bpi)
U9H	9-track tape unit, 800 bpi density
U9V	9-track unit, 1600 bpi density
U9S	9-track unit, 6250 bpi density
26N	UNISERVO 26 nine-track tape unit
28N	UNISERVO 28 nine-track tape unit
32N	UNISERVO 32 nine-track tape unit
34N	UNISERVO 34 nine-track tape unit
36N	UNISERVO 36 nine-track tape unit
45N	2145 nine-track tape unit
22D	UNISERVO 22 dual-density 9-track tape unit
24D	UNISERVO 24 dual-density 9-track tape unit
30D	UNISERVO 30 dual-density 9-track tape unit
HIC	All half-inch cartridge tapes (any available 5073, 4780, or USR5073 18- or 36-track drive).
HICL	All 18-track half-inch cartridge tapes (any available 5073, 4780, or USR5073 18-track drive).
U47	4780 18-track stand alone or ACS units

Table D-3. Tape Assign Mnemonics

Mnemonic	Device
U47L	4780 18-track ACS units
LCART	4780 18-track ACS units
NLCART	4780 18-track stand alone
U47NL	4780 18-track stand alone
HICM	All 36-track half-inch cartridge tapes (any available 4780, 5136, or 5236 36-track drive).
U47M	4780 36-track stand alone
U47LM	4780 36-track ACS units
HIC40	UNISERVO 40 18-track tape cartridge unit
HIC51	SCSI 36-track tape cartridge unit (OST4890, OST5136)
HIC52	SCSI or SBCON 36-track tape cartridge unit (5236)
HICSL	Half-inch cartridge tape (SCSI channel low density)
HICSL	SCTAPE
DLT70	SCSI DLT U7000 cartridge tape
HIS98	Half-inch serpentine tape (CTS9840)
HIS98C	Half-inch serpentine tape (U9840C)
HIS98D	Half-inch serpentine tape (U9840D)
HIS99B	Half-inch serpentine tape (CTS9940B)
T10KA	Half-inch serpentine tape (T10000A)
VT	VTH virtual tape unit
VTH	VTH virtual tape unit
DVDTP	DVD
LTO	All LTO cartridge tape devices
LTO3	Generation 3 Linear Tape Open cartridge tape
LTO4	Generation 4 Linear Tape Open cartridge tape

Table D-4 shows the tape selection sequences supplied by Unisys. Assign mnemonics may result in free standing (FS) drives, Automated Cartridge System (ACS) drives, or Cartridge Tape Library (CTL) drives.

Table D-4. Default Tape Assign Mnemonic Selection Sequences

Type	Order of Satisfying Request
T	U9840C, U9840, U9940B;U5236, U5136, U47M, U47LM;CT0899, SCTAPE, U47, U47L, U7000, ULTO4, ULTO3; U36N, U34N, U32N, U28N, U26N;U45N, U30D,U24D, U22D
U9	36N, 34N, 32N, 28N, 26N, 45N, 30D, 24D, 22D, 18N
U9S	36N, 34N, 32N, 28N, 26N, 45N
U9V	36N, 34N, 32N, 28N, 26N, 45N, 30D, 24D, 22D,
U9H	30D, 24D, 22D
U47LM	U47LM (4780 36-track ACS unit)
U47M	U47M (4780 36-track stand-alone unit)
U47NL	U47 (4780 18-track stand-alone unit)
HIC	CT0899, SCTAPE, U47, U47L, U47M, and U47LM
HIC40	HIC40, SCTAPE
HICM	U47M, U47LM, all 36-track tapes, (any available 4780 36-track drive)
LCART	U47L (4780 18-track ACS unit)
HIC51	U5136
HIC52	U5236
DLT70	U7000
LTO	ULTO4, ULTO3
LTO3	ULTO3
LTO4	ULTO4
HIS98	U9840
HIS98C	Half-inch serpentine tape (U9840C)
HIS98D	Half-inch serpentine tape (U9840D)
NLCART	U47 (4780 18-track stand-alone unit)
ACS Name	ACS (CTL 18- or 36-track ACS unit)
HIS99B	U9940B
T10KA	Half-inch Serpentine tape (T10000A)
VT	VTH virtual tape
VTH	VTH virtual tape
DVDTP	DVD

Compatibility Issues with 18- and 36-Track Tapes

Here are 36-track tape considerations:

- A cartridge tape written on a 36-track cartridge drive can be read only on a 36-track capable cartridge drive.
- A cartridge tape written on an 18-track cartridge drive can be read on a 36-track cartridge drive.
- A labeled cartridge tape written in 36-track format will appear as an unlabeled cartridge tape when mounted on an 18-track capable cartridge drive. If this happens, the Exec displays the console message:

```
IS TAPE LABELED? Y,N
```

If the operator answers N to this message, the tape can be overwritten without tape labeling protection.

- In some cases, an 18- or 36-track tape cannot be appended. It can only be written from loadpoint.
- *E-carts* can be used only on 36-track cartridge tapes. E-carts are the size of standard half-inch cartridge tapes, but the actual tape length is thinner than one 1100ths of an inch.

Caution

When you use T and HIC general assign mnemonics, be sure that you determine the exact device type that was allocated to you so that you can later use the required mnemonic to read the tape. For example, if you begin with "T" and are allocated a 36-track device, you must use HICM (or another half-inch 36-track cartridge mnemonic) to reassign and read the tape.

- A cartridge tape written on a 9840C drive cannot be read on either a 9840A or 9840B drive.
- A cartridge tape written on a 9840D drive cannot be read on a 9840A, 9840B, or 9840C drive.
- In some cases, 9840 media cannot be appended because it cannot contain mixed density. It can only be written from loadpoint if the write is on a drive with different density than information previously written to the tape.

Appendix E

Job and Summary Accounting Report (Tail Sheet)

This appendix describes the Job Accounting and Summary Accounting Report for an Exec-produced tail sheet. The report is generated after termination of a run.

There is a configuration parameter, `site_formatted_tailsheet`, displayed in the "Run Scheduling" area that is entered from the initial menu displayed by COMUS. If the value of the parameter is TRUE, the Exec does not produce a tail sheet.

The following is a report sample and a description of fields:

```
RUNID: XXXXXX      ACCT: XXXXXXXXXXXX  PROJECT: XXXXXXXXXXXX  HOST: X
SITE:  XXXXXX      USERID: XXXXXXXXXXXX SYSTEM: XXXXXXXXXXXX
```

CONSOLE MESSAGE TEXT LOCATION

```
DEVICE USAGE:      REQUESTS      TRANSFERRED
      DISK:                0          0 WORDS
      TAPE:                0          0 WORDS
MAX TAPES ASG:      0
RESOURCE WAIT:      00:00:00
```

```
MEMORY USAGE:      0 PAGE-SECONDS
MAX MEMORY:        0 PAGES
```

```
TIME:    TOTAL:  00:00:00.000    ELAPSED:   00:00:00
      CPU:    00:00:00.000    I/O:    00:00:00.000
      CC/ER:  00:00:00.000    WAIT:   00:00:00.000
```

```
SUAS USED:          0.00    SUAS REMAINING:          0.00
IMAGES READ:        0    PAGES:        0
START:   yyyy-mm-dd hh:mm:ss zone    FIN:   yyyy-mm-dd hh:mm:ss zone
```

The following are contained in the report.

RUNID

The ID of the run for which accounting data was gathered.

ACCT

The account to which this run was assigned.

PROJECT

The project to which the run was assigned.

HOST

The host-id (such as A, B, C, D, ...) appears if MHFS is installed, UP, and not reemploying.

SITE

The site identification number.

USERID

The user-id used to execute the run.

SYSTEM

The system identification number.

Console Message Area

This area contains the console messages directly issued by the run, or console messages issued by the Exec in response to a service request issued by the run.

This area contains the following message types:

- Run termination message
- Tape/disk mount messages
- Operator message responses
- Log messages
- Nonrecoverable device error messages
- Any console message issued by the run

DEVICE USAGE

REQUESTS

The number of physical I/O operations to tape and disk issued on behalf of the run.

TRANSFERRED

The total number of words transferred to tape and disk issued on behalf of the run.

MAX TAPES ASG

The maximum number of tapes concurrently assigned by the run.

RESOURCE WAIT

The time consumed in the run waiting for resources to become available. The value does not include voluntary waiting time.

MEMORY USAGE

MEMORY USAGE PAGE-SECONDS

The number of memory pages in use multiplied by the number of seconds in use during the life of the run.

MAX MEMORY

The maximum number of memory pages in use during the life of the run.

TIME

Values in these fields are expressed in standard units of processing (SUP) except where noted. See the *Exec Administration Reference Manual (7831 0323)* for a detailed description of SUPs.

TOTAL

Total of CPU time, CC/ER time, and I/O time.

ELAPSED

The wall clock time taken by the run, that is FIN - START, expressed in hours, minutes, and seconds.

CPU

Actual CPU time consumed while the run was active.

CC/ER

The time based on standard SUP charges for control cards and Executive requests.

I/O

The time required for data transfer between the CPU and peripheral devices based on standard SUP charges.

WAIT

The wait time due to conditions other than system busy.

SUAS USED

The total number of standard units of accounting (SUA) consumed by the run. See the *Exec Administration Reference Manual* (7831 0323) for a detailed description of these units.

SUAS REMAINING

SUAs remaining in the account after the run.

IMAGES READ

The number of input images or images included in the batch.

PAGES

The number of output pages (print or terminal) at the default number of lines per page (66) or the configured number of lines per page.

START

The date, start time, and zone of the run.

FIN

The date, end time, and zone of the run.

Index

A

absolute device assignment
 description, 7-53
 devices other than mass storage or
 tape, 7-55
 mass storage, 7-53
 tape, 7-54
 absolute elements
 description, 3-25
 executing, 6-1, 6-3, 6-6, 7-270
 produced by the Collector, 6-6
 rules for locating, 6-1
 absolute F-cycle, 3-12, 3-13
 access control record (ACR)
 @ASG statement, 7-19, 7-47, 7-78
 attaching to files, 4-8
 creating, 4-8
 definition, 4-6
 effect of changes to, 4-9
 modifying, 4-9
 on @CHG, 7-83
 recovering, 4-8
 removing, 4-9
 specifying on @CAT statement, 7-65,
 7-78
 TIP files and, 4-9
 access list
 changing for a file, 4-10
 choosing for a file, 4-5
 giving to a file, 4-5
 account number
 supplying during demand sign-on, 5-9
 supplying on @RUN statement, 7-231,
 7-238
 ACR, *See* access control record
 ADD statement
 demand runs, 7-236
 description, 7-5
 ER QECL\$ and, 7-6
 example, 7-7
 format, 7-5
 options, 7-5
 adding partial canned runstreams and
 data, 7-5

alternate print, punch, and read files, 7-58
 application modes, 5-3
 application-private feature, @ASG
 statement, 7-20
 ASG statement
 absolute devices
 mass storage, 7-53
 other than mass storage or tape, 7-55
 tape, 7-54
 acr-name field, 7-19, 7-47
 assigning
 mass storage files, 7-9
 tape files, 7-24
 virtual tape files, 7-51
 buffered-write field, 7-41
 changing assignment fields for mass
 storage files, 7-22
 data-converter field, 7-38
 description
 absolute devices, 7-54
 changing assignment fields for mass
 storage files, 7-22
 tape files, 7-24
 virtual tape files, 7-51
 examples
 mass storage files, 7-20
 tape files, 7-39, 7-43, 7-48, 7-77
 expanded buffer field, 7-42
 expiration field, 7-45
 filename field, 7-23
 format
 absolute devices, 7-53, 7-54
 field, 7-37
 mass storage files, 7-10
 tape files, 7-51, 7-66, 7-152
 general description, 7-8
 granule field, 7-16
 log field, 7-36
 mass storage files, 7-9
 maximum field, 7-17, 7-23
 noise field, 7-36
 options
 absolute devices, 7-55
 mass storage files, 7-10, 7-22

- order of satisfying device requests, D-1, D-2
- pack-id field, 7-18
- placement field, 7-17, 7-23
- processor/tape field, 7-37
- reel-n field, 7-43, 7-78
- releasing assigned files, 7-155
- reserve field, 7-15, 7-23
- ring-indicator field, 7-47
- tape files, 7-24
- type field, 7-15, 7-23
- units field, 7-36
- virtual tape files, 7-51
- ASIS, using passwords, 7-185
- assign mnemonics, *See also* equipment mnemonics
 - @CAT statement, 7-71
 - mass storage, 7-15, 7-63
 - removeable disk, 7-15, 7-63
 - tape, 7-34, 7-71
- assigning absolute devices, 7-53
- assigning files and peripheral devices, *See also* @ASG statement
 - absolute devices, 7-53
 - access validation, 7-19
 - file sharing, 7-20
 - mass storage, 7-9, 7-22
 - overview, 7-8
 - tape, 7-24
- assignment fields, mass storage files, 7-15, 7-22, 7-23*
- authentication and session initiation
 - subsystem (ASIS), 5-11
- automatic @RUN generation, 5-4
- Automatic Cartridge System (ACS), 7-34

B

- basic file format, 3-28
- Basic keyin, console modes, 7-98
- basic mode programs, *See* traditional programming environment
- basic sign-on mode
 - description, 5-4
 - examples, 7-235, 7-239
- batch output, 3-21
- batch runs
 - description, 7-225
 - run priority in, 7-222, 7-227, 7-236, 7-246
- block numbering
 - @ASG statement, 7-38, 7-173

- @MODE statement, 7-172
- block sequence number, 3-23
- bound object module (BOM), 3-25, 6-8
- branching from a runstream (@JUMP), 7-166
- breakpoint output, 3-22
- breakpointing
 - alternate symbiont files, 7-58
 - primary print files, 7-55
- BRKPT statement
 - alternate symbiont files, 7-58
 - example, 7-57
 - formats, 7-55, 7-56, 7-58
 - primary print files, 7-55
 - use with batch runs, 3-21
 - use with demand runs, 3-21
 - using with @SYM statement, 7-59
- buffered-write*
 - @ASG statement, 7-41
 - @CAT statement, 7-76
 - @MODE statement, 7-174
- buffers, queuing input, 7-131, 7-165

C

- calling compilers
 - traditional programming environment, 6-10
 - Universal Compiling System, 6-13
- canned runs
 - adding partial, 7-5
 - description, 7-225
 - starting full, 7-245
- card files
 - creating, 7-150, 7-152
 - terminating, 7-146
- card reader
 - creating a mass storage card file, 7-150
 - creating a tape file from card images, 7-152
 - terminating card files, 7-146
- cards, on @RUN statement, 7-234
- cartridge tape library*
 - @ASG examples, 7-50
 - CTL-pool field, 7-48, 7-54, 7-78, 7-95
- CAT statement
 - acr-name field, 7-65, 7-78
 - buffered-write field, 7-76
 - description, 7-60, 7-66
 - examples
 - mass storage files, 7-65
 - tape files, 7-75, 7-78

- format field*, 7-73
- format, mass storage files, 7-60
- granule field, 7-64
- maximum field, 7-64
- noise field*, 7-72
- options
 - mass storage files, 7-60
 - tape files, 7-67
- pack-id field*, 7-65
- processor/tape field*, 7-72
- reel field, 7-78
- reserve field, 7-63
- tape assign mnemonics, 7-71
- type field*, 7-71
- cataloged files
 - deleting, 7-137
 - listing, 7-192
 - undeleting, 7-141
- cataloging files, *See also* CAT statement
 - F-cycles, 3-12
 - mass storage files, 7-60
 - tape files, 7-66
- changing
 - a file's protection, 4-10
 - DCT terminal characteristics, 7-136
 - element names, 7-88
 - insert point on display, 7-166
 - mode settings for tape file, 7-169
 - timeout period, 7-263
 - TTY terminal characteristics, 7-264
 - version names, 7-88
- checkpoint number, 7-224
- checkpoint/restart program, 7-93
- checksum number, 3-23
- CHG statement, 4-10, 4-11
- CKPT statement
 - description, 7-93
 - format, 7-93
 - options, 7-93
- clearance level
 - as a mandatory access control, 4-3
 - description, 5-6
 - signing on to the system and, 5-6
 - supplying on @PASSWD statement, 7-185
- CLIST mode, terminating, 7-147
- closing a session, 5-19
- CM statement, 7-96
- COL statement, label use, 2-2
- Collector, 6-5
- comments, control statements, 2-3
- communications console, sending
 - messages to, 7-178
- compartment sets
 - @PASSWD statement, 7-185
 - as a mandatory access control, 4-3
 - definition, 5-7
 - entering in batch, 7-185
 - entering in demand, 5-7
 - compiler call statement options, 6-10
 - compiler statement, examples, 6-12, 6-14
 - condition word
 - storing a value (@SETC), 7-242
 - testing (@TEST), 7-260
 - CONS statement
 - description, 7-96, 7-97
 - format, 7-96
 - Console Mode Keyins, 7-101
 - console mode statements
 - description, 7-96, 7-97
 - format, 7-96
 - Console modes, 7-98
 - consoles, sending messages to, 7-96, 7-178
 - CONT statement
 - definition, 7-108
 - format, 7-108
 - continuing a control statement, 2-4
 - control statements
 - comment, 2-3
 - continuing, 2-4
 - definition, 2-1
 - field, 2-2
 - label field in, 2-1
 - length, 2-4
 - mnemonic, 2-2
 - Conversational Time Sharing (CTS), 3-21
 - copying
 - program files to tape, 7-111
 - tape files to program files, 7-108
 - correcting symbolic elements
 - deleting lines, 6-20
 - editing statements, 6-23
 - inserting new lines, 6-21
 - replacing old lines with new, 6-20
 - symbolic input field, 6-18
 - correction images, *See* correcting symbolic elements
 - correction lines, *See* correcting symbolic elements
 - CQUE statement, 7-131
 - creating and using print files, @@CONS Statement, 7-96
 - cycles, file, *See* F-cycles

D

DAC, *See* discretionary access control

data compression

- @ASG statement, 7-38
- @MODE statement, 7-173

data conversion

- @ASG statement, 7-38
- @MODE statement, 7-172

data transfer format

- @ASG statement, 7-37
- @CAT statement, 7-73
- @MODE statement, 7-172

DCP, *See* distributed communications processor, 7-101

DCT statement, 7-136

deadline batch runs

- examples, 7-233

default printer, 3-20

deferred control statement execution,

- @ADD, 7-6

deleting

- elements, 7-139
- files, 7-138

demand runs

- application modes for, 5-3
- description, 7-236
- starting, 7-224
- transparent statements, 2-5

devices, *See also* ASG statement

- absolute, assigning, 7-53
- order of satisfying requests, 7-63, D-1, D-2
- releasing from your run, 7-155

DIAG\$ file, 3-20

diagnostic messages, *See* error message

directory, specifying implied or default, 7-215

directory-id

- @QUAL statement, 7-216
- in display of master file directory, 7-202
- naming files, 3-8
- shared files, 3-8

disabled files, 7-144

discretionary access control (DAC)

- access list, 4-5
- read-only/write-only mechanism, 4-9

disk packs, deleting files on, 7-138

Display and Keyin, 7-99

DVD tape, assigning, 7-52

E

EDIT 1100

- for canned runs, 7-225
- for correcting symbolic elements, 6-24

editing statements, 6-23

element name table

- description, 3-26

element subtypes, error messages, B-3

element table, program file, contents, 3-26

elements

- @CYCLE statement, 7-135
- changing element cycle, 7-135
- changing name, 7-88
- deleting, 7-139
- listing text, 7-192
- locating on tape, 7-154
- naming, 3-10
- removing deleted, 7-181
- subtypes, 3-25
- types, 3-25

END statement

- @@CQUE statement, 7-131
- @@FUL statement, 7-158
- description, 7-145

ENDF statement

- description, 7-146
- label use, 2-2

end-of-file marks, 3-2

- moving tape, 7-177
- with @FIND statement, 7-154
- writing, 7-167

end-of-record sentinel, moving tapes, 7-177

end-of-reel configuration, multireel

- tapes, 3-6

end-of-volume, multireel tape files, 3-5

ENDX statement

- description, 7-147

entry points

- adding, 3-27

entry-point table

- creating, 7-190
- description, 3-27
- destroying with relocatable elements, 7-108
- entry-point handling, 3-27
- recreating with @PREP statement, 7-108

EOF statement

- description, 7-147
- example, 7-147
- format, 7-147
- label and, 2-2

equipment mnemonics
 mass storage, D-1
 tape (table), D-5
 ER QECL\$, @ADD files and, 7-6
 ERA, *See* end-of-record sentinel
 erasing files, 7-148
 error messages
 element subtypes, B-3, B-4
 Prep, B-4
 quarter-inch tape subsystems, B-3
 with I/O error codes, B-1
 without I/O error codes, B-3
 ESC statement, 7-150
 examples
 @COPIN, 7-110
 @COPOUT, 7-113
 @COPY, 7-128
 @CYCLE, 7-136
 @DELETE, 7-140
 @PCH, 7-189
 @PRT, 7-211
 execution sign-on mode
 definition, 5-4
 Executive Control Language (ECL)
 console mode statements, 7-97
 control statement format, 2-1
 definition, 2-1
 for assigning files and peripheral
 devices, 7-8
 for getting on the system, 5-3
 for starting a demand run, 7-224
 processor call statements, 6-1
 expanded buffer, @ASG statement, 7-42
expiration period, files, 7-45
 extended mode
 calling UCS compilers, 6-13
 generating and executing programs, 6-7
external file names
 @QUAL statement, 7-216
 @USE statement, 7-264

F

F-cycles
 absolute, 3-13
 changing, 7-132
 creating, 3-12
 deleting, 3-13
 description, 3-12
 examples, 3-14
 file sharing, 3-7
 internal name, 3-14
 maximum, 3-13
 referencing, 3-12
 relative, 3-12
 temporary files, 3-14
 with file names, 3-9
 fields, control statements, 2-2
 file continuation indicator, *See* swap
 indicator
 file cycles, *See* F-cycles
 file formats
 basic, 3-28
 element, 3-24
 omnibus, 3-7
 program file, 3-24
 SDF, 3-22
 track-binary, 3-23
 file names
 on processor call statements, 6-1
 referencing a file by another name, 7-264
 selecting defaults, 7-217
 specifying an implied or default qualifier
 or directory, 7-215
 when using file sharing feature, 7-218
 FILE statement
 format, 7-150
 label use, 2-2
 files
 @CYCLE statement, 7-132
 assigning
 mass storage, 7-9
 tape, 7-24
 virtual tape, 7-51
 cataloging
 mass storage, 7-60
 tape, 7-66
 changing F-cycle, 7-132
 deleting, 7-138
 enabling disabled files, 7-144
 erasing, 7-148
 listing text of symbolic elements, 7-192
 master file directory information, 3-3
 measuring, 3-4
 naming, 3-7
 protecting, 4-2, 4-9
 referencing by another name, 7-264
 releasing
 from your run, 7-155
 initial reserve area, 7-149
 run and file concepts, 7-240
 shared, 3-7
 system created, 3-19
 temporary, 3-3

FREE statement
 assigning files, 7-8
 description, 7-155
 format, 7-155
 options, 7-155
FUL statement, 7-158
full canned runstreams
 description, 7-225
 See also canned runs., 7-225
 starting, 7-245
Full Keyin, 7-98
full-screen mode, 7-158
functional messages, *See* error messages
 FURPUR, B-1
FURPUR
 error messages, B-1
 relationship to ECL, 2-1

G

granule size
 @ASG statement, 7-16
 @CAT statement, 7-64

H

Half-inch Serpentine(HIS), 7-35
hardware-confidence console, sending
 messages to, 7-178
HDG statement
 description, 7-159
 examples, 7-163
 format, 7-159
 options, 7-159
 print control functions, 7-160
heading, printer output, 7-159
HOLD statement, 7-164
holding terminal messages and print
 output, 7-164

I

ignore escape, end-of-transmission
 characters, 7-264
initial reserve area, 7-149
input
 entering a full screen, 7-158
 passing unaltered input to a
 program, 7-150

 queuing in main storage buffer, 7-131
 queuing in mass storage buffer, 7-165
Input/Output rules, console mode, 7-100
INQ statement, 7-165
INS statement
 description, 7-166
 examples, 7-166
 format, 7-166
insert point, screen display, 7-166
inserting new lines in symbolic
 elements, 6-21
Interactive Processing Facility (IPF), 5-4
interlock, with @REWIND statement, 7-220
internal file names, 7-264
invalid element table entry count, 7-210

J

JUMP statement, 7-166

L

label statement, 2-2
labeled tapes, end-of-reel configuration, 3-6
labels
 branching from within a runstream, 7-166
 on control statements, 2-2
 security labels on printouts, 7-258
 used to pass control, 2-2
large element program file, 3-28
large program files, 3-4
LEPF, 3-28
Limited keyin, console modes, 7-98
Linear Tape Open (LTO), 7-24
LIST processor, 7-192
load point, 3-2
local printer
 starting, 7-191
 terminating, 7-181
LOG statement, 7-167

M

MAC, *See* mandatory access control
machine-generated passwords
 @@PASSWD and, 5-15
 changing, 5-14
 defined, 5-14
main storage buffer, queuing input, 7-131

mandatory access control (MAC)
 clearance levels, 4-3
 compartment sets, 4-3
 mandatory access validation rules, 4-4
 MAP processor, creating absolute
 elements, 3-25
 MAP statement, 6-6
 mass storage assign mnemonics, D-1
 mass storage buffer, queuing input, 7-165
 mass storage files
 absolute device assign, 7-53
 assigning (@ASG), 7-9
 breakpointing, 3-22
 cataloging (@CAT), 7-60
 changing assignment fields (@ASG), 7-22
 master file directory (MFD)
 description, 3-3
 directory-id, @QUAL statement, 7-216
 listing, 7-192
 obtaining display with @PRT, 7-202
 masterspace (@)
 control statements, 2-1
 transparent statements, 2-5
 messages
 ECL, A-1
 inserting in system log, 7-167
 terminal, holding, 7-164
 MFD, *See* master file directory
 mode settings for tape files, 7-169
 MODE statement
 example, 7-175
 options, 7-169
 modifying mandatory security
 attributes, 4-10
 moving tape files, 7-176
 MSG statement
 description, 7-178
 examples, 7-179
 format, 7-178
 multiframe volumes, 3-2
 multireel tape files
 FURPUR handling of, 3-5
 using @FIND on, 7-154
 with @MOVE statement, 7-177

N

naming elements
 description, 3-10
 examples, 3-11
 naming files

 description, 3-7
 examples, 3-11
 New Programming Environment
 compilers, 6-7
 static linking, 6-8
noise constant
 @ASG statement, 7-36
 @CAT statement, 7-72

O

object module
 bound object module, 6-8
 compiling programs, 6-7
 creating entry point tables, 7-190
 entry-point table, 3-26
 static linking, 6-8
 zero overhead object module (ZOOM), 6-8
 omnibus elements, definition, 3-25
 omnibus file format, 3-7
 opening a session, 5-3
 operator's console, sending messages
 to, 7-96
 options, control statements, 2-2
 output
 batch, 3-21
 breakpoint, 3-22
 continuing to send, 7-108
 print, holding, 7-164
 printer, providing a heading, 7-159
 sent to printer connected to
 terminal, 7-190
 skipping lines, 7-244
 stopping and placing file in queue, 7-221
 owned files, protecting
 discretionary access controls, 4-5
 mandatory access controls, 4-3

P

pack-id
 @ASG statement, 7-18, 7-54
 @CAT statement, 7-65
 @FILE statement, 7-151
 pages, batch run, 7-234
 partial canned runstreams
 adding, 7-5
 description, 7-225
 See also canned runs, 7-225
 PASSWD statement

- demand, 5-16, 7-186
- examples, 5-16, 7-186
- format, 5-16, 7-185, 7-186
- restrictions on use of, 5-15
- passwords
 - changing, 5-5, 5-14, 5-16, 7-186
 - configuration options, 5-14
 - demand sign-on and, 5-5
 - guidelines for choosing, 5-15
 - initial, 5-5
 - machine-generated, 5-5, 5-14
 - response to invalid, 5-6
 - security levels and, 5-14
 - supplying in remote batch runs, 7-184
 - user specified, 5-14, 5-15
- periods, in naming files, 3-7
- peripheral devices, releasing from your run, 7-155
- permanent files
 - definition, 3-3
 - run and file concepts, 7-240
- PMOD statement, 7-190
- position granularity, 7-183
- position, definition, 3-4
- PostMortem Dump (PMD) processor
 - @RUN options for postmortem dumps, 7-230, 7-237
 - DIAG\$ file dump, 3-20
- pound sign (#), as directory-id, 3-8
- print control functions, 7-160
- print files
 - alternate, breakpointing, 7-58
 - description, 3-20
 - primary, breakpointing, 7-55
 - queued to your user-id, receiving, 7-241
 - queuing, 7-255
- print output, holding, 7-164
- PRINT\$ file
 - batch runs, 7-225
 - description, 3-20
 - diverting output to, 7-55
- printer
 - default, 3-20
 - local, starting (@@PRNT), 7-191
 - local, terminating (@@NOPR), 7-181
 - See also* print files, 7-191
- printout security labels, 7-258
- private files
 - creating, 7-11, 7-27, 7-61, 7-70
 - owned, 4-5
 - unowned, 4-3
- private mode*, 7-81, 7-145
- PRNT statement, 7-191
- procedure name table
 - adding entries, 7-124
 - description, 3-26
- procedure tables, program file, contents, 3-26
- processor call statements
 - definition, 6-1
 - description, 6-1
 - fields*, 6-2
 - format, 6-1
 - options, 6-1, 6-2, 6-4
- processor control statement, unformatted, 6-3
- processor/tape mnemonics*
 - @ASG statement, 7-37
 - @CAT statement, 7-72
- processors
 - calling, 6-1
 - definition, 6-1
 - halting, 7-267
 - indicating input to, 7-147
 - language, 6-10, 6-13
 - using correction images, 6-18
- program control table (PCT)
 - assigning files, 7-9
 - run and file concepts, 7-240
- program file
 - copying from tape, 7-108
 - copying to tape, 7-111
 - description, 3-24
 - element table, 3-26
 - element types, 3-25
 - elements in, 3-25
 - entry point table, 3-27
 - format, 3-24
 - listing table of contents, 7-192
 - naming, 3-7
 - naming elements, 3-10
 - object module entry point table, 3-26
 - omnibus elements, 3-25
 - procedure tables, 3-26
 - relocatable elements, 3-25
 - symbolic elements, 3-25
 - table of contents, 3-26
- programs
 - writing basic mode programs, 6-5
 - writing extended mode programs, 6-7
- project-id
 - supplying in demand sign-on, 5-10
 - supplying on the @RUN statement, 7-224, 7-232
- protecting owned files, 4-3
- protecting unowned files, 4-2

PRT statement, 3-4, 3-26, 7-265
public mode, 7-81, 7-145

Q

QUAL statement
 description, 7-215
 examples, 7-217, 7-219
 file sharing, 7-218
 format, 7-216
 initial values, 7-217
 options, 7-216
 selection of filename default, 7-217
 specifying an implied or default qualifier, 7-216
 systems with file-sharing feature, 7-218
 qualifier, in naming files, 3-9
 quarter-inch cartridge tapes
 error messages, B-3
 multireel files, 3-5
 using @MOVE statement, 7-177
 queuing input in main storage buffer, 7-131
 queuing input in mass storage buffer, 7-165
 queuing print files
 See print files, 7-255

R

read key
 protecting unowned files, 4-2
 READ\$ file
 batch runs, 7-225
 description, 3-19
 read/write keys, 4-2
 read-only files
 assigning, 7-13, 7-26, 7-29
 creating, 7-11, 7-61, 7-70
 owned, 4-9
 unowned, 4-3
 reel identifier
 @ASG statement, 7-43
 @CAT statement, 7-78
 reels, releasing from your run, 7-155
 relative F-cycle, 3-12
 relocatable elements
 adding to program file, 7-108
 compiling programs, 6-5
 copying, 7-124
 creating entry point table, 7-190
 definition, 3-25
 description, 3-25
 relocatable output, 6-12
 remote batch runs
 definition, 7-226
 examples, 7-235
 passwords in, 7-184
 starting, 7-229, 7-237
 remote response console, 7-99
 removable disk file assignments, 7-15, 7-63
 removable packs, deleting files from, 7-138
 requeuing an output file, 7-221
 Response and keyin, 7-99
 restarting runs, 7-222
 RLD statement
 description, 7-220
 examples, 7-220
 format, 7-220
 RLU statement
 description, 7-221
 examples, 7-221
 format, 7-221
 RQUE statement, 7-221
 RSTRT statement
 description, 7-222
 format, 7-222
 options for, 7-222
 run sign-on mode, 5-4
 RUN statement
 acct-id field, 7-231, 7-238
 basic sign-on mode, 5-4
 bypass automatic statement, 5-4
 cards field, 7-234
 deadline field, 7-233
 description, 7-224
 example, 7-235, 7-239
 execution sign-on mode, 5-4
 formats, 7-226, 7-236
 options, 7-229, 7-237
 pages field, 7-234
 priority field, 7-222, 7-227, 7-236, 7-246
 project-id field, 7-232, 7-239
 run sign-on mode, 5-4
 run-id field, 7-231, 7-238
 run-time field, 7-232
 starting a demand run, 7-224
 start-time field, 7-235
 user-id field, 7-239
 run termination, identifying the end of a
 run, 7-153
 run-id
 supplying in batch run, 7-231
 supplying in demand run, 7-238
 runs

- branching from, 7-166
- canned, 7-225
- definition, 7-224
- full canned, starting, 7-245
- partial canned adding, 7-5
- remote batch, 7-101
- restarting, 7-222
- run and file concepts, 7-240
- saving complete or partial copy, 7-93
- storing a value in a condition word, 7-242
- testing a condition word to control execution, 7-260

runstreams, *See* runs

run-time, 7-232

S

- saving runs, 7-93
- screen, entering input, 7-158
- scrolling screen images
 - down, 7-220
 - up, 7-221
- S-cycles*
 - definition, 3-10
 - naming elements, 3-10
- sector, 3-7
- sector, definition, 3-4
- sector-addressable files
 - @ASG statement, 7-17
 - @CAT statement, 7-15, 7-62, 7-64
- sector-formatted mass storage
 - copying into, 7-108
 - copying out of, 7-111
 - track size, 3-23
- security level, giving to a file, 4-3
- security management product, 4-1
- security system
 - description, 4-2
 - passwords, 5-14
 - security levels, 4-1
- semiprivate access, 4-5
- SEND statement, 7-241
- sending messages
 - to another user, 7-262
 - to I/O, communications, or hardware-confidence, 7-178
 - to system console, 7-96, 7-178
 - to system log, 7-167
- SETC statement
 - description, 7-242
 - examples, 7-243
 - format, 7-242
 - options, 7-242
- SHARED
 - directory-id mnemonic, 3-8
 - with @PRT statement, 7-202
- shared files, 3-7
- signing off from system, 7-260
- signing on to the system
 - account number, 5-9
 - changing your password, 5-14, 5-16
 - clearance level, 5-6
 - compartment information, 5-7
 - format for, 5-5
 - illegal password, 5-6
 - illegal user-id and, 5-6
 - in demand, 5-5
 - passwords and, 5-5
 - project-id and, 5-10
 - starting a demand run, 7-224
 - TIP user, 5-13
- sign-on modes, 5-4
- sign-on security measures, 5-6
- SIMAN
 - attaching ACRs, 4-8, 4-9
 - console modes, 7-98
- site-id
 - disabling after invalid sign-on attempts, 5-6
 - receiving print files queued to, 7-241
 - sending a message to, 7-262
- SKIP statement
 - description, 7-244
 - examples, 7-244
 - format, 7-244
- skipping lines of output, 7-244
- slashes, control statements, 2-3
- SOLI statement
 - definition, 7-244
 - format, 7-244
- START statement
 - canned runs, 7-225
 - description, 7-245
 - examples, 7-254
 - formats, 7-246
 - options, 7-247
- start time, @RUN statement, 7-235
- starting full canned runstreams, 7-245
- statements
 - @@CM, 7-96
 - @@CONT, 7-108
 - @@CQUE, 7-131
 - @@DCT, 7-136
 - @@END, 7-145

- @@ESC, 7-150
- @@FUL, 7-158
- @@HOLD, 7-164
- @@INQ, 7-165
- @@INS, 7-166
- @@NOPR, 7-181
- @@PMOD, 7-190
- @@PRNT, 7-191
- @@RLD, 7-220
- @@RLU, 7-221
- @@RQUE, 7-221
- @@SEND, 7-241
- @@SKIP, 7-244
- @@SOLI, 7-244
- @@TERM, 7-260
- @@TM, 7-262
- @@TOUT, 5-13, 7-263
- @@TRNS, 7-264
- @@TTY, 7-264
- @@X, 7-267
- @ADD, 7-5
- @ASG/@@ASG
 - mass storage files, 7-9
 - tape files, 7-24
 - virtual tape files, 7-51
- @BRKPT/@@BRKPT, 7-55
- @CAT
 - mass storage files, 7-60
 - tape files, 7-66
- @CKPT, 7-93
- @CLOSE, 7-95
- @COPIN, 7-108
- @COPOUT, 7-111
- @CYCLE, 7-132
- @DELETE, 7-137
- @ENABLE, 7-144
- @ENDF, 7-146
- @ENDX, 7-147
- @EOF, 7-147
- @ERS, 7-148
- @FILE, 7-150
- @FIN, 7-153
- @FIND, 7-154
- @FREE/@@FREE, 7-155
- @HDG/@@HDG, 7-159
- @JUMP, 7-166
- @LOG/@@LOG, 7-167
- @MARK, 7-167
- @MODE/@@MODE, 7-169
- @MOVE, 7-176
- @MSG/@@MSG, 7-178
- @PACK, 7-181
- @PCH, 7-188
- @PREP, 7-190
- @PRT, 7-192
- @QUAL/@@QUAL, 7-215
- @REWIND, 7-220
- @RSTRT, 7-222
- @RUN, 7-224
- @SETC, 7-242
- @START/@@START, 7-245
- @SYM/@@SYM, 7-255
- @TEST, 7-260
- @USE/@@USE, 7-264
- @XQT, 7-270
 - undelete, 7-141
- static linking, 6-8
- STD
 - directory-id mnemonic, 3-8
 - with @PRT statement, 7-202
- storing a value in a condition word, 7-242
- subsystem definition elements,
 - description, 3-25
- subtypes
 - error messages, B-3, B-4
 - of elements, 3-25
- swap indicator
 - moving tape files, 7-177
 - multireel files, 3-5
- SYM statement
 - description, 7-255
 - example, 7-259
 - formats, 7-255
 - options*, 7-255
 - security labels, 7-258
 - using with @BRKPT statement, 7-59
- symbolic elements*
 - correcting, 6-18
 - definition, 3-25
 - description, 3-25
 - inserting new lines, 6-21
 - listing text, 7-192
 - S-cycles*, 3-10
- symbolic input, 6-18
- symbolic output*, 6-12
- system data format (SDF), file format, 3-22
- system log, inserting information
 - (@LOG), 7-167
- system-created files
 - description, 3-19
 - DIAG\$, 3-20
 - PRINT\$, 3-19
 - READ\$, 3-20
 - TPF\$, 3-19

T

- table of contents
 - element name table, 3-26
 - entry point table, 3-27
 - listing, 7-192
 - procedure name table, 3-26
 - program file
 - contents, 3-26
- tape cassette, printing output, 7-190
- tape files
 - absolute assign, 7-54
 - assigning, 7-24
 - breakpointing, 3-22
 - cataloging, 7-66
 - changing mode settings, 7-169
 - closing, 7-95
 - copying to program files, 7-108, 7-111
 - creating from card images, 7-152
 - description, 3-2
 - locating elements, 7-154
 - moving, 7-176
 - multireel, 3-5
 - rewinding, 7-220
 - writing EOF mark, 7-167
- TAVR feature, 7-35
- temporary files
 - definition, 3-3
 - F-cycle, 3-14
 - listing, 7-192
 - releasing, 7-155
 - run and file concepts, 7-240
 - TPF\$, 3-19
- Temporary Sector-Addressable, 7-200
- terminal display
 - changing
 - DCT terminal characteristics, 7-136
 - insert point on display, 7-166
 - timeout period, 7-263
 - TTY terminal characteristics, 7-264
 - continuing to send output, 7-108
 - holding messages and print output, 7-164
 - passing
 - unaltered input to a program, 7-150
 - unaltered output to a terminal, 7-150
 - request for a traditional sign-on, 7-244
 - scrolling screen images
 - down, 7-220
 - up, 7-221
- terminating
 - a local printer, 7-181
 - a run, 7-153
 - special modes, 7-145
 - your terminal, 7-260
- terminating CLIST mode, 7-147
- TEST statement
 - description, 7-260
 - examples, 7-261
 - format, 7-260
- testing a condition word to control execution, 7-260
- timeout period, changing, 7-263
- TIP
 - changes to ACR and, 4-9
 - session control, 5-13
 - signing on to, 5-13
- TOUT statement
 - description, 7-263
 - examples, 7-263
 - format, 7-263
- TPF\$ file
 - description, 3-19
 - executing absolute elements, 7-273
- track, 3-7
- track granularity, 7-183
- track, definition, 3-4
- track-binary format, 3-23
- traditional programming environment, *See also* programs
 - calling compilers, 6-5
 - generating and executing programs, 6-5
 - using Collector, 6-5
- transparent statements
 - definition, 2-5
 - demand symbiont control, 2-6
 - ECL statements, 2-5
 - entering, 2-5
 - guidelines, 2-1, 2-6
- TRNS statement, description, 7-264
- TTY statement, 7-264
- TTY terminal, changing characteristics, 7-264

U

- U option with @START, 7-250
- UCS compilers, 6-7
- undeleting files and elements, 7-141
- unlabeled tapes
 - end-of-reel configuration, 3-6
 - writing EOF mark, 7-167
- unowned files
 - ACRs, 4-3
 - private by account or project-id, 4-3

- protecting, 4-2
- read-only/write-only mechanism, 4-3
- USE statement
 - description, 7-264
 - examples, 7-266
 - format, 7-264
 - options, 7-264
- user-id
 - @RUN statement, 7-224, 7-232
 - application modes and, 5-3
 - description, 5-5
 - file ownership, 7-238
 - receiving a print file queued to user-id, 7-241
 - supplying on @RUN statement, 7-239
 - TIP access and, 5-13
- UV option, with @START, 7-251
- UZ option, with @START, 7-252

V

- V option, with @CHG, 7-125
- V option, with @COPY, 7-125
- V option, with @START, 7-248, 7-250
- version names
 - changing, 7-88
 - in naming elements, 3-10
- virtual tape files, assigning, 7-51
- V-option, with @CHG, 7-90

W

- word, 3-4
- word-addressable files
 - @ASG statement, 7-16
 - @CAT statement, 7-15, 7-62, 7-64
- write key
 - protecting unowned files, 4-2
- write-only files
 - assigning, 7-12, 7-26, 7-29
 - creating, 7-12, 7-62, 7-71
 - owned, 4-9
 - unowned, 4-3

X

- X option, with @COPY, 7-127
- X statement, 7-267
 - data field, 7-268

- examples, 7-269
- format*, 7-267
- XQT statement, 7-270
 - description, 7-270
 - examples, 7-273
 - format, 7-270

Y

- Y option, with @CHG, 7-90
- Y option, with @COPY, 7-114, 7-116, 7-125

Z

- Z option, with @START, 7-251
- zero overhead object module (ZOOM)
 - definition, 6-8
 - description, 3-25

A

- # (pound sign) as directory-id, 3-8
- \$\$CLOSE statement, 5-19
- \$\$OPEN statement, format, 5-3
- / (slash), in element-id, 3-10
- @@CONS Statement, 7-96
- @@PASSWD, visibility of password in, 7-187
- @@TERM operator command, 7-184
- @@X T operator command, 7-184
- @ASG, with directory-id, 3-8
- @CAT, with directory-id, 3-8
- @CHG
 - element and version names, 7-88
- @CLOSE, closing tape files, 7-95
- @COPIN
 - copy from tape to program files, 7-108
 - examples, 7-110
 - used with @FIND, 7-154
- @COPOUT
 - copy program files to tape, 7-111
 - description, 7-111
 - element file format, 3-24
 - examples, 7-113
- @COPY
 - description, 7-124
 - examples, 7-128
 - omnibus file format, 3-7
 - SDF file format, 3-22
 - track-binary format, 3-23

- @CYCLE
 - alter cycle retention limit, 7-132
 - changing element cycles, 7-135
 - examples, 7-136
- @DELETE
 - deleting files and elements, 7-137
 - description, 7-138
 - examples, 7-140
- @ENABLE
 - description, 7-145
 - enable disabled files, 7-144
- @ERS
 - description, 7-148
 - erasing files, 7-148
- @FIND
 - multireel file restriction, 3-6
 - positioning tape, 7-154
- @FREE, with directory-id, 3-8
- @MARK
 - description, 7-168
 - marking EOF on tape, 7-167
- @MODE, with directory-id, 3-8
- @MOVE
 - description, 7-177
 - multireel file restriction, 3-6
 - positioning tape, 7-176
 - positioning tape for copy, 7-124
- @PACK
 - description, 7-183
 - removing deleted elements, 7-181
 - with prep operation, 7-191
- @PASSWD, batch run and, 7-226
- @PCH
 - examples, 7-189
 - punch program file elements, 7-188
- @PREP
 - adding entry points, 3-27
 - creating entry points, 7-190
 - error messages, B-4
 - recreating entry-point table, 7-124
 - recreating relocatable entry-point table, 7-108
- @PRT
 - description, 7-202
 - examples, 7-211
 - obtaining listings, 7-192
- @QUAL, with directory-id, 3-8
- @REWIND
 - multireel files, 3-6
 - rewinding tape files, 7-220
- @USE, with directory-id, 3-8

